



Fishing boats moored at Carey's Bay, Otago Peninsula (*Raewyn Peart*)

10

Aquaculture





Contents

203	Introduction
205	Aquaculture methods
205	Mussels
206	Oysters
207	Salmon
208	Indigenous finfish
208	Hāpuku
208	Yellowtail kingfish
209	Eels
209	Potential environmental effects – general
210	Biosecurity
211	Marine mammals and seabirds
212	Seabed habitats
212	Landscape, natural character and amenity
212	Restoration and replenishment
213	Potential environmental effects – shellfish
213	Seabed habitat
213	Water column
214	Wild species
214	Abandonment of farm structures
214	Siltation and alteration of natural currents
215	Potential environmental effects – finfish
215	Seabed habitat
215	Water column
215	Local biodiversity
215	Toxic chemicals
216	Decision-making process
216	Allocation of space
217	Undue Adverse Effects test
218	New Zealand Coastal Policy Statement
218	Protecting outstanding natural features and landscapes
219	Protecting indigenous vegetation and fauna
219	Harmful aquatic organisms
219	Protecting water quality
220	Existing marine farming rights
220	Elements of best practice
223	Case studies
223	Guidance for aquaculture monitoring in the Waikato Region
224	Revive Our Gulf
225	Using DNA technology to monitor environmental change around farms
226	Further reading and references
227	Endnotes

Introduction

New Zealand's aquaculture sector is a significant primary industry, currently estimated to have revenue in excess of \$400 million, of which about 75 per cent is exported to 79 countries.¹ Māori own an estimated 50 per cent of the aquaculture industry. The annual aquaculture production and revenue is shown in Figure 10.1.

The most commonly farmed species in New Zealand are the indigenous green-lipped mussel, which is not farmed anywhere else in the world, the introduced Chinook salmon (also called king salmon) and the Pacific oyster. Mussels are predominantly farmed in the Marlborough Sounds, as well as at Golden Bay, the Coromandel Peninsula, Great Barrier Island, Canterbury and Stewart Island. King salmon are farmed in the Marlborough Sounds, Canterbury and Stewart Island. Pacific oysters are mainly farmed in the Far North, as well as at Coromandel, Auckland and Marlborough. New Zealand also has growing industries in pāua and seaweed. Other species, including kingfish, eels, geoduck clams and hāpuku have been farmed or trialled as possible species of the future.² Land-based aquaculture is also a rising sector. The location of aquaculture production in New Zealand is shown in Figure 10.2.

	Mussels	Salmon	Oysters
Harvested product (greenweight tonnage)	101,311	14,037	1804
Export revenue NZ\$ (millions)	218.1	63.4	16.6
Domestic revenue NZ\$ (millions) (estimated)	35.0	65.0	8.0

Figure 10.1 New Zealand aquaculture production and revenue for 2011 (*Aquaculture New Zealand*)³

The high quality of New Zealand's coastal waters, the abundance of plankton, and the prevalence of sheltered harbours and inlets means that the country has considerable aquaculture potential. The consumer demand for seafood (including shellfish) is rising internationally at a time when global wild stocks are under pressure, creating growing markets for farmed seafood products. In 2006, members of the industry working with the former Ministry for Economic Development released a strategy aimed at making aquaculture a billion-dollar industry in New Zealand by 2025.⁴

The Government's 'Aquaculture Strategy and Five-year Action Plan to Support Aquaculture', developed in 2012, sets out a 'whole-of-government pathway' to support growth in the aquaculture industry. This includes developing new farming space, improving the use of existing space, and increasing the value gained from production.⁵



Oysters such as those shown here on racks in the Whangaroa Harbour contributed over \$16 million in export revenue to New Zealand during 2011 (*Raewyn Peart*)

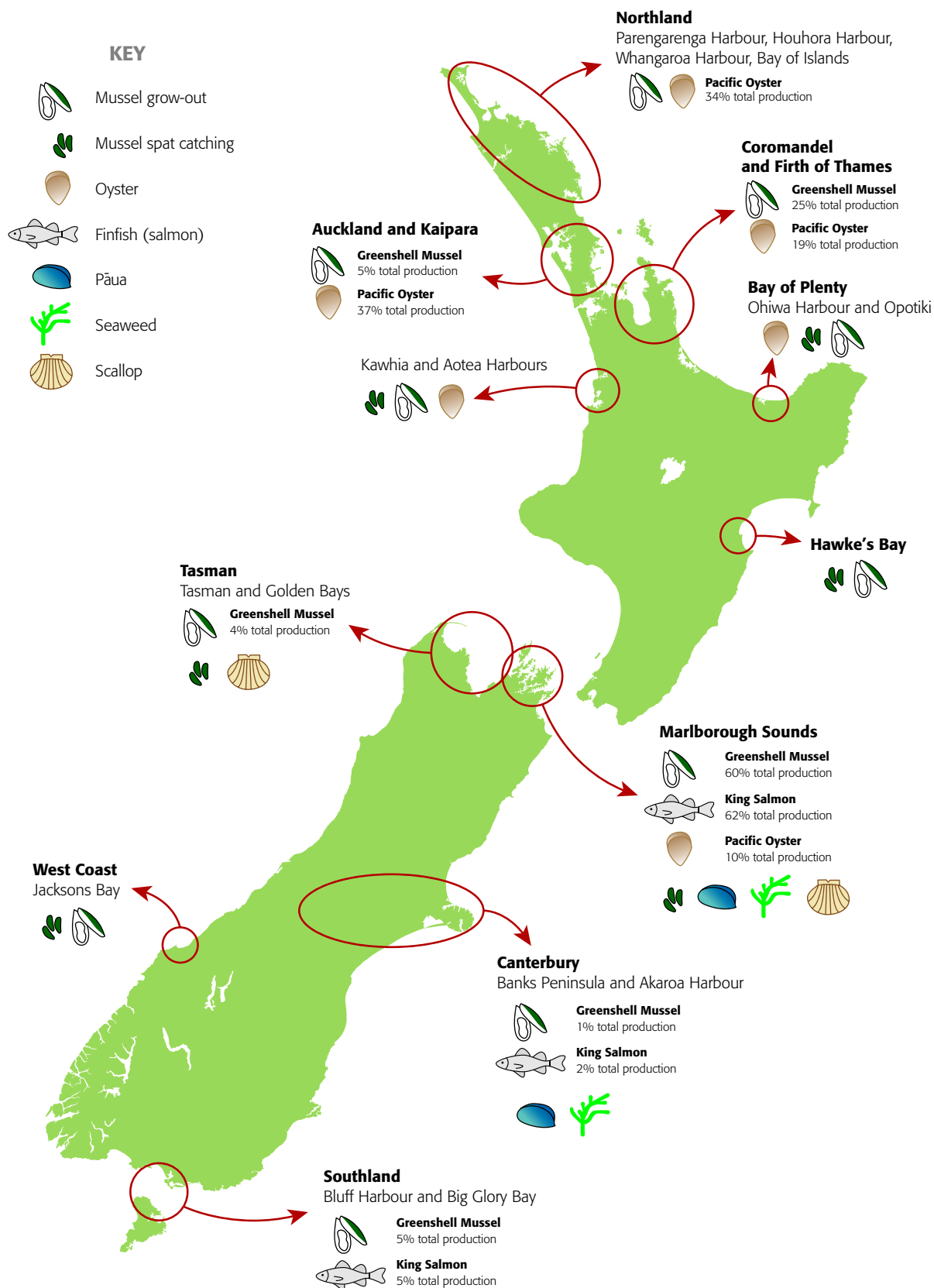


Figure 10.2 Location of aquaculture in New Zealand (*Aquaculture New Zealand*)

Aquaculture methods

Mussels

The main harvestable mussel species is the native New Zealand green-lipped mussel (*Perna canaliculus*), the shell of which can be up to 260 millimetres long, 110 millimetres wide and 90 millimetres deep.⁶ These mussels have been harvested for eating since humans first settled in the country.⁷ However, wild populations of the mussels were significantly reduced after heavy dredging during the 1960s. Commercial production then focused on farming the mussels.

A mussel farm consists of a series of rope 'backbones' suspended in, or on top of, the water column by a series of buoys. A continuous rope (the 'longline'), on which the mussels are grown, is suspended from the backbone line.⁸ The line is anchored to the seafloor at either end. A 3 hectare farm would typically have nine 110 metre lines. Each line is supported by 50 to 70 large plastic floats. Each float may support one tonne of mussels. One 110 metre line will generally support 3,500 to 4,000 metres of crop rope.

Most crop lines are seeded with mussel spat (juvenile mussels), with the majority sourced from Ninety Mile Beach in the Far North. Here, at irregular times throughout the year, large quantities of spat attached to seaweed are washed up on the beaches. This spat is collected and transported by air or truck to farmers in other parts of the country. The collection of this spat is managed under the QMS based on a management plan prepared in association with local iwi. Local spat catching also occurs on a smaller scale at some mussel farm sites in Golden and Tasman Bays, in Jackson Bay and in some bays in Marlborough.

Traditionally, virtually all mussel stock farmed in New Zealand was collected as spat from the wild. However, in recent years there has been an increasing focus on the opportunities offered by the use of hatcheries, where mussels can be selectively bred for specific qualities such as higher meat yield or faster growth. Hatchery spat are grown from a 'family' where the offspring are from two known parents. Each family may consist of several thousand siblings, grown together until they are large enough to transfer to the farm, with about 50 to 60 families per cohort needed to establish an effective breeding programme.⁹

A team of scientists from Shellfish Production and Technology New Zealand Limited and the Cawthron Institute have been rearing 80 families as part of a Primary Growth Partnership programme between the company, Sanford Limited and the Ministry for Primary Industries.¹⁰ Grow-out trials commenced in April 2013, with about 60,000 mussels from 50 families being sent to several sites to assess how they perform. Work has now begun on building a pilot-scale mussel hatchery, with the first commercial spat scheduled to be produced from 2015.¹¹

Once the spat are at the mussel farm, approximately 1,000 to 5,000 juveniles are wrapped onto each metre of rope using biodegradable cotton stocking mesh. This mesh gradually disintegrates, by which time the spat have attached themselves to the rope. The mussels are left to grow for three to six months, and then these nursery lines are lifted. The young spat are stripped from the ropes, reseeded onto a final production rope at approximately 150 to 200 per metre, and then returned to the sea to grow. Mussels take between 12 and 18 months to grow to a harvestable shell size of 90 to 100 millimetres.¹² The mussels are harvested by pulling the ropes up onto barges and removing the mussels.



Mussel farming, such as shown here in the Waiheke Channel, Hauraki Gulf, is the most common form of aquaculture in New Zealand (Raewyn Peart)

Offshore extensive farms hold some interest for the aquaculture industry. These can be several miles offshore (although still within the coastal marine area) and several thousand hectares in size. Due to the more exposed conditions offshore, the mussel lines must be spaced further apart to avoid entanglement. Such farms are still very much at the experimental stage, are untested commercially, and rely on substantial infrastructure investment before becoming viable.

Offshore mussel farming

In 2006, Eastern Sea Farms Limited, which is majority-owned by the Whakatōhea Māori Trust Board, was granted a licence for a 3,800 hectare marine farm to be established off the coast of Ōpōtiki.¹³ The farm is sited three nautical miles offshore where the water depth and currents are suitable for mussel growing. The site has been chosen to minimise visual, navigational and environmental effects. It is away from sensitive habitats and important fishing areas. The planned marine farm will be the largest aquaculture venture in New Zealand, with the potential to produce approximately 20,000 tonnes of mussels per annum by the year 2025.¹⁴ There is also potential for further aquaculture development off the Ōpōtiki coastline.

Oysters

Pacific oysters (*Crassostrea giga*) have been farmed in New Zealand since the 1960s. It is unclear as to when the Pacific oyster species arrived in New Zealand from Asia, but it is thought to have been in the late 1950s.¹⁵ They soon started displacing the New Zealand native rock oyster and so farmers adopted the new arrival as their principal commercial species.

Today Pacific oysters are the main farm-raised oyster in New Zealand. They are grown on wooden racks and baskets in sheltered and shallow intertidal bays around the northern North Island coast and in the Marlborough Sounds. The seed used to stock most farms is caught from the wild, predominantly from the Kaipara Harbour. Oyster larvae drifting in the water settle on bundles of wooden sticks which are placed in the water column. The sticks are then transported to marine farms, where the larvae are transferred to racks for on-growing the baby oysters. Sometimes, the oysters are transferred to mesh bags, hanging baskets or wire racks for growing. Using baskets suspended on wires can be an effective method of farming, allowing oyster farms to be located in more exposed locations because in rough weather the baskets swing and thereby resist the damage that would be caused to the traditional fixed racks. Pacific oysters grow to marketable size within 12 to 18 months.¹⁶

A growing proportion of oyster spat is obtained from hatcheries where it has been selectively bred. Cawthron's Pacific oyster breeding programme has seen growth rates more than 20 per cent higher than those normally achieved from a crop of wild spat.¹⁷ Hatchery-bred oysters have become much more important to the industry during the past couple of years due to the devastating



Oysters can be raised in baskets in shallow intertidal bays such as those shown here in Te Matuku Bay, Waiheke Island, Hauraki Gulf (Raewyn Peart)

impact of the Ostreid herpesvirus (OsHV-1). Aquaculture New Zealand and the New Zealand Oyster Industry Association, through the Ministry for Primary Industries' Sustainable Farming Fund, are undertaking a selective breeding programme which has been designed to assist the oyster industry overcome the virus. The programme is using selective breeding, and survival trials, to enable hatchery supply of reliably resistant juvenile oysters.¹⁸ The oysters will be on-grown to bigger sizes, before they are transferred to locations where the water temperatures and environmental conditions create a risk of small oysters succumbing to the virus.

The endemic flat oyster (*Tiostrea chiensis*) is referred to by a number of names, including Chilean oyster, dredge oyster and Bluff oyster. These oysters are found all over the coast, but particularly in extensive beds in the Foveaux Strait, Golden Bay and Tasman Bay, where they have been dredged since the late 1800s.¹⁹ They generally reach the harvestable size of 58 millimetres within four to six years. The Cawthron Institute is undertaking breeding programmes for the species.

Salmon

New Zealand is responsible for approximately 70 per cent of the global king salmon (*Oncorhynchus tshawytscha*) production. Sea pen farming of finfish has become widespread internationally, driven by a growing demand for fish and global pressure on wild stocks.

New Zealand has no native salmon and only king salmon, which is native to the cool waters of the northern Pacific rim, have been successfully farmed here. After their introduction, king salmon became established on the east coast of the South Island, and to a lesser extent also on the South Island's west coast. Commercial farming of salmon started in the 1970s, both in inland ponds and in coastal waters. Today, most commercially-sold salmon are grown in pens in the sea.²⁰

No imports of live salmon have been allowed into New Zealand for the last 50 years – instead eggs and milt are collected from captive broodstock. Fertilised eggs are incubated in a freshwater hatchery, where newly hatched fry are reared for another six to 12 months. The fish are then transferred to the sea pens, where they remain until they reach harvestable size, in about two to three years.²¹

Whereas shellfish feed on naturally occurring plankton, farmed fish such as salmon are top-end carnivores. They are fed an artificial diet of food pellets. Historically, much of the protein and oil in the feed was sourced from wild fisheries. However, in recent years, land-based proteins have increasingly been used to reduce the wild fish component. For example, Skretting Australia now produces fish pellets where only 28 per cent of the protein is from wild fisheries and 60 per cent is from land-animal proteins, with the balance derived from vegetables. Of the wild fishery component, just under half is sourced from fish processing wastage, with the rest consisting of wild-caught Peruvian anchovy.²² Fish pellets are not currently manufactured in New Zealand because of the small size of the finfish aquaculture industry here.



Salmon are reared in sea pens such as shown here in Akaroa Harbour, Banks Peninsula (Raewyn Peart)

Indigenous finfish

The farming of indigenous finfish in New Zealand, such as hāpuku (*Polyprion oxygeneios*) and yellowtail kingfish (*Seriola lalandi*), has yet to occur on a commercial scale. Interest in the development of finfish aquaculture in the Firth of Thames and southern Hauraki Gulf led the Government to legislate a zone of approximately 390 hectares for the culture of yellowtail kingfish and hāpuku in the Waikato regional coastal plan.²³ A resource consent is still required before such farming can proceed.

Currently there is limited information available about the specific finfish farming methods which might be adopted. However, assessments of the anticipated impacts of such farms have assumed that the two finfish species would be grown in floating sea-pens, with juvenile stock sourced from land-based seawater hatcheries. Plastic circular pens (typically 28 to 51 metres in diameter) are anticipated because the wave energy is expected to be too great for the square metal pens used at the South Island salmon farms.²⁴

Hāpuku

Hāpuku is a member of the *Polprionidae* or wreckfish, which is comprised of nine species of large fish that live and feed on or near the bottom of the sea. Hāpuku is large and thick-bodied, reaches 150 centimetres in length, and can vary in colour from silver and blue, to a pink, brown and silver combination. Most often found in deep waters over a depth range of 50 to 854 metres, this species is found throughout southern waters, and is common in the waters around New Zealand and Australia. Where stocks are still abundant (such as around the Chatham Islands) the species is known to frequent shallow waters. Unlike kingfish, hāpuku is a winter-spawning species producing eggs in late July to early September.²⁵ Scientists at NIWA have been undertaking detailed research on growing hāpuku in controlled environments and these early trials indicate that the species could potentially be farmed in New Zealand.

Yellowtail kingfish

Yellowtail kingfish are widely distributed throughout the warm, temperate waters of the Southern Hemisphere. New Zealand kingfish, also known as kingi or yellowtail, are found from the Kermadec Islands to Banks Peninsula during the summer months. In the wild they can reach 1.7 metres in length and weigh 56 kilogrammes. Wild populations have been significantly reduced due to fishing pressure and commercial catches of New Zealand kingfish are small, seasonal and unpredictable. Closely-related species are commonly farmed overseas, in countries such as Japan and Australia.

Research undertaken by NIWA indicates that kingfish could be an attractive aquaculture species for a number of reasons including:²⁶

- It is highly valuable (earning up to \$17 per kilogramme on the European market)
- It has a rapid growth rate (reaching marketable size of three kilogrammes in 12–15 months)
- It is amenable to aquaculture conditions
- It has excellent flesh quality for a range of product options (such as whole fillets, sushi and the highly valued sashimi)
- It has significant domestic and international market opportunities

Around 500,000 kingfish fingerlings per year could be produced from NIWA's facility in Northland, which would meet the needs of the early stages of an industry in New Zealand. Research is now focused on selective breeding and the on-growing technology, including the setting up and running of sea pens to grow the hatchery-produced fingerlings to marketable size.²⁷



Kingfish and hāpuku are being bred and raised at the NIWA hatchery in Bream Bay, Whangarei (Raewyn Peart)

Eels

Freshwater eels are found in many river systems and lakes throughout New Zealand. Worldwide eel aquaculture is estimated to be worth over US\$1 billion, accounting for 65 per cent of total eel production. Asia and Europe are the largest eel markets, but a decline in glass eel (returning juvenile stage) stocks in both regions has opened up opportunities for exploitation of species in other areas.²⁸

Eel farming involves catching juvenile eels when they enter freshwater from offshore spawning areas and on-growing them to a marketable size. Glass eels can be grown to maturity in two years in captivity. Eel aquaculture requires knowledge of upstream glass eel migration patterns, together with effective collection methods and appropriate water quality and temperature conditions for culture.²⁹

As there is also a commercial and customary eel fishery in New Zealand, there is some opposition to the farming of eels if it is to be at the expense of the wild fishery, the management of which is currently being reviewed by the Ministry for Primary Industries.

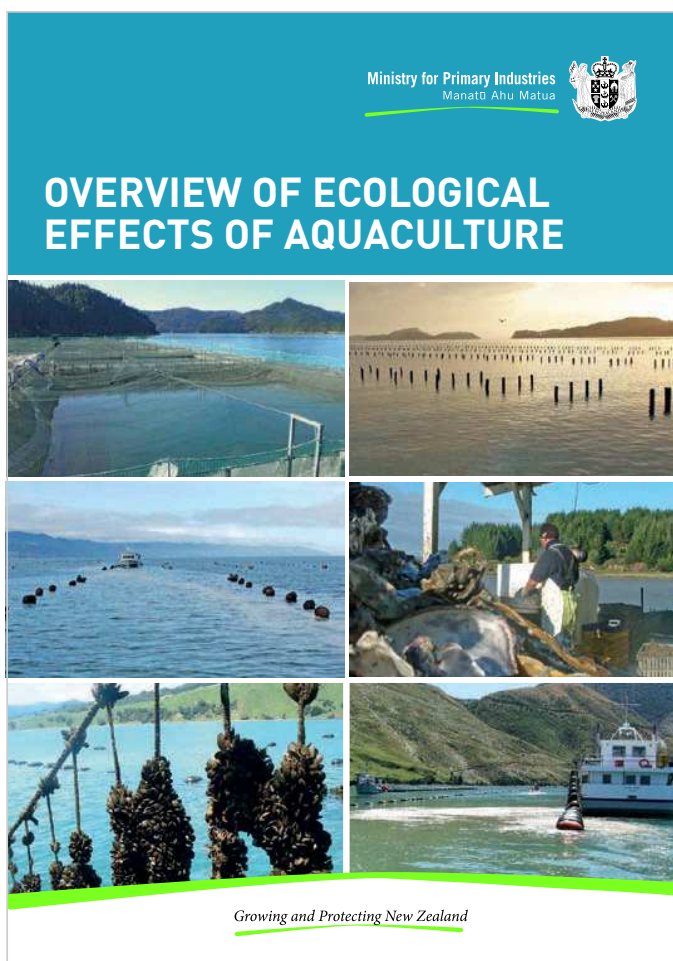
Potential environmental effects – general

There are a number of potential environmental effects that can result from aquaculture operations, both positive and negative, which are discussed below. Management options for mitigating the negative environmental impacts are described at the end of the chapter. The aquaculture industry has developed Environmental Codes of Practice specific to each species, which are designed to minimise potential effects on the environment. These are discussed further in the section on best practice.³⁰

In 2012 a risk assessment workshop run by the Ministry for Primary Industries produced a ranking of relative ecological risks from aquaculture activities. This involved an expert panel approach using an independent facilitator and internationally-recognised methods, with input from scientists, regional council and industry. The produced ranking of the potential risks included:

1. Biosecurity threats
2. Pelagic (water column) effects
3. Marine mammal interactions
4. Benthic (seabed) effects
5. Seabird interactions
6. Additive effects
7. Escapee effects
8. Wildfish interactions
9. Hydrodynamic alteration of flows

An 'Aquaculture Ecological Guidance Package'³¹ has been developed by the Ministry for Primary Industries in association with the Cawthron Institute, NIWA, Department of Conservation, regional councils, the aquaculture industry and others. This package provides 'information and advice on the ecological effects of marine-based aquaculture to assist in planning and managing aquaculture development'. It contains science-based information on the ecological effects of marine-based aquaculture to assist local authorities, the aquaculture industry and other stakeholders with their planning for and management of aquaculture activities.



This report, published by the Ministry for Primary Industries, provides useful information about the potential ecological effects of aquaculture

Biosecurity

Biosecurity pathways are human activities that, intentionally or unintentionally, may move a harmful organism from one place in New Zealand to another.³² Figure 10.3 below highlights some of the pathways related to marine farming.

	Transfer mechanism for both finfish and shellfish
Farm structures and activities	Provide a vector for marine pathogens and diseases Facilitate the movement of species attached to equipment around the coast, including: - Fouled sea pens or harvesting gear - Fouled hulls or contaminated bilge water Can create environmental conditions that facilitate the establishment of pest species
Wider impacts	Can spread disease to wild populations Can breed with wild fish, altering the wild stock. Can out-compete wild stock Can act as a reservoir from which marine pests or disease agents spread to the environment, potentially leading to irreversible regional-scale effects

Figure 10.3 Marine pest pathways related to aquaculture

The fixed structures found in marine farms can provide ideal habitats for some marine pest species, particularly biofouling species such as barnacles, tube worms and seaweed that grow on submerged structures and on the shellfish themselves. A marine farm provides a substrate on which these organisms can settle and multiply when there might not naturally have been such an opportunity, hence increasing the chance of establishment in an area.

Such biofouling can include invasive species such as the Japanese kelp (*Undaria pinnatifida*) which has been associated with mussel farms in the Hauraki Gulf. The nitrogen discharges produced by finfish farming may increase the growth of such unwanted species.³³

As well as potentially increasing biosecurity risks, biofouling is of direct concern to marine farmers, as it can lead to lower yields through poor growth of the shellfish. Removal of biofouling organisms – which is usually done by hand – can be a very expensive and labour-intensive job for marine farmers. Biofouling can also result in higher maintenance costs, as a result of equipment becoming damaged or unusable because of the growths. When biofouling organisms grow on finfish nets, they can prevent water flow through the nets and increase their weight, so that they can be pulled from their moorings. Nets must therefore be changed or cleaned regularly.

The transportation of spat, gear, stock and vessels around the country has the potential to spread pests and diseases. It is therefore important to ensure that all stock and spat comes from healthy sources and that all equipment is cleaned properly before transportation. International transfers of finfish feed and used aquaculture farm equipment are controlled by import health standards that outline stringent quarantine and control measures to minimise the risk of disease transfer.

Most marine farmers follow a code of practice in order to minimise the risk of spread of pests and diseases. The industry has also become involved in joint biosecurity efforts, including the long-term management plan for the invasive sea squirt *Styela clava* (clubbed tunicate). In addition, marine farmers play an important role in biosecurity monitoring, due to their frequent surveillance of marine stock and equipment.

Marine biosecurity is discussed further in Chapter 5.



It is important that aquaculture equipment is kept clean to reduce the risk of spread of invasive species. These salmon nets at Ruakākā Bay, Marlborough Sounds, are being cleaned by hand (Raewyn Peart)

Marine mammals and seabirds

The effects of marine farming on marine mammals and seabirds are not fully understood. There is concern that marine farms could adversely affect such species if they reduce or block access to food sources, limit the space available for hunting, or interfere with normal behaviour such as migratory routes. In addition, marine mammals can become entangled in marine farming gear. In addition, underwater noise generated by activities can attract or exclude marine mammals, with whales and some species of dolphins being more sensitive to this kind of disturbance.³⁴ Currently, the limited extent of marine farming in New Zealand means that only a few such effects have been observed here, but this may become an issue as the industry grows.

In other countries, where marine farming is much more widespread, more serious problems have been experienced.³⁵ High stocking densities for finfish, in particular, has been identified as a potential risk to marine mammals.

In New Zealand, there have been no reported cases of dolphins or seals becoming entangled in mussel farm lines. Between 1996 and 2003, one Bryde's whale was reported as a casualty of entanglement in mussel spat catching lines.³⁶ Entanglement for whales may become more of a problem if large offshore marine farms are established in areas which overlap with the migratory paths of humpback and southern right whales.³⁷

New Zealand fur seals are attracted to salmon farms. Most farms use predator exclusion nets to reduce the risk of entrapment. In approximately 25 years of sea pen salmon farming in New Zealand, there have been five reported cases of marine mammals becoming trapped in farm nets, two cases involving seals and the other three involving dusky and Hector's dolphins.³⁸ Improvements in net design, and refined net changing and other practices, can reduce the risk of entanglement.

In the case of mussel farms, small cetaceans (whales and dolphins) can enter farm areas and swim between lines, and have often been observed to do so.³⁹ However problems can occur when marine farms are located in areas which are important habitats for specific marine mammal species. For example, research has indicated that mussel farms located in Admiralty Bay in the Marlborough Sounds likely interfere with the co-operative hunting of dusky dolphins, with the dolphins observed to avoid the mussel farms. This is probably because the floats and lines physically obstruct the ability of the dolphins to jointly herd fish. In addition, the structures may interfere with the dolphins' echolocation capabilities and thus their ability to 'see' and navigate within their surroundings. These concerns resulted in the Environment Court turning down applications to expand marine farming in the bay.⁴⁰

The main negative impacts for seabirds from marine farms are entanglement, habitat exclusion, displacement from feeding areas, increased potential for pathogens and changes to the food web. The way that marine farms attract small fish means that the structures may enhance feeding opportunities and enable more efficient foraging.⁴¹

Compared to mussel farms, habitat exclusion resulting from fish farms is more absolute and care is needed to ensure that they are not located in areas which are important habitats for marine mammals and important roosting areas for seabirds.



Mussel farms may disrupt the co-operative feeding behaviour of dusky dolphins, shown here off Kaikōura (Raewyn Peart)

Seabed habitats

The effects on the benthic community from marine farms are well understood and can be summarised as:⁴²

- Enrichment of the organic content of the seabed in the immediate area beneath the farm
- Smothering of benthic organisms
- Debris changing the composition of the benthic community
- Biofouling drop-off
- Shading of the seabed which may impact on algal production

These potential impacts are described in more detail in the sections below addressing specific farmed species.

Landscape, natural character and amenity

Locations for siting marine farms are typically restricted by suitable environmental parameters. For example, shellfish filter feed, taking particulates of food (plankton) from the water column, and can concentrate toxins from the water in their flesh.⁴³ Thus for successful production of key New Zealand aquaculture species, unpolluted and sheltered waters are required, away from centres of high population density. This is to ensure that runoff from land-based activities does not affect the marine environment where the farm is located.

There are relatively few places which are appropriate sites for aquaculture, and these tend to be places which are also valued for other activities, and for their unspoilt landscape values and natural character.

Shellfish farms usually have elements that are visible from the shore. Mussel farms have lines of buoys holding up the mussel lines which are visible as horizontal structures on the water plane. Some lines are deliberately suspended below the surface to avoid swells. Although not permanent fixtures, barges working the farms may be perceived as having an impact on amenity values when stationary. In addition, navigational lighting marking the farms is typically visible from 2 nautical miles away at night time. In the case of oyster farms, rows of wooden racks can be visible in the intertidal zone during low tide.



Finfish farms can impact on landscape values because of infrastructure such as nets and sheds constructed above the water surface as shown here with the salmon farm in Ruakākā Bay, Marlborough Sounds (Raewyn Peart)

Finfish farms are usually visible as vertical structures on the surface of the water. They can be particularly noticeable as they typically use high nets to keep predators out. Similar to mussel farms, service structures and craft working on the farm will also be visible.

In terms of location, several key variables affect the ability of the marine environment to visually absorb marine farming activities. These are the visual scale of the affected coastal environment, the degree of visual interest in the view, and the elevation and distance from which the farm is likely to be viewed.⁴⁴

Where an area has outstanding natural character, or is an outstanding natural landscape, Policies 13 and 15 of the NZCPS require any adverse effects to be avoided. In practice, this means that consent should not be granted for marine farms which have adverse effects on the landscape values of these areas.⁴⁵ This issue is discussed further in the decision-making framework section below.

Restoration and enhancement

Aquaculture can play a role in fisheries enhancement. For example, the salmon industry provides smolt to recreational fisheries and marine farmers provide pāua to enhance recreational, customary and commercial pāua fisheries. A case study on the way the aquaculture industry is helping to replenish mussel beds is provided at the end of this chapter.

Potential environmental effects – shellfish

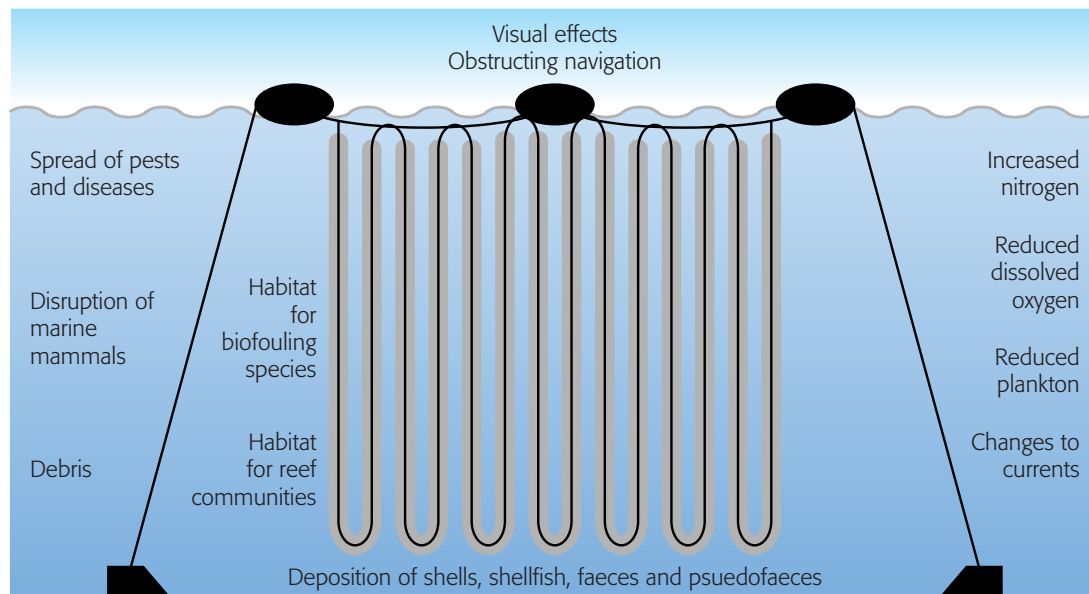


Figure 10.4 Summary of potential environmental effects of mussel farming

Seabed habitats

The principal ecological effect of mussel and oyster farming is localised changes to the seabed caused by shells, shellfish, faeces and pseudofaeces (particles which are not digested bound in mucous) dropping from the mussel lines and oyster racks. These items can accumulate on the seabed and alter its physical, chemical and biological nature. The waste products are nutrient-rich and thus increase microbial activity, which can lead to oxygen depletion within the sediment, affecting species that live there and reducing biodiversity.

Changes to the seabed are particularly pronounced where the seabed is sand or pebble, rather than mud. In these areas, benthic species are less able to cope with smothering by fine material from the farm. The extent of these effects will largely depend on the type of substrate under the farm, the flushing capacity of the site, and the depth between farm structure and substrate. Shellfish farms are less likely to be situated over or near to rocky reefs where such effects could be even more pronounced.

The creation of a nutrient-rich environment may give rise to future opportunities to farm other species (polyculture) underneath such farms. Species such as sea cucumbers have been the subject of trials. A community-based aquaculture research project between the Hongoeka Development Trust and NIWA focused on using the waste products from one species as a food input for another, thereby reducing waste and maximising energy flows throughout an integrated system. In this particular project, the polyculture was investigated to see whether the waste produced by a pāua farm could be recycled to produce other high-value foodstuffs relevant to Māori.⁴⁶

Water column

Farmed shellfish extract phytoplankton and organic particulates from the water column, and can thereby cause their depletion, particularly in sheltered bays. This depletion tends to be localised and its significance will depend on the density of the farms and the level of flushing in the area. Phytoplankton are the building blocks of the marine food web, and the impact of their depletion on the broader marine ecosystem is poorly understood.⁴⁷ Farmed shellfish release ammonia and organic particles into the water column which promote some additional phytoplankton and macro-algal growth. In addition, the shellfish eat fish eggs so could potentially have an impact on fish stocks if very large areas of shellfish farms are established.

The structures required for mussel farming create obstructions in the water column that can alter the direction and speed of currents in the area, which can in turn alter the rate at which plankton and nutrients are replenished and waste dispersed. The extent of these effects will depend on the particular characteristics of the area, the scale and density of the farm, and the orientation

of the lines relative to the predominant direction of the currents. The effects can be minimised through good design, with larger, denser and poorly orientated farms having greater effects. Research suggests that dispersed development of marine farms in New Zealand is unlikely to significantly impact on the hydrodynamics over a wider area.⁴⁸

Limits and trigger points for phytoplankton depletion

Work has been underway for a number of years looking at the pelagic impacts from marine farms in the Firth of Thames. An adaptive management framework was recommended in 2005 whereby Limits of Acceptable Change would be developed and agreed upon. The limits agreed to were that phytoplankton depletion shall not exceed 20 per cent over 10 per cent or more of the Firth of Thames and phytoplankton depletion shall not exceed 25 per cent over an area twice the size of the Aquaculture Management Area.^{49,50} 'Trigger points' were used to determine when limits have been reached. These can then be reviewed and modified as necessary. In addition, these trigger points enable marine farm managers to respond with a range of management actions.⁵¹

Wild species

Material falling from mussel farms can attract small fish and encourage the development of reef-type communities including sponges, crabs, starfish and tube worms. Juvenile fish commonly use artificial structures placed in the sea for shelter, which can lead to the creation of new communities and increased biodiversity.

The overall effect of marine farms on fish abundance is not well understood. Research on farms in the Marlborough Sounds indicates that there is more abundance and diversity of fish adjacent to the farms than in surrounding areas. Anecdotal evidence from recreational fishers indicates that species such as snapper and kingfish can be found in higher numbers around marine farms, attracted to the food supply provided by the mussels and juvenile fish, especially during harvesting. An increasingly popular charter fishing industry is developing around the Wilson Bay mussel farms in the Firth of Thames.

Shellfish farms can also change the community structure in their vicinity because of the way they change the seabed substrate, increasing the amount of shell and nutrients. This results in a shift towards species better adapted to the new conditions. For example, studies of farms in the Marlborough Sounds found that heart urchins and brittle stars declined in the locality of mussel farms and were replaced by other species.⁵²

Spat for mussel farms is predominantly sourced from Ninety Mile Beach in Northland and then moved to the rest of New Zealand to seed the farms. There is currently no evidence that wild green-lipped mussels are declining in population fitness due to reliance on spat from one region, although mixing with farmed mussels has probably already occurred. This could be due to the long-lived dispersive larval stage of green-lipped mussels, which facilitates wide genetic mixing.

Pacific oyster farming in areas where they don't already exist could result in the establishment of populations of a non-native oyster in the wild. However, wild populations of Pacific oysters are already widespread around the coastline, and are present in areas where there are no marine farms.

Increasingly, selective breeding is being used to supply spat for oyster farms and, with new research underway, is likely to become more common for mussel farms. Trials have started where selectively bred hatchery-produced spat mussels have been transferred to a marine farm.⁵³ There is little information available about the impact of selective breeding on wild populations of oysters or mussels.

Farm structures

Abandonment of marine farms can result in degrading structures being left in the marine area. The structures required for finfish and mussel farming are generally of a high enough value to incentivise farmers to remove them, whereas oyster racks are not. If oysters are left behind on the racks they can outgrow the structures and cause the farms to collapse into the mud and build up on the seafloor. The abandoned structures can be very visible for many years. This problem can be addressed by conditions of resource consents and bonds to ensure structures are removed on the cessation of farming, or by the requirement for reasonable assurance (such as a fidelity fund).

Siltation and alteration of natural currents

Oyster farms can cause siltation on the seabed by disturbing the natural flow of water in the area, and changing the height of the seabed beneath and around the farm through scouring or silt build-up. These effects can be minimised by ensuring that oyster racks are oriented parallel with, rather than at right angles to, currents and wave action. The growing of oysters in baskets, which is becoming more common, minimises the trapping of sediment.

Potential environmental effects – finfish

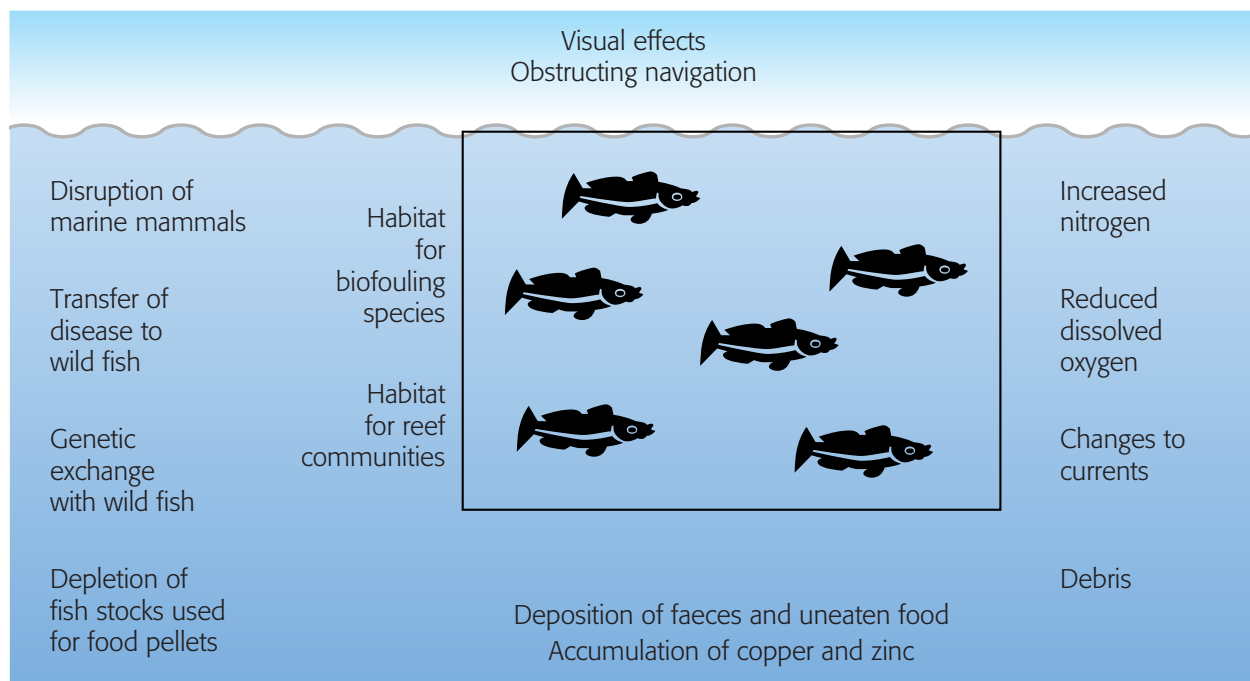


Figure 10.5 Summary of potential environmental effects of finfish farming

Seabed habitats

Deposition of faeces and uneaten feed causes changes to the seabed beneath finfish farms. These particles are organic-rich and can alter the physical, chemical and biological nature of the seabed, increasing microbial activity. This in turn can decrease oxygen levels in the sediment and lead to changes to the biodiversity of the area.

These effects tend to be greater in finfish farming than in shellfish farming. However, they can be minimised by siting farms in deep, well-flushing locations and by good farm management such as using higher-quality feed to gain an improvement in feed conversion rates (meaning that less feed is needed to grow a fish), and using automated feeders with shut-off signals linked to underwater cameras that detect waste feed.⁵⁴ Farmed areas can also be rotated so that areas of the seabed are regularly fallowed to allow recovery. Such approaches are also important for reducing the effects that uneaten food and fish waste have on the water column.

Water column

Finfish farming results in the release of nutrients into the water column from uneaten feed and fish wastes. These can stimulate phytoplankton growth and at excess levels may result in algal blooms and oxygen depletion. Nutrient enrichment can also cause changes in the mix of phytoplankton species with potential impacts on food webs. Such impacts can be reduced through siting farms in deep and well-flushed waters.⁵⁵

Local biodiversity

Finfish farms can cause changes to local biodiversity, in particular a decrease in those benthic species that are intolerant of the enriched conditions and a corresponding increase in species that are adapted to them. Wild fish and other species can be attracted to finfish farms because they feed on the dropped fish feed and use the structures for shelter.

Toxic chemicals

Therapeutants (vaccines, antifouling paints and feed) are all potential sources of chemical pollutants to the marine environment. Farmed finfish are very susceptible to diseases and large numbers of chemical treatments can be needed to combat these.

Currently in New Zealand, feed for farmed salmon does not contain antibiotics, vaccines, steroids or colourants. However the feed contains zinc, necessary to prevent cataract formation and other health problems, which can accumulate in sediments beneath salmon farms at levels potentially toxic to benthic organisms. The zinc now used in salmon feed is organic which has reduced impacts on the benthic environment. Any effects are thought to be limited to the footprint of the farm.⁵⁶

Therapeutants for treating parasites and diseases have not been required in New Zealand salmon farms, due to the relatively low density and small scale at which they are farmed and absence of diseases. However, the expansion of finfish farming into new areas and other use of types of fish may require new methods of disease prevention. Kingfish trials undertaken in the Marlborough Sounds indicate that there will be a need for disease treatments because kingfish are unusually susceptible to some parasites. It is proposed that these parasites will be managed by low-concentration in-water baths of hydrogen peroxide.

Finfish farmers, have in the past, used antifouling paint to prevent the build-up of biofouling organisms on structures. These paints gradually wash off the structures and can accumulate on the seabed under the farms. Historically, many of these paints contained copper, which in high concentrations is toxic. The extent to which such concentrations of copper may impact on marine life is not fully understood, although it may be toxic to organisms living in the sediment.⁵⁷ The effects are normally confined to the boundary of the farm.

Although manual removal of fouling is possible for farm nets, there are few alternatives to putting antifouling paint on items such as predator nets which must be kept clear of debris to allow water to flow through. Where the use of antifouling paints cannot be avoided, copper-free antifouling paints are now available, and there has been a significant reduction in the use of copper in antifoul paints by the aquaculture industry.

Decision-making process

Aquaculture reforms, which took place in 2004, placed all aquaculture planning and consent processing under the RMA. Previously marine farmers had to obtain separate consents from the regional council and the former Ministry of Fisheries. The reforms were intended to address the 'gold-rush' situation which had developed in some areas, where numerous consent applications for marine farms were lodged in an attempt to occupy areas of the seabed in a 'first-in, first-served' system. The reforms placed a moratorium on the processing of such applications to give councils time to undertake proper planning for the activity through the creation of aquaculture management areas.

Further aquaculture reforms came into effect on 1 October 2011 designed to free up marine space for farming, as a result of slow progress in creating operative aquaculture management areas. These removed the requirement for such an area to be in place before a resource consent application could be lodged. As a result, the normal RMA decision-making framework now applies to marine farming, subject to the following unique provisions:

- Permitted activity status is not allowed – regional coastal plans are not able to classify aquaculture activities as permitted activities.⁵⁸ This is because the 'Undue Adverse Effects' test (described below) is carried out alongside the consent process.
- A 20 year minimum consent term – coastal permits for aquaculture are required to have a minimum term of 20 years, unless a shorter term is requested by the applicant or a shorter term is required to manage the effects of the activity⁵⁹
- A three-year lapse period – generally resource consents must be exercised within five years of being granted otherwise they will lapse. The reforms reduce the lapse period for aquaculture consents to three years.⁶⁰

The 2011 RMA reforms now give the Minister responsible for aquaculture the power to recommend regulations to amend provisions in a regional coastal plan related to the management of aquaculture activities.⁶¹ This process allows changes to be made directly to regional coastal plans without following the normal plan-making process set out in Schedule 1 of the RMA.

Allocation of space

New allocation mechanisms enable councils to better manage a 'gold rush' situation, should it arise under the 2011 rules. Figure 10.6 lists the tools that are now in place.

Method	Details
Addressing the effects of occupation of the coastal marine area	• A regional coastal plan may include provisions to address the effects of occupation of the coastal marine area and to manage competition for the occupation of space (including for aquaculture activities) • Plans may include rules which: - specify that no application can be made before a certain date - state that the consent authority may process and hear applications relating to the same space or spaces in close proximity together - place limits on the character, intensity, scale or size of activities
Alternative allocation mechanisms	• The default allocation mechanism under the RMA is 'first in, first served' • This means that a complete resource consent application lodged with the consent authority must be determined before any resource consent applications lodged later in time for the same or overlapping space • Alternative allocation mechanisms may be provided for through a regional coastal plan or through <i>Gazette</i> notice
Regional coastal plan	• Alternative allocation mechanisms (public tender of authorisations or any other method of allocating authorisations) can be set out in a regional coastal plan • Before setting out an alternative allocation mechanism, the council must be satisfied that such a rule is necessary or desirable in the circumstances of the region and that the proposed method is the most appropriate for allocation of space in the circumstances of the region, having regard to its efficiency and effectiveness, compared to other methods of allocating space
<i>Gazette</i> notice	• As a result of the 2011 reforms, regional councils may also ask the Minister of Conservation to approve an alternative allocation method for activities in the coastal marine area through <i>Gazette</i> notice • The regional council will need to provide evidence demonstrating high or competing demands for coastal permits • While a <i>Gazette</i> notice is in place, no person can apply for a coastal permit authorising occupation of space unless the person is the holder of an authorisation that relates to that space and activity
Suspension of applications	• Regional councils may ask the Minister of Aquaculture for a stay on new applications for consents to occupy space for specified aquaculture activities • The regional council will need to demonstrate that there is actual or anticipated high or competing demand • The suspension must be for no more than 12 months
Refusal of similar applications	• A consent authority can refuse to receive an application for a coastal permit to occupy space in the coastal marine area if, within one year before the application is made, the consent authority has refused to grant an application for a permit for a similar or same activity in respect of the same space or space in close proximity

Figure 10.6 Tools available to councils to allocate marine space

Undue Adverse Effects test

The effects of proposed marine farm areas on fishing are assessed through the Undue Adverse Effects test which is undertaken by the Ministry for Primary Industries.⁶² A proposed marine farm cannot proceed if it would have undue adverse effects on recreational or customary fishing, or commercial fishing for non-QMS stocks. In addition, unless an aquaculture agreement or compensation declaration is registered, a proposed marine farm cannot proceed if it would have undue adverse effects on commercial fishing for QMS stocks.

There are a number of matters that the Ministry for Primary Industries considers in applying the test, being:⁶³

- The location of the proposed marine farm in relation to fishing areas

- The likely effect of the proposed marine farm on fishing, including the proportion of any fishery that would be affected
- The degree that the proposed marine farm would exclude fishing
- The extent that fishing for a species in the proposed marine farm area could occur in other areas
- The extent that occupation of the coastal marine area by the proposed marine farm would increase the cost of fishing
- The cumulative effect on fishing of any authorised aquaculture, including any structures authorised before the introduction of any relevant stock to the QMS

The test is limited to the effects on the practicalities of catching, taking and harvesting fish.

New Zealand Coastal Policy Statement

The NZCPS contains a number of policies relevant to aquaculture developments and operations. These are described more fully in Chapter 2: Marine management. The policies attempt to strike an effective balance between providing for such activities in the coastal environment and ensuring that important natural resources and community and cultural values are protected in doing so.⁶⁴ The Department of Conservation has provided some guidance to planners in interpreting the application of the NZCPS policies to aquaculture planning and decision-making, including a checklist of important matters to consider.⁶⁵

Policy 8 recognises the significant existing and potential contribution of aquaculture to the social, economic and cultural well-being of people and communities. This is done through directing that aquaculture activities are provided for in appropriate places within regional planning documents. It recognises that aquaculture activities require high water quality and land-based facilities. In addition, the policy directs that development in the coastal environment does not make water quality unfit for aquaculture activities in areas approved for that purpose.

NZCPS, Policy 8: Aquaculture

Recognise the significant existing and potential contribution of aquaculture to the social, economic and cultural well-being of people and communities by:

- a. including in regional policy statements and regional coastal plans provision for aquaculture activities in appropriate places in the coastal environment, recognising that relevant considerations may include:
 - i. the need for high water quality for aquaculture activities; and
 - ii. the need for land-based facilities associated with marine farming;
- b. taking account of the social and economic benefits of aquaculture, including any available assessments of national and regional economic benefits; and
- c. ensuring that development in the coastal environment does not make water quality unfit for aquaculture activities in areas approved for that purpose.

Protecting outstanding natural landscapes and natural character

Policies 13 and 15 of the NZCPS require adverse effects on outstanding natural features and landscapes and areas of the coastal marine area with outstanding natural character to be avoided. This effectively means that marine farms will not be able to proceed in sensitive areas of the coastal marine area and should instead be directed to localities where the natural character or landscape values of the area have already been compromised or are not outstanding. A recent Supreme Court decision on a salmon farm proposal in the Marlborough Sounds confirmed that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes.⁶⁶

Case law on marine farms in outstanding natural landscapes

New Zealand King Salmon Company Limited applied for plan changes and resource consents for salmon farms at nine sites in the Marlborough Sounds. The Board of Inquiry found that an industrial-scale salmon farm within Port Gore site would adversely affect an outstanding natural landscape, but the farm was still approved. The Supreme Court overturned the decision, making it clear that a plan could not provide for the establishment of a marine farm in a location where it results in adverse effects on an outstanding natural landscape.⁶⁷

Case law on marine farm conversion in an outstanding natural landscape

The Environment Court recently considered whether a resource consent application to convert a mussel farm (located in Pelorus Sound) to a salmon farm should be granted.⁶⁸ The site was identified as an 'Area of Outstanding Landscape Value' in the Marlborough Sounds Resource Management Plan and therefore section 6(b) RMA and Policy 15 NZCPS were applicable. Section 6(e) regarding Māori interests was also applicable. The application was turned down on the basis that section 6(b) and (e) (as particularised by the NZCPS) were 'sticking points' (at [215]) and '[p]lacing all relevant considerations in the balance ... we consider the purpose of the RMA, and the instruments made under it, is better achieved if consent is refused (at [217])'.

Protecting indigenous vegetation and fauna

Under Policy 11, a marine farm applicant will need to show that there will be no adverse effects on indigenous vegetation and fauna which are threatened, at risk or naturally rare. In relation to other indigenous species and ecosystems, an applicant must show that significant adverse effects have been avoided and other adverse effects have been avoided, remedied or mitigated. These matters are discussed further in Chapter 2: Marine management.

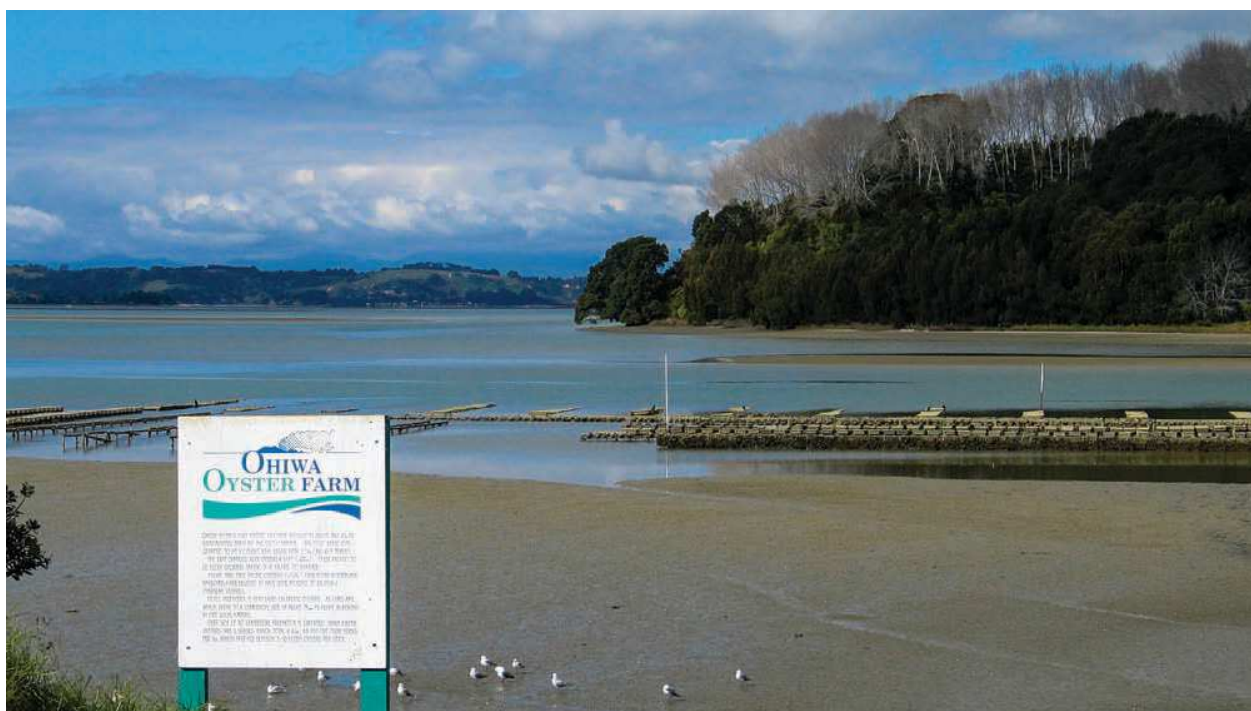
Harmful aquatic organisms

Policy 12 addresses the management of harmful aquatic organisms and requires councils to provide in regional policy statements and plans, as far as practicable, for the control of activities that could cause harmful aquatic organisms to be released or otherwise spread including 'the establishment and relocation of equipment and stock required for or associated with aquaculture'. This is discussed further in Chapter 5: Marine biosecurity.

Protecting water quality

Policy 23 of the NZCPS addresses the discharge of contaminants into the marine environment, which includes those by aquaculture. It directs particular regard to be had to a number of matters including the sensitivity of the receiving environment, the nature of the contaminants to be discharged and the capacity of the receiving environment to assimilate the contaminants.

In addition, Policies 8(c) and 21 of the NZCPS require councils to consider the effects of other resource use on the water quality required for aquaculture, and to work to improve coastal water quality where aquaculture growing areas are significantly adversely affected.⁶⁹



The NZCPS requires councils to consider the effects of other activities on water quality required for aquaculture, such as for the oyster farm in Ōhiwa Harbour shown here (Raewyn Peart)

Existing marine farming rights

Existing aquaculture consent holders have two key protections, which apply even if the consent holder is applying to establish a different aquaculture activity, although it must be within the same space:

- When a resource consent is due to expire the consent holder can apply for a new consent for an aquaculture activity in the same space (at least six months before the expiry of the existing consent) and can continue to exercise the existing consent until a decision is made to approve or decline the application
- If a consent is due to expire and an application has been lodged for an aquaculture activity in the same place, the existing consent holder is notified and thus has an opportunity to lodge a competing application which is given priority in terms of processing

It is important to note that these provisions do not amount to a right of renewal, or even a presumption that the existing consent will be renewed.

Case law on renewal of aquaculture coastal permits in an outstanding natural landscape

Sanford Limited and Port Gore Marine Farms applied to renew coastal permits for three mussel farms in Port Gore, in the Marlborough Sounds. The Marlborough District Council declined one application and granted the other two applications for limited terms. The applicants appealed the decision to decline/grant for limited terms. The marine farms were within a zone where new farms were prohibited. An appendix to the plan allowed existing farms to apply for consent as a discretionary activity. The Court determined that the landscape of the inner Port Gore is an outstanding natural landscape. Policy 15(a) of the NZCPS requires adverse effects on outstanding natural landscape areas to be avoided. The Court considered the existing environment of Port Gore as if the existing farms were not present (because the coastal permits had expired). The Court determined that each farm would, in a more than minor way, diminish the natural character of the locality. The Court refused the appeals by Sanford and Port Gore Marine Farms.⁷⁰

Elements of best practice

Some of the risks from aquaculture activities discussed in the previous sections can be minimised through good siting of farms and good design. These include marine mammal interactions, adverse impacts on landscape and natural character, and alterations of water flows. Other risks, such as those associated with marine biosecurity, can be managed through good on-farm practices. Water-quality impacts and cumulative risks are harder to predict and are therefore best managed adaptively, by phasing development within initially precautionary principles. Figure 10.7 below provides guidance on best practice responses to potential impacts.

Effects	Best practice response
General aquaculture activities	
Transporting spat and machinery used on marine farms, around the country, can spread diseases and pests	Good management, including careful washing and drying of equipment before transportation, should be adopted to prevent spread of disease and pests
Fixed structures used by a marine farm can attract biofouling species which can spread to the surrounding environment	Fixed structures should be regularly cleaned to prevent the build-up of biofouling species
Copper from antifouling paints can build up on the seabed to toxic levels	Marine farms should either manually defoul equipment or use paints which are free of copper
Alteration of the behaviour of marine mammals as a result of marine farms obstructing access to, or use of, feeding grounds and other areas of their habitat	Marine farms should be carefully located in order to ensure that access by marine mammals to habitual feeding, haul out, migratory routes and other areas is not obstructed Marine farms should not be located in areas which are used by dolphins for co-operative hunting
Potential for marine mammals to become caught in longline or finfish nets used on the marine farm	Marine farms should use the best available technology and techniques to minimise the risk that marine mammals will become trapped in nets (such as adequate net tension, enclosing anti-predator nets at the bottom, eliminating food waste and reducing the mesh size of nets) ⁷¹

Effects	Best practice response
Debris from poorly maintained farms can be a hazard to other water users and to marine fauna	All non-biodegradable waste material should be removed from the coastal marine area, and structures should be adequately maintained to ensure their integrity. Lines should be kept under tension (this reduces likelihood of nuisance for other users and reduces likelihood of marine mammal entanglement).
Degradation of landscape values and natural character of the area	Marine farms should be sited away from areas with outstanding natural landscape or character values Marine farms should be sited away from areas where the visual effects of the farm could be significant on landscape or natural character values
Marine farms can pose a hazard to watercraft and navigation	Marine farms should be adequately lit at night to ensure safety of navigation Marine farms should not be located in areas which obstruct navigation or safe anchorage, such as shipping routes and popular recreational vessel routes and anchorages
Shellfish	
Effects on water quality through excretion of faeces and alteration of naturally occurring currents	Marine farms should be designed to minimise effects on currents, should be sited in areas with adequate flushing, and should not be stocked at densities that result in significant build-up of faecal material in the water column
Depletion of naturally occurring plankton leaving less available for wild species	The effects of shellfish farming on the amount of plankton available should be carefully managed, including the cumulative effects of having more than one shellfish farm in the same area
Effects on light-sensitive marine flora as a result of shading by the farm structure	Marine farms should be located away from areas where they can have negative impacts on seagrass and other light-sensitive species such as seaweeds
Changes to the seabed through deposition of shell, shellfish and faeces	Marine farms should be sited in areas where benthic habitats are less sensitive to impacts such as areas with a muddy substrate rather than rocky or sandy areas
Effects on wild species as a result of changes to their environment	Marine farms should be located away from areas where they can have negative impacts on species or habitats with special conservation value or of ecological importance
Changes to patterns in the current and natural flow of water as a result of farms, particularly oyster racks, causing sand build-up under the farm	Proposals for marine farms should be accompanied by an assessment of the hydrodynamic patterns at the subject site, with current speeds and directions, flushing times and water-dispersion patterns and modelling of how this will be affected by the proposed farm Marine farms should not be sited in poorly flushing areas
Abandonment of marine farm sites and structures when farming ceases	If marine farming activity ceases, all structures associated with its use should be removed and the site rehabilitated. The consent authority should require this as a condition of consent or require a bond or payment to cover the cost of removal by the local authority.
Finfish	
Increased likelihood of disease from high-density stocking rates for fish farming	Good husbandry, stress reduction and appropriate nutrition should be used to help prevent disease outbreaks Fish should not be farmed at high intensity to minimise the likelihood of disease There should be single age-class fish farms Farm spacing should be at least 1 kilometre within the same management plan and at least 2.5 kilometres if not within the same management plan Equipment should be regularly checked
Depletion of fish stocks when farmed fish are fed wild-captured fish or feed pellets made from wild-captured fish	The proportion of wild-captured fish used to provide protein and oils in fish feed should be minimised Feed which maximises the conversion rate between the amount of wild fish consumed and the amount of farm fish produced should be used Only wild fish from sustainable capture fisheries should be used

Effects	Best practice response
Deposition of faeces and uneaten food causing changes to the physical, chemical and biological nature of the seabed and water quality	High-density stocking rates should be avoided, high-quality feed used to reduce the amount required, and automated feeding systems deployed which stop dispensing when they detect uneaten food in the water Farms should be located in areas which flush well Adopt integrated co-culture/multi-trophic aquaculture where possible (where finfish, mussels, seaweed and other species are grown in an integrated farm where one species counteracts the negative effects of another)
Effects on local biodiversity as a result of changes to their environment	Marine farms should be located away from areas where they can have negative impacts on species or habitats with special conservation value or of ecological importance
Escape of farmed fish and spread of diseases to wild fish	Fish farms should be designed to minimise escapees
Breeding of escaped fish with wild fish that adversely affects the genetic robustness of the wild population	All reasonable efforts should be made to prevent the escape of farmed fish, including utilising new technology where available
Harm to the marine environment and to humans consuming the products from chemicals used to treat the fish	Finfish should be farmed at a density which limits the likelihood of disease and therefore does not require the use of antibiotics and other therapeutants, as has been the case up to this point in New Zealand

Figure 10.7 Best practice considerations for aquaculture activities

The aquaculture industry has developed Environmental Codes of Practice designed to promote the adoption of best industry practices, throughout the growing and harvesting cycles, in order to minimise potential effects on the environment.⁷² These Codes have been independently benchmarked against similar international standards. They are specific to three species:

- Greenshell™ Mussel Industry Environmental Code of Practice, 1999 (Revised 2007)
- Oyster Industry Code of Practice, 2007
- Finfish Aquaculture Environmental Code of Practice, 2012

Aquaculture New Zealand is currently redrafting the Environmental Codes of Practice to reflect current industry practices, changes to the regulatory framework and the latest market conditions, as well as to incorporate any new global standards. Key performance indicators will be set for each Code. An online sustainability assessment and reporting tool is also being considered. This is intended for use by marine farm operators, third-party auditors and the industry body to help rate and consequently improve environmental performance against the standards. One of the aims of this work is that individual operators can use the environmental management system as a pathway to become fully accredited with an international certification.



Aquaculture New Zealand has developed codes of practice for marine farms, such as the salmon farm shown here in the Marlborough Sounds, in order to promote the adoption of best industry practice (Raewyn Peart)

Guidance for aquaculture monitoring in the Waikato region

The Waikato Regional Council, with funding from the Ministry for Primary Industries, is developing a regional environmental monitoring framework to integrate consent-related environmental monitoring with state of the environment monitoring. The aim of this initiative is to rationalise and improve all environmental monitoring conducted in the Waikato coastal marine area. Aquaculture has been chosen as a first case study, in anticipation of the development of fish farming and increase in shellfish farming in the region.⁷³

The purpose of the framework is to achieve a more consistent and transparent approach to environmental monitoring that better integrates consent-related environmental and state of the environment monitoring, and accounts for cumulative effects. A regional monitoring framework has the potential to facilitate these types of improvements, and to provide a broader context for understanding the effects of existing and future anthropogenic activities against the background of natural variability.

The framework is based on the understanding that the environmental effects of consented activities are best managed if environmental monitoring is aligned with broader state of the environment monitoring. In addition, it is accepted that environmental quality standards should be established prior to the commencement of the consented activity, so that consent monitoring can be designed to verify compliance with these standards.⁷⁴

The first phase of this project focused on reviewing current monitoring approaches, reviewing relevant consents, assessing locally relevant ecological effects and their significance, and developing the structure for the monitoring framework. The second phase focused on developing the detailed framework, identifying priority issues that need to be addressed either through industry best practice and reporting or through monitoring, and identifying environmental indicators and indicative standards. All information is currently being collated in a series of three technical reports. These will be distributed to the aquaculture industry for feedback prior to being finalised.

Ultimately this project will produce a guidance document for the aquaculture industry and other stakeholders which provides clear information on environmental monitoring requirements for aquaculture consents in the Waikato region. As a result, the outcome of this project should be practicable and directly applicable to the resource consenting process.

Within an integrated monitoring framework, state of the environment monitoring and consent-related environmental monitoring should be more effective in improving understanding of the overall state of the environment (including cumulative effects). This project supports the concept that having well-designed environmental quality standards and monitoring approaches at the time of consent will enable meaningful adaptive management strategies to be designed for marine farms.⁷⁵



This mussel farm in the Coromandel Harbour will be part of the Waikato Regional Council's aquaculture monitoring programme (Raewyn Peart)

Revive Our Gulf

Subtidal green-lipped mussels once covered over 500 square kilometres of reefs throughout much of the Firth of Thames and Tāmaki Strait, forming one of New Zealand's most productive shallow-water ecosystems.⁷⁶ From the early 1940s, these mussel beds were dredged for around five decades, until the commercial fishery collapsed. The damaged mussel beds never recovered, resulting in the loss of habitat and alteration of food webs as they gradually turned to less productive mud.⁷⁷

Mussels are powerful filter feeders. A single mussel is capable of passing 350 litres or a bathtub of water a day through its gills, secreting nutrients that support other marine life at the same time.⁷⁸ In their original numbers, the waters of the Firth would have been filtered daily, but with the few mussels left, filtering is anticipated to take two years. Without the filtering abilities of the mussels, the waters of the Hauraki Gulf have become murkier and more vulnerable to the effects of increasing nutrient run off.

In an effort to restore the mussel beds of the Hauraki Gulf, the Mussel Reef Restoration Trust set up the 'Revive Our Gulf' initiative in 2012. In December 2013 the first seven tonnes of mussels, stock donated by North Island Mussels Limited, were deposited in the Gulf. They were used to create 7 'living room'-size trial plots in a sheltered bay off eastern Waiheke Island.⁷⁹ The plots are being monitored by a student from the University of Auckland as part of work towards a doctorate. Within the first six months, there have been reports of spat starting to settle, an early signal of a self-sustaining reef.⁸⁰ Snapper and starfish have been attracted to the plots as the mussels provide a valuable source of food. In addition, new habitat has been provided to invertebrates and fish which have started to colonise the beds.

With the assistance of New Zealand's aquaculture industry, the Hauraki Gulf's mussel reefs are being re-introduced and re-established. It is hoped that water quality can be improved, and fish life will once again flourish amidst the newly-created habitat. Revive Our Gulf Chairman John Laurence said the group planned partnerships with mussel farms to utilise reject mussels for further field trials and initiatives to build awareness about the project's potential: 'Our vision is to prove a technology and then provide it to communities around the Gulf to enhance environmental quality'. This project highlights how valuable industry support can be to delivering marine restoration projects.



The Hauraki Gulf's mussel reefs are being re-introduced and re-established and it is hoped that water quality can be improved and fish life will once again flourish amidst the newly-created habitat (*Shane Kelly and Shaun Lee*)

Using DNA technology to monitor environmental change around marine farms

Monitoring the environment in the vicinity of marine farms is a requirement of resource consents, but the challenge has always been having the technology to do this in an effective way that supports adaptive management. Traditional methods, such as chemical analysis of ecological changes, can be time-consuming, highly skilled and expensive, and the results can often take months to be delivered. This means that often the marine farm is continuing to operate in a void of information.

New technology developed by the Cawthron Institute called 'Next-Generation Sequencing' will enable the aquaculture industry to monitor the environment around marine farms in near real time. This means that there will be almost instantaneous knowledge about any changes in the marine ecosystems. From this information marine farm managers will be able to quickly adapt their management practices.

This project has been in the 'proof of concept' stage for the last two years. The study has been funded by the Ministry of Business, Innovation and Employment, with in-kind support from New Zealand King Salmon Company Limited through trialling around some salmon farms in the Marlborough Sounds and at Stewart Island. These trials have been completed, alongside the traditional monitoring methods to enable comparison and to assess the effectiveness of the tool. Scientists have now proven that with the new technology, using a liquid workstation 'robot' they can potentially analyse hundreds of species at the same time using a group of microscopic single-celled organisms known as foraminifera, which are highly responsive to environmental change.

With such fast results being possible, the aquaculture industry will be in a much better position to make rapid changes to management systems, ultimately minimising environmental impacts. 'Marine industries are very excited about this – they need a tool that allows them to provide an almost real-time result, reducing the cost and demonstrating environmental sustainability', said Cawthron Institute scientist Dr Xavier Pochon.

Phase two of this research will see monitoring results being compared annually over a three-year period. It will also explore the potential to look at other types of aquaculture such as mussel farming, as well as other marine industries like the oil and gas sector.



New monitoring technology is being trialled around some salmon farms in the Marlborough Sounds to provide near real-time information (Raewyn Peart)

Further reading and references

- Aquaculture New Zealand, 2012, *New Zealand aquaculture: A sector overview with key facts, statistics and trends*, Aquaculture New Zealand, Nelson
- Bernard Brown Associates Limited, 2008, *Visual impact assessment of potential finfish farming development*, Environment Waikato, Hamilton, available at <http://www.ew.govt.nz/PageFiles/9863/TR08-24.pdf>
- Burrell M and L Meehan, 2006, *New Zealand aquaculture strategy*, New Zealand Aquaculture Council, Nelson, available at http://www.seafood.co.nz/f431,15491/15491_Aquaculture_Sector_Strategy.pdf
- Department of Conservation, 2013a, *NZCPS 2010 Guidance note Policy 3: Precautionary approach*, Department of Conservation, Wellington, available at <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/coastal-management/guidance/policy-3.pdf>
- Department of Conservation, 2013b, *NZCPS 2010 Guidance note Policy 6: Activities in the coastal environment*, Department of Conservation, Wellington, available at <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/coastal-management/guidance/policy-6.pdf>
- Department of Conservation, 2013c, *NZCPS 2010 Guidance note Policy 8: Aquaculture*, Department of Conservation, Wellington, available at <http://doc.org.nz/Documents/conservation/marine-and-coastal/coastal-management/guidance/policy-8.pdf>
- Du Fresne S, 2008, *Evaluation of the impacts of finfish farming on marine mammals in the Firth of Thames*, Environment Waikato, Hamilton, available at <http://www.ew.govt.nz/PageFiles/9912/TR0827.pdf>
- Forrest B, N Keeley, P Gillespie, G Hopkins, B Knight and D Govier, 2007, *Review of the ecological effects of marine finfish aquaculture*, Cawthron Institute, Nelson, available at http://www.aquaculture.govt.nz/files/pdfs/Review_of_ecological_effects_of_finfish_farming.pdf
- Keeley N, B Forrest, G Hopkins, P Gillespie, B Knight, S Webb, D Clement and J Gardner, 2009, *Sustainable aquaculture in New Zealand: Review of the ecological effects of farming shellfish and non-fish species*, Cawthron Institute, Nelson, available at http://www.aquaculture.govt.nz/effects_of_shellfish.php
- Kelly S, 2008, *Potential environmental effects associated with the proposed shift from mussel to finfish farming in the Firth of Thames*, Environment Waikato, Hamilton, available at <http://www.ew.govt.nz/PageFiles/12521/TR2008-38.pdf>
- Kemper C, D Pemberton, M Cawthorn, S Heinrich, J Mann, B Wursig, P Shaughnessy and R Gales, 2003, 'Aquaculture and marine mammals: Co-existence or conflict?', in N Gales, M Hindell and R Kirkwood (eds), *Marine mammals: Fisheries, tourism and management issues*, CSIRO Publishing, Victoria
- Lloyd B D, 2003, *Potential effects of mussel farming on New Zealand's marine mammals and seabirds: A discussion paper*, Department of Conservation, Wellington
- McShane P E, 1997, 'Environmental effects of sea-based farming of paua (*Haliotis spp.*) and kina (*Evechinus chloroticus*)', *Conservation Advisory Science Notes No. 143*, Department of Conservation, Wellington
- Maritime New Zealand, 2005, *Guidelines for aquaculture management areas and marine farms*, Maritime New Zealand, Wellington, available at <http://www.maritimenz.govt.nz/Publications-and-forms/Commercial-operations/Ports-and-harbours/Guideline-for-Aquaculture-Management-areas-and-Marine-Farms.pdf>
- Ministry for Primary Industries, 2013a, *Literature review of ecological effects of aquaculture*, Ministry for Primary Industries, Wellington, available at: <http://www.fish.govt.nz/NR/rdonlyres/9AB1802D-8EE0-4C65-BA20-581D00C0111C/0/2013chapter1introduction.pdf>
- Ministry for Primary Industries, 2013b, *Comparison of the international regulations and best management practices for marine finfish farming*, MPI Technical Paper No. 2013/47, Ministry for Primary Industries, Wellington
- Ministry for Primary Industries, 2013c, *Overview of ecological effects of aquaculture*, Ministry for Primary Industries, Wellington, available at: <http://www.mpi.govt.nz/Default.aspx?TabId=126&id=1950>

Ministry of Fisheries, 2008, *The Ministry of Fisheries assessment of effects of aquaculture management areas on fishing*, Ministry of Fisheries, Wellington, available at http://www.aquaculture.govt.nz/files/pdfs/UAD_Test.pdf

Peart R, 2013, *Dolphins of Aotearoa: Living with New Zealand dolphins*, Craig Potton Publishing, Nelson

Plew D R, C L Stevens, R H Spiegel and N D Hartstein, 2005, 'Hydrodynamic implications of large offshore mussel farms', *IEEE Journal of Oceanic Engineering* 30, 95–108

Skretting Australia, 2013, *Annual sustainability report*, Skretting Australia, Hobart

Wursig B and G A Gailey, 2002, 'Marine mammals and aquaculture: Conflicts and potential resolutions', in R R Stickney and J P McVey (eds), *Responsible marine aquaculture*, CABI, Oxfordshire,

Zeldis J, N Broekhuizen, A Forsythe, D Morrissey and J Stenton-Dozey, 2010, *Waikato marine finfish farming: Production and ecological guidance*, Prepared for Ministry of Fisheries Aquaculture Unit, NIWA, Christchurch

Zeldis J, M Felsing and J Wilson, 2005, 'Limits of acceptable change – a stakeholder-collaborative framework for managing environmental performance of New Zealand marine farming', *Coastal News*, 30

Endnotes

- 1 Aquaculture New Zealand, 2012
- 2 <http://www.seafood.co.nz/aquaculture>
- 3 <http://aquaculture.org.nz/industry/overview/>
- 4 http://www.aquaculture.govt.nz/government_response.php
- 5 <http://www.fish.govt.nz/NR/rdonlyres/20A0ED89-A20B-4975-9E63-6B302187840D/0/AQUAStrat5yrplan2012.pdf>
- 6 http://www.fao.org/fishery/culturedspecies/Perna_canaliculus/en
- 7 http://www.fao.org/fishery/culturedspecies/Perna_canaliculus/en
- 8 <http://www.seafood.co.nz/greenshell>
- 9 <http://www.cawthron.org.nz/aquaculture/services/breeding-improvements/>
- 10 <http://aquaculturedirect.blogspot.co.nz/2014/04/spatnz-breeding-better-mussels-for-new.html#sthash.mXBLndEQ.dpuf>
- 11 <http://www.mpi.govt.nz/agriculture/funding-programmes/primary-growth-partnership/spatnz>
- 12 <http://www.nzmfa.co.nz/faq.asp>
- 13 <http://www.odc.govt.nz/harbour-project/fact-sheet/>
- 14 <http://www.whakatohea.co.nz/aquaculture>
- 15 Keeley N et al., 2009
- 16 <http://www.niwa.cri.nz/ncfa/aquaspecies/pacific>
- 17 <http://www.nzaquaculture.co.nz/AQUACULTURE%2011.pdf>
- 18 <http://maxa.maf.govt.nz/sff/about-projects/search/12-006/index.htm>
- 19 <http://www.teara.govt.nz/en/shellfish/page-5>
- 20 Some salmon are also farmed in freshwater locations. Ocean ranching was another method tried in New Zealand in the 1980s. This involved releasing large numbers of smolts into the sea to fend for themselves until adulthood and then relying on their homing ability to guide them back to their point of release to be harvested. This method proved inefficient as marine survival rates were too low.
- 21 <http://www.salmon.org.nz/aboutsalmon.shtml#production>
- 22 Skretting Australia, 2013
- 23 <http://www.legislation.govt.nz/bill/government/2010/0239/21.0/DLM3342378.html> – see clause 101
- 24 <http://www.waikatoregion.govt.nz/PageFiles/20527/TR201122.pdf>
- 25 <http://www.niwa.co.nz/our-science/aquaculture/aquaculture-species/hapuku>
- 26 <http://www.niwa.co.nz/our-science/aquaculture/aquaculture-species/kingfish>
- 27 <http://www.niwa.co.nz/our-science/aquaculture/aquaculture-species/kingfish>
- 28 <http://www.niwa.co.nz/our-science/aquaculture/aquaculture-species/eel>
- 29 <http://www.niwa.co.nz/our-science/aquaculture/aquaculture-species/eel>
- 30 <http://aquaculture.org.nz/environment/sustainability/>
- 31 Ministry for Primary Industries, 2013a and 2013c
- 32 <http://www.marinebiosecurity.co.nz/downloads/2216309/Pest+Pathways.pdf>
- 33 Du Fresne S, 2008, 2
- 34 Ministry for Primary Industries, 2013c, 20
- 35 Kemper C et al., 2003
- 36 Lloyd B D, 2003
- 37 Du Fresne S, 2008, 13
- 38 Forrest B et al., 2007, 28–29; Du Fresne S, 2008, 14

39 <http://www.waikatoregion.govt.nz/PageFiles/9912/TR0827.pdf>

40 Peart R, 2013, 214–215

41 Ministry for Primary Industries, 2013a, 6-2

42 Ministry for Primary Industries, 2013c

43 <http://www.greenshell.com/safe.html>

44 Bernard Brown Associates Limited, 2008

45 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited and Ors*, SC 82/2013 [2014] NZSC 38

46 <http://www.niwa.co.nz/aquaculture/research-projects/accessible-aquaculture-hongoeka-development-trust>

47 Ministry for Primary Industries, 2013c, 13

48 Plew D R et al., 2005

49 Aquaculture Management Areas were introduced in 2005 as a tool to address cumulative effects of aquaculture by including in planning documents specific zones for aquaculture. The 2011 aquaculture law reforms removed the requirement for these to be established.

50 Zeldis J R, Felsing M and J Wilson, 2005

51 <http://www.niwa.co.nz/sites/niwa.co.nz/files/import/attachments/lac.pdf>

52 www.aquaculture.govt.nz

53 <http://www.mpi.govt.nz/news-resources/news/spatnz-breeding-better-mussels-for-new-zealand>

54 Forrest et al., 2007, para 3.3.2

55 Ministry for Primary Industries, 2013c, 34–36

56 Kelly S, 2008, 7

57 Forrest B et al., 2007, 35

58 RMA, section 68A

59 RMA, section 123A

60 RMA, section 125

61 RMA, sections 360A to 360C

62 <http://www.fish.govt.nz/en-nz/Commercial/Aquaculture/Marine-based+Aquaculture/Undue+Adverse+Effects+test/default.htm>

63 <http://www.fish.govt.nz/en-nz/Commercial/Aquaculture/Marine-based+Aquaculture/Undue+Adverse+Effects+test/default.htm>

64 NZCPS, Policy 6(1)(c)

65 Department of Conservation, 2013c

66 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited and Ors*, SC 82/2013 [2014] NZSC 38

67 <http://www.eds.org.nz/content/documents/pressreleases/2014/140417%20KS%20media.pdf>

68 *KPF Investments Limited v Marlborough District Council* [2014] NZEnvC at [152]

69 Department of Conservation, 2013c

70 *Port Gore Marine Farms v Marlborough District Council* [2012] NZEnvC at [72]

71 Kemper C et al., 2003

72 <http://aquaculture.org.nz/environment/sustainability/>

73 <http://www.fish.govt.nz/NR/rdonlyres/B19A19C5-6B76-4BE7-881E-B5CA5177122F/0/APF1204WebSummary.pdf>

74 <http://www.fish.govt.nz/NR/rdonlyres/B19A19C5-6B76-4BE7-881E-B5CA5177122F/0/APF1204WebSummary.pdf>

75 <http://www.fish.govt.nz/NR/rdonlyres/B19A19C5-6B76-4BE7-881E-B5CA5177122F/0/APF1204WebSummary.pdf>

76 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=10860099

77 [http://www.reviveourgulf.org.nz/resources/Paul-\(2012\)-FoT-Mussels-AEBR-94.pdf](http://www.reviveourgulf.org.nz/resources/Paul-(2012)-FoT-Mussels-AEBR-94.pdf)

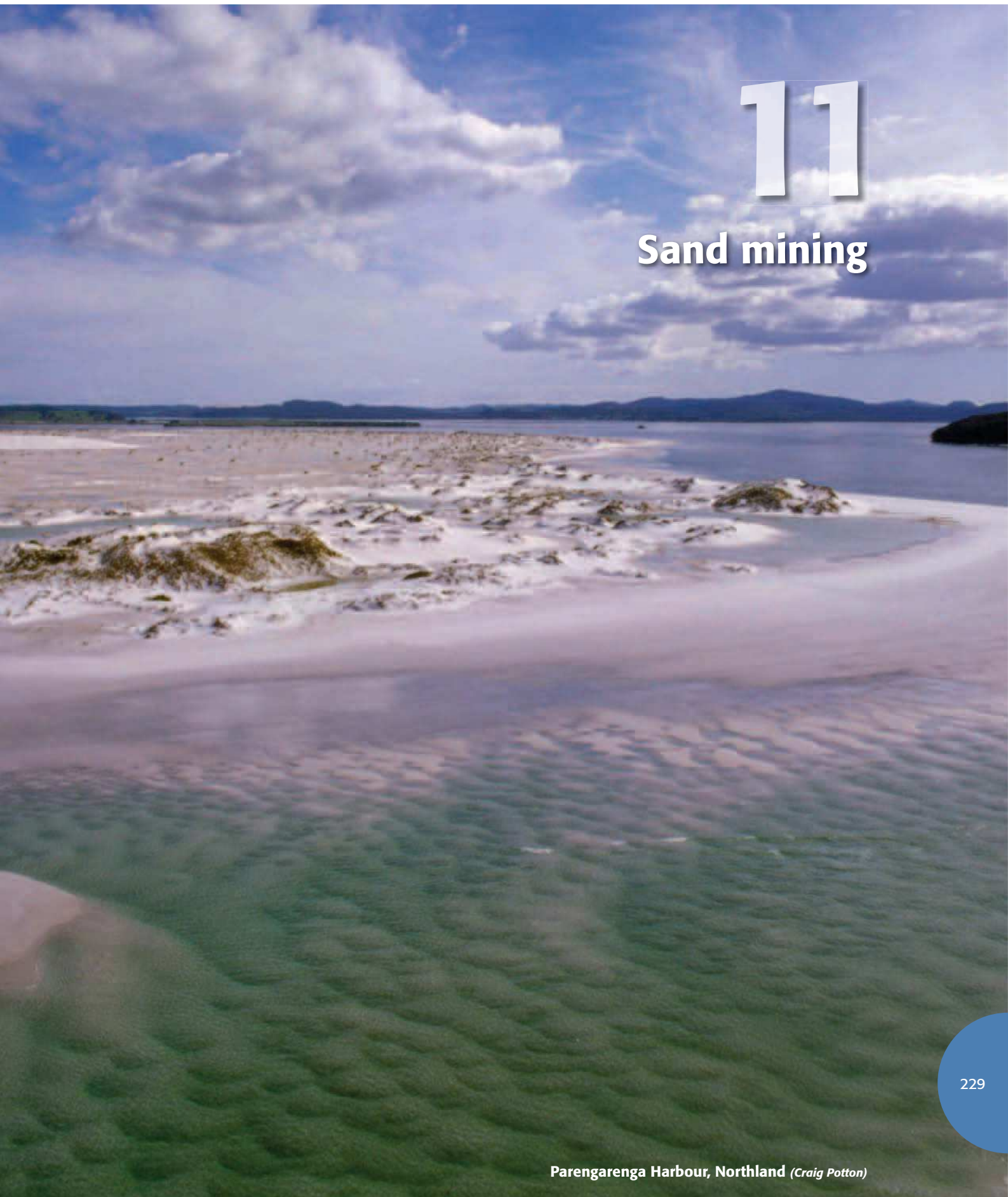
78 <http://www.marine.auckland.ac.nz/en/about/our-research/featured-research/reestablishing-mussel-beds.html>

79 <http://www.reviveourgulf.org.nz/resources/Aquaculture%20NZ%20June%202014%20ROG.pdf>

80 <http://www.reviveourgulf.org.nz/resources/Aquaculture%20NZ%20June%202014%20ROG.pdf>

11

Sand mining





Contents

231	Introduction
231	Inshore sand mining
232	Offshore sand mining
232	Mining methods
232	Prospecting and exploration
234	Active mining
234	Offshore sand mining methods
234	Inshore sand mining methods
235	Potential environmental effects
236	Seabed habitat
237	Sedimentary and coastal processes
238	Marine mammals, fish and seabirds
239	Natural landscapes, character and amenity values
240	Other
241	Decision-making framework
241	Allocation of mining rights
241	Management under the RMA
242	Management under the EEZ Act
243	Elements of best practice
245	Case study
245	South Taranaki Bight iron sand extraction
248	Further reading and references
249	Endnotes

Introduction

Mining of sand for commercial purposes has been occurring in New Zealand at a number of locations for many years. Extraction generally targets grain sizes of between 0.063 and 2 millimetres. Sand is commonly used as a component of aggregate in the production of concrete and as an ingredient for other cement-based products. The titanomagnetite contained in the black iron sand off the west coast is used for the manufacture of iron and steel. Sand is also used to restore popular beaches, such as Auckland's St Heliers and Mission Bay. More recently there has been a focus on the opportunities to mine iron sand from offshore areas.

This chapter discusses mining of sand from both inshore and offshore areas. Other forms of marine mining are discussed in Chapter 12: Deep-sea mineral exploration and mining, and Chapter 13: Oil and gas exploration and mining.

The two areas which are distinct in terms of the regulatory framework which governs sand mining are:

- Inshore sand mining – within the coastal marine area out to the 12 nautical mile limit (including onshore beach and dune deposits)
- Offshore sand mining – within the EEZ from the 12 to the 200 nautical mile limit

Inshore sand mining

New Zealand has extensive sand deposits spread over the continental shelf, particularly close to major rivers. In many areas, the sand is derived from land-based erosion, with the sediment being carried by rivers out to the sea. But some sand comes from other sources, such as shell-based sands made up of the crushed remnants of molluscs and bryozoan sands which are found in Northland. In the early days of European settlement, many local beaches were mined for sand until concerns about erosion stopped the practice. For example, Limeburners Bay in the upper Waitematā was where shells mined from beaches around the Hauraki Gulf were processed to make lime and cement. In addition, sand from Whiritoa was mined for over 50 years, with a total of more than 180,000 cubic metres of sand removed.¹



Mined sand has been used to replenish beaches such as at Point Chevalier, Auckland, shown here, which have been mined in the past (Raewyn Peart)

In more recent years, quartz-rich silica sand has been mined from sandbars off Mangawhai Heads and Pākiri Beach. The sand found at Parengarenga is very pure silica. Titanomagnetite iron sand (sand that contains both titanium and iron oxides) is found along the west coast of the North Island between Kaipara Harbour and Whanganui. This sand has been mined from the beach, from the adjacent coastal dunes, and from the near-shore marine area.² New Zealand Steel mines up to 1.2 million tonnes of sand each year from the north head at Port Waikato.³ The Taharoa iron sand mining site, located just south of the Kawhia Harbour, has the largest deposits of iron sand in New Zealand with an estimated 300 million tonnes of concentrate available at commencement of mining.⁴ There are also the rutile and ilmenite sand deposits of Westland, and the platinum, uranium and thorium-bearing sands of the South Island.

Inshore sand mining is regulated under the Crown Minerals Act for allocation and under the RMA for managing environmental effects, which is discussed in the section below on decision-making framework and in Chapter 2: Marine management.

Offshore sand mining

Offshore deposits of sand, associated with beaches formed during previous ice ages when the sea level was lower, occur off the west and east coasts, including Taranaki and the Bay of Plenty. Iron sand deposits are found in the northern and southern Taranaki Bights with localised concentrations of greater than 10 per cent titanomagnetite.⁵ An iron sand mining project to excavate up to 50 million tonnes of seabed material per year from the South Taranaki Bight was recently declined consent (this is discussed further in the case study at the end of this chapter).

Sand mining from offshore areas is regulated under the Crown Minerals Act for allocation and the EEZ Act for managing environmental effects, which is discussed in the section below on decision-making framework and in Chapter 3: EEZ legislation.

Mining methods

Both inshore and offshore sand mining follow a staged process from prospecting through to active mining, as shown in Figure 11.1. During each phase, different methods are used to assess or extract minerals from the seabed or land adjacent to the coast. Quantifying potential resources through prospecting and exploration typically takes around seven years before the active mining phase can begin.⁶

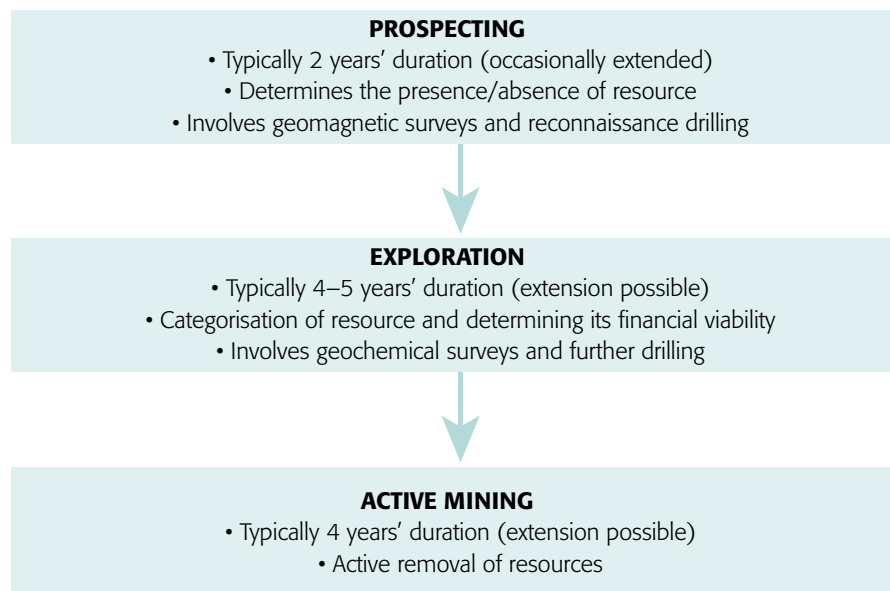


Figure 11.1 Phases of mineral extraction (Thompson K, 2012)

Prospecting and exploration

Prospecting generally involves reconnaissance drilling to recover sediment samples for analysis onshore, airborne geomagnetic studies and seismic surveys, as summarised in Figure 11.2 below.⁷ A limit on the volume of sediment which can be removed, and on the total number of core samples which can be taken, is normally set in the prospecting permit. For example, the Trans-Tasman

Resources Limited prospecting permit sets a limit of 600 cores in total over an area of 6,319 square kilometres.⁸ The intensity of sampling will typically increase when the exploration phase starts.

Method	Summary
Boat-towed magnometer systems	• Useful for mapping placer deposits or diffuse deposits where the magnetic minerals are spread over a wide area • Recent research has demonstrated that titanomagnetite can be mapped well using a seabed sledge developed by the University of Bremen that measures magnetic fields over a sediment thickness of a metre or so. This work is still in progress⁹
Vibracore sampling	• Undertaken in water depths of up to 200 metres and to a target depth of 10 metres below the seafloor • A single 2 to 10 metre long core tube (deployed and operated by a cable) is driven into the subsoil • A vibration allows the tube to penetrate the surface • When the core is removed, the hole created collapses in on itself, leaving a slight depression in the seabed • Sampling sites are generally discrete and around 200 metres from each other • A vibraborer does not work well if the material is too coarse, well-consolidated or cohesive
Reverse circulation sampling	• Used for unconsolidated sediment samples • Tube is generally around 70 to 80 millimetres in diameter • A series of water jets drive a hole into the substrate • Often uses a drilling 'mud' which is mixed with water and pumped down into the hole • Samples withdrawn using an air-lift system • Hole created collapses in on itself, leaving a slight indentation in the seabed around 250 millimetres deep and 400 millimetres in diameter • Used for sampling along the Taranaki coast where six discrete samples per hole are taken at a depth of approximately 6 metres
Sonic drilling	• High-frequency vibrations sent down a drill 'string' (or sampling rod) to the drill bit • Vibrations resonate and magnify the amplitude of the drill bit, which then fluidises the seafloor • Has the ability to recover samples to 100 metres deep • Typically uses a 100 millimetre drill bit within a 150 millimetre external diameter sleeve • No materials discharged at sea • Sampling generally occurs at discrete distances between 100 to 200 metres • Drilling 'muds' are sometimes used
Side-scan sonar and multi-beam sonar	• Maps topography and sedimentary environment of seafloor • Provides information about the surface sediments in terms of texture and reflectivity rather than the minerals that are present • Side-scan sonar uses a towed sonar device to emit a conical or fan-shaped pulse of sound through the water across a wide angle • Device can be towed or mounted to the hull of the vessel • Water-depth data is collected using multi-beam sonar • The frequency of the sound is generally between 100 and 500 kilohertz, but closer to 100 kilohertz
Low-frequency echo sounder	• Used to map sand in harbours and on the continental shelf • Systems such as a CHIRP Subbottom Profiler can be used • These generally operate at between 3.5 and 15 kilohertz, which gives a maximum range of between 20 and 50 metres depending on the type of sediment • This corresponds to a peak sound output of 149 to 158 decibels (compared to 176 decibels for a standard 200 kilohertz echosounder)

Figure 11.2 Main methods for sand prospecting and exploration (*Adapted from Thompson K, 2012, and De Lange W, 2014*)

Active mining

Active mining of the seabed is generally undertaken using suction pipes to pump sand that has been mixed with seawater from the seabed. Iron ore is magnetically separated from the sand whilst other minerals are extracted by sieving, before returning the residue back to the sea.¹⁰ The sand mining methods used will vary, depending on the depth of the seafloor where the sand is being extracted from, and the distance from shore.

Offshore sand mining methods

Figure 11.3 shows the process used for mining offshore sand resources.

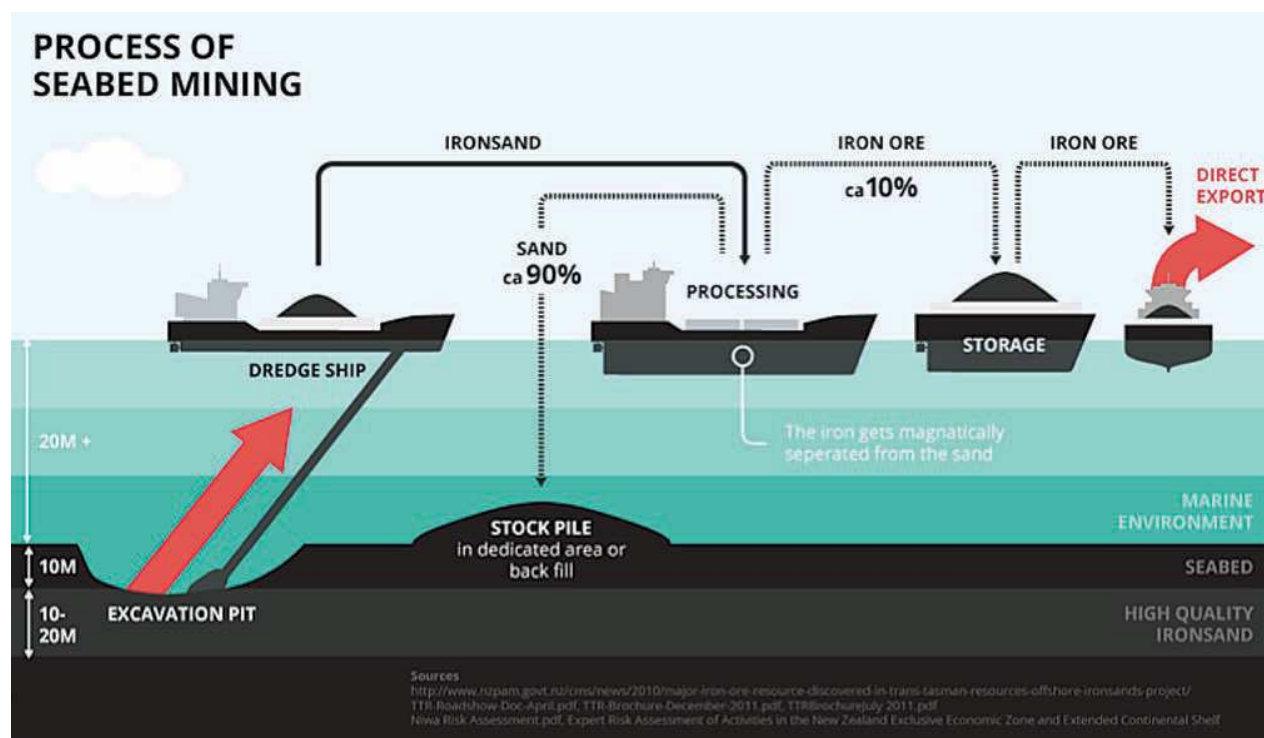


Figure 11.3 Process used for offshore sand mining (KASM¹¹)

Process proposed for offshore iron sand mining

Trans-Tasman Resources Limited applied to mine iron sands in the South Taranaki Bight using a subsea sediment extraction device called a 'Crawler'. This would create a mining cut approximately 12 metres wide and up to 11 metres deep. The seabed material would then be pumped by a slurry delivery pipe to a 'Floating Processing Storage and Offloading Vessel' where it would be processed to separate the iron ore (approximately 10 per cent yield). Processing would involve screening to remove material greater than two millimetres in size, and grinding and magnetic separation. The de-ored sand would have been redeposited into areas which had been mined.¹² The iron ore concentrate would be transferred to a Floating, Storage and Offloading Vessel where dewatering and storage would occur. It would then be transferred to bulk carrier vessels for export.

Inshore sand mining methods

Different methods are used for inshore mining, depending on the location of the sand resource. For example, at the Waikato North Head mine site, the iron sand is processed at an on-site concentration plant, where a magnetic concentrate is extracted through a series of separation processes producing a slurry. The slurry is pumped



to the Glenbrook Mill through an 18 kilometre long underground pipe. Freshwater used during this process is drawn from a lagoon adjacent to the Waikato River. The by-products from these processes are used for site rehabilitation after they have been dewatered.¹³

At the Taharoa mining site, sand is extracted from a pond by a floating dredge and then, because the area has no natural harbour, pumped as a slurry to a bulk cargo ship moored 3 kilometres from the shore plant to be processed and then exported.¹⁴ Between 200 and 300 tonnes of magnetic concentrate is produced per hour.¹⁵ The removal of sand from the Mangawhai-Pākirī embayment is generally completed using a suction dredge. This is typically a rotary cutter suction dredge where a rotating head disturbs the sediment, allowing it to then be entrained into the inflow of the suction pipe. Sand is then screened using mesh sizes of 3 millimetres, which removes gravel, shell and debris.

Potential environmental effects

There are a number of potential environmental impacts resulting from both inshore and offshore sand mining which are summarised in Figure 11.4 below and discussed in further detail following. Whether such effects will occur in practice, and their significance, is dependent on a number of factors including the equipment and methods used, the dredging frequency, the volume being extracted, and the physical characteristics of the area.¹⁶ Over the past two decades a number of Environment Court cases on seabed mining,¹⁷ and the recent Trans-Tasman Resource Marine Consent Decision (June 2014), have provided some important guidance on the significance of environmental impacts. Ways of reducing potential environmental effects are discussed below in the section on best practice.

Area	Potential impact
Destruction of benthic communities	• Mortality of organisms living on the seafloor, and within the seabed sediment, in the area mined • Smothering of benthic organisms inside and outside the mining area as a result of sediment plumes • Sediment in water column reducing light penetration, potentially affecting food webs
Disruption of sedimentary and coastal processes	• Physical changes to the habitat within the mined area • Changes to the seabed bathymetry affecting wave patterns, surf breaks, coastal stability and the sediment budget – dependent on vicinity of the mine site to shore, existing processes and the level of mining proposed
Disturbance of marine mammals, fish and birds	• Disruption to marine mammals that are dependent on sound for locating prey, navigating and communicating, from the acoustic impact of seismic surveys and use of mining equipment • Entanglement, injury or drowning in equipment • Birdstrike from the introduction of man-made light above the surface • Subsurface activity and lighting potentially interfering with the feeding and reproductive behaviour of marine mammals and seabirds¹⁸ • Increased risk of ship strike
Changes to natural character, landscape and amenity	• Reduction in the ‘naturalness’ of landscapes and the coastal environment due to the introduction of man-made structures, vessels and equipment • Reduction in recreational amenity
Other	• Increased biosecurity risks through overseas equipment acting as a vector for marine pest species • Reduction in fish stocks and spatial displacement of commercial fisheries • Disruption of marine traffic routes through exclusion zones around the mining area • Degradation of areas and resources of importance to iwi and hapū • Destruction or degradation of historic heritage (such as shipwrecks and other marine archaeology)

Figure 11.4 Potential environmental impacts from sand mining (Adapted from Secretariat of the Pacific Community, 2013, various Environment Court cases¹⁹ and Trans-Tasman Resource Marine Consent Decision, June 2014)

Seabed habitat

Any sampling of the seafloor during prospecting and exploration will disturb the benthic community to some extent. In addition, research has suggested that during drilling or suction sampling a 'sand plume' has the potential to be formed, risking the smothering of benthic life as suspended material settles.²⁰ Sand from a plume quickly drops onto the seabed due to its weight, but fine sediment (silt and particularly clay) stays suspended for longer, and therefore can travel further afield before it settles, causing wider impacts.

Seabed mining causes considerable disruption to aquatic habitats on the seabed as large amounts of material are removed, filtered and then returned, once the mineral has been extracted.²¹ Within the dredged area itself, benthic communities will be significantly disturbed and may be destroyed. These can include shellfish beds and other benthic fauna living within the sand being extracted.

Bulk extraction of sand and redeposition of tailings on the seabed cause large sediment plumes which can travel many kilometres. The significance of their impacts will depend on the communities affected and whether they are adapted to regular disruption. For example, they would be greater if the plume travels outside the mobile sandy habitat and settles onto rocky or biogenic reefs. These impacts will move as the site is progressively mined and new areas disturbed, and are likely to be cumulative over time and space.²²

Some coastal and continental shelf areas are used by fish as nursery grounds and feeding areas. Changes to the benthic fauna following large-scale removal of sand may result in cumulative impacts higher up the trophic levels, including changes to feeding patterns of fish.²³ For example, research has shown that juvenile red snapper use sandy shoal areas to feed on small crustaceans commonly found in such material.²⁴ In addition, destruction of benthic communities in one area may have implications for those elsewhere. A 2008 report on the Kaipara Harbour raised concerns about the impact that mining may be having on tuatua populations there. In particular, it was thought that the tuatua beds located in and around the sand extraction area may be an important brood-stock for other populations found in the harbour.²⁵

The recovery of benthic habitats post-exploration and mining will depend on the extent of impact and the species affected, but in general those living in mobile sand environments are adapted to frequent disruptions and have the ability to rapidly repopulate areas.



Sand mining can change coastal dynamics, thereby affecting surf breaks such as this one shown at Wainui Beach, Gisborne (Raewyn Peart)

Sedimentary and coastal processes

Dredging may affect the physical environment by altering the bathymetry, which in turn can change the velocities of currents and sedimentary processes.²⁶ In what are known as 'closed' systems, there may be little or no new sand entering the sediment cycle, depending on the size of the system. If sand is removed from these systems, there can be less available to be deposited on the beach, and erosion of the coastline may result as seabed reserves are depleted.



If sand is removed from closed systems, coastal erosion can result, as occurred at Ōmaha where expensive groynes were required to remedy the problem (*Raewyn Peart*)

Case law on sustainability of sand mining

In 2009 the High Court upheld a decision by the Environment Court to reverse the former Auckland Regional Council's decision to decline consent to McCallum Bros Limited. The consent was to renew dredging consents in the Mangawhai-Pākirī embayment at a rate of 76,000m³ per annum, an increase of 6,000 cubic metres per year above the previous 20 years. The Environment Court had granted consent for 14 years on the basis that the total input of sediment into the system meant that the proposed extraction was not unsustainable. It also concluded that shoreline retreat and erosion could not be attributed to sand extraction over the last 80 years.²⁷ This indicates that sand mining activities should be directed to areas where the amount to be extracted is less than new inputs into the system.

In 'open' systems new sand is constantly entering the cycle. This can include sand that is being transported along the coast from other areas by marine currents, sand being carried down rivers into the sea, and sand being produced by erosion of nearby coastal cliffs and headlands. In some areas, such as along the west coast of New Zealand, sand may also be transported into coastal areas from further out on the continental shelf. If the amount of sand removed from open systems by mining is less than the quantity of new sand entering the system from outside sources, coastal erosion is much less likely to occur. Ultimately, to avoid erosion it is desirable to maintain a positive sediment budget.

Cessation of sand mining due to erosion

The extraction of sand from the Pouto shoreline in the Northland Region has occurred for many decades, but has now been abandoned due to possible effects of the mining on shoreline erosion.²⁸ Several companies extracted sand from Pouto, particularly during the late 1970s. In 1992, Mt Rex was granted the sole resource consent for the continued extraction of sand, and this expired in 2004 and was not renewed. The resource consent allowed for a maximum of 60,000 cubic metres to be extracted per year. Monitoring of the Pouto shoreline began in 1990, at the request of the Department of Conservation. The monitoring confirmed that the shoreline was eroding and therefore that it was not appropriate to continue to extract large quantities of sand immediately adjacent to the shoreline.²⁹



Sand mining was stopped at Poutu in the Kaipara Harbour due to concerns about the eroding shoreline (Raewyn Peart)

Marine mammals, fish and seabirds

Seabed sand mining, through the introduction of equipment, increased underwater noise levels and potential ecological changes to the marine environment, can have a range of impacts on marine mammals, fish and seabirds. Small coastal cetaceans, such as Hector's and Maui's dolphins, are at particular risk from human activities such as sand mining, due to their restricted ranges and the cumulative impacts of many activities affecting one location.³⁰ Figure 11.5 provides some examples of the potential impacts of various sand mining activities on marine mammals.

Mining activity	Example of potential impact	Potential level of impact
Boat-towed surveying equipment	• Behavioural changes	Low – for example, no behavioural changes were observed during research work in the Bay of Plenty and Hauraki Gulf ³¹
Reconnaissance drilling and sampling	• Behavioural changes	Proximity dependent

Mining activity	Example of potential impact	Potential level of impact
Suction sampling	• Avoidance • Habitat degradation	Proximity dependent
Sonar mapping	• Habitat degradation • Behavioural changes • Avoidance of the area	Low ³²
Shallow seismic surveying (low-intensity subbottom profiling)	• Behavioural changes • Displacement from habitat • Changes in vocalisation	Low
Vessel activity	• Avoidance • General habitat degradation • Behavioural changes	Potentially high
Active mineral extraction	• Habitat degradation • Avoidance • Behavioural responses • Potential for indirect effect due to the negative effects on prey	Potentially high

Figure 11.5 Examples of the potential impacts of sand mining off the North Island west coast on marine mammals
(Adapted from Table 5, Thompson K, 2012)

Noise is a concern with marine mining activities, particularly in relation to its potential effects on marine mammals. Seismic surveys are known to impact on marine mammals. In addition, any drilling activities, suction and pumping, and routine maintenance will produce underwater noise. Helicopter overflight also has a potential to disturb marine mammals. Noise from vessel traffic, particularly that associated with active mining, is highly likely to significantly raise the local ambient noise levels.³³ The consequences on marine mammals and other fauna will vary from none to behavioural (may leave the area) to acute injury (ear drum damage) to serious (death) depending on noise level encountered, species and habitat. There may also be cumulative effects from repeated exposure.³⁴

Survey and support vessels create a risk of ship strike. The risk will depend on the speeds of the vessel involved, with greater speeds creating greater risk. During surveying, the vessels tend to move slowly (typically operating at around 5 knots) and special flags are flown to warn other vessels of the slow movement.³⁵

Dolphin conditions on sand mining consent

In the final decision of the Kaipara Harbour sand mining application in 2006 a new condition was inserted relating to increased vessel traffic in the area: 'Vessels used by the Consent Holder will, to the extent practicable, travel at a speed that results in no, or minimal, wake when within 300 metres of dolphins'.³⁶ In addition an Advice Note was inserted: 'Sightings of Maui's dolphins will be reported by the Consent Holder to the WWF Sightings Network. Where practicable detail on the sighting including GPS coordinates, number of dolphins, and date and time of sighting, will be provided'.³⁷ This new condition recognises the impact sand mining activities can have on marine mammals, particularly rare species.

Large-scale increases in water turbidity have been shown to impact some marine fish species inhabiting these areas, by lowering feeding rates.³⁸

Lights at night on vessels can attract and disturb seabirds and may cause them to collide with the above-sea-surface structure. Subsurface activity and lights may affect marine mammals by interfering with their navigation.³⁹

Natural landscapes, character and amenity values

Seabed sand mining can potentially impact on natural landscapes, character and amenity, particularly when it is close to shore or in the foreshore area. This includes unwelcome visual impacts from barges and tugs operating in the marine area, particularly in those

areas which are undeveloped and normally little used by vessels. However, the significance of these impacts is directly related to the location of the sand mining operation, and the level of 'naturalness' of the background environment.

Findings on effects of sand mining on natural character and amenity

In 2006 there were appeals against the former Auckland Regional Council's decision to refuse consent for the continuing extraction of sand from near-shore areas of Mangawhai-Pākirī Bay by Sea-Tow Limited. Sea-Tow had applied for consents to enable extraction of 27,000 cubic metres of sand per year for 20 years. McCallum Bros applied for consents to enable extraction of 49,000 cubic metres of sand per year for 20 years. Both extractions were to continue activities authorised under previous expired consents and were for discretionary activities.⁴⁰ Parties within the meaning of section 274 of the RMA variously claimed that extraction of sand from a closed system such as this would result in degradation of the natural character of the beach. The Court held that the Hauraki Gulf is not a wilderness waterway. It is frequented by pleasure boats, shipping and other craft. Although a barge and tug passing regularly may not be an attractive view to residents or leisure users of the beach, their presence at sea could not be deemed markedly out of place. It held there were no more than minor adverse visual effects. The Court also held that there would not be any significant adverse effect on natural character. This was largely related to the expectation that the resource would not be depleted because the system was not closed.⁴¹

Other

Equipment and vessels brought into New Zealand from overseas can act as vectors for non-indigenous species.⁴² These impacts are discussed in detail in Chapter 5: Marine biosecurity.

Mining may negatively impact on areas and resources of importance to Māori. These could include kaimoana areas, wāhi tapu or urupa sites which lie on or under the seabed. This may be through the direct impacts of the mining activity on the seabed, through accelerated coastal erosion caused by the removal of sand from the system, or through the wider impacts of the sediment plume.

Consideration of impacts of sand mining on Māori cultural and spiritual values

The Environment Court heard appeals against the former Auckland Regional Council's decision to grant replacement consents for sand extraction in the Kaipara Harbour in 2006. The Oruawharo Marae Trust opposed the applications on the grounds that continued sand extraction would have a negative adverse effect on Māori cultural and spiritual values, and that the extraction site was wāhi tapu, among other matters. The Court held that the concerns raised by the Trust did not of themselves provide a basis for declining or recommending against the granting of consents, and that monitoring and review conditions were satisfactory.⁴³



The Environment Court found that mining from near-shore areas of Mangawhai-Pākirī Bay, shown here, would not have more than minor visual effects and would not have significant effects on natural character (Raewyn Peart)

Decision-making framework

Allocation of mining rights

Any party wishing to undertake prospecting, exploration or mining activities must obtain a permit issued under the Crown Minerals Act. Applications are processed by New Zealand Petroleum and Minerals which is a branch of the Ministry of Business, Innovation and Employment. Under the Act, the Crown creates minerals programmes.⁴⁴ The programme that applies to the exploitation of sand resources is the 'Minerals Programme for Minerals (excluding Petroleum) 2013'. This sets out the policies and procedures to be followed for the allocation of mineral resources (including consultation with iwi and hapū). The requirements to be met by permit holders are defined in the regulations. It does not address environmental impacts, which are addressed under other legislation.

The current minerals programme provides that mining for sand in the coastal marine area does not require a permit under the Crown Minerals Act, but mining for minerals that are a component of sand, such as titanomagnetite, does require a permit. Hence iron sand mining is governed by a different regime to the mining of sand for aggregate.

Management under the RMA

The environmental effects of sand mining within the coastal marine area are managed under the RMA. The regional coastal plan will determine if a resource consent is required, and if so, what kind of consent. In practice, a proponent will normally need to obtain a resource consent to undertake sand mining. When an application is lodged, the consent authority will determine whether or not it should be publicly notified. A consent application must be publicly notified if the consent authority determines that its adverse effects on the environment are likely to be more than minor.⁴⁵

When considering an application for consent, the consent authority must also have regard to the relevant provisions of the applicable statutory instruments made under the Act, for example the NZCPS (shown in Figure 11.6), regional policy statements and the regional coastal plan.



The impacts of sand mining on seabirds, such as those shown here at Te Arai, Auckland, need to be considered as part of sand mining consent applications (Raewyn Peart)

The NZCPS contains a number of policies that are relevant to applications for minerals activities, including:

- It is a national objective to recognise that some uses and developments which depend upon the use of natural and physical resources in the coastal environment are important to the social, economic and cultural well-being of people and communities⁴⁶
- It is recognised that the extraction of minerals is an activity which is important to the social, economic and cultural well-being of people and communities⁴⁷
- A precautionary approach is to be adopted towards proposed activities whose effects on the coastal environment are uncertain, unknown, or little understood, but potentially significantly adverse⁴⁸
- Adverse effects on certain indigenous biodiversity, such as taxa which are threatened or at risk or naturally rare, must be avoided⁴⁹
- Adverse effects must be avoided in areas with outstanding natural character⁵⁰
- Adverse effects must be avoided in outstanding natural landscapes⁵¹
- Requiring that subdivision, use or development will not result in a significant increase in sedimentation in the coastal marine area, or other coastal water⁵²
- Protecting surf breaks of national significance by ensuring that activities in the coastal environment do not adversely affect the surf breaks⁵³
- In managing discharges to water in the coastal environment have particular regard to avoiding significant adverse effects on ecosystems and habitats after reasonable mixing⁵⁴

Figure 11.6 NZCPS policies relevant to sand mining applications

Often the scientific evidence supporting an application contains considerable uncertainties, because the marine environment is a dynamic system subject to constant change. Obtaining scientific data can be expensive and challenging because of the difficult environment within which the research is being undertaken. A lack of certain scientific information means that there is often disagreement about how coastal sand systems work.

Adaptive management is an approach which has developed to address scientific uncertainty. It is about informed trial and error, essentially learning as the mining progresses. There is an emphasis on monitoring and measurement (against pre-defined thresholds), learning what works best, and then adapting management processes accordingly. Monitoring needs to be long-term, and is a major commitment.⁵⁵ Increasingly, consent authorities deal with the lack of information by granting consent accompanied by strict conditions providing that a programme of adaptive management and monitoring must be followed. Conditions can provide that reviews of the consent must be undertaken once particular extraction quantities have been reached. This approach is discussed further in Chapter 3: EEZ legislation.

Case law on need for adaptive management and monitoring

In 2006, the Royal Forest and Bird Society sought the imposition of a scientifically-credible monitoring regime to protect conservation values, when replacement consents were granted for sand extraction in the Kaipara Harbour. This included an 'adaptive management regime' involving the undertaking of baseline studies, particularly in relation to Maui's dolphin, as a pre-condition of increases in extraction rates.⁵⁶ In relation to marine mammals, the Environment Court concluded that there was no evidence that harm had arisen from the current high level of operations, and that the risk of adverse effects was less than minor and probably negligible. The Environment Court also rejected a sophisticated monitoring programme proposed by a submitter in relation to shorebirds on the basis that the effects would be less than minor. Essentially, this means there is a need to demonstrate a more than minor risk of harm before the Court is likely to require an adaptive management regime.

Management under the EEZ Act

Under the EEZ Act, prospecting and exploration for sand are permitted, subject to conditions. Mining is a discretionary activity. This means that a marine consent is required for sand mining, and the application for consent is notified, allowing the public to make submissions and participate in the hearing. To date, no minerals activities have been classified as non-notified discretionary. Further details about requirements for sand mining under the EEZ Act are outlined in Chapter 3: EEZ legislation, and in the case study at the end of this chapter.

Elements of best practice

As described above, sand mining can potentially have a number of negative impacts on the environment; as a result, it is important that it is only undertaken in appropriate areas. Sand mining activities should avoid, where possible:

- Closed sand systems – depending on how big the closed system is and how much sand is being extracted over what time period
- Sediment with a high silt content
- Benthic habitats of high ecological value
- Significant shellfish beds
- Fish spawning areas and areas of high fish density (can be time dependent)
- Areas frequented by threatened or endangered marine mammals including whales, dolphins and sea lions
- Important seabird areas
- Areas of high cultural or archaeological significance
- Areas of recreational significance including areas near high-quality surf breaks



Sand mining should not be undertaken in areas frequented by threatened or endangered marine mammals such as the New Zealand sea lions shown here at Sandfly Bay, Otago Peninsula (Raewyn Peart)

When sand mining is authorised, strict conditions are generally included as part of the consent, including a requirement for a robust monitoring regime to identify any negative environmental effects. Some of the Environment Court cases on sand mining found that the precautionary approach was appropriate, considering that the extractions were to occur within dynamic systems subject to constant change. However, these cases generally took the view that monitoring and other conditions could encompass this precaution.⁵⁷

With respect to offshore sand mining applications, there is limited knowledge of the EEZ, as only a small proportion of this expansive area has been scientifically investigated. Therefore, applicants will need to undertake significant scientific investigations before lodging a marine consent application. The applicant must have detailed information about the receiving environment, the proposed operation, and existing interests. But even given such baseline information, there are still likely to be considerable uncertainties about impacts. This means that the caution requirement contained in the EEZ Act will be a key consideration for decision-makers.

Decision on sand mining application within the EEZ

The Trans-Tasman Resources Decision Making Committee refused the marine consent application and stated, 'Overall, we think this application was premature. More time to have better understood the proposed operation and the receiving environment and engage more constructively with existing interests and other parties may have overcome many of the concerns we have set out in this decision ... Ultimately, the information we have to make our decision, while voluminous, was too uncertain and inadequate, and we did not have sufficient confidence in the adaptive management approach proposed to address that uncertainty to enable the activity to be undertaken. For all of these reasons, the application as presented to us does not meet the sustainable management purpose of the EEZ Act'.

Decision-makers will need to assess the extent to which an adaptive management approach can diminish the environmental risk and the uncertainty, prior to deciding if the activity can proceed. Decision-makers should consider whether:

- There will be good baseline information about the receiving environment
- The conditions provide for effective monitoring of adverse effects using appropriate indicators

- Thresholds are set to trigger remedial action before the effects become overly damaging
- Effects that might arise can be remedied before they become irreversible

In addition to the areas to be avoided (discussed above) the following best practice considerations outlined in Figure 11.7 can be applied to sand mining activities, both offshore and inshore.

Area	Detail
Destruction of benthic communities	• Mine in a chequerboard or strip pattern to allow recolonisation from adjacent populations • Ensure that sufficient research has been undertaken to identify what species are likely to be present in the area to be mined and their distribution • Minimise the area which is directly affected by mining activity • Assess options for biodiversity offsetting as appropriate to protect or restore seabed • Adopt mining methods which minimise the amount of sediment which is released into the water column
Disruption of sedimentation and coastal processes	• Monitor changes in shoreline position • Complete beach profiles prior to mining commencing • Minimise the area which is directly affected by mining activity
Disruption of marine mammals, fish and seabirds	• Ensure a comprehensive understanding of the presence, distribution, abundance and density of marine mammals in the area both pre-operation and during operation to assist to determine whether and to what extent mining operations and associated activities may be impacting on marine mammals • Make sure all personnel on board project vessels receive training in regards to monitoring and management of any impacts on marine mammals and seabirds, such as identification and behaviour observation • Limit unnecessary lighting and ensure that lights that are required are directed vertically onto work surfaces • Monitor bird strike and review mitigation measures in response to bird mortality above an identified level • Minimise underwater noise levels • Reduce the speeds of vessels in the vicinity of marine mammals
Changes to natural character and amenity	• Assess options for biodiversity offsetting as appropriate to protect or restore inshore area
Other	• Regularly check, clean and apply antifouling paints (preferably copper-free) to infrastructure that is brought into New Zealand's marine area from overseas. Biosecurity New Zealand has prepared a craft risk management standard – discussed in Chapter 5: Marine biosecurity. • Adopt an adaptive management approach, where the impacts of a pilot operation are monitored and assessed before considering whether expanding to full production can be allowed or whether amendments are required • Undertake rigorous baseline monitoring prior to lodging application so that sufficient information is available during the hearing and so that ongoing monitoring can be compared to the baseline • Develop and undertake a robust monitoring regime – this should include biological and geomorphological monitoring as well as baseline monitoring, so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner

Figure 11.7 Best practice considerations for sand mining activities

South Taranaki Bight iron sand extraction

Trans-Tasman Resources Limited holds a mining permit to undertake iron ore extraction in the South Taranaki Bight. In October 2013 the company applied for marine consent under the EEZ Act to undertake iron ore extraction in a 66 square kilometre area in the South Taranaki Bight. In June 2014 the EPA Decision-Making Committee refused the marine consent application.

The project would have involved the excavation of up to 50 million tonnes (27 million cubic metres) per year of seabed material containing iron sand for processing on an offshore floating vessel. Approximately 10 per cent of the material would have been processed offshore into iron ore for export, with the remaining material returned to the seabed.⁵⁸

The company prepared an environmental impact assessment, as well as more detailed technical reports, to support its application for marine consent. The Decision-Making Committee considered the effects on the environment as detailed in Figure 11.8.

Area	Detail
Sediment plume	The sediment plume was an important consideration as many effects of the proposal would result from it. A sediment plume model was provided and this was relied on to analyse other environmental effects. The Committee accepted that the plume model was fit for purpose and had an accuracy of -50 per cent to +100 per cent.
Primary productivity	The sediment plume would reduce light penetration in the water column and therefore reduce primary production. Modelling predicted that there would be a three- to four-fold reduction in light penetration at the mining site and a two-fold change in light penetration 10 kilometres east of the mining site. It predicted that this would result in a 10 percent reduction in total primary productivity in the South Taranaki Bight. The Committee concluded that there was considerable uncertainty as to the effects of changes in primary productivity on the wider ecosystem.
Benthic environment	The application area is a high-energy environment characterised by soft-sediment habitats subject to regular disturbance and colonised by fast-growing, opportunistic species. The area to be mined would be significantly disturbed, with near total mortality of benthic fauna, and with recovery taking approximately 10 years. The benthic environment outside the mining area would be subjected to sedimentation and reduced light levels as a result of the sediment plume.
Marine mammals	A number of dolphins, whales and seals could be present in the proposed project area, including a number with threat classifications (such as Maui's dolphin, Southern/Antarctic blue whale and orca). Despite finding that the application area is unlikely to be primary habitat for Maui's dolphin, the Committee determined that the threat status of the Maui's dolphin meant that a very cautious approach was required. The Committee reached a similar conclusion in relation to blue whales. The activity had the potential to cause the following adverse effects on marine mammals: loss of food resources, displacement, noise effects, risk of collision and entanglement with vessels. The Committee considered that a comprehensive and longer-term baseline study of the marine mammals present in the South Taranaki Bight would have assisted it to understand the importance of the area to various species and what they use the area for. The absence of such information meant the Committee was uncertain as to the significance of the area to cetaceans. However, the consent conditions proposed by the company were comprehensive and may have addressed those concerns if consent was granted.
Fish	The species richness of the reef, demersal and pelagic fish assemblages in the South Taranaki Bight is moderate on a national scale. The activity had the potential to cause the following adverse effects on fish: effects of suspended sediment and sedimentation, effect of heavy metals released into the water column, loss of benthic food sources, displacement of fish and loss of habitat, loss of spawning nurseries, noise, entrainment of fish in mining apparatus, and light spill from mining operations. The Committee accepted that the effects on fish were unlikely to be significant, provided no significant food web effects arose due to primary production changes.
Seabirds	No field surveys were undertaken to inform the report on seabirds. As a result, the Committee could not reach a conclusion as to the importance of the South Taranaki Bight for seabirds. The activity had the potential to cause the following adverse effects on seabirds: direct effects of vessel lighting and noise, effects of sediment on visual foragers and effects on the food web, including habitat loss and the loss of foraging areas. The Committee concluded that it lacked an understanding of how important the South Taranaki Bight is for seabirds and therefore the significance of the potential effects.

Area	Detail
Biosecurity	There is a risk that non-indigenous organisms could be introduced to New Zealand by ships transiting into New Zealand to support the proposed activities. The Committee accepted that these would be managed under the Biosecurity Act.
Human health	The Committee considered four issues: air discharges from the combustion of heavy fuel oil in the engines on the vessels, contamination to seafood from the proposed operation, risk to other users of the South Taranaki Bight (e.g. recreational) and health and safety of workers. The Committee concluded that the effects on human health would have been manageable through consent conditions.
Physical environment	The Committee considered four issues: seabed bathymetry, waves and surf, coastal stability and the sediment budget and freshwater springs. It concluded that there would not be significant adverse effects on the physical environment. There was no evidence that changes to the seabed bathymetry would cause more than minor effects on waves, surf, coastal stability or the sediment budget.
Visual, natural features, natural landscapes and natural character	The mining activity, and the sediment plume created, had the potential to cause adverse visual effects. The Committee found that visual effects from the mining activity would be minor and visual effects from the plume would be moderate. The impact on natural features and landscapes, outside the mining area itself, would be minor. The effects on natural character would be minor to moderate at the mining site and minor in coastal waters.
Archaeology	The Committee considered the potential for shipwrecks or other archaeological artefacts to be present in the proposed mining site and for these to be destroyed by the mining operations. It was satisfied that a proposed 'Discovery Protocol for Shipwreck Finds' would have appropriately addressed archaeological effects.

Figure 11.8 Summary of decision on application to extract sand from the South Taranaki Bight

The Decision-Making Committee also considered effects on existing interests:

- Iwi – iwi were concerned that they had not received adequate information to allow them to make an informed response to the proposal. The Committee found that iwi's existing interests would potentially be adversely affected in more than a minor way by the project. It acknowledged that the proposed consent conditions sought to involve iwi in a substantive way. However, iwi were not engaged in drafting those consent conditions and had not agreed to them.
- Fishing – the Committee heard evidence regarding potential effects on commercial, recreational and customary fishing. It concluded that the 'effects [on commercial and recreational fisheries] are at best uncertain'. It also concluded that there was 'considerable uncertainty' with respect to customary fishing.
- Recreation and tourism – the Committee found that it was unlikely that the proposal would harm New Zealand's international reputation
- Origin Energy Resources – the proposed mining area is adjacent (1.14 kilometres) to the existing Kupe platform. The Committee found that Origin Energy Resources' interests could be satisfactorily addressed, predominately through existing Crown Minerals Permit conditions.
- Marine traffic – the proposed mining area is removed from regular marine traffic routes and therefore no conflict was considered likely



The Decision-making Committee considered that the effects on recreational users of the Taranaki Bight of the sand mining proposal would have been manageable through consent conditions (Raewyn Peart)

The Committee determined that there was considerable uncertainty in the information provided to it as to both the nature of the environment and the way the mining operation might affect it. It also found information on how Māori existing interests would be addressed to be inadequate.⁵⁹ As a result, under the legislation, the Committee was required to favour caution and environmental protection and to consider whether an adaptive management approach would allow the activity to be undertaken (section 61 of the EEZ Act).

Trans-Tasman Resources did not propose an adaptive management approach that would enable the activity 'to commence on a small scale or for a short period so that its effects on the environment and existing interests can be monitored' for commercial reasons. Instead, the applicant proposed an approach which involved:⁶⁰

- Qualitative environmental performance objectives (specified in the consent conditions)
- Pre-operational baseline monitoring
- A process for developing quantitative 'trigger indicators or values' which would inform compliance with environmental performance objectives
- Operational monitoring
- Management of operations to achieve the identified environmental performance objectives

The Decision-Making Committee assessed the approach against the following criteria:

- The extent of the environmental risk
- The importance of the activity
- The degree of uncertainty
- The extent to which an adaptive management approach will sufficiently diminish the risk and uncertainty

The Trans-Tasman Resources' adaptive management approach prescribed a process for the establishment of quantitative triggers based on qualitative 'Environmental Performance Objectives'. This meant that the Environmental Performance Objectives were the 'cornerstone pillars' of the adaptive management approach.⁶¹ The Decision-Making Committee was concerned that the Environmental Performance Objectives were 'very general and non-specific'.⁶² They contained terms such as 'no more than moderate', 'moderate impacts' or 'no more than moderate and temporary adverse effects'. The Committee was not confident that these could be reasonably translated into trigger values by a third party. The Decision-Making Committee was also concerned 'that, without the option of a scale or staged implementation, the applicant has not provided any compelling adaptive management alternatives should the environmental objectives not be met'.⁶³

The Decision-Making Committee concluded, 'Ultimately, the information we had to make our decision, while voluminous, was too uncertain and inadequate, and we did not have sufficient confidence in the adaptive management approach proposed to address that uncertainty to enable the activity to be undertaken. For all of these reasons, the application as presented to us does not meet the sustainable management purpose of the EEZ Act'



The Trans-Tasman Resources proposal was found to be unlikely to have an effect on waves, surf, coastal stability or the sediment budget off the Taranaki coast (*Neale Wills*)

Further reading and references

Allsopp M, C Miller, R Atkins, S Rodcliffe, I Tabor, D Santillo and P Johnston, 2013, *Review of the current state of development and the potential for environmental impact of seabed mining operations*, University of Exeter, Exeter, available at <http://www.greenpeace.to/greenpeace/wp-content/uploads/2013/07/seabed-mining-tech-review-2013.pdf>

Batson, P, 2003, *Deep New Zealand: Blue water, black abyss*, Canterbury University Press, Christchurch

Badesab F, T von Dobeneck, K Bryan, H Müller, R Briggs, T Frederichs and E Kwohl, 2012, 'Formation of magnetite-enriched zones in and offshore of a mesotidal estuarine lagoon: An environmental magnetic study of Tauranga Harbour and Bay of Plenty, New Zealand', *Geochemistry Geophysics Geosystems*, 13, Q06012

Bollmann M, T Bosch, F Colijn, R Ebinghaus, R Froese, K Güssow, S Khalilian, S Krastel, A Körtzinger, M Langenbuch, M Latif, B Matthiessen, F Melzner, A Oschlies, S Petersen, A Proelß, M Quaas, J Reichenbach, T Requate, T Reusch, P Rosenstiel, J O Schmidt, K Schrottke, H Sichelschmidt, U Siebert, R Soltwedel, U Sommer, K Stattegger, H Sterr, R Sturm, T Treude, A Vafeidis, C van Bernem, J van Beusekom, R Voss, M Visbeck, M Wahl, K Wallmann and F Weinberger, 2010, *World ocean review: Living with the oceans*, Maribus gGmbH, Hamburg

Craw A, 2013, *Deep seabed mining: An urgent wake-up call to protect our oceans*, Greenpeace International, Amsterdam, available at <http://www.greenpeace.org/canada/en/campaigns/ocean/Seafood/Resources/Reports/Deep-Seabed-Mining/>

De Lange, W, 2014, pers. comm., 6 August

Environmental Protection Agency, 2014, *EEZ004 South Taranaki Bight offshore iron sand project Trans-Tasman Resources Limited*, EPA Staff Report, Environmental Protection Authority, Wellington

Haggitt T, S Mead and M Bellingham, 2008, *Review of environmental information on the Kaipara Harbour marine environment*, Prepared by ASR/CASL for Auckland Regional Council, Auckland Regional Council Technical Publication No. 354, Auckland Regional Council, Auckland

Hume T, R Ovenden and I MacDonald, 2012, *Coastal stability in the South Taranaki Bight – Phase 1: Historical and present day shoreline change*, Prepared for Trans-Tasman Resources Limited, NIWA, Hamilton

International Marine Minerals Society, 2001, *Code for environmental management of marine mining*, International Marine Minerals Society, Honolulu, available at http://www.immsoc.org/IMMS_downloads/codefeb2002.pdf

Ministry for the Environment, 2005, *Offshore options: Managing environmental effects in New Zealand's Exclusive Economic Zone*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/offshore-options-jun05/index.html>

Ministry for the Environment, 2007, *Improving regulation of environmental effects in New Zealand's Exclusive Economic Zone: Discussion paper*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/nz-exclusive-economic-zone-discussion-paper-aug07/index.html>

Ministry for the Environment, 2008, *Cabinet Paper: Proposal for Exclusive Economic Zone environmental effects legislation*, Unpublished paper available at <http://www.mfe.govt.nz/issues/oceans/current-work/cabinet-paper-eez-effects-legislation.html>

Ministry for the Environment, 2012, *Managing our oceans: A discussion document on the regulations proposed under the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Bill*, Ministry for the Environment, Wellington

Ministry of Economic Development, 2008, *Minerals programme for minerals (excluding petroleum) 2008*, Ministry of Economic Development, Wellington, available at <http://www.crownminerals.govt.nz/cms/pdf-library/minerals/minerals-programme-for-minerals-2008>

NIWA, 2012, *Expert risk assessment of activities in the New Zealand Exclusive Economic Zone and extended continental shelf*, Prepared for the Ministry for the Environment, NIWA, Wellington

Parliamentary Commissioner for the Environment, 1998, *Compliance monitoring of a coastal permit for sand extraction, Otamarakau Beach, Bay of Plenty*, Parliamentary Commissioner for the Environment, Wellington, available at <http://www.pce.parliament.nz/assets/Uploads/Reports/pdf/sand.pdf>

Pearl R and K Mulchay, 2010, *Managing the marine environment: An EDS community guide*, Environmental Defence Society, Auckland

Secretariat of the Pacific Community, 2013, *Deep sea minerals and the green economy*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 2, Secretariat of the Pacific Community, Noumea

Sinclair Knight Merz, 2013, *Review of technical reports relating to TTR marine consent application – oceanographic processes and the physical environment*, Prepared for the Environmental Protection Authority, Sinclair Knight Merz, Auckland, available at http://www.epa.govt.nz/Publications/Trans-Tasman%20Resources%20Irons%20Sand%20Project_Review%20of%20technical%20reports%20on%20physical%20environment.pdf

Suhr L, 2008, 'Environmental protection in deep seabed mining: international law and New Zealand's approach', *New Zealand Journal of Environmental Law*, 12, 97–144

Thompson K, 2012, *Maui's and mining: A review of marine mineral mining activity on the West Coast of New Zealand and its potential impacts*, Prepared for Department of Conservation Auckland Area Office, Department of Conservation, Auckland

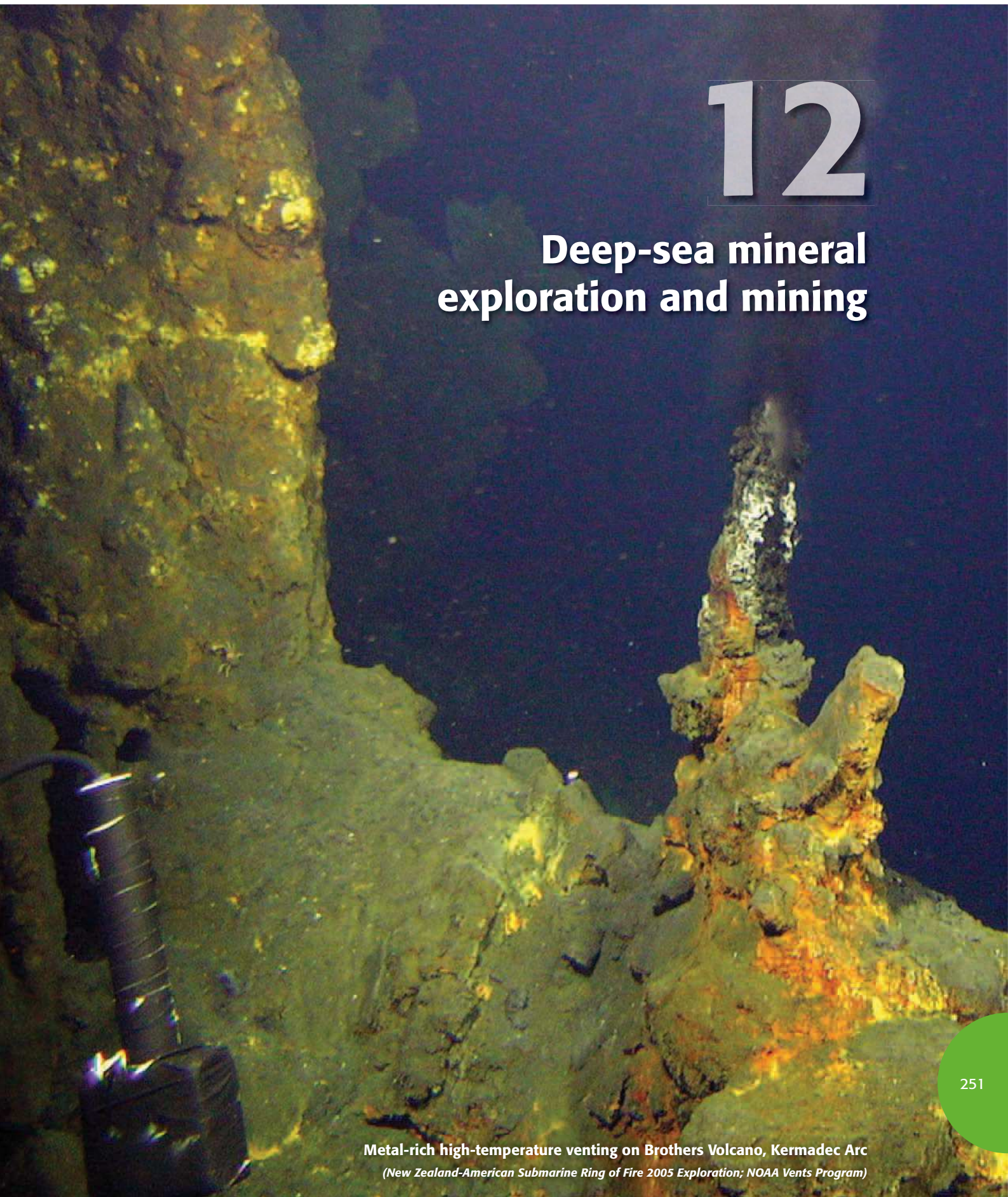
Endnotes

- 1 <http://kasm.org.nz/seabed-mining/seabed-mining-new-zealand-case-study/>
- 2 <http://www.teara.govt.nz/en/marine-minerals/page-2>
- 3 <http://www.nzsteel.co.nz/new-zealand-steel/the-story-of-steel/the-mining-operations/waikato-north-head-mine-site/>
- 4 <https://www.ipenz.org.nz/heritage/itemdetail.cfm?itemid=134>
- 5 <http://www.teara.govt.nz/en/marine-minerals/page-2>
- 6 Thompson K, 2012
- 7 Thompson K, 2012
- 8 <http://kasm.org.nz/stopsandmining/assets/Permit-50383-Trans-Tasman-Resources-Ltd-Crown-Minerals-Permits-Issued.pdf>
- 9 Badesab F et al., 2012
- 10 https://www.niwa.co.nz/our-science/freshwater/tools/kaitiaki_tools/land-use/mining/activities
- 11 <http://kasm.org.nz/seabed-mining/what-is-seabed-mining/>
- 12 http://www.epa.govt.nz/Publications/TTR_Impact_Assessment_Summary_FINAL_21_October.pdf
- 13 <http://www.nzsteel.co.nz/new-zealand-steel/the-story-of-steel/the-mining-operations/waikato-north-head-mine-site/>
- 14 <http://www.ipenz.org.nz/heritage/itemdetail.cfm?itemid=134>
- 15 <http://www.nzsteel.co.nz/new-zealand-steel/the-story-of-steel/the-mining-operations/taharoa-mine-site/>
- 16 Haggitt T, S Mead and M Bellingham, 2008
- 17 Pearl R and K Mulchay, 2010, Appendix One
- 18 Secretariat of the Pacific Community, 2013
- 19 Pearl R and K Mulchay, 2010, Appendix One
- 20 Thompson K, 2012
- 21 https://www.niwa.co.nz/our-science/freshwater/tools/kaitiaki_tools/land-use/mining/activities
- 22 <http://www.ttrl.co.nz/assets/Documents/Project-overview-and-effects-web-version-20-Nov-2013.pdf>
- 23 Thompson K, 2012, 27
- 24 Thompson K, 2012, 27
- 25 Haggitt T, S Mead and M Bellingham, 2008
- 26 Haggitt T, S Mead and M Bellingham, 2008
- 27 *The Friends of Pakiri Beach v The Auckland Regional Council and Anor* HC AK CIV-2006-404-3544 [26 March 2009]
- 28 Haggitt T, S Mead and M Bellingham, 2008
- 29 <http://www.nrc.govt.nz/upload/3465/Natural%20Character%20of%20the%20Coast.pdf>
- 30 Thompson K, 2012, 29
- 31 De Lange W, 2014
- 32 De Lange W, 2014
- 33 Thompson K, 2012
- 34 NIWA, 2012
- 35 De Lange W, 2014
- 36 *Oruawhoro Marae Trust v Auckland Regional Council* NZEnvC A113/2006
- 37 *Oruawhoro Marae Trust v Auckland Regional Council* NZEnvC A113/2006
- 38 Thompson K, 2012, 27
- 39 NIWA, 2012
- 40 *Sea-Tow Limited v Auckland Regional Council* A066/06
- 41 Pearl R and K Mulchay, 2010, Appendix One

- 42 NIWA, 2012
- 43 *Oruawhoro Marae Trust v Auckland Regional Council* A083/06
- 44 Crown Minerals Act 1991, section 13
- 45 RMA, section 95(2)(a)
- 46 NZCPS, Objective 6
- 47 NZCPS, Policy 6
- 48 NZCPS, Policy 3
- 49 NZCPS, Policy 11
- 50 NZCPS, Policy 13
- 51 NZCPS, Policy 15
- 52 NZCPS, Policy 22(2)
- 53 NZCPS, Policy 16
- 54 NZCPS, Policy 23(1)(d)
- 55 <http://www.niwa.co.nz/publications/wa/water-atmosphere-2-february-2011/oceans-treasure>
- 56 Peart R and K Mulchay, 2010, Appendix One
- 57 Peart R and K Mulchay, 2010, Appendix One
- 58 <http://www.ttrl.co.nz/assets/Documents/Project-overview-and-effects-web-version-20-Nov-2013.pdf>
- 59 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [138]
- 60 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [793]
- 61 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [816]
- 62 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [820]
- 63 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [850–851]

12

Deep-sea mineral exploration and mining





Contents

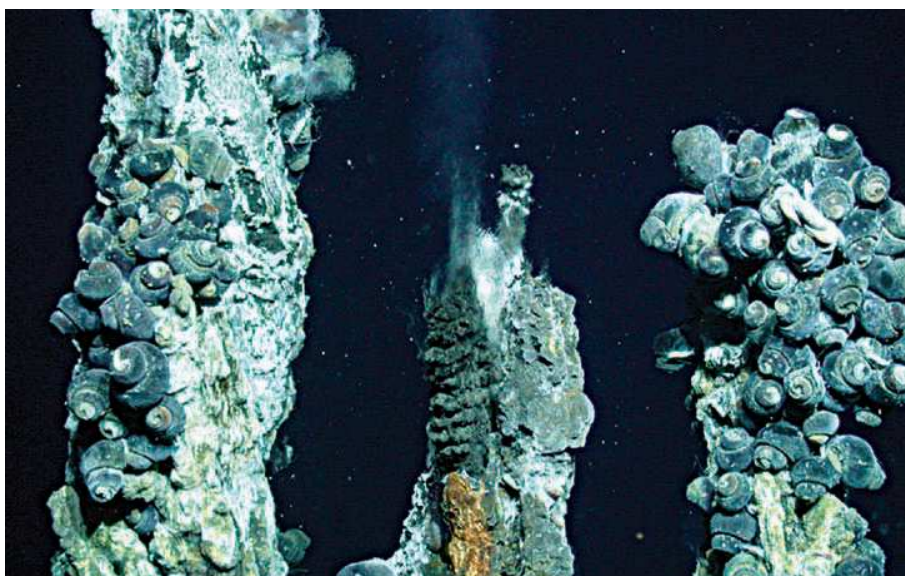
253	Introduction
253	Mineral resources
255	Mining methods
255	Stages of work
256	Phosphates
257	Seafloor massive sulphides
257	Manganese nodules and cobalt crusts
258	Methane hydrates
259	Potential environmental effects
259	Seabed and water column
260	Marine mammal and seabirds
260	Biosecurity
261	Other impacts
261	Potential environmental impacts specific to types of minerals
261	Phosphates
261	Seafloor massive sulphides
262	Manganese nodules
263	Cobalt crusts
263	Methane hydrates
264	Decision-making framework
264	Allocation of mining rights
264	Environmental management under the EEZ Act
265	Elements of best practice
267	Case study
267	Chatham Rock phosphate mining
271	Further reading and references
273	Endnotes

Introduction

Natural gas and oil have been extracted from the sediments beneath the seafloor for decades, but the ores and minerals that lie under the ocean have been considered too difficult to extract except in shallow waters. However, as resource prices rise and deep-sea technology develops, the attention paid to these minerals has dramatically increased. The excavation of phosphate nodules, seafloor massive sulphides and manganese nodules is expected to be of increasing commercial interest within the next few years.¹

New Zealand's EEZ and extended continental shelf cover around 6 million square kilometres. Northwest, south and east of New Zealand are large areas of relatively shallow sea underlain by plateaus and tectonic ridges that border the deep ocean basins of the Pacific Ocean and Tasman Sea.² To the northeast there are volcanic arcs, including the Kermadec Arc. These areas offer significant potential as sources of minerals.

Sand mining and extraction, both inshore and offshore, are covered in Chapter 11. Oil and gas exploration and mining are covered in Chapter 13.



Aggregations of the snail *Ifremeria nautilai* on active hydrothermal chimneys (NSF Ridge 2000 Program)

Mineral resources

Figure 12.1 below highlights the potential large-scale sites of mineral resources found within New Zealand's marine area.

Mineral mining of New Zealand's deep-water seabed is an emerging industry, with no mineral mining yet being undertaken on a commercial basis within the EEZ or extended continental shelf. The Trans-Tasman Resources Limited marine consent application was the first EEZ seabed mining application to be processed (see Chapter 11: Sand mining). In June 2014, Chatham Rock Phosphate Limited lodged a marine consent application for phosphate mining on the Chatham Rise. In February 2015 this application was declined. This project is discussed in further detail in the case study section below. Figure 12.2 provides detail about the main deep-sea mineral resources found within New Zealand's marine area.



Fauna on an undisturbed hydrothermal vent (National Science Foundation Ridge 2000 Program and Charles Fisher, Penn State University)

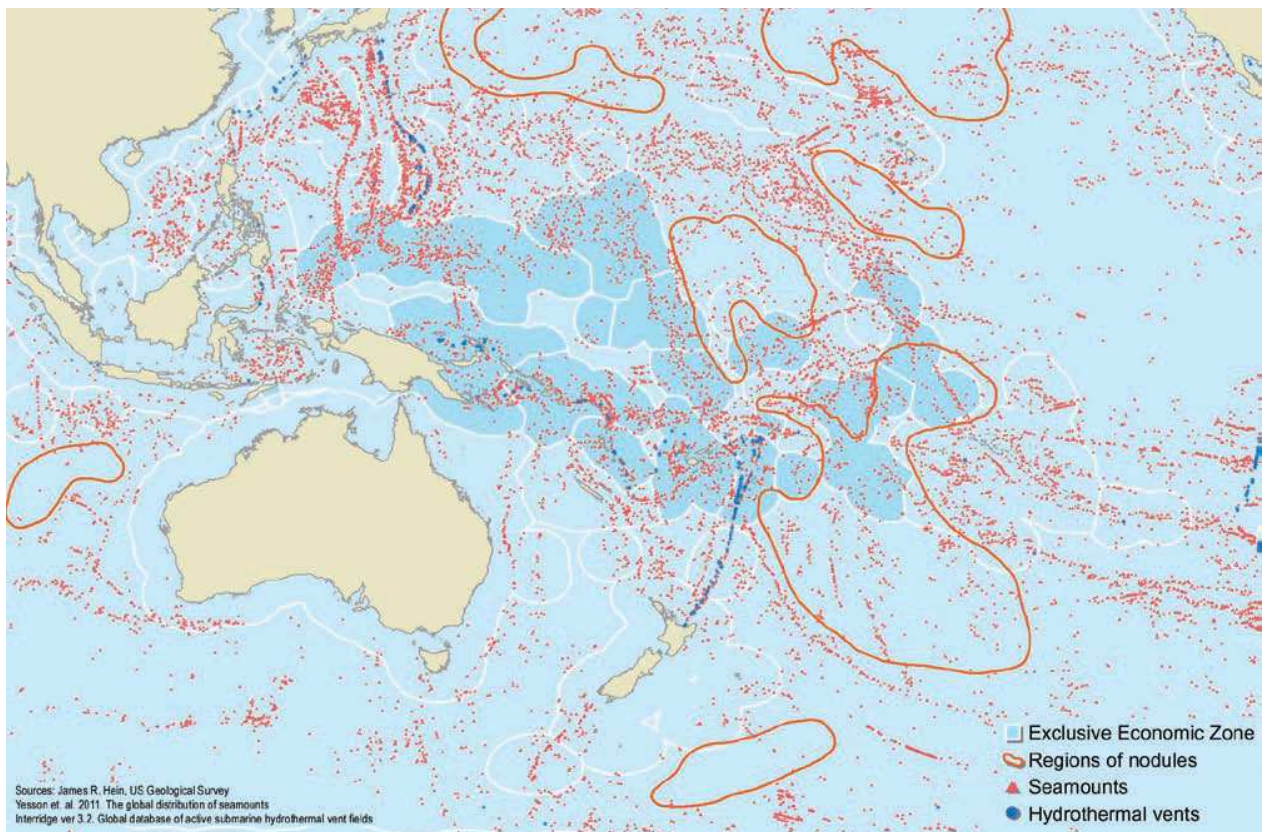


Figure 12.1 Distribution of sedimentary basins and some mineral resources, including those within New Zealand's marine jurisdiction (NIWA, 2012)

Mineral	Detail
Phosphate nodules	• They are formed in highly-productive areas of upwelling • The nodules are found within a 1 metre deep seabed layer consisting of fine sandy silt³ • The nodules that are economically viable to recover range in size from 2 to 150 millimetres • The phosphorite nodules are ground up, turning them into rock phosphate • Phosphate nodules are found on the Chatham Rise east of Christchurch at water depths of about 400 metres⁴ • Phosphate is an essential ingredient in fertilisers, and is therefore potentially commercially valuable, particularly as the majority of New Zealand's fertiliser is currently imported • Seafloor mining for phosphorite nodules is unprecedented at the moment, thus New Zealand could be the first country to permit this type of mining activity
Seafloor massive sulphides	• These are deep undersea deposits of high-grade polymetallic sulphides rich in copper, zinc and lead, and with a high gold and silver content • The largest of these deposits form by mineralisation of fragmented rocks • Chimney-like deposits are particularly rich • They are produced in underwater volcanic regions, usually on seamounts, by the deposition of metals from hydrothermal fluids • Hydrothermal vents, which contain these minerals, have been discovered on the Kermadec Ridge off the north-east coast of the North Island of New Zealand. The locations of these hydrothermal vents in the Kermadec Arc are known and mapped by researchers.⁵ • Both inactive and active vents with a low level of hydrothermal activity have the potential to be mined

Mineral	Detail
Manganese nodules and cobalt crusts	• Manganese nodules are rocky lumps found on the seabed in areas where sedimentation rates are very slow • They are created when manganese dissolved in seawater reacts with oxygen to create insoluble manganese oxide • They often contain metals such as cobalt, nickel and copper • Crusts that are rich in several elements, including cobalt, are found on the summits and flanks of seamounts, whereas manganese nodules are spread out over the abyssal plains and cover around 10 to 30 per cent of the seafloor • Some of the most extensive deposits of cobalt crusts in the world are found in the Pacific Ocean • Manganese nodules are abundant in an area over 250,000 square kilometres in extent, at water depths between 4,000 and 5,000 metres south-east of the Campbell Plateau and in the vicinity of Bollon's Seamount, 600 kilometres south of the Chatham Islands⁶ • The composition of nodules found within New Zealand's EEZ has been found to be relatively low in valuable metals; the value of the constituents varies with time • Since 2001 a number of exploratory permits have been issued by the International Seabed Authority for government institutions to survey manganese fields in areas of the deep sea outside national jurisdiction⁷
Methane hydrates	• Are a crystalline form of methane trapped in water, occurring in continental shelf sediments in many parts of the world⁸ • New Zealand has some large offshore gas hydrate provinces located along the east coast of the North Island and the south-west coast of the South Island⁹ • The methane hydrates in New Zealand are found in relatively shallow water¹⁰ • Research is currently underway to explore the most economical and safest way to mine methane hydrates in New Zealand¹¹

Figure 12.2 New Zealand's deep-sea mineral resources

Mining methods

Stages of work

Generally, there are some key stages of work involved in mining for marine minerals. These include:^{12,13}

- **Prospecting** – the purpose of this stage is to identify areas that are likely to contain mineral deposits. The work includes geological, geochemical and geophysical surveys (e.g. seismic surveying), and aerial surveys, including the taking of samples by low-impact mechanical methods.¹⁴ For example, multi-beam swath bathymetry can be used to map the seafloor, and remotely-operated vehicles or submersibles can be used to collect images of the seafloor, take core samples of the seabed at low concentrations, and collect targeted samples.



Manganese nodules are rocky lumps found on the seabed where sedimentation rates are low and they often contain metals such as cobalt, nickel and copper (*Secretariat of the Pacific Community, 2013b*)

- Exploration – the purpose of this stage is to evaluate the feasibility of mining identified deposits. This stage can include any drilling, dredging or excavations needed to assess the nature and size of a mineral deposit.¹⁵ The exploration phase may involve core sampling at higher concentrations (than undertaken in the prospecting phase) in small areas, test pit excavation, test drilling, trials of extraction methods and bulk sampling. This exploration could involve the removal of larger volumes of deposits.
- Production – the purpose of this stage is to actively extract and process the mineral resource. Techniques for mining minerals in the marine environment may involve seafloor suction dredging, seafloor slurry pipes, use of tracked vessels on the seafloor and seafloor cutting/fragmentation. Following extraction and processing, tailings (unwanted material) are deposited back onto the seafloor. Some production operations involve a processing vessel that moors to ‘permanent’ anchor blocks or anchor moorings.

Due to the high variation in the geological processes governing resource formation, and concentration associated with the different mineral deposits, there is no single, universal method which can be deployed. A variety of methods will be required to mine these deposits. Some of the tools used are shown in Figure 12.3. Technology that can be used to mine minerals from the seabed efficiently in deeper waters is only just being developed, and work is still ongoing to determine what the most effective methods of mining may be, both in terms of resource recovery and for minimising environmental impacts.

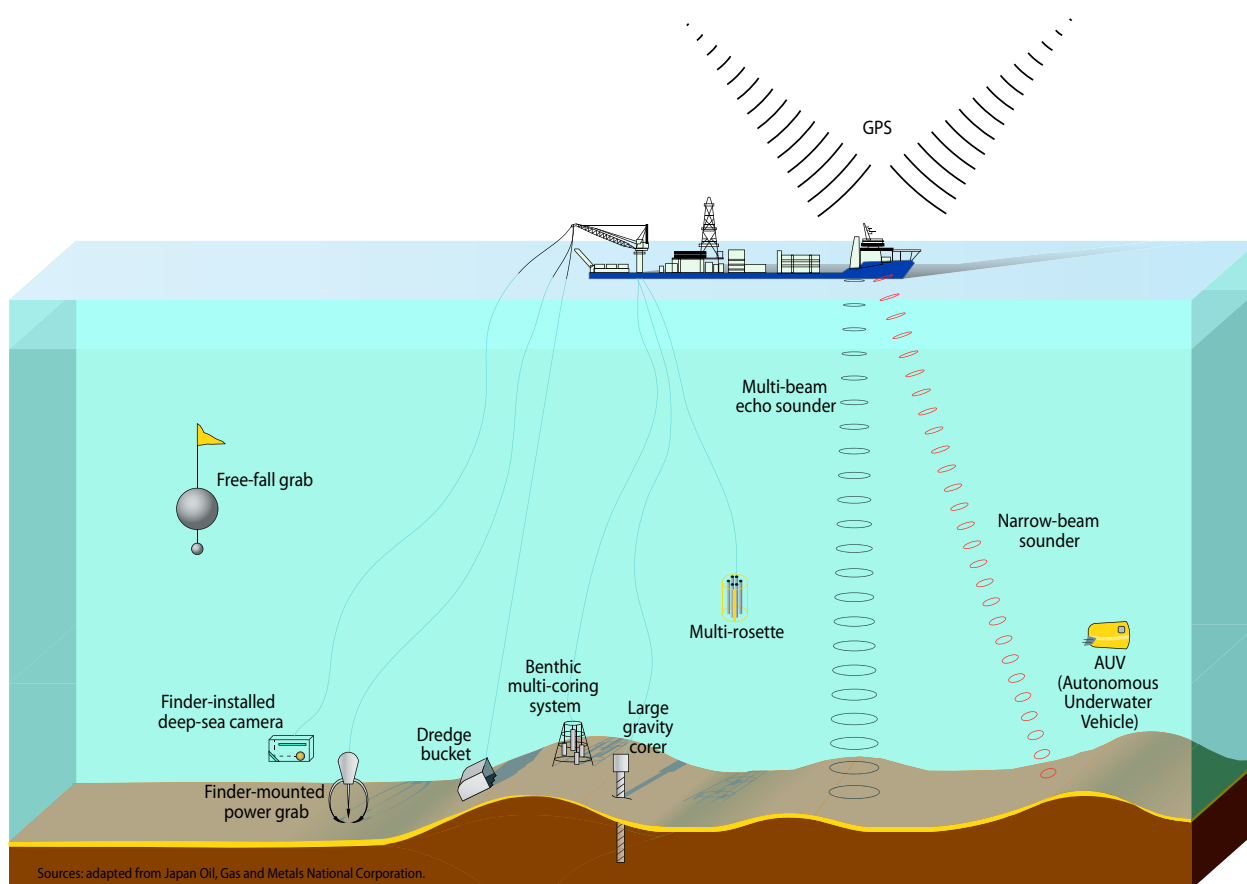


Figure 12.3 Examples of deep-sea minerals exploration tools (*Secretariat of the Pacific Community, 2013a*)

Phosphates

Mining for phosphate generally requires a mining vessel that deploys a trailing suction draghead that moves along the seabed. Water jets and possibly cutting teeth are used to loosen the top centimetres of seabed sediment, and the resulting slurry of sediment and seawater is pumped to the mining vessel. The material will be mechanically processed on board the ship to separate the coarse phosphatic material from the finer non-phosphatic sediments. The fine non-phosphatic sediment and seawater will be discharged to the seafloor or above the seafloor.¹⁶

In 2014 Chatham Rock Phosphate Limited applied for marine consent to mine phosphate nodules from the crest of Chatham Rise, about halfway between Banks Peninsula and Chatham Islands. The application was declined in February 2015 despite the company holding a mining permit for 820 kilometres on the crest of the Chatham Rise, and two prospecting licences for surrounding areas. The process proposed for the mining operation is discussed further in the case study at the end of this chapter.

Seafloor massive sulphides

Seafloor massive sulphide deposits are found at water depths of up to 4,000 metres, and the technology to extract minerals from the deposits is as yet unproven. Steps involved in the exploration and extraction of these deposits include:¹⁷

- Identifying potential targets
- Testing potential targets and delineating prospects
- Evaluating the resource
- Extracting minerals from the seafloor using remotely-operated mining equipment
- Transporting the ore, via a lifting system, from the seafloor to a production support vessel at the water surface
- Dewatering the ore on board a production support vessel or platform
- Transferring the ore from the production facility to a transport barge or bulk carrier
- Transporting the ore to land for treatment and/or processing
- Transporting concentrate and saleable products to market

Currently the only model for mining is that proposed by Nautilus Minerals Limited at Solwara 1, a high-grade copper-gold mining project which is located 30 kilometres off the coast, and about 1,600 metres deep, in the territorial waters of Papua New Guinea.¹⁸ This company proposes to use a 'Seafloor Mining Tool', envisioned to be a large-tracked remotely operated vehicle, incorporating tools for cutting up the vent chimneys and hydrothermal deposits, and suctioning the material to a Mining Support Vessel at the surface.¹⁹ The ore will be dewatered on the vessel, and transferred to a barge for transport to shore for processing.

Manganese nodules and cobalt crusts

Manganese nodules sit on a soft-sediment surface, and can be recovered by dredging through trawling or by suction. There is currently no known active extraction of manganese nodules anywhere in the world.²⁰ Since 2001, several permits have been issued to governmental institutions and private companies by the International Seabed Authority to survey manganese fields. These permits are not for mining, but for a detailed initial investigation of the potential mining areas. In 2013, several companies also submitted applications for the exploration of manganese nodule fields in the high seas in co-operation with developing countries (Kingdom of Tonga and Republic of Nauru).²¹ There is now interest in manganese nodule fields inside the EEZ of countries such as the Cook Islands.

Exploration and mining of manganese nodules and cobalt crusts are expected to involve the following basic processes:²²

- Recovery of minerals from the seafloor using remotely operated seafloor production equipment



Exploration of seafloor massive sulphides requires remotely operated mining equipment (MARUM, University of Bremen (Secretariat of the Pacific Community, 2013a))

- Transport of a slurry (ore and seawater) vertically from the seafloor to a vessel or platform on the sea surface
- If substrate rock is collected, separation of ore and gangue (the material in which ore is found that has no economic value) using froth flotation on board a vessel
- Dewatering of the ore on board a vessel or platform
- Transfer of the ore from the vessel to a transport barge or bulk carrier/storage facility
- Disposal of the separated seawater

Technologically, the mining of cobalt crusts is considerably more complex than that of manganese nodules, as the crusts form a hard substrate. Ideally, a mining operation would only remove the crust, and not the underlying volcanic rocks. In addition, the slopes of the underwater volcanoes where the crusts are located are very rugged, which can make the use of excavation equipment more difficult. Cobalt crust mining is only at the conceptual stage at present. It is believed that the greatest barrier to mining both manganese and cobalt resources is the processing stage, with viable processes to extract the minerals from the ore once it has been removed from the seabed yet to be developed.



Manganese nodule collector (COMRA (Secretariat of the Pacific Community, 2013b))

Methane hydrates

Mining for methane hydrates is still in its early stages. In March 2013 a Japanese company became the first to extract gas from offshore methane hydrates. It is aiming to commence commercial production by early 2019.²³

In 2008, a joint Canadian-Japanese expedition in Mallik in the Northwest Territories, Canada, established that methane hydrates could be collected by using a water pump to depressurise a well already drilled. This process involved lowering the pressure by pumping out the water that naturally accumulates in the well. Standard oil and gas drilling equipment was used to re-enter the existing well drilled to a depth of just over 1,000 metres and then to 'refurbish' it by casing the entire well with lengths of steel tubing cemented into place to prevent it from collapsing.²⁴

Subsequently, gas hydrate resources on the Alaskan North Slope have been classified as technically recoverable, and the first production tests offshore in Japan suggest production of gas from offshore hydrates is now also technically feasible.²⁵

Exploitation in New Zealand remains a major challenge for a number of factors:²⁶

- The logistical cost of exploration is high because of the offshore location of potential resources
- New Zealand's geology is complex, meaning the geological parameters such as source rock, migration pathways and reservoir rock are highly variable, both within a basin and between basins
- New Zealand has few exploration wells to date, so that the volumes of undiscovered methane hydrate reserves are difficult to predict
- The methane hydrate is under immense pressure, thus there are significant concerns about being able to bring the resource safely to the surface

Potential environmental effects

Deep-sea mining can cause a range of both direct and indirect impacts on the marine environment. Some of these potential impacts are insignificant and/or reversible. But others are potentially significant and could cause irreversible impacts on different components of the marine ecosystem. Figure 12.4 summarises the full range of potential impacts which need to be considered, but which will not necessarily result from any individual mining operation. Ways to mitigate these potential effects are discussed in the best practice section below.

	Potential environmental impact
Benthic and water column impacts	• Mortality of organisms on the seafloor, and within the seabed, in the direct mining area • Loss of sensitive and/or unique habitats if these exist in the area to be directly mined • Removal of habitat which underpins marine communities such as through the direct removal of a vent • Permanent change in the composition of marine habitats, such as through the removal of hard substrate, which species such as cold-water corals require for attachment • Smothering of benthic organisms outside the mining area by a sediment plume generated by tailings deposition • Reduced ability of visual predators to feed due to increased turbidity of seawater • Clogging of gills of fish and disruption of filter feeders from high levels of sediment in the water column • Reduction in primary productivity through reduced light levels
Marine mammals, seabirds	• Acoustic impacts from seismic surveys and other production equipment interfering with the way marine mammals locate prey, navigate and communicate • Lights causing seabirds to collide with vessels • Increased risk of ship strike
Biosecurity	• Increased biosecurity risk from introduction of overseas equipment acting as vectors for marine pest species
Other	• Other ecological impacts resulting from increased light, noise and vibration in the deep ocean environment. These are poorly understood and dependent on the spatial scale of the mining and the transient nature of methods used. • Accidents, such as oil spills and loss of equipment, having adverse impacts on the marine environment • Exclusion zones around mining areas disrupting shipping and fishing activity

Figure 12.4 Potential environmental impacts from deep-sea mining (*Secretariat of the Pacific Community, 2013a, b and c*)

Seabed and water column

Seabed mining inevitably results in significant physical changes to the area that is being mined. The removal of rock or sediment from the seabed will result in the mortality of all species living on or in the sediment layer, and those living immediately above.

Mining may also change the topography of the seabed, and associated hydrodynamics of the area, as well as the composition of the substrate. As a consequence, the original community may be unable to re-establish after mining activity ceases, and be replaced by different species better adapted to the new environment. The significance of this change is dependent on whether the affected species and communities are unusual, rare and/or play a particularly important role in the food web and broader ecosystem functioning.



Crab shown on the flange of a hydrothermal area associated with a deep-sea vent
(Secretariat of the Pacific Community, 2013a, photo courtesy of Chuck Fisher)

Deposition of sediment tailings into the marine environment is likely to extend the environmental footprint of the mining activity over a wider area than just the active mine site. Smaller particles may become suspended in a plume, which can travel some distance with ocean currents before settling out. Water-borne sediment is known to be a particular problem for suspension-feeding organisms, such as sponges and corals, as the fine particles can clog their filtering apparatus. It can also affect the gills of fish. In addition, such sediment can affect primary productivity by reducing light levels and thereby the ability of phytoplankton to grow.

When sediment settles, it can smother benthic organisms, and also change the nature of the substrate making it unsuitable for attachment by invertebrates such as corals. The distance and direction the sediment plume will travel depends on its composition (i.e. grain size) and on oceanographic conditions (e.g. direction, frequency and velocity of currents).

Mining activity has the potential to impact on fish stocks in the longer term, if it disrupts food supplies or spawning and nursery areas on which the fishery is dependent.

These impacts will be sequential during the long production phase, as adjacent areas are progressively mined. Recovery potential, once mining has ceased, will depend on a range of factors. These include whether the affected species are reliant on the specific features of the habitat which have been removed and whether there are remaining communities in adjacent areas that provide a source for recolonisation. As many deep-sea species are slow-growing and long-lived, recovery of the benthic communities could take considerable time; in some cases recovery may never occur.

The major challenge for deep-sea mining is the lack of baseline data on what marine species are associated with prospective mining areas. There is even less understanding of the interactions between different species and trophic levels, the ecological significance of any impacts, and recovery and recolonisation potential.

Marine mammals and seabirds

Underwater sound may have acoustic impacts on marine mammals as well as other fauna.²⁷ Vessel and mining equipment, including underwater pumps, riser and sinker activity and the vessel-based processing plant, produce varying levels of underwater noise. Generally, such activities are of a short duration. The possible consequences vary from none (affected animals may leave the area), to acute injury (ear drum damage), to serious death, which depends on the noise level encountered, species and habitat.²⁸ There may also be cumulative effects from repeated exposure. In addition, the use of mining vessels can increase the risk of ship strike, and the use of lights at night can attract seabirds, causing them to collide with ships.²⁹

Biosecurity

Vessels and mining equipment brought into New Zealand from overseas can act as vectors for non-indigenous pest species. Non-indigenous species are unlikely to survive independently of the vessel in deep offshore waters, but may be transferred to and invade shallow coastal habitats.³⁰ (This topic is discussed in more detail in Chapter 5: Marine biosecurity.)

Other impacts

There may be other ecological impacts, which are not yet fully understood, resulting from increased light, noise and vibration in the deep ocean environment.³¹ There is also the risk of accidents leading to spillages and loss of equipment within the marine environment.

Mining activities can displace other marine users, such as shipping and commercial fishing, as there are likely to be exclusion zones around the area of mining activity. This is only likely to cause significant problems if large areas are excluded, or if there are multiple mining operations undertaken in one region at the same time. If fishers are excluded from a significant portion of a quota management area, this could increase fishing pressure on the balance of the area. In addition, fish behaviour may be affected, with fish moving away from the area of mining influence.³²

Potential environmental impacts specific to types of minerals

Phosphates

Phosphate mining involves the removal of hard phosphate nodules from an otherwise (mostly) sedimentary environment. This means that suitable hard habitat, needed by sessile organisms such as corals, is greatly reduced. In addition, as a number of these organisms create habitat themselves (biogenic habitat), it has direct repercussions for associated species potentially resulting in a significant shift in community structure.

Once the phosphate nodules have been removed, the finer non-phosphatic sediments and seawater used to produce the slurry for the phosphorite nodule extraction will be discharged on or near the seafloor, resulting in a sediment plume. The range of potential impacts from phosphate mining is described in more detail in the case study below.

Seafloor massive sulphides

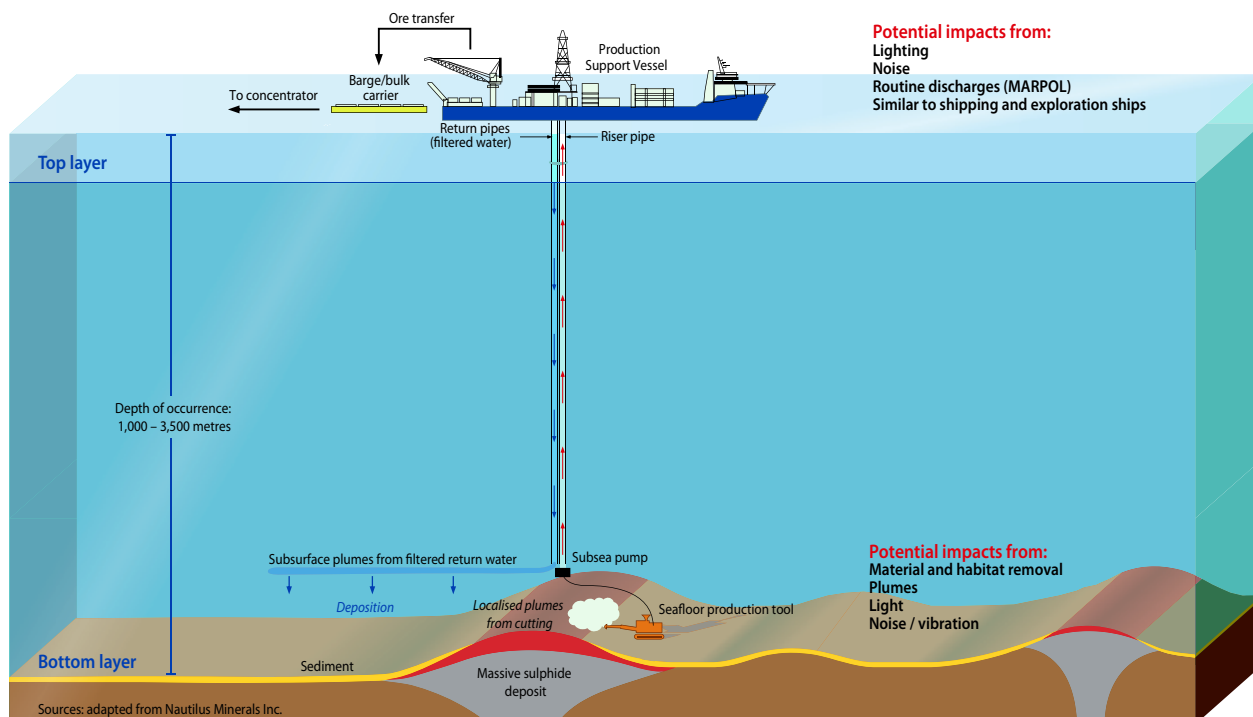


Figure 12.5 Example of seafloor massive sulphide mining system and related sources of potential environmental impacts
(Secretariat of the Pacific Community, 2013a)

Active deep-sea vents, where seafloor massive sulphides are found, are associated with highly unusual and productive communities that rely on sulphur compounds as an energy source. These communities were only discovered in 1977³³ and remain poorly understood. The environments around active vents, and the communities they support, are dynamic and subject to irregular flows of hot fluid.

Mining will leave permanently altered areas at the mine site, with surrounding areas potentially impacted by the sediment plume. The spatial scale of the impacted zone is dependent on the volumes of mined material, and current regime near the seafloor.³⁴ In areas of active hydrothermal venting, species may be adapted to natural changes in the environment, and therefore be able to recolonise more rapidly than other deep-sea communities, dependent upon there being a viable source of recruits nearby.³⁵

A major minerals research programme in New Zealand, led by GNS Science, aims to gain a better understanding about how these minerals are formed. The programme is also considering the development of effective exploration and extraction practices. This research is currently examining seafloor massive sulphide mineralisation along the Kermadec Arc and in Frontier regions, including the Colville Ridge and Kermadec Ridge.³⁶ The work should assist with building a deeper understanding of how these minerals form and the environments surrounding them.

Manganese nodules

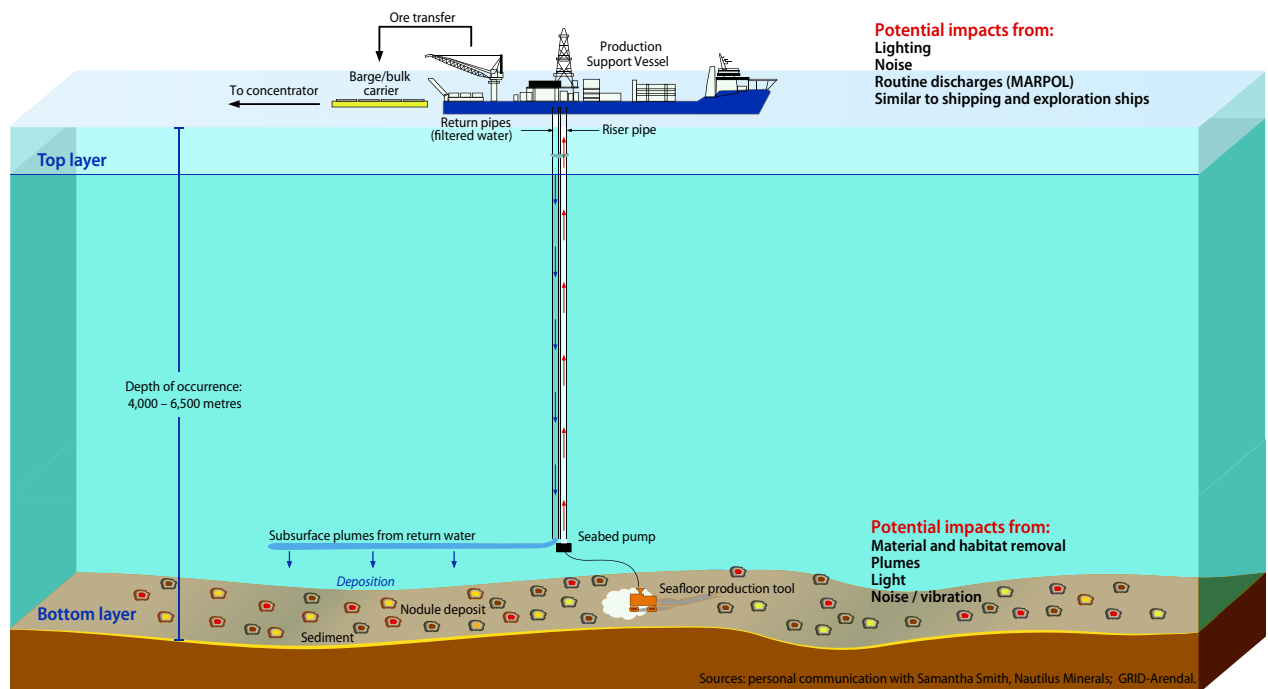


Figure 12.6 Example of seafloor manganese nodule mining system and potential environmental impacts (*Secretariat of the Pacific Community, 2013b*)

The excavation of manganese nodules would disturb parts of the seabed. Potentially large amounts of sediment, water and many organisms would be dug up with the nodules (depending on the method proposed), and the destruction of the deep-sea habitat could be substantial.³⁷ Nodules take millions of years to form, so these habitats would be permanently lost from large areas of seabed following mining.³⁸

The potential environmental impacts from exploration and mining for manganese nodules are highlighted in Figure 12.6 and summarised as:³⁹

- Destruction of the benthic community in the mining area including organisms living in the crevices of the nodules which would be permanently lost
- A sediment plain left remaining after the removal of the nodules which is suitable only for a subset of the original benthic community. Nodules will not reform for millions of years.
- Mobilisation of sediment into the water column, which is potentially a major concern for the affected ecosystem as currents at that depth are often very slow, as is the settling velocity of the disturbed material

Cobalt crusts

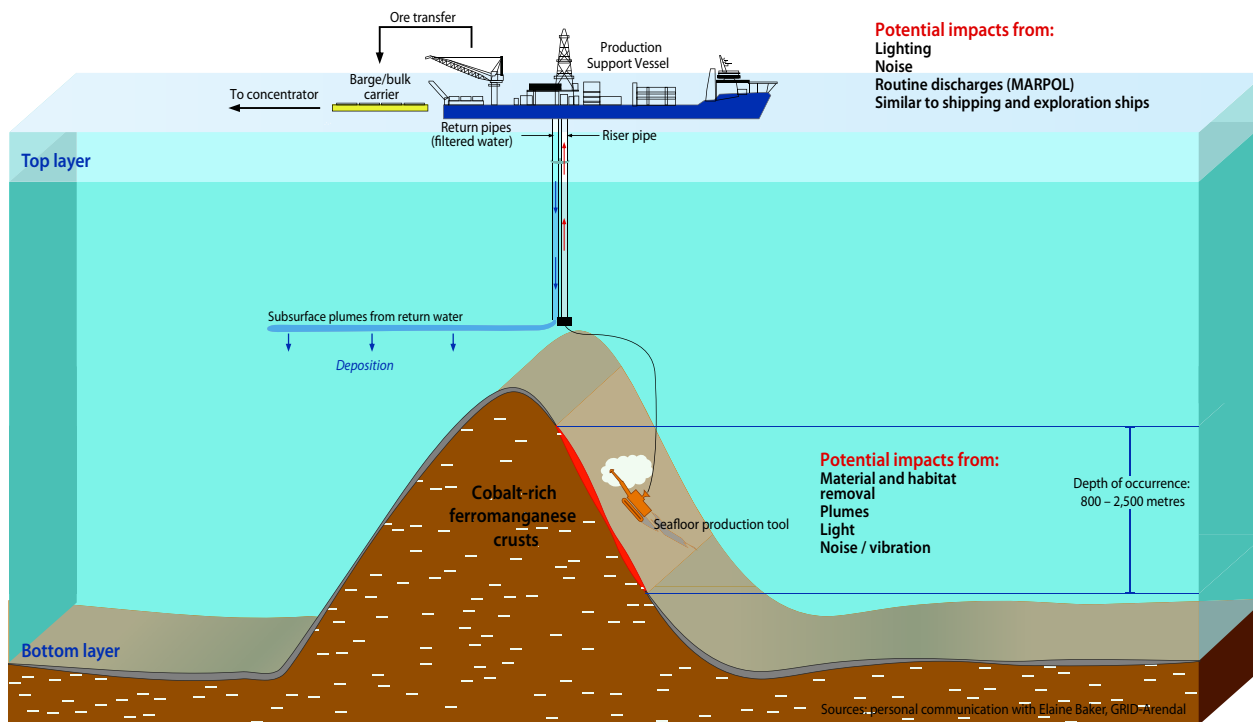


Figure 12.7 Example of seafloor ferromanganese crust mining system and related sources of potential environmental impacts (Secretariat of the Pacific Community, 2013c)

Seamounts studied to date show that there is an abundance of life associated with some of them, including corals and sponges and huge aggregations of fish. Seamounts have been described as ‘underwater oases’, and they can be important habitats for migrating species. Mining in the vicinity of these structures will destroy any corals and sponges which grow on the seamounts. This removal of habitat-forming species is significant, and recent studies indicate that recovery times would be decades to centuries.⁴⁰

The potential environmental impacts from exploration and mining for cobalt crusts are highlighted in Figure 12.7 and summarised as:⁴¹

- Destruction of sessile marine life due to the removal of material from the summit of seamounts, and the scraping of the crusts of underlying rocks
- Plumes from the mobilised sediment impacting life in the surrounding area
- Rock surfaces left behind will be recolonised over timeframes that are largely unknown (but likely to be slow)

Methane hydrates

The methane gas is trapped at such pressure that when released at the surface it expands 160 times, presenting both a massive technological challenge for extraction and a hazard.⁴² There is the potential for collapsing of areas of the seabed following the extraction of the methane. Such areas of collapse would devastate any marine life in the mining area.

There is still much work to be done on the potential environmental impact from methane hydrate extraction activities. Scientists from GNS Science, NIWA, and the Universities of Otago and Auckland have begun a Government-funded project for the Hikurangi Margin. NIWA will provide detailed information on the stratigraphic and geological architecture of the reservoirs and surroundings, and establish an ecological and biological baseline of potential exploration sites. These studies will be critical for management and environmental protection of any mining sites. The aim is to assess optimal sites by 2015, with test drilling scheduled by 2018.⁴³

Decision-making framework

Allocation of mining rights

Applications for prospecting and mining licences are processed by New Zealand Petroleum and Minerals. The 'Minerals Programme for Minerals (Excluding Petroleum) 2013' sets out the policies and procedures to be applied to the allocation of mineral resources (including consultation with iwi and hapū by the Crown prior to allocation). The requirements to be met by permit holders are set out in regulations.

The Crown Minerals Act separates activities into Tier 1 and Tier 2. Tier 1 includes high-return, high-risk projects, and most, if not all, marine minerals operations will fall within this category. The Act requires a high-level preliminary assessment of the applicant's health, safety and environmental policies, and technical and financial capabilities to be undertaken for all applications for Tier 1 activities. The purpose is to ensure that the applicant is able to undertake the work proposed, and that the applicant does have a satisfactory history and hence will not likely cause health, safety or environmental issues in New Zealand.

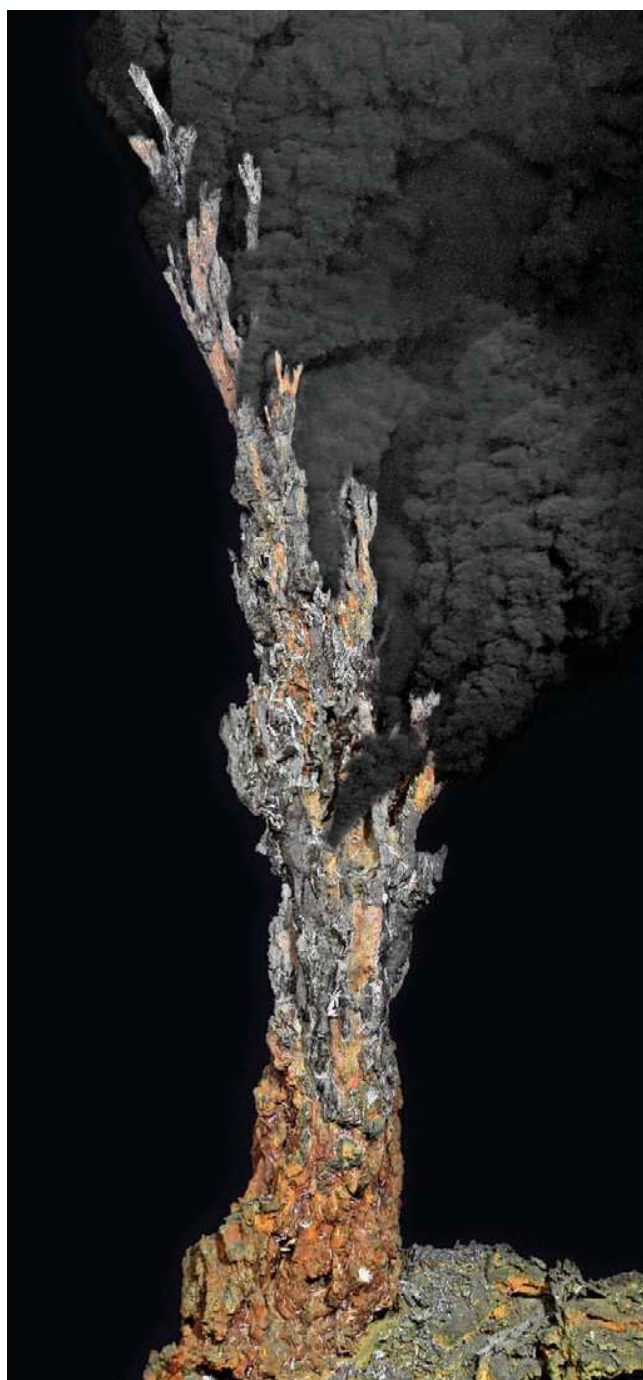
Before granting a permit for Tier 1 exploration or mining, the Minister of Energy and Resources must be satisfied that the proposed permit operator has, or is likely to have by the time the relevant work in any granted permit is undertaken, the capability and systems that are likely to be required to meet the environmental requirements of all specified Acts for the types of activities proposed under the permit. The Minister may seek the views of other regulatory agencies, including local authorities, the EPA and Maritime New Zealand, in satisfying himself or herself of these matters.⁴⁴

Environmental management under the EEZ Act

The EEZ Act regulates the environmental effects, and effects on existing interests, of minerals mining (and other seabed activities) beyond the territorial sea. Under the EEZ Permitted Activity Regulations prospecting and exploration for minerals are permitted activities, subject to conditions.

Minerals production mining in the EEZ is a discretionary activity. This means a marine consent is required, and the application is notified, allowing the public to make submissions and participate in the hearing. To date, no minerals activities have been classified as non-notified discretionary.

Discharges and dumping from seabed mining vessels and installations are also regulated under the EEZ Act. At the time of writing the relevant regulations were still being developed. The provisions of the EEZ Act which apply to mining activities are described in more detail in Chapter 3: EEZ legislation.



Seafloor massive sulphides are rich in copper, zinc and lead and also have a high gold and silver content (GEOMAR)

Elements of best practice

The best practice considerations identified in Figure 12.8 can be applied to deep-sea minerals exploration and mining.

	Detail
Marine mammals	• Avoid areas frequented by threatened or endangered marine species, such as Hector's and Maui's dolphins, Bryde's whale and Southern right whale, as well as those identified on the 'Species of Concern' list within the Department of Conservation's 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals for Seismic Survey Operations'⁴⁵ • Use marine mammal observers • Use passive acoustic monitoring • Investigate options to cease operation if marine mammals are present • Limit unnecessary underwater sound generation and lighting • Reduce vessel speeds in areas known to be frequented by marine mammals
Seabirds	• Limit unnecessary lighting, and ensure that lights that are required are directed vertically onto work surfaces and/or utilise green lighting • Monitor bird strike and review mitigation measures in response to bird mortality above an identified level • Implement measures to reduce the likelihood of oil spills that may affect seabirds
Biodiversity	• Undertake sufficient research, to identify what species are likely to be present in the area to be mined, including their distribution • Avoid areas of high ecological value, including areas of high endemism, such as active vent sites • Only mine a small portion of any specific habitat • Consider the cumulative impacts of mining activity, and other activities such as deep-sea trawling, on benthic habitats • Consider trialling artificial hard surfaces on the seafloor to act as settlement and attachment surfaces for corals, sponges and similar organisms • Regularly check, clean and apply antifouling paints (preferably copper-free) to infrastructure that is brought to New Zealand from overseas (see 'Craft Risk Management Standard for Biofouling on Vessels Arriving to New Zealand' discussed in Chapter 5: Marine biosecurity) • Assess options for biodiversity offsetting, as appropriate, to protect seabed habitats
Water column	• Adopt mining methods that minimise the amount of sediment that is released into the water column⁴⁶ • Avoid areas where the sediment plume would disrupt areas with high ecological value • Deposit tailings close to the seafloor to minimise extent of plume
Monitoring	• Develop and undertake a robust monitoring regime – including biological and geomorphological monitoring • Undertake rigorous baseline monitoring, prior to mining commencing, so that ongoing monitoring data can be compared to the baseline • Undertake sufficient monitoring so any significant adverse effects on the marine environment can be identified and addressed in a timely manner • Publicly release monitoring information in a timely fashion • Consider involving tangata whenua and stakeholders in overseeing monitoring programmes
Other	• Avoid significant fish spawning and nursery areas, and areas of high fish density • Adopt an adaptive management approach where appropriate • Support the efficient use and recycling of minerals to minimise the need for mining activity in the marine area

Figure 12.8 Best practice considerations for deep-sea minerals exploration and mining activities

There are a number of mineral-specific best practices that were identified by the Secretariat of the Pacific Community's 2013 reports on deep-sea mineral mining. These are summarised in Figure 12.9 as they provide some additional guidance for specific mineral mining activities.

Seafloor massive sulphide mining operations	
Area	Action
Seafloor	• Setting aside part of the habitat, with representative animals, to serve as reference areas and sources of animals to assist recolonisation of the impacted site • Relocating some of the representative animals • Installing artificial substrates to help recolonisation • Considering location, positioning and method of any subsea discharge to limit its impact • Employing technologies/methodologies so that sediment resuspension is minimised during cutting • Limiting the number and/or size of sites
Mid-water column	• Developing and implementing emergency response procedures in the event of accidents leading to spills that might affect water quality • Using fully-enclosed ore delivery systems
Surface	• Releasing discharge and water from dewatering plant where possible, at least below the top 200 metres of the water column (i.e. below the light zone) • Developing and implementing environmental management plans that specifically cover waste minimisation and loss prevention, in order to reduce impacts on water quality • Developing and implementing emergency response procedures in the event of accidents leading to spills that could affect water quality • Adopting effective mitigation measures to minimise the risk of injury to marine animals from ship strike or collision • Installing approved sewage treatment plants, certified through relevant international standards, to handle such normal ship discharges as treated sewage • Maintaining surface deck lights at the lowest levels needed to ensure safe working conditions • Developing safety, health and environmental policies and plans for all offshore operations
Manganese nodule and cobalt crust mining operations	
Area	Action
Operational	• Developing and implementing environmental management plans that cover waste minimisation and loss prevention to minimise impacts on water quality – a working example of a management plan was prepared by the International Seabed Authority in 2012 • Developing and implementing emergency response procedures in the event of accidents leading to spills to the environment • Using effective mitigation measures to minimise the risk of injury to marine animals from ship strike or collision • Installing an approved sewage treatment plant, certified to meet relevant international standards and/or other relevant regulations • Developing and implementing safety, health and environmental policies and plans for all offshore operations
Spatial	• Setting areas aside for conservation, possibly within mine site/mine lease areas. Depending on the site, it may not be possible to find an appropriate site within the mine lease that will not be impacted by the mining (e.g. by sediment plumes), in which case a similar site nearby may be nominated as the conservation area instead. Attention should be given to the representativeness, adequacy, resilience and connectivity of a network of areas. • Establishing marine management areas, which involve zoning of different areas for different uses or intensities of use. Such an approach may designate areas that are acceptable for total mining, areas that can only be partially mined, or areas set aside for conservation. • Evaluating other generic measures, if feasible, including installation of artificial substrates to encourage more rapid colonisation by fauna impacted by mining, and animal relocation if there are populations of rare, endemic or highly endangered species • Evaluating the location of discharges to ensure minimal impact on ecosystems • Considering the use of diffusers to aid dispersal • Considering animal relocation, if there are populations of rare, endemic or highly endangered species • Employing reserve networks, such as marine protected areas
Temporal	• The time scales of nodule and crust growth and faunal recovery in the abyssal deep sea almost certainly make consideration of short time-scale measures impractical • Measures to rehabilitate degraded areas or encourage longer-term faunal recruitment are much less likely to be effective than spatial management approaches
Figure 12.9 Possible mitigation and management strategies specific to seafloor massive sulphide, manganese nodule and cobalt crust mining operations (<i>Secretariat of the Pacific Community, 2013a, b and c</i>)	

Chatham Rise phosphate mining

In June 2014 Chatham Rock Phosphate Limited lodged an application under the EEZ Act for a marine consent to mine phosphorite nodules from the crest of Chatham Rise, about halfway between Banks Peninsula and the Chatham Islands. In February 2015, the Decision-Making Committee appointed to consider the application released its decision, refusing consent.

The application

The crest of the Chatham Rise is characterised by low sedimentation rates (because of the distance from land and prevailing currents) and high productivity which makes it favourable for both nodule formation and fish production.

Chatham Rock Phosphate holds a mining permit extending over 820 square kilometres of seabed on the crest of the Chatham Rise. It also holds two prospecting licences for surrounding areas. The original application covered the entire area where the company held prospecting or mining permits, in total 10,192 square kilometres of the Chatham Rise. Chatham Rock Phosphate subsequently revised its application to reduce the proposed mining area to 5,207 square kilometres by excluding the eastern prospecting area.

The phosphate nodules were proposed to be mined using a trailing suction hopper dredger or drag head connected to the mining vessel by a flexible pipe. The nodules were to be sucked up the pipe using a series of pumps, and separated from the other seabed material on board the mining vessel. Waste tailings were to be discharged, using a sinker pipe with a diffuser, 10 metres above the seafloor.

Phosphorite nodules contain phosphate, one of the main components of commercial fertiliser. In its proposal, Chatham Rock Phosphate stated that the project could reduce phosphate imports by \$85 million per year and increase GDP by \$280 million per year. The company also stated that the phosphate has low cadmium content and could reduce nutrient runoff if applied as a direct application fertiliser (as it is less soluble than superphosphate).

The potential effects of the proposal on the marine environment, as identified in the Chatham Rock Phosphate application, are shown in Figure 12.10.

	Potential effect
Benthic environment	The mining operations will completely remove the upper layer of the seabed, including resident benthic fauna. Fauna directly above the seafloor – hyperbenthos – will also be removed/killed. There will be a long-term change in habitat from mixed nodule/soft sediment habitat to soft sediment habitat. Benthic fauna which live in soft sediments are likely to recover, taking at least 10 years to attain community complexity. Benthic fauna which depend on hard substrate are unlikely to recover due to the removal of the hard material. This fauna includes cold-water coral species and associated organisms. After the phosphoric material is extracted, the non-phosphoric material (approximately 85 per cent of the material extracted) will be returned to the seabed using a pipe. These tailings will be discharged approximately 10 metres above the seabed in the area that is being mined. The sediment will form a plume of suspended sediment. Depending on currents, the sediment plume will be transported from the site of discharge (with distance depending on the sediment particle size) and sediment will subsequently settle on the seabed. The suspended sediment plume will have sedimentation effects on the benthic environment adjacent to the mining site (deposition of 1 to 2 centimetres could extend 3 kilometres from the mining blocks). Chatham Rock Phosphate states that the suspended sediment concentration is predicted to be near background levels (around 1 milligram per litre) within about 15 kilometres of the mining site. Approximately half of the marine consent area overlaps with approximately 60 per cent of a Benthic Protection Area (the overlap is 5,236 square kilometres). These are areas which are closed to bottom trawling in order to protect seabed habitat. Chatham Rock Phosphate states that it has identified mining exclusion areas (1,822 square kilometres) for protection from seabed disturbance which better recognise the values associated with the central crest of Chatham Rise.

	Potential effect
Water quality	The mining activity will result in the formation of a suspended sediment plume reducing water clarity. The disturbance of sediment will result in the release of chemical elements (nutrients such as nitrogen and phosphorus and trace elements such as arsenic, nickel and uranium) found in the sediments into the water column. Chatham Rock Phosphate states that the elements released are likely to result in the 99 per cent ANZECC (Australian and New Zealand Guidelines for Fresh and Marine Water Quality) trigger value being exceeded, but that following dilution, concentrations will not exceed the trigger value. In the case of uranium, there will be a near-field concentration increase of 10 per cent.
Sediment quality	There will be discharge of organic matter after mining due to the incorporation of benthic biota in the mined sediment. This discharge is likely to increase oxygen demand. If organic enrichment occurs, then the oxygen demand and consumption by bacteria decomposing the organic matter will increase, possibly leading to hypoxia or anoxia. Chatham Rock Phosphate states that it will not result in anoxia at the water–sediment interface.
Food web	Chatham Rock Phosphate has commissioned the development of a trophic model of the food web on the Chatham Rise. The model indicates that much of the Chatham Rise fisheries resource is reliant on primary production through phytoplankton and the buildup of food resource in the zone between 200 and 1,000 metres below sea level. It suggests that there is limited food dependency between fisheries and benthic fauna because the proposed mining is below the photic zone. Chatham Rock Phosphate concludes that the trophic impacts of removing benthic fauna in the mining area will be minor in the broader context.
Fish	More than 200 species of fish have been identified on the Chatham Rise. Key commercial species include hoki, hake and ling. Fish may be impacted by increased suspended sediment, changes in water quality and noise.
Marine mammals	Based on data sets of opportunistic sightings, Chatham Rock Phosphate states that 12 species of whales or dolphin and one species group (beaked whales) have been observed in Chatham Rise waters between 1981 and 2007. These sightings include two threatened species (killer whale and the southern right whale) and three International Union for Conservation of Nature endangered species (sei, blue and humpback whales). Marine mammals may be impacted by noise created by mining activities, interactions with the mining vessel (e.g. ship strike), disruption of food sources, and introduction of pollutants into the environment. Chatham Rock Phosphate states that noise effects may occur within 2 kilometres of the vessel for some marine mammal species. It states that the sound levels produced by an operating dredger (similar to the equipment to be used on the mining vessel) are comparable to the sound levels produced by a similar-sized ship transiting across the Chatham Rise. It proposes consent conditions to avoid any impacts of noise on marine mammals.
Seabirds	Chatham Rise is one of the most important areas for seabirds in New Zealand, with the majority of seabirds being present over the course of a year. The high marine productivity in this area attracts foraging birds. The company has identified that approximately 28 species are ‘key members of the Chatham Rise avifauna’, most of which are threatened or at risk. Seabirds may be impacted by interaction with the mining vessel (particularly lighting impacts), disruption of food sources, and the introduction of pollutants into the environment.
Cumulative impacts	Currently, the only human use of natural resources on the Chatham Rise is fishing, which results in damage to benthic environments and bycatch of marine mammals, seabirds and other non-target species. Chatham Rock Phosphate has identified that approximately 92,000 square kilometres of the Chatham Rise (under 1,000 metres depth) was trawled between 1989 and 2011. The company states that the impacts of mining on the crest of the Chatham Rise (450 square kilometres over 15 years) will be proportionally very small compared to the area on the flanks of the Chatham Rise that has been and continues to be impacted by commercial fishing.

Figure 12.10 Potential environmental effects of the Chatham Rise phosphate mining
(Adapted from the Chatham Rock Phosphate marine consent application)

The decision

The Decision-Making Committee acknowledged that Chatham Rock Phosphate had gone to considerable lengths to provide information about the proposed mining operation. However, it was left with a number of uncertainties, many stemming from the fact that the proposal was the first of its kind in the world to be undertaken at such depths. Of particular concern was the reliance on modelling which had not been empirically and *in situ* validated. A staged adaptive management approach requiring extensive data collection and *in situ* validation was rejected as economically unviable by Chatham Rock Phosphate. Ultimately the Committee declined consent.

Benthic communities

The Committee found that the most significant adverse effect on the environment would be caused by the extractive part of the mining operation itself. This would result in the physical removal of the phosphate nodules and associated surface sediment along with all benthic biota living within and on the seabed. There was no dispute that this habitat and associated benthic communities would be destroyed over a large area. This included rare and vulnerable stony coral communities where were potentially unique to the crest of the Chatham Rise. The Committee found there was insufficient information verifying the existence of equivalent coral communities within other areas of the Chatham Rise outside the mining area. Referring to the Food and Agriculture Organization International Guidelines the Committee found these coral communities to constitute a rare and vulnerable ecosystem requiring protection.

The Committee concluded that recolonisation of the mined seabed by stony coral and other sessile organisms was unlikely because of their reliance on a hard substrate habitat. Re-establishment of benthic communities following mining was uncertain and at best would take decades. Further, the re-established communities would be based on soft sediment as opposed to a hard substrate and would likely be made up of different species. In addition, the benthic communities surrounding the worked mining block would be impacted by deposition of the returned sediment discharge. Although the extent of the sediment plume had been modelled, it had not been validated *in situ*. This meant that there was considerable uncertainty as to the extent of the area over which discharged sediment would disperse.

Benthic protection area

Part of the area in which Chatham Rock Phosphate proposed to mine the seafloor fell within an established benthic protection area where ground trawling and dredging by fishing vessels is prohibited. The Committee concluded that allowing mining of the seafloor in an area in which a directly comparable activity was prohibited was, at the very least, contradictory. The resultant effects of allowing mining to take place in the benthic protection area (or directly contiguous to it) were considered to be clearly contrary to its purpose, which was to protect the benthic environment, and would compromise its integrity.

The Committee did not accept that proposed mining exclusion areas could be considered compensation for the damage from mining because areas within the benthic protection area were already protected from bottom trawling and shellfish dredging. If the proposed exclusion areas were outside the benthic protection area there was no ability to ensure that they would be protected from activities other than the currently proposed mining activities.

Adaptive management

Chatham Rock Phosphate had proposed an adaptive management approach, restricting mining to a 820 kilometre area of the Chatham Rise for the first five years of mining operations, with the requirement that a set of conditions be met before mining could be expanded. Under the proposed approach, mining would be able to commence in the additional mining area only if:

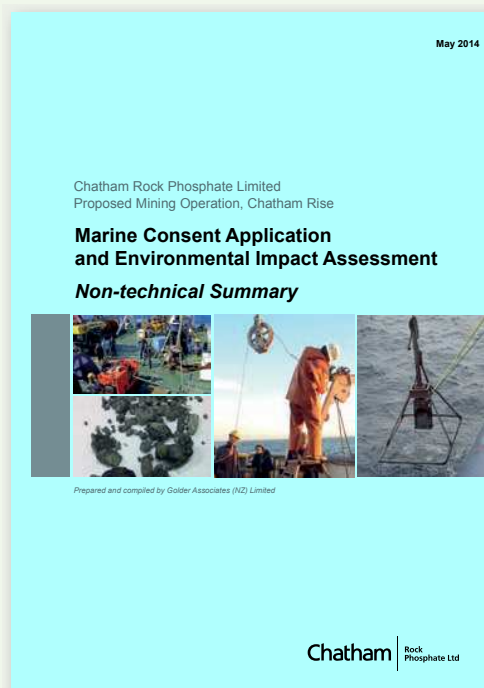
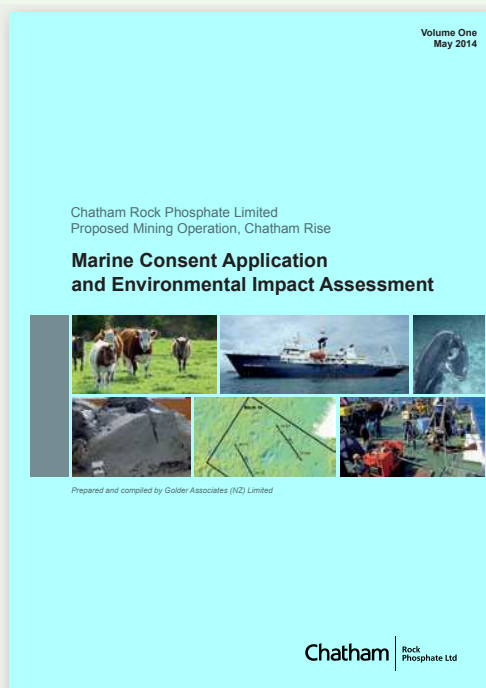
- The applicant had investigated the bathymetry, physical characteristics of the seabed, and benthic ecology of the additional mining area
- The plume modelling results had been confirmed based on current data for the additional mining area
- The applicant had identified mining exclusion areas in the additional mining area
- The Environmental Reference Group (made up of representatives of key stakeholders and scientists) had reviewed information gathered
- The applicant had been granted a mining permit under the Crown Minerals Act 1991

- Where total suspended solids concentrations exceed 50 milligrams per litre at a point 5 kilometres or greater away from the mining operations, or at a point 50 metres or greater above the seabed, the applicant had:
 - Undertaken additional monitoring to confirm that the threshold was exceeded
 - If the threshold was exceeded, advised the EPA
 - Undertaken investigations to identify the point at which total suspended solids reach 50 milligrams per litre and the point at which they reach 'background' levels
 - Within three months of advising the EPA, completed an assessment of responses which would avoid, remedy, or minimise total suspended solid levels. If a solution was identified, the applicant must identify timeframes for implementation.
 - If an adaptive management approach was to be implemented, it had to be implemented in accordance with the timeframe identified

The Committee found that the adaptive approach proposed by the applicant was inadequate because it did not provide for the validation of the modelling undertaken regarding, *inter alia*, the existence of equivalent habitat and benthic ecology outside the mining area and the extent of sediment dispersal. The Committee considered that this would require a three-stage adaptive management strategy. This would begin with a no-mining stage to allow for the collection of sufficient data regarding coral communities. If the uniqueness of the communities in the mining area was disproven, a trial mining stage to validate the sedimentation plume model over a small area could be authorised. Only then (provided necessary adjustments were made) would fully operational mining be permitted. The first two stages would contain a 'go/no go trigger', which could terminate the consent if the field evidence was sufficiently contrary to the applicant's prediction.

The staged adaptive management approach was rejected by Chatham Rock Phosphate on the basis that it would render the project unviable. As a result, it could not be imposed without frustrating the consent.

The Committee concluded that the proposal would create significant and permanent adverse effects which were incapable of being avoided, remedied or mitigated and this aspect of sustainable management was of paramount concern in this instance. With a staged adaptive management approach requiring considerable pre-mining research being dismissed as unviable by the applicant, the Committee declined consent.



A sample of the detailed documentation supporting the Chatham Rock Phosphate Limited's marine consent application

Further reading and references

Allsopp M, C Miller, R Atkins, S Roccliffe, I Tabor, D Santillo and P Johnston, 2013, *Review of the current state of development and the potential for environmental impact of seabed mining operations*, University of Exeter, Exeter, available at <http://www.greenpeace.to/greenpeace/wp-content/uploads/2013/07/seabed-mining-tech-review-2013.pdf>

Batson, P, 2003, *Deep New Zealand: Blue water, black abyss*, Canterbury University Press, Christchurch

Bollmann M, T Bosch, F Colijn, R Ebinghaus, R Froese, K Güssow, S Khalilian, S Krastel, A Körtzinger, M Langenbuch, M Latif, B Matthiessen, F Melzner, A Oschlies, S Petersen, A Proelß, M Quaas, J Reichenbach, T Requate, T Reusch, P Rosenstiel, J O Schmidt, K Schrottke, H Sichelschmidt, U Siebert, R Soltwedel, U Sommer, K Stattegger, H Sterr, R Sturm, T Treude, A Vafeidis, C van Bernem, J van Beusekom, R Voss, M Visbeck, M Wahl, K Wallmann and F Weinberger, 2010, *World ocean review: Living with the oceans*, Maribus gGmbH, Hamburg

Castle C, 2008, 'Interest soars in widespread Chatham Rise mineral prospect application', *Widespread Portfolios News*, 28 July, available at <http://widespread.co.nz/2008/7/27/interest-soars-in-widespread-s-chatham-rise-minerals-prospect>

Centre for Advanced Engineering, 2003, *Economic opportunities in New Zealand's oceans: Informing the development of oceans policy*, Centre for Advanced Engineering, Christchurch, available at <http://www.mfe.govt.nz/publications/oceans/economic-opportunities-oceans-jun03/economic-opportunities-oceans-jun03.pdf>

Centre for Advanced Engineering, 2006, *A report on possible government interventions to promote the sustainable development of New Zealand's ocean resources*, Centre for Advanced Engineering, Christchurch, available at <http://www.mfe.govt.nz/publications/oceans/government-interventions-ocean-resources-may06/index.html>

Collins P C, P Croot, J Carlsson, A Colaco, A Grehan, K Hyeong, R Kennedy, C Mohn, S Smith, H Yamamoto and A Rowden, 2013, A primer for the environmental impact assessment of mining at seafloor massive sulfide deposits, *Marine Policy*, 42, 198–209

Craw A, 2013, *Deep seabed mining: An urgent wake-up call to protect our oceans*, Greenpeace International, Amsterdam, available at <http://www.greenpeace.org/canada/en/campaigns/ocean/Seafood/Resources/Reports/Deep-Seabed-Mining/>

Funnell R, P King, V Stagpoole, C Uruski, B Field and M Isaac, 2009, *Potential undiscovered oil and gas resources of New Zealand, Phase II*, prepared for the Ministry of Economic Development, GNS Science, Wellington

Halfar J and R M Fujita, 2001, 'Precautionary management of deep sea mining', Unpublished paper, available at http://www.edf.org/documents/736_DeepSeaMining.pdf

International Marine Minerals Society, 2001, *Code for environmental management of marine mining*, International Marine Minerals Society, Honolulu, available at http://www.immsoc.org/IMMS_downloads/codefeb2002.pdf

International Seabed Authority, n.d., *Mining code: Comprehensive set of rules, regulations and procedures issued by the International Seabed Authority to regulate prospecting, exploration and exploitation of marine minerals in the international seabed area*, International Seabed Authority, Jamaica, available at <http://www.isa.org.jm/en/mcode>

MacDiarmid A, D Bowden, V Cummings, M Morrison, E Jones, M Kelly, H Neil, W Nelson and A Rowden, 2013, *Sensitive marine benthic habitats defined*, Prepared for the Ministry for the Environment, NIWA, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/managing-our-oceans/sensitive-marine-benthic-habitats-defined.pdf>

MacDiarmid A, R Boschen, D Bowden, M Clark, M Hadfield, G Lamarche, S Nodder, M Pinkerton and D Thompson, 2014, *Environmental risk assessment of discharges of sediment during prospecting and exploration for seabed minerals*, Prepared for Ministry for the Environment, NIWA, Wellington

Ministry for the Environment, 2005, *Offshore options: Managing environmental effects in New Zealand's Exclusive Economic Zone*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/offshore-options-jun05/index.html>

Ministry for the Environment, 2007, *Improving regulation of environmental effects in New Zealand's Exclusive Economic Zone: Discussion paper*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/nz-exclusive-economic-zone-discussion-paper-aug07/index.html>

Ministry for the Environment, 2008, *Cabinet paper: Proposal for Exclusive Economic Zone environmental effects legislation*, Unpublished paper, available at <http://www.mfe.govt.nz/issues/oceans/current-work/cabinet-paper-eez-effects-legislation.html>

Ministry for the Environment, 2012, *Managing our oceans: A discussion document on the regulations proposed under the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Bill*, Ministry for the Environment, Wellington

Ministry of Economic Development, 2008, *Minerals programme for minerals (excluding petroleum) 2008*, Ministry of Economic Development, Wellington, available at <http://www.crownminerals.govt.nz/cms/pdf-library/minerals/minerals-programme-for-minerals-2008>

Neptune Minerals, 2008, *Environmental strategy and management: Proposed SMS environmental baseline approach*, Unpublished report, Neptune Minerals, Wellington

NIWA, 2012, *Expert risk assessment of activities in the New Zealand Exclusive Economic Zone and extended continental shelf*, Prepared for the Ministry for the Environment, NIWA, Wellington

Scott S, 2006, *The dawn of deep ocean mining*, Unpublished paper, Natural Sciences and Engineering Research Council of Canada, Ottawa, available at http://www.eurekalert.org/pub_releases/2006-02/nsae-tdo021706.php

Secretariat of the Pacific Community, 2013a, *Sea-floor massive sulphides: A physical, biological, environmental, and technical review*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 1A, Secretariat of the Pacific Community, Noumea

Secretariat of the Pacific Community, 2013b, *Manganese nodules: A physical, biological, environmental, and technical review*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 1B, Secretariat of the Pacific Community, Noumea

Secretariat of the Pacific Community, 2013c, *Cobalt-rich ferromanganese crusts: A physical, biological, environmental, and technical review*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 1B, Secretariat of the Pacific Community, Noumea

Secretariat of the Pacific Community, 2013d, *Deep sea minerals and the green economy*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 2, Secretariat of the Pacific Community, Noumea

Secretariat of the Pacific Community, 2013e, *Summary highlights*, Deep Sea Minerals Volume SH, Secretariat of the Pacific Community, Noumea

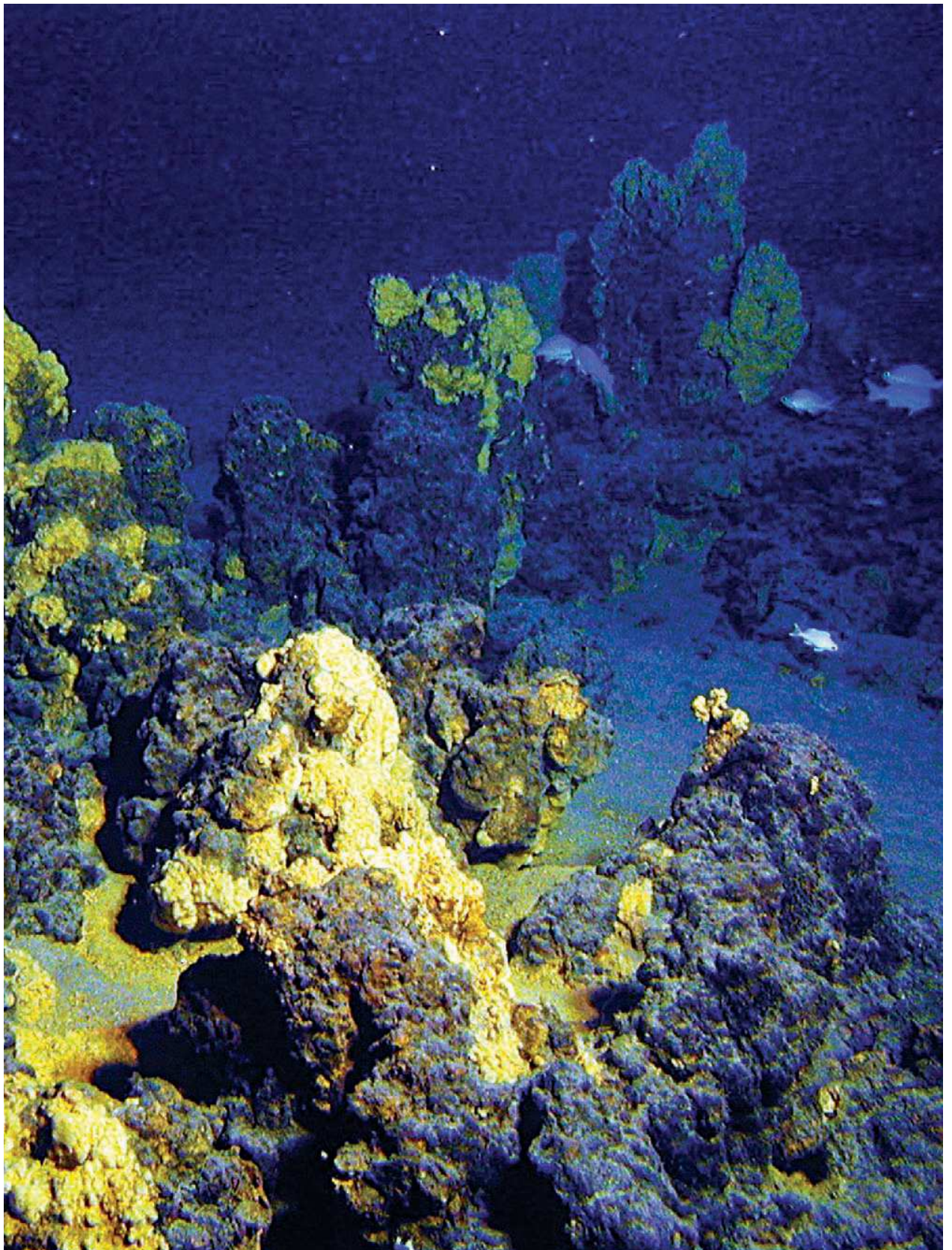
Steiner R, 2009, *Independent review of the environmental impact statement for the proposed Nautilus Minerals Solwara 1 seabed mining project, Papua New Guinea*, Conducted for the Bismarck-Solomon Seas Indigenous Peoples Council, available at <http://canaryinstitute.org/Documents/PNG.nautilus.indReview2009.PDF>

Stirling M, 2002, 'New Zealand's offshore mining regime – rights and responsibilities beyond the 12 mile limit', *New Zealand Journal of Environmental Law*, 6, 139–176

Suhr L, 2008, 'Environmental protection in deep seabed mining: International law and New Zealand's approach', *New Zealand Journal of Environmental Law*, 12, 97–144

Endnotes

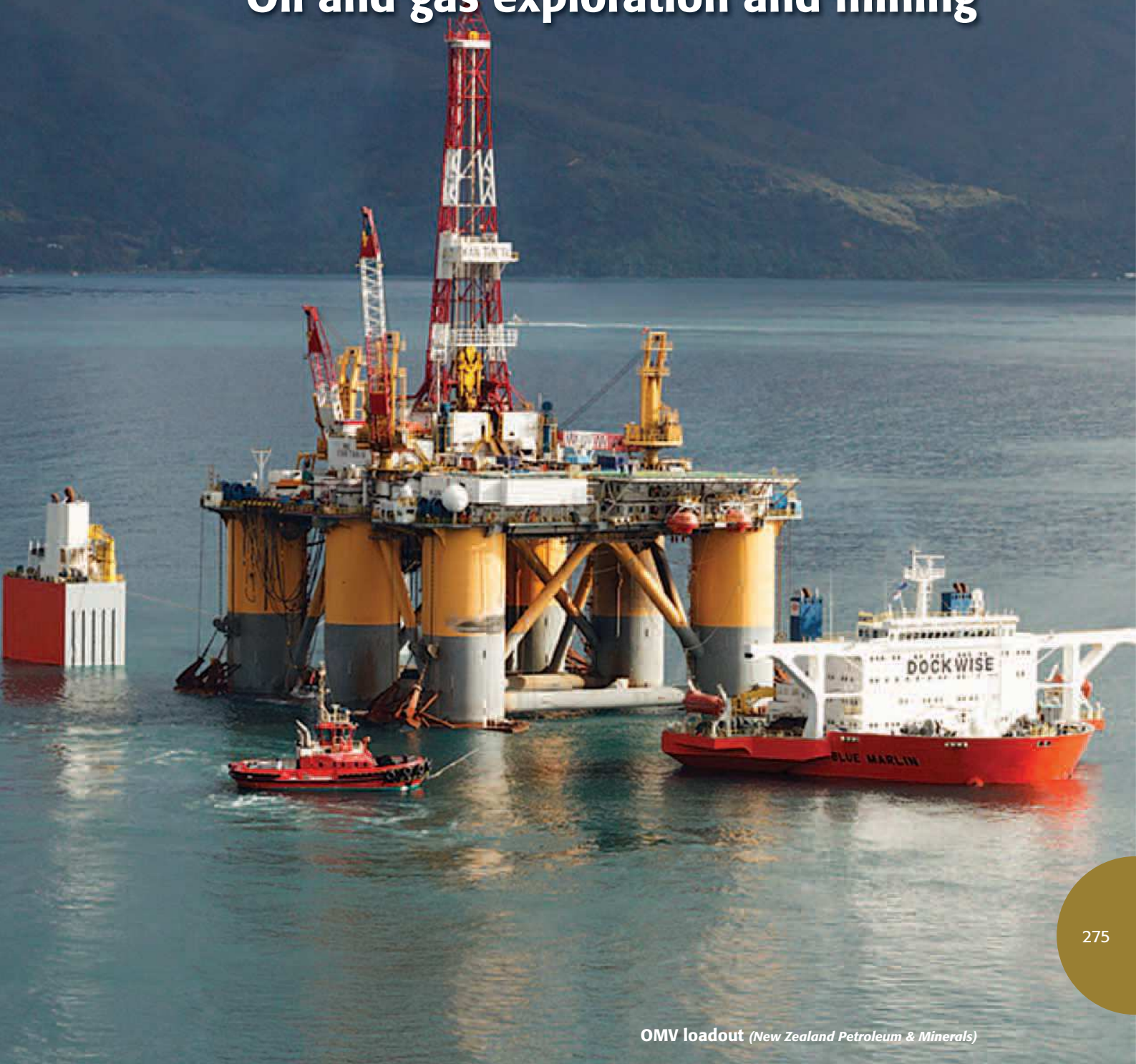
- 1 <http://worldoceanreview.com/en/wor-1/energy/marine-minerals/>
- 2 <http://www.gns.cri.nz/Home/Our-Science/Energy-Resources/Oil-and-Gas/NZs-Sedimentary-Basins>
- 3 Ministry for the Environment, 2012
- 4 NIWA, 2012
- 5 Ministry for the Environment, 2012
- 6 NIWA, 2012
- 7 <http://worldoceanreview.com/en/wor-1/energy/marine-minerals/>
- 8 <http://www.straterra.co.nz/21st%20century%20mining>
- 9 NIWA, 2012
- 10 <http://www.straterra.co.nz/21st%20century%20mining>
- 11 <http://www.gns.cri.nz/Home/Our-Science/Energy-Resources/Gas-Hydrates/Current-Research/Gas-hydrates-as-an-energy-resource>
- 12 NIWA, 2012
- 13 Ministry for the Environment, 2012
- 14 Crown Minerals Act, section 2(1)
- 15 Crown Minerals Act, section 2(1)
- 16 NIWA, 2012
- 17 Secretariat of the Pacific Community, 2013a
- 18 <http://www.nautilusminerals.com/s/Projects-Solwara.asp>
- 19 Collins P C et al., 2013
- 20 Collins P C et al., 2013
- 21 Bollmann M et al., 2010
- 22 Secretariat of the Pacific Community, 2013b and 2013c
- 23 http://www.worldenergy.org/wp-content/uploads/2013/10/WER_2013_3_Natural_Gas.pdf
- 24 <http://www.technologyreview.com/news/413730/mining-ice-that-burns/>
- 25 <http://www.gns.cri.nz/Home/Our-Science/Energy-Resources/Gas-Hydrates/Current-Research/Gas-hydrates-as-an-energy-resource>
- 26 Funnel R et al., 2009
- 27 NIWA, 2012, 27
- 28 NIWA, 2012
- 29 NIWA, 2012
- 30 NIWA, 2012
- 31 Suhr L, 2008, 113
- 32 NIWA, 2012
- 33 Secretariat of the Pacific Community, 2013a
- 34 Secretariat of the Pacific Community, 2013e
- 35 Secretariat of the Pacific Community, 2013e
- 36 <http://www.gns.cri.nz/index.php/Home/Our-Science/Earth-Science/Ocean-Floor-Exploration/Oceans-Research/Geological-Exploration-of-the-EEZ>
- 37 Bollmann M et al., 2010
- 38 Allsopp M et al., 2013
- 39 Secretariat of the Pacific Community, 2013e
- 40 Allsopp M et al., 2013
- 41 Secretariat of the Pacific Community, 2013e
- 42 <http://www.niwa.co.nz/publications/wa/water-atmosphere-2-february-2011/oceans-treasure>
- 43 <http://www.niwa.co.nz/publications/wa/water-atmosphere-2-february-2011/oceans-treasure>
- 44 <http://www.nzpam.govt.nz/cms/contact/ask-a-question/crown-minerals-act-regime-changes#4>
- 45 <http://www.doc.govt.nz/Documents/conservation/native-animals/marine-mammals/seismic-survey-code-of-conduct.pdf>
- 46 MacDiarmid A et al., 2014



Iron and silica rich chimneys on the summit of Giggenbach volcano, Kermadec Arc
(New Zealand-American Submarine Ring of Fire 2005 Exploration; NOAA Vents Program)

13

Oil and gas exploration and mining



Contents

277	Introduction
279	Methods used for oil and gas exploration and mining
279	Prospecting and early exploration
280	Exploration
281	Production
281	Decommissioning
282	Potential environmental effects
282	Blowout and oil spills
283	Other potential impacts
284	Fisheries
285	Seabed habitats
285	Wildlife
286	Biosecurity
286	Amenity
286	Other impacts
287	Decision-making framework
287	Allocation of mining rights
287	Resource Management Act
287	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act
288	Marine Mammals Protection Act
291	Maritime Transport Act
292	Health and safety management
292	Oil spill response
293	Elements of best practice
295	Case study
295	International recommendations for improving well incident prevention, intervention and response capacity
297	Further reading and references
298	Endnotes

Introduction

Oil and gas are currently an essential part of this country's energy mix. They are used in transportation, production, manufacturing, construction, domestic and international travel and leisure. Oil is now New Zealand's fourth-largest merchandise export (after dairy, meat and wood), with a value in 2011 of nearly NZ\$2.2 billion.¹ This is despite the fact that production is limited to a relatively small number of fields and only one petroleum basin.² New Zealand has rights to exploit oil and gas over an area of seabed equivalent in size to the European Union, the North Sea and a quarter of the Mediterranean combined.³ There are multiple sedimentary basins with known or potential hydrocarbons onshore and underlying the extensive continental shelf of New Zealand, as well as several deepwater basins within the EEZ (shown in Figure 13.1 which has been prepared by the Ministry of Business, Innovation and Employment).⁴

While New Zealand has multiple basins with hydrocarbon potential, only one, the Taranaki Basin, is currently producing. Over 400 onshore and offshore exploration and production wells have been drilled to date in the Taranaki Basin and there are currently seven producing installations.⁵ No wells have yet been drilled beyond the continental shelf edge.

There are four other basins in the country which are known to hold petroleum – the Reinga-Northland Basin, East Coast Basin, Canterbury Basin and Great South Basin. New Zealand has at least 14 sedimentary basins of various ages with hydrocarbon potential.⁶

Most of New Zealand's marine area remains unexplored, but our knowledge of its geology suggests that over 20 per cent of it has the potential to hold petroleum resources.⁷ Currently, just over 30 offshore exploration permits have been issued by New Zealand Petroleum and Minerals.⁸ With the historical increase in petroleum prices, and the decline in existing fields, there was (until recent price declines) increasing interest in petroleum exploration within New Zealand's marine environment.⁹

The burning of oil and gas creates greenhouse gas emissions contributing to global climate change. For this reason, some argue that no further oil and gas exploration should be undertaken in New Zealand or elsewhere. This chapter focuses on the environmental impacts arising from oil and gas exploration and mining activity itself, rather than addressing the wider global impacts of burning the resulting product.



The Taranaki Basin is the only basin in New Zealand currently producing oil and gas, although four other basins are known to hold petroleum (Raewyn Peart)



NEW ZEALAND
PETROLEUM & MINERALS

New Zealand Petroleum Basins

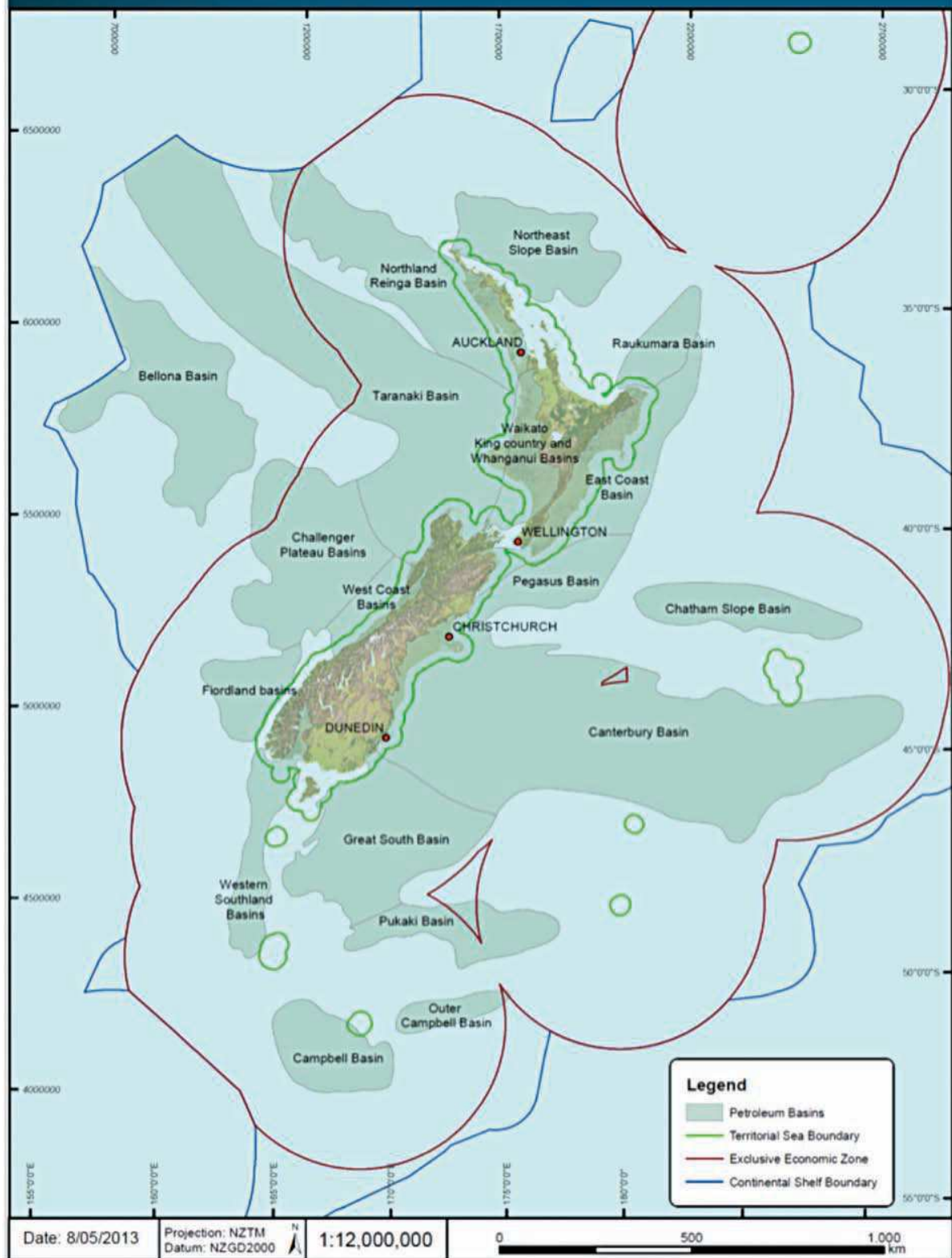


Figure 13.1 Locations of New Zealand's petroleum basins (Ministry for the Environment, 2013)

Methods used for oil and gas exploration and mining

There are some key stages involved in oil and gas exploitation as outlined in Figure 13.2.



Figure 13.2 Main stages in the exploration and mining of oil and gas resources (www.cairnenergy.com)¹⁰

Prospecting and early exploration

Identifying a petroleum prospect to be drilled involves various steps, the most significant and costly of which is seismic surveying over a large area to obtain data about the topography of the seafloor and geology of the subsurface. Seismic surveying involves sending acoustic sound waves into the rock layers beneath the seafloor and then recording the time it takes for each wave to bounce back, as well as measuring the strength of each returning wave.¹¹ The various layers of rock reflect back different signals. Figure 13.3 illustrates the seismic survey at work. Underwater remotely operated vehicles or submersibles and camera systems can also be used to capture a wide range of data through both images and the collection of samples.

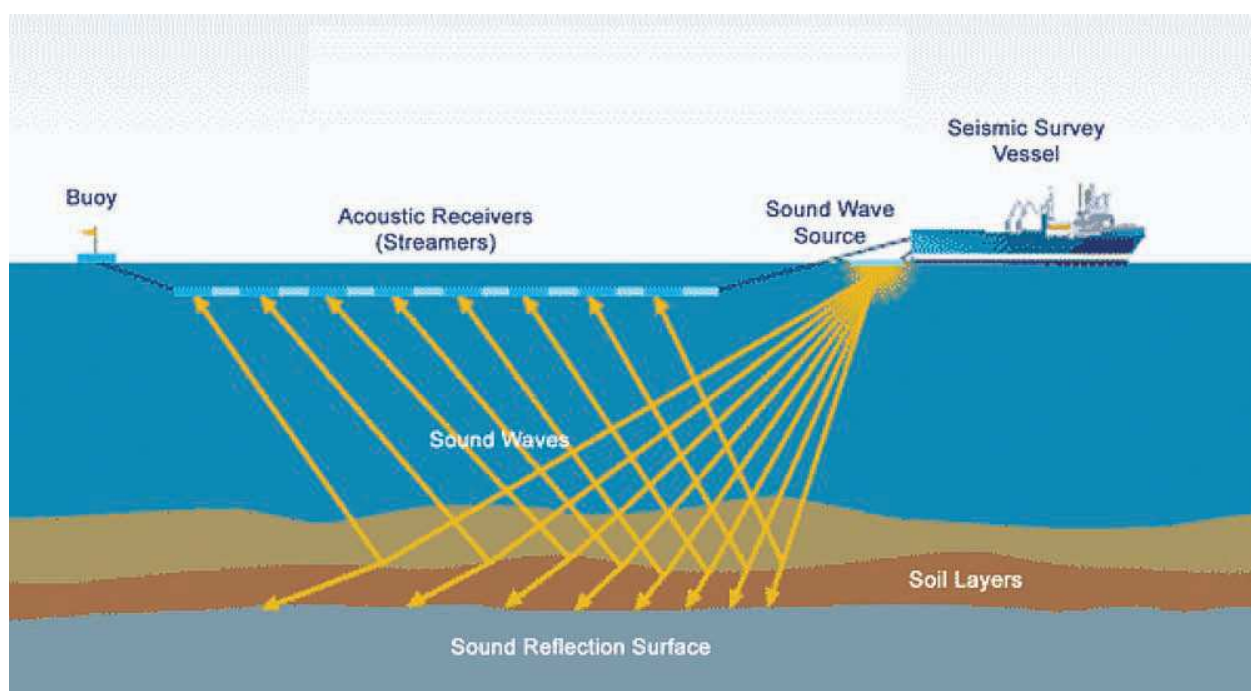


Figure 13.3 Seismic surveying (*Oil Spill Solutions*)¹²

Exploration

If the initial phase delivers some positive results, then well drilling to discover and define potential oil and gas deposits may take place. Exploration typically focuses on a smaller spatial area of interest than prospecting, which then becomes the focus of more intense investigation. Activities authorised by an exploration permit can also include further more detailed seismic surveying (usually three-dimensional).

A mobile offshore drilling unit is used to drill a well in order to assess whether oil and/or gas is located in a specific geological zone, and in the specific location that has been identified as a 'prospect' based on seismic surveying and other analysis. The type of rig used will depend on the depth of water, ocean conditions and other factors. The three main types of rigs used for exploration are:¹³

- Jack-up rigs – used in shallower water up to around 120 metres deep. The rig is towed to the location, and its legs are lowered to the bottom to enable the rig to be raised above the water surface.
- Semi-submersibles – used in depths of 100 to 3,000 metres. These consist of a floating, column-stabilised vessel, with a submerged hull. The rigs can be towed and anchored, or are able to maintain their own position with thrusters.
- Drillships – used in depths of up to 3,000 metres. These are purpose-built ships with drilling and other equipment installed on board. They function as a ship until they reach the drill site, where they are either anchored or maintain their own position with thrusters while undertaking drilling operations.

Drilling a single well may take weeks to a few months to reach the target depth, depending on various factors including the depth of the target, the rock types encountered and weather conditions. One or more wells may be drilled under a single exploration permit. Exploration permits require a certain number of wells to be drilled within a specified time frame.

In order to create an offshore well, an oil rig drills through the seabed and into the subsurface reservoir. There are a number of steps that are needed for drilling a well which are described below:¹⁴

1. Spudding – a metal tube is forced into the ground (this tube can be up to 100 metres long by 92 centimetres in diameter). This provides a casing which is the backbone of the well. The casing is lowered onto the seafloor by a 'drill pipe'. More drill pipe is added over time to force the casing into the seabed until it reaches the targeted petroleum prospect. Some rigs use large hammers to push the casing deeper. Once the right depth is reached the drill pipes are pulled back to the surface.
2. Drilling a hole for a second string of casing – a drill bit is lowered inside the casing. This is generally drilled between 300 and 1,000 metres into the seabed. High-pressure drilling fluids called 'mud' is pumped down the centre as the drill is rotated. This fluid helps to keep the drill cool, provides lubrication, and carries chipped rock pieces away from the well. This mud is also the primary barrier for keeping hydrocarbons in the well during drilling. A second smaller casing (often 56 centimetres in diameter) is lowered down the hole.
3. Cementing the casings in place – cement is pumped down the drill pipe and back up between the two casings. This helps to cement the casing to the rock in the seafloor and is a vertical barrier on the outside of the casing.
4. Connecting the blowout preventer and marine riser – a large underwater control valve known as the 'blowout preventer' is placed on top of the wellhead. It is a critical piece of equipment, that is generally fixed directly on top of the wellhead on the ocean floor, and functions as the secondary barrier to controlling hydrocarbon pressures. The blowout preventer is connected by a marine riser to the drilling rig or drillship. Once these are installed, drill pipe is lowered through the marine riser and the blowout preventer. The marine riser allows the mud and rock cuttings to be pumped back to the surface and onto the drilling rig for processing.
5. Drilling the remaining sections of the well – steps 2 and 3 can be repeated to drill the remaining well sections. The rock cuttings and mud are circulated back up to the drilling rig.
6. Conducting test logs – once the correct depth has been reached, tests are conducted to see whether oil and gas is present, and if so the nature and extent of it. The information is logged. In some cases hydrocarbons will be brought to the surface to flow test the well.
7. Once an exploration well is completed, it will usually be plugged with a cement plug and abandoned. In some situations, it may be temporarily abandoned so that it can be subsequently tested or produced from. Once the well is abandoned, the drilling rig will leave the drill site.

If the exploratory phase is successful, then the operator will usually return later to drill appraisal wells to further understand the nature and extent of the field that has been discovered. In parallel to this, the operator will be working on development scenarios for the field, which will involve considering engineering, commercial, environmental and other factors.

Production

The development phase leading to production generally involves well drilling, construction of a platform structure or floating production vessels, and laying of underwater pipelines. Examples of production from existing offshore operations in New Zealand are:

- Māui A – two fixed platforms offshore with pipelines to shore and further processing onshore
- Tui field – the Umuroa floating production, storage and offloading facility
- Maari field – one fixed platform connected to the Raroa floating production, storage and offloading facility

Gas can be injected into an oil reservoir, which is known as re-injection. This process increases the pressure within the reservoir to help maintain pressure and production levels. In addition, it can assist with stimulating more oil recovery from a mature oil field. This has been done at the Kapuni field to extend its economic life.¹⁵



Māui A platform off the Taranaki coast (Shell Todd Oil Services Limited)

Figure 13.4 illustrates different kinds of production facilities.

Commissioning the Māui platform

To enable the development of the Māui gas field in 1973, one of the largest in the world at the time, a drilling platform had to be found which could withstand one of the most hostile environments of any offshore discovery. No existing designs were suitable for the extreme wind and waves. One of the largest steel manufacturers in Japan began construction in 1974. The tower was built at the Tsu shipyard before crossing more than 8,400 kilometres of ocean to New Zealand. Bad weather delayed the project for several months before the platform was officially handed over to the drilling teams in December 1977.¹⁶

There is a range of support vessels and helicopters which are located in the vicinity of the oil and gas field providing assistance to the drilling rig itself. In addition, production facilities and related infrastructure are required. These may be located on or offshore depending on the nature of the field (oil or gas) and the distance away from the shore. At the Tui and Maari facilities the oil is loaded directly onto oil tankers for export.

Decommissioning

Once all the oil and gas has been extracted, the wells and facilities will all be decommissioned. This



Figure 13.4 Various kinds of rigs and platforms above and below the water (Oil Spill Solutions¹⁶)

involves plugging the well with cement and dismantling removing of equipment, plant and machinery. All New Zealand's offshore fields remain in production so decommissioning is yet to take place here.

Prior to decommissioning, surveys are usually undertaken to map the location and quantity of debris, pipelines, power cables and natural marine environments. Post-decommissioning surveys can identify any debris left behind during the removal process and confirm any environmental damage. Remote underwater vehicles and divers can be used to remove any debris and to ensure the area is free from any of the operational equipment.

Abandonment of oil rig infrastructure has become a major issue in the North Sea where there are many fields reaching the end of their producing phase. Best practice is to remove as much of the infrastructure as safely as possible. Another option being investigated is the storage of CO₂, where some of the existing infrastructure of depleted offshore oil and gas fields might be used for CO₂ sequestration after adaptation.¹⁷

Potential environmental effects

Blowouts and oil spills

The risk and consequences of oil spills are probably the greatest environmental concern associated with oil and gas operations. While most oil enters the oceans unseen, massive oil spills highlight the catastrophic effects that oil can have on marine wildlife and human communities.¹⁸ There has been only one major deepwater blowout, which was the Deepwater Horizon well in the Gulf of Mexico, which blew catastrophically in 2010. So far, there have been no major well blowouts in New Zealand, but the risks may increase as oil and gas activity expands. Because this country is so far away from any offshore oil and gas servicing ports, it would take a long time for international aid to respond.¹⁹

Formations deep in the ground, where hydrocarbons are located, are under intense pressure. Controlling these pressures is fundamental to successfully drilling wells. Various 'barriers' are employed to ensure pressures are controlled throughout the life of the well, including the initial drilling phase. The key principle is that there are always two tested barriers in place ('the two barrier rule').

The high pressure drilling fluids called 'mud', which are pumped down the centre of the drill string during drilling, create a primary barrier which exerts hydrostatic pressure to stop hydrocarbons coming to the surface. Mud weights are adjusted during drilling to ensure balance is maintained. The blowout preventer generally acts as the second barrier if the mud fails when drilling an exploration well.²⁰

Well blowouts occur when the pressure in the reservoir being drilled overwhelms the primary mud barrier, and the blowout preventer and other secondary barriers are unable to stop the flow. This can be a result of faults in the safety device which is intended to seal the well, poor well design, and/or human error. Often it is the result of a combination of these factors. Blowouts can sometimes be brought under control quickly, but in other situations they can last for a prolonged period. In 1979, when the Ixtoc 1 drilling rig sank as the result of a blowout, it took engineers nearly a year to stem the flow of oil into the Gulf of Mexico.²¹

Deepwater Horizon blowout

A report by the Chemical Safety Board, an independent investigative arm of the USA federal government, concluded that the 2010 Deepwater Horizon disaster was caused by the blowout preventer failing to properly close the well when a surge of natural gas from the deep formation caused a blowout. The blowout preventer itself punctured the riser pipe, allowing oil to start leaking into the Gulf of Mexico close to the seabed. This triggered the worst offshore oil spill in history. About five million barrels of oil flowed into the Gulf before the well was closed off, 87 days later.²²

The impact of such oil spills can vary from minimal to large-scale mortality of marine life, depending on the sensitivity of the environment and its ability to recover. The Ixtoc 1 disaster contaminated the Gulf of Mexico waters with millions of barrels of crude. Fisheries had to close, birds and other wildlife perished, and vast lengths of coastline were soiled.²³

Figure 13.5 highlights some of the key potential environmental impacts resulting from oil spills caused by oil and gas drilling, including well blowouts. Oil spills from vessels and their impact on the environment are discussed in Chapter 15: Vessels.

Area	Potential impact from oil spill
Habitats	- Habitats become unsuitable for feeding, nesting and other services they provide for marine life - Intertidal areas, wetlands and mangroves become inhabitable - Nursery areas for fish and shellfish species become damaged
Animals	- Direct mortality of marine life - Oil coating limits the ability to swim or fly, and to maintain body temperature, feed properly and even reproduce - Oil harms the eyes, mouth, nasal tissue, immune system, red blood cells and organs of marine animals - Greater death of young animals which are usually the most vulnerable to the effects of oil
Toxic compounds	Release of toxic and carcinogenic compounds from the oil into the marine environment including trace metals and polycyclic aromatic hydrocarbons
Social/cultural	- Significant loss to customary and recreational fishing - Stress on subsistence communities which depend on fish and shellfish for food - Reduction in amenity and recreational values, particularly when oil reaches the shoreline
Economic	- Loss of incomes to tourism-based businesses and communities - Financial loss to the commercial fishing industry, both because of impacts on the fisheries themselves, and on market perceptions - High costs of clean-up which can reach billions of dollars, as seen in the Gulf of Mexico spill

Figure 13.5 Potential environmental impacts from an oil spill (*Adapted from Oceana, 2009*)



Petroleum offloading off the Taranaki coast (*New Zealand Petroleum and Minerals*)

Other potential impacts

Other environmental impacts can result from equipment and ship movements, the use of artificial light, increased noise levels and air emissions. Figure 13.6 identifies some of these potential direct and indirect environmental impacts, which are then discussed in further detail below. Many of these impacts are confined to a small area surrounding the activity. Ways of addressing them are discussed in the best practice section below.

Activity	Potential impact
Ship movements	- Noise, lights and the physical presence of ships affecting the movement of sensitive species - Dredging of navigation channels if necessary, disturbing and modifying the seafloor
Seismic surveys	Underwater noise generated by air-based energy sources in seismic arrays disturbing marine animals during the duration of their deployment, especially marine mammals
Drilling rig placement	Localised physical damage to the seafloor where the rig is placed
Anchoring	Localised physical damage to the seafloor where anchors are lodged
Drilling	- Localised smothering and toxic impacts on marine life from drill cuttings if they are discharged overboard - Release of toxic substances into the marine environment from clay-based drilling fluids which include metals such as barium and a range of additives, depending on the type of rock being drilled - Reduction of light in the water column due to suspended sediment which may reduce plant growth until the particulate material settles
Platform placement	- Localised physical damage to the seafloor where the platform is placed - Enhanced marine habitat through the creation of artificial reefs - Provision of important resting places for seabirds and haul outs for seals and/or sea lions
Produced formation water	- Localised contamination of seawater through the discharge of low levels of hydrocarbons contained in the produced formation water. Long exposures to low levels of particular hydrocarbons can result in cumulative and/or sublethal impacts. - Localised warming of seawater from the discharge of warmer produced formation water - Localised changes in salinity levels as produced formation water may have a salt content greater or less than seawater
Greenhouse gas emissions	Contributes to global greenhouse concentrations through flaring and use of fuel for vessels and power generation

Figure 13.6 Potential environmental impacts from offshore oil and gas exploration and mining

(Adapted from the World Petroleum Organization)²⁴

Fisheries

Permanent oil and gas production activities may affect fisheries through direct effects on fished species. Fish may move away from the area of influence due to increased levels of activity.²⁵ In addition, the new structures will alienate a small area of seafloor, so that it no longer provides foraging habitat for demersal fishes (which live and feed on or near the seafloor). However, these same structures may provide new artificial reef-like surfaces and increase reef habitat in the area.²⁶

There may be impacts on fisheries management as production platforms and pipelines invariably have a zone of restricted access around them that restricts fishing access in these areas. Unless commercial fish quota is bought by the petroleum company, and retired for the duration of the petroleum activities, commercial fishing will be displaced into the remainder of the quota management area, increasing pressures on fish populations there. The impact of a single production facility is unlikely to be great, but the cumulative impact could be significant if multiple production facilities and pipelines are located within a single quota management area.²⁷ There are fisheries exclusion zones around rigs and pipelines in the North Sea. However, it has been estimated that despite more than 500 platforms and thousands of kilometres of pipelines being in place there, the loss of fishing area was only around one per cent.²⁸



Fisheries may be impacted by oil and gas production activities (Daniel Sharp)

Seabed habitats

Marine life present on the seabed where drilling occurs will be directly impacted, in particular by the legs of a jack-up rig or fixed platform, the well bore and wellhead itself, and any mooring chains and anchors.

The type of drilling lubricant used (oil-based, synthetic or water-based) has a large bearing on the toxicity of drill cuttings and the speed of impact if cuttings from the well are discharged into the water column and then settle on the seabed, rather than being collected and disposed of on land.²⁹ If a submerged pipeline is built to bring the oil or gas onshore, then the seabed will be damaged during construction and any associated trenching.

The impacts of an individual platform and associated infrastructure are localised, but there may be more significant cumulative impacts from multiple production facilities and pipelines in a single region. The level of this impact will depend on whether the area contains vulnerable benthic species, and whether these are limited in range or are rare.

The effects during decommissioning will depend on whether the infrastructure is dismantled and removed, sunk to the seafloor and abandoned, or left intact for another use such as carbon sequestration. If left intact, the effects of the structure being present will continue as for the production phase. Removal of the structure will have some impact on the seafloor and benthic organisms, as well as on the creatures which have become attached to the structure. In some cases the environmental impacts may be less if it is left *in situ*. If the infrastructure is abandoned, the oils and toxic substances will need to be cleaned off the structure to ensure that they do not dissipate into the marine environment. The platform will slowly rust and disintegrate, and this may cause negative effects over a long term.³⁰

Wildlife

Seismic surveys may have an acoustic impact on marine fauna. The acoustic impact of seismic surveys on marine mammals is particularly well documented. The consequences will vary from none to behavioural (may leave the area) to acute injury (ear drum damage) to serious (death) depending on the noise level encountered, the species and the habitat. There may also be cumulative effects from repeated exposure.³¹

Seismic streamers (as shown in Figure 13.3) may attract fur seals and sea lions during deployment and retrieval, creating a risk of entanglement and injury or drowning.

Lights at night (such as those on a drilling platform) typically attract and disturb seabirds and may cause them to collide with the above sea-surface structure. Subsurface activity and lights may affect marine mammals. Fauna may also interact with the infrastructure. For example, seal haul outs occur on the leg bracing struts of some platforms.³²

Survey and support vessels create a risk of ship strike. In addition, the installation of the production platform, pipeline construction and trenching, underwater inspection using remotely controlled vehicles or submersibles, drilling activities, and routine maintenance and pumping will produce some underwater noise that could affect marine mammals, mainly causing them to avoid the area.³³



Seals may become entangled in oil and gas exploration and mining equipment and may haul out on the struts of some platforms (Raewyn Peart)

Biosecurity

Oil drilling platforms create biosecurity risks as they are large structures which lie stationary in coastal waters for long periods of time, which enables organisms to grow on the immersed parts. Drillships and platforms brought in from overseas can potentially bring with them invasive marine pests. In the past rigs have been towed to New Zealand, but recently, they have been transported on heavy lift vessels, so biosecurity can be well managed.

The South African brown mussel (*Perna perna*) was detected in Tasman Bay following the cleaning of a semi-submersible oil rig's submersed surfaces that had arrived from Australia in November 2007.³⁴ In total, 10 non-indigenous species were found on the rig. This resulted in a clean-up operation to remove the risk of the exotic pests establishing in New Zealand. Transported species are unlikely to survive independently of the rig in deep offshore waters – the risk is that the platform acts as a stepping stone for non-indigenous species to invade shallow coastal habitats. The risk can be lowered by cleaning and/or fresh antifouling if necessary.³⁵

Biosecurity risks and impacts are discussed further in Chapter 5: Marine biosecurity.

Amenity

Near-shore oil and gas activities can potentially have the following amenity effects:

- Noise from the operation could affect residents and visitors to the area, if operations are close to the shoreline
- Vessels passing regularly through the marine area can create an unwelcome visual impact, particularly in areas which are normally little used by vessels
- The presence of oil rigs in the sea can industrialise the natural ocean landscape

Other impacts

Near-shore oil and gas activities may negatively impact on areas and resources of importance to iwi and hapū. These could include kaimoana areas and wāhi tapu or urupa sites which lie on or under the seabed.

There may be other ecological impacts resulting from increased light and noise in the deep-sea environment. These are poorly understood but could impact on fish and marine mammals that are dependent on sound for locating prey, navigating and communicating.³⁶ There is also the risk of accidents leading to spillages and loss of equipment within the marine environment.



Māui B platform off the Taranaki coast (Shell Todd Oil Services Limited)

Decision-making framework

Allocation of mining rights

Applications for prospecting, exploration and mining licences are processed by New Zealand Petroleum and Minerals. The Minerals Programme for Petroleum sets out the policies and procedures which are followed for the allocation of mineral resources (including consultation with iwi and hapū). The requirements to be met by permit holders are defined in the regulations.

The Crown Minerals Act separates activities into Tier 1 and Tier 2. Tier 1 includes high-commercial-return, high-commercial-risk projects, and most, if not all, marine petroleum operations will fall within this category. The Act requires a high-level preliminary assessment of the applicant's health, safety and environmental policies and capabilities to be undertaken for all applications for Tier 1 activities. This is not required to duplicate any assessment process under other legislation, nor does it affect the requirements under any health and safety or environmental legislation. Permits cannot be granted over marine reserves as these are in Schedule 4 of the Crown Minerals Act.

Resource Management Act

The RMA regulates the environmental effects of oil and gas activities within the territorial sea. The relevant regional coastal plan determines whether a resource consent is required. An activity cannot occur if the relevant regional coastal plan classifies it as a prohibited activity, unless the plan is changed. Generally, petroleum prospecting is classified as a permitted activity and mining as a discretionary activity, except in high-value areas where non-complying or prohibited activity status may apply.

When an application is lodged, the consent authority will determine whether or not it should be publicly notified. A consent application must be publicly notified if the consent authority determines that its adverse effects on the environment are likely to be more than minor.³⁷ This is likely to be the case for most production drilling proposals, so they should in most cases be publicly notified, enabling wide public involvement in the decision-making process.

When considering an application for consent, the consent authority must also have regard to the relevant provisions of the applicable statutory instruments made under the Act, for example the NZCPS, regional policy statements and the regional coastal plan. The NZCPS contains a number of policies that are relevant to applications for petroleum activities, including:

- Adopt a precautionary approach towards proposed activities whose effects on the coastal environment are uncertain, unknown, or little understood, but potentially significantly adverse³⁸
- Adverse effects on certain indigenous biodiversity, such as taxa which are threatened or at risk or naturally rare, must be avoided³⁹
- Adverse effects must be avoided in areas with outstanding natural character⁴⁰
- Adverse effects must be avoided in outstanding natural landscapes⁴¹
- In managing discharges to water in the coastal environment, have particular regard to [the need to] avoid significant adverse effects on ecosystems and habitats after reasonable mixing⁴²

Often the scientific evidence supporting an application contains considerable uncertainties because the marine environment is a dynamic system subject to constant change, and obtaining scientific data can be expensive and difficult. Demonstrating a cause and effect linkage can be problematic. In such cases the NZCPS dictates that a precautionary approach should be adopted.

The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

The EEZ Act regulates the environmental effects of petroleum activities beyond the territorial sea. The EEZ Permitted Activities Regulations permit seismic surveying if the person undertaking the seismic surveying complies with the Department of Conservation's 2013 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations',⁴³ which is discussed further below.

The regulations also permit prospecting and exploration for petroleum (except for any drilling for petroleum). Petroleum exploration drilling is non-notified discretionary meaning that consent is required but there is no opportunity for public submissions. Petroleum drilling for mining is a discretionary activity and applications must be publicly notified.

No prohibited activities are identified in the regulations, so a marine consent can be sought to authorise petroleum drilling and mining activity anywhere within New Zealand's EEZ and extended continental shelf.

The EEZ Act will also regulate discharges and dumping from offshore installations, as well as discharges from ships if it is a mining-related discharge, once regulations are made. Discharges and dumping from vessels continues to be regulated under the Maritime Transport Act. The Exclusive Economic Zone and Continental Shelf (Environmental Effects – Discharge and Dumping) Regulations 2014 classify discharge and dumping activities.⁴⁴ At the time of writing these were still in draft format and had not yet been promulgated.

Figure 13.7 sets out the activity status of discharge and dumping activities that specifically relate to oil and gas activities. These were not final at the time of writing.

Activity	Activity status
Discharge of harmful substances from petroleum exploration drilling	Non-notified discretionary
Discharge of harmful substances from petroleum production drilling	Discretionary
Discharge of oily waste from machinery waste	Permitted, subject to conditions Prohibited, if the oil content exceeds 15 parts per million
Discharge of contaminated seawater from operations	Permitted, subject to conditions Prohibited, if the oil content exceeds 15 parts per million
Discharge of drilling fluids	Non-notified discretionary
Discharge of food waste	Permitted, subject to conditions
Discharge of garbage	Prohibited, unless necessary for safety reasons or accidental
Dumping of fish waste, sewage sludge, organic material or mineral exploration structures	Non-notified discretionary
Dumping of dredged material or vessels in authorised dumping grounds	Non-notified discretionary
Dumping of other dredged material, other vessels, other structures, CO ₂ streams from CO ₂ capture processes or inert inorganic geological material	Discretionary
Dumping of waste or other material not otherwise classified	Prohibited

Figure 13.7 Proposed activity status of discharge and dumping activities that specifically relate to oil and gas activities

The marine consent process is described in full in Chapter 3: EEZ legislation.

To date the EPA has considered two non-notified applications for marine consent for oil and gas exploration wells from Shell Todd Oil Services Limited and OMV New Zealand Limited. Both applicants commenced drilling under the transitional provisions of the EEZ Act but were required to seek a marine consent because drilling was not able to be completed before 28 June 2014. The OMV New Zealand application was to drill two exploration wells near the Maari platform in the South Taranaki Bight. The wells will be drilled in a water depth of 98 metres and to a total depth of 2,839 metres.⁴⁵ The Shell Todd application was to drill two exploration wells near the Māui field in the South Taranaki Bight. The wells will be drilled in a water depth of 105–108 metres and to a total depth of 3,370–5,183 metres.⁴⁶ Both applications were granted.

Marine Mammals Protection Act

Marine mammal sanctuaries can contain restrictions on extractive industry-related activities. For example, the West Coast North Island Marine Mammal Sanctuary restricts seabed mining and puts stringent conditions on seismic surveying.⁴⁷

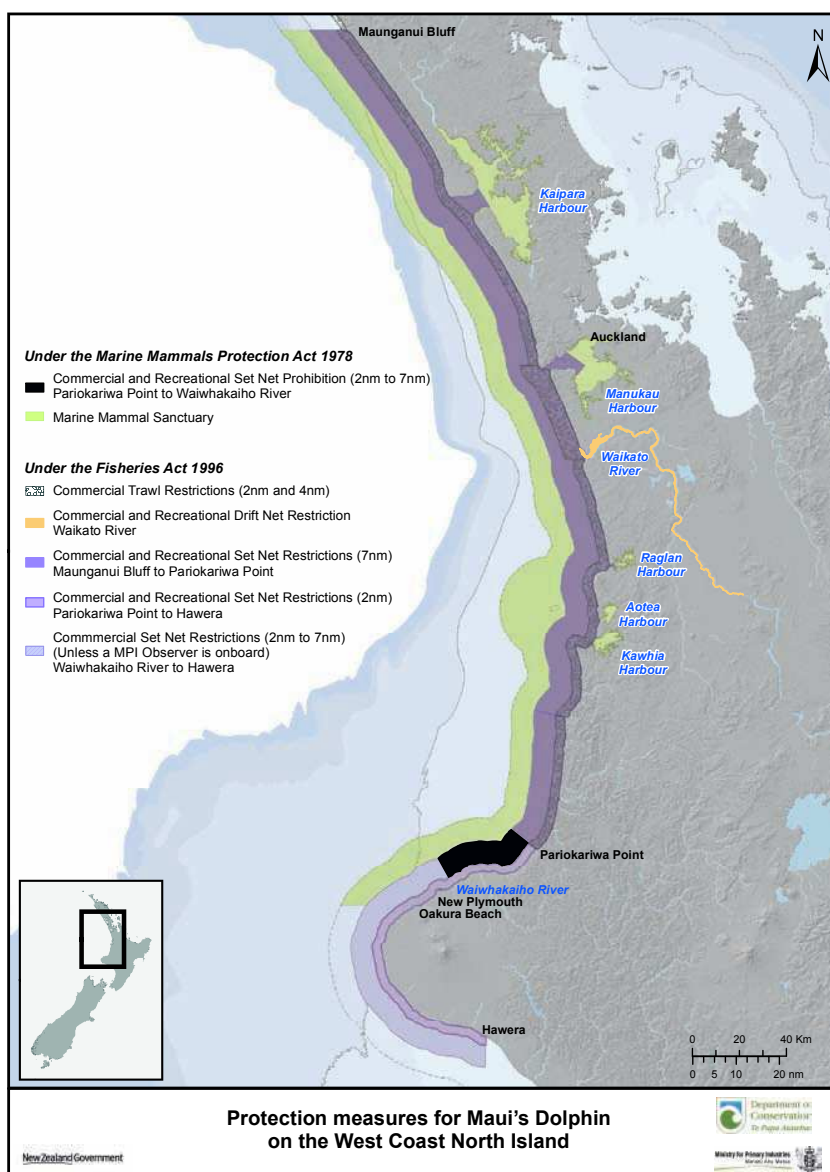
The Department of Conservation's 2013 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations' was developed in conjunction with international and domestic stakeholders representing industry, operators, observers and marine scientists. Its purpose is to provide effective and practical mitigation measures for minimising acoustic

disturbance of marine mammals during seismic surveys. The Code has been endorsed as industry best practice by the Petroleum Exploration and Production Association of New Zealand.⁴⁸ This Code is discussed further in Chapter 3: EEZ legislation.

There are requirements in this Code for carrying out seismic surveying in a Marine Mammal Sanctuary (as defined in section 22 of the Marine Mammals Protection Act), where compliance with the Code is compulsory. The Code states that ‘particular attention needs to be given to minimising effects where surveys are planned in any of the Areas of Ecological Importance (which includes the marine mammal sanctuaries)’ as shown in Figure 13.8. In addition, no person may carry out a marine seismic survey within a Marine Mammal Sanctuary

unless he or she has, at the earliest opportunity but not less than three months before commencing the survey:

- *Notified the Director-General in writing of his or her intention to carry it out*
- *Submitted a written environmental impact assessment, and subsequently*
- *Agreed to comply with any additional conditions, such as increasing the mitigation zones or number of qualified observers required, imposed by the Director-General for operating in a Marine Mammal Sanctuary.*



Stringent conditions are placed on seismic surveying in marine mammal sanctuaries including the West Coast North Island sanctuary shown here (Department of Conservation⁴⁹)

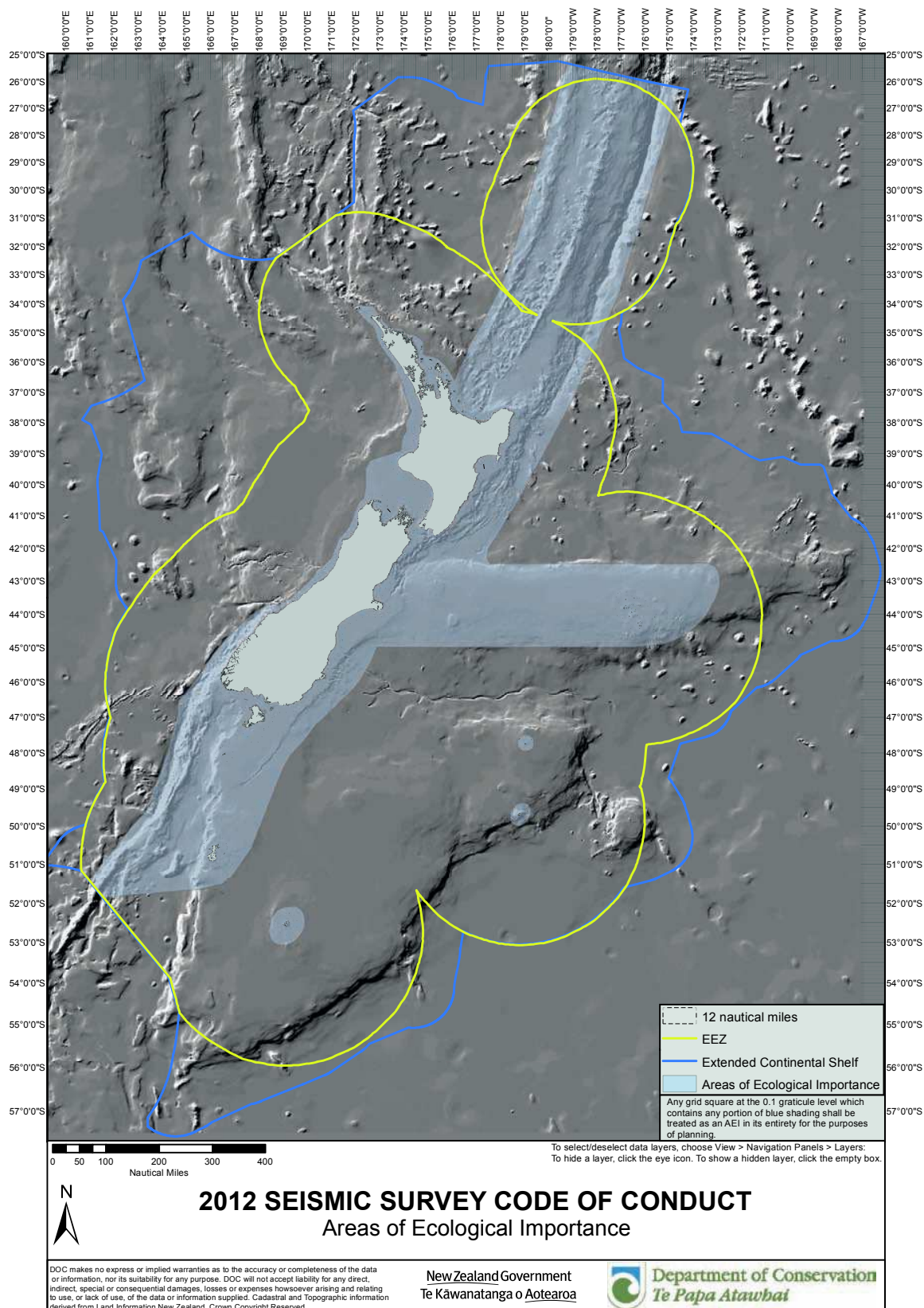


Figure 13.8 Map of Areas of Ecological Importance in relation to seismic surveying (Department of Conservation⁵⁰)

Maritime Transport Act

Section 226 of the Maritime Transport Act states that ‘harmful substances shall not be discharged or escape, otherwise than in accordance with the marine protection rules, from any ship, offshore installation or pipeline into the sea within the exclusive economic zone or over the continental shelf of New Zealand’.

Part 200 of the Marine Protection Rules requires operators of offshore installations to develop a discharge management plan which must be approved by Maritime New Zealand. The purpose of a discharge management plan is to establish procedures and practices that will minimise environmental impacts from operational or accidental discharges of any harmful substances, including oil.

The discharge management plan must set out a number of elements as shown in Figure 13.9 and outlined below:

- Steps to control operational activities that may harm the marine environment
- Steps to minimise activities that present a risk of accidental discharge of harmful substances, including oil
- An emergency response plan that demonstrates that the operator can respond rapidly to minimise the extent and impact of a spill of harmful substances

Part of plan	Elements for inclusion
In relation to a potential oil spill	A risk assessment of the likelihood and consequences of a marine oil spill An inventory of all oils stored or produced at the installation Oil spill trajectory modelling results to indicate the likely fate of spilled oil A description of the surrounding environment highlighting sensitive areas Emergency response procedures to address oil spills in the marine environment
Emergency response procedures	Ensure the safety of personnel Stop the discharge and prevent further discharges Contain and prevent the spill from entering the sea if possible Notify relevant parties and authorities of the spill Evaluate the extent of the spill and appropriate response options Clean up spilled oil where possible Dispose of waste responsibly Review the plan and restock response equipment

Figure 13.9 Elements that need to be included in a discharge management plan

There is no public submission process for the approval of the discharge management plan, which is undertaken in-house. A discharge management plan may be approved for a period of up to three years. This means that a new approval is required at least every three years. Maritime New Zealand generally audits installations annually. Offshore installations must hold a valid International Oil Pollution Prevention Certificate which is issued by Maritime New Zealand.

Under the EEZ Act, the EPA is required to take into account ‘the nature and effect of other marine management regimes’ when considering a marine consent application.⁵¹ This includes those established under the Maritime Transport Act. So although Maritime New Zealand approves discharge management plans, the EPA can take them into account when determining if marine consent should be granted or refused.

Part 21 of the Maritime Transport Act provides for regulation of deliberate disposal into the sea of platforms or other man-made structures. Maritime New Zealand’s position on this matter mirrors Annex II of the London Protocol from which the regulation comes, which notes a hierarchy of preferred waste management options in increasing order of environmental impact:

- Re-use
- Off site recycling
- Destruction of hazardous constituents
- Treatment to reduce or remove the hazardous constituents

- Disposal on land, into air and in water

Marine disposal is considered the last resort where no other viable option exists, and in the specific circumstances of decommissioning this implies that all practically recoverable material will be removed.⁵² In 2010, Maritime New Zealand began a process intended to develop a national policy on offshore decommissioning in consultation with the industry. This is still underway.

Health and safety management

Although health and safety regulations are directed at protecting employees from hazards, they can provide some environmental protection. This is because avoiding hazards to people, and avoiding environmental damage such as by minimising the risk of oil spill, often involve the same technical steps (e.g. keeping the hydrocarbons controlled). For example, the 2014 Ministry for the Environment Guidelines for onshore petroleum development states, 'Well design and integrity are important matters to consider for managing risk to both health and safety and the environment. The regulation of well design and integrity to manage risk to health and safety is the responsibility of the High Hazards Unit (part of WorkSafe New Zealand)'.⁵³

The High Hazards Unit within WorkSafe New Zealand (a Crown Agent with an independent Board appointed by the Minister of Labour) was established to enforce health and safety legislation in New Zealand's mining, petroleum and geothermal sectors, recognising the inherent and significant hazards in these industries. However, WorkSafe New Zealand has no legal mandate to protect the environment.⁵⁴

The Health and Safety in Employment Act 1992 is the overarching legislation relevant to workplace health and safety. The object of the Act is to promote the prevention of harm to all persons at work or in a place of work. It requires employers to take all practicable steps to ensure the safety of employees while at work.

The Health and Safety in Employment (Petroleum Exploration and Extraction) Regulations 2013 establish requirements specific to the oil and gas industry. The regulations contain a general duty to take all practicable steps to ensure that the installation is safe for any person on or near it and that all work and other activities carried out on the installation are carried out in a manner that is safe for any person on or near the installation. There are also specific duties pertaining to safe operations.

These regulations also require:

- Each offshore installation to have a safety case approved by the High Hazards Unit
- Each offshore installation and its key systems to be annually certified by an independent certification agency
- An independent well examiner to examine every exploration or production well over its full lifecycle (i.e. from design to abandonment)

An emergency response plan for the installation must be prepared. This is a plan for responding to emergencies that occur while petroleum workers are working on an installation. The emergency response plan must identify:

- People authorised to set emergency procedures in motion
- Person in charge of co-ordinating the emergency response
- Foreseeable conditions or events that could bring about a major accident
- Actions that could control those conditions or events or limit their consequences
- Arrangements for limiting the risks to persons on or near the installation
- Arrangements for training petroleum workers in the duties they will be expected to perform in an emergency

Oil spill response

The Maritime Transport Act requires Maritime New Zealand and regional councils to respond to a marine oil spill (Part 23). New Zealand has a three-tiered approach to marine oil spill preparation and response. Industry (Tier 1), regional councils (Tier 2) and Maritime New Zealand (Tier 3) have clear roles and responsibilities under the Act.

Tier 1 plans are prepared by industry and are site-specific. Tier 2 plans are the regional marine oil spill contingency plans that regional councils must produce. Regional councils must respond to marine oil spills within their jurisdiction that exceed the clean-up capability of the Tier 1 response. Regional marine oil spill contingency plans are approved and audited by Maritime

New Zealand. Maritime New Zealand is responsible for the national marine oil spill contingency plan. Tier 3 is the highest level of response and occurs when the marine oil spill exceeds the capacity of the Tier 1 and 2 responses. If additional response resources are required, international assistance will be requested.

Maritime New Zealand maintains a response capability of sufficient size to counter an oil spill of 3,500 tonnes, which is deemed to be a 'one in a hundred year' event. The Deepwater Horizon oil disaster in the Gulf of Mexico released around five million barrels of oil (each barrel contains just over one tonne of oil). The 2011 *Rena* disaster in Tauranga spilled approximately 350 tonnes of heavy fuel oil.

If the scale of an incident is beyond the nation's domestic capability, arrangements are in place to secure overseas assistance. This relationship is reciprocal — New Zealand will be expected to assist its neighbours if requested. Each region has resources to deal with the smaller spills that they are most likely to experience, calling on national or international assistance in the case of larger spills.

Oil spill preparedness is funded by an industry levy, the Oil Pollution Levy, which is required under the Maritime Transport Act. The Levy is collected to run New Zealand's maritime oil pollution preparedness and response system. The Oil Pollution Levy applies to all commercial vessels over 100 gross tonnes and 24 metres or more in length (except those operating in freshwater), offshore oil installations, exploration wells and oil pipelines.⁵⁵ The intention behind Maritime New Zealand's oil pollution preparedness and response is that the polluter pays. In circumstances where the polluter is not identified, the Levy is used to pay expenses from financial reserves set aside for that purpose.⁵⁶



Onwater oil recovery vessel using booms to contain oil on the stranded *Rena* off Tauranga (Maritime New Zealand)

Elements of best practice

The best practice guidelines prepared by the Ministry for the Environment and Maritime New Zealand in 2006 recommend that, while undertaking petroleum exploration, development and production activities in the EEZ and extended continental shelf, industry should adopt and apply the best practicable options to minimise or prevent adverse effects on the environment.⁵⁷ Best practice considerations for mitigation measures identified in Figure 13.10 can be applied to oil and gas activities in the marine area.

Area	Options for mitigation
Exploration	Ensure that sufficient research has been undertaken to identify what species are likely to be present in the area and their distribution Avoid areas frequented by threatened or engendered marine species such as the Māui's dolphin and New Zealand sea lion Follow the Department of Conservation's Code of Conduct for minimising acoustic disturbance to marine mammals from seismic survey operations Adapt or schedule seismic surveys to avoid seasonal migrations or key breeding locations Avoid anchoring on seabed structures of environmental or other significance Check, clean and apply non-toxic antifouling paints to infrastructure that is brought in from overseas Undertake sufficient monitoring so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner For any exploratory well drilling, consider the options outlined for production well drilling
Production	Choose a site for platform placement after an environmental impact assessment has been undertaken Locate drilling platforms as far away as possible from sensitive species or habitats, avoiding areas of high ecological value including areas of high endemism Avoid areas with outstanding biodiversity, natural character or landscape values Avoid significant shellfish beds, fish spawning areas and areas of high fish density Avoid areas of recreational significance, including nationally important surf breaks Ensure well integrity is regulated for the purpose of protecting the environment Purchase and retire appropriate commercial fish quota for the duration of the production phase Limit unnecessary lighting and ensure that lights that are required are directed vertically onto work surfaces Reduce levels of toxic components in fluids by removing or recycling them before discharge Ensure any dredge spoil is only dumped in locations where impacts will be minimal In sensitive environments, collect all waste from drilling operations, and store on board for disposal on land Re-inject gas to decrease the volume of gas that might be otherwise flared (reducing emissions of greenhouse gases) Deploy robust safety features, such as automatic shut-down valves, to reduce the risk of a significant oil spill Ensure adequate systems and training are in place to minimise oil spill risk Treat sewage prior to discharge Check, clean and apply non-toxic antifouling paints to infrastructure that is brought in from overseas Comply with the requirements of the craft risk management standards Minimise greenhouse gas emissions through careful management of sources, including power generating equipment, flaring and fugitive fuel emissions Undertake sufficient monitoring so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner Involve tangata whenua and stakeholders in overseeing monitoring programmes Make monitoring information publicly available in a timely fashion
Decommissioning	Leave a clean seafloor following decommission Ensure that the well is decommissioned properly and formations are sealed off permanently

Figure 13.10 Possible mitigation options for managing the potential environmental impacts of oil and gas exploration
(Adapted from the World Petroleum Organization)⁵⁸

International recommendations for improving well incident prevention, intervention and response capability

Following a number of international well accidents, the International Association of Oil and Gas Producers established the Global Industry Response Group to learn from the accidents that had happened and identify recommendations for applying these learnings within the industry. The Group was divided into three core areas:⁵⁹

- Prevention – developing better capabilities and practice in well engineering design and well operations management
- Intervention – improving well capping response readiness in the event of an incident and undertaking further study into the need for, and feasibility of, global containment solutions
- Response – delivering effective and fit-for-purpose oil spill response preparedness and capability

In May 2011 the Group released its report. There were a number of key findings which are outlined below. Integral to these was the setting up of three groups:⁶⁰

- A Wells Expert Committee tasked with looking at all the data on well incident reports, and identifying opportunities for developing consistent standards, and promoting good practice
- The Subsea Well Response Project (which includes a group of nine major oil companies) with the purpose of designing a capping toolbox and additional hardware, and looking at global containment solutions
- A Joint Industry Project to develop new recommended practices, improve understanding of oil spill response tools and methodologies, and enhance co-ordination between key stakeholders internationally

The key findings from the investigation and recommendations are outlined in Figure 13.11.

Prevention: Well engineering design and well operations management recommendations

- Creation of a new, permanent, Wells Expert Committee
- Introduction of a three-tier review process
- The promotion of human competency management systems to ensure individual staff and management teams always have the skills they need
- The promotion of a culture that fosters the adherence to standards and procedures
- Recognition of existing internationally and nationally agreed standards as a baseline for industry improvements
- New and improved technical and operational practices for the overall governance of well construction
- Recommend to industry and regulators that a 'two (independent and physical) barrier' policy is in place during the life of the well
- During the drilling, completion, and abandonment phases of a well a blowout preventer is regarded as a barrier for the purposes of such a policy

Intervention: Capping and containment recommendations

- Creation of a consortium to investigate and deliver improved capping response
- Consortium to manage the concept selection and design of caps and associated equipment that could be deployed internationally
- Consortium to manage the selection and design of subsea dispersant hardware that could be deployed internationally
- Equipment to supplement existing intervention capability
- Consortium to study further the need for, and feasibility of, global containment solutions
- Consortium to investigate and develop procedures for specific operational issues related to the capping and containment of hydrocarbons released from a well blowout event
- Activity to be co-ordinated with the other initiatives

Response: Oil spill preparedness and response recommendations

- Creation of a joint industry project to improve co-ordination between key stakeholders (oil spill response organisation, governments and companies) internationally and with all responsible representative bodies
- Establish principles on dispersant use with key stakeholders
 - document the effectiveness and the value of the use of surface, aerial and, in particular, subsea dispersants
 - recommend how best to deploy dispersants
 - obtain regulatory pre-approval for their use
- Promote research to advance understanding and response methodologies and risk assessment models
- Enhance existing recommended practices for *in situ* burning (controlled burning of oil that has spilled)

Final conclusions

- Action on prevention will yield the most effective outcomes
- Implementation of the following recommendations by member companies will improve industry prevention and intervention capabilities
 - Rigorous adherence to standards and practices is central to prevention
 - With specific enhancements, current standards will be effective if applied and observed rigorously. We strive to ensure greater alignment between standards internationally
 - Verification of adherence to standards and practices is required
 - Transparency in sharing and expediting learning from well incident analyses is required
- An ongoing commitment from the industry is required to work together for continuous improvement in skills development, competency and a culture of compliance
- Implementation of proposed response research and development activities will lead to the development of new response technologies and techniques, as well as a better understanding of and more effective application of existing technologies and techniques
- Active co-ordination and collaboration with international initiatives and national oil industry associations is essential

Figure 13.11 Key recommendations of the Global Industry Response Group (*International Association of Oil and Gas Producers, 2011*)



The 2011 report by the Global Industry Response Group provided important recommendations for the oil and gas industry (*International Association of Oil and Gas Producers*)

Further reading and references

Batson, P, 2003, *Deep New Zealand: Blue water, black abyss*, Canterbury University Press, Christchurch

Bollmann M, T Bosch, F Colijn, R Ebinghaus, R Froese, K Güssow, S Khalilian, S Krastel, A Körtzinger, M Langenbuch, M Latif, B Matthiessen, F Melzner, A Oschlies, S Petersen, A Proelß, M Quaas, J Reichenbach, T Requate, T Reusch, P Rosenstiel, J O Schmidt, K Schrottke, H Sichelschmidt, U Siebert, R Soltwedel, U Sommer, K Stattegger, H Sterr, R Sturm, T Treude, A Vafeidis, C van Bernem, J van Beusekom, R Voss, M Visbeck, M Wahl, K Wallmann and F Weinberger, 2010, *World ocean review: Living with the oceans*, Maribus gGmbH, Hamburg

Centre for Advanced Engineering, 2003, *Economic opportunities in New Zealand's oceans: Informing the development of oceans policy*, Centre for Advanced Engineering, Christchurch, available at <http://www.mfe.govt.nz/publications/oceans/economic-opportunities-oceans-jun03/economic-opportunities-oceans-jun03.pdf>

Centre for Advanced Engineering, 2006, *A report on possible government interventions to promote the sustainable development of New Zealand's ocean resources*, Centre for Advanced Engineering, Christchurch, available at <http://www.mfe.govt.nz/publications/oceans/government-interventions-ocean-resources-may06/index.html>

International Association of Oil and Gas Producers, 2011, *International recommendations on well incident prevention, intervention and response – Global Industry Response Group recommendations*, International Association of Oil and Gas Producers, London, available at <http://www.ogp.org.uk/global-insight/well-safety/>

Minister for the Environment, 2008, *Cabinet paper: Proposal for Exclusive Economic Zone environmental effects legislation*, Unpublished paper, available at <http://www.mfe.govt.nz/issues/oceans/current-work/cabinet-paper-eez-effects-legislation.html>

Ministry for the Environment, 2005, *Offshore options: Managing environmental effects in New Zealand's Exclusive Economic Zone*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/offshore-options-jun05/index.html>

Ministry for the Environment, 2007, *Improving regulation of environmental effects in New Zealand's Exclusive Economic Zone: Discussion paper*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/nz-exclusive-economic-zone-discussion-paper-aug07/index.html>

Ministry for the Environment, 2012, *Managing our oceans: A discussion document on the regulations proposed under the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Bill*, Ministry for the Environment, Wellington

Ministry for the Environment, 2013, *Managing environmental effects of petroleum development activities (including hydraulic fracturing): Guidelines for local government*, Ministry for the Environment, Wellington

Ministry for the Environment and Maritime New Zealand, 2006, *Environmental best practice guidelines for the offshore petroleum industry*, Ministry for the Environment and Maritime New Zealand, Wellington

Ministry of Business, Innovation and Employment, 2012, *Economic contribution and potential of New Zealand's oil and gas industry*, Occasional Paper 12/07, Ministry of Business, Innovation and Employment, Wellington

NIWA, 2012, *Expert risk assessment of activities in the New Zealand Exclusive Economic Zone and extended continental shelf*, Prepared for the Ministry for the Environment, NIWA, Wellington

Oceana, 2009, *Toxic legacy: Long-term effects of offshore oil on wildlife and public health*, Oceana, Washington, DC

Stirling M, 2002, 'New Zealand's offshore mining regime: Rights and responsibilities beyond the 12 mile limit', *New Zealand Journal of Environmental Law*, 6, 139–176

Suhr L, 2008, 'Environmental protection in deep seabed mining: International law and New Zealand's approach', *New Zealand Journal of Environmental Law*, 12, 97–144

Endnotes

- 1 Ministry of Business, Innovation and Employment, 2012
- 2 Ministry of Business, Innovation and Employment, 2012
- 3 <http://www.nzpam.govt.nz/cms/petroleum/petroleum-basins>
- 4 <http://www.nzpam.govt.nz/cms/petroleum/petroleum-basins>
- 5 <http://www.nzpam.govt.nz/cms/petroleum/petroleum-basins>
- 6 NIWA, 2012
- 7 <http://www.nzpam.govt.nz/cms/pdf-library/petroleum-publications/nz-petroleum-basins-part-one>
- 8 NIWA, 2012
- 9 Ministry for the Environment, 2007
- 10 http://www.caimenergy.com/assets/images/cms/1_4_Oil_and_gas_exploration.gif
- 11 <http://www.pepanz.com/news-and-issues/issues/offshore-drilling/>
- 12 <http://www.oilspillsolutions.org/offshore.htm>
- 13 <http://www.pepanz.com/news-and-issues/issues/offshore-drilling/>
- 14 <http://www.pepanz.com/news-and-issues/issues/deep-sea-drilling/>
- 15 <http://www.pepanz.com/assets/Uploads/NZ-Petroleum-Industry-2009.pdf>
- 16 <http://www.stos.co.nz/maui.asp>
- 17 http://www.imo.org/blast/mainframe.asp?topic_id=1548
- 18 Oceana, 2009
- 19 [http://www.edsconference.com/content/docs/2011_papers/Barton,%20Barry%20\(Paper\).pdf](http://www.edsconference.com/content/docs/2011_papers/Barton,%20Barry%20(Paper).pdf)
- 20 <http://www.pepanz.com/news-and-issues/issues/deep-sea-drilling/>
- 21 <http://www.scientificamerican.com/article/catastrophic-thinking/>
- 22 <http://online.wsj.com/articles/investigators-fault-blowout-preventer-in-deepwater-horizon-oil-spill-1401984643>
- 23 <http://www.scientificamerican.com/article/catastrophic-thinking/>
- 24 <http://www.world-petroleum.org/index.php?/Environment/exploration-a-production-in-the-marine-environment.html>
- 25 NIWA, 2012
- 26 NIWA, 2012
- 27 NIWA, 2012
- 28 http://www.cefas.defra.gov.uk/media/20461/tr_003.pdf
- 29 NIWA, 2012
- 30 NIWA, 2012
- 31 NIWA, 2012
- 32 NIWA, 2012
- 33 NIWA, 2012
- 34 <http://www.biosecurity.govt.nz/files/pests/salt-freshwater/oil-rig-enclosure-report.pdf>
- 35 NIWA, 2012
- 36 Suhr L, 2008, 113
- 37 RMA, section 95(a)(2)
- 38 NZCPS, Policy 3
- 39 NZCPS, Policy 11
- 40 NZCPS, Policy 13
- 41 NZCPS, Policy 15
- 42 NZCPS, Policy 23
- 43 <http://www.doc.govt.nz/Documents/conservation/native-animals/marine-mammals/2013-seismic-survey-code-of-conduct.pdf>
- 44 <http://www.mfe.govt.nz/issues/oceans/current-work/draft-regulations-eez-discharge-dumping-document.pdf>
- 45 http://www.epa.govt.nz/EEZ/applications_EEZ/Non-notified_consents/Documents/EEZ0202OMV_Whio1_FINAL_decision.pdf
- 46 http://www.epa.govt.nz/Publications/141013_EZ0201STO_Marine_Consent_Decision_Revision_1.pdf
- 47 <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/marine-protected-areas/marine-mammals-protection-westcoastnorthisland.pdf>
- 48 <http://www.doc.govt.nz/Documents/conservation/native-animals/marine-mammals/2013-seismic-survey-code-of-conduct.pdf>
- 49 <http://www.doc.govt.nz/nature/native-animals/marine-mammals/dolphins/maui-dolphin/current-protection-measures/>
- 50 <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/areas-of-ecological-importance.pdf>
- 51 EEZ Act, section 59(2)(h)
- 52 <http://www.nzpam.govt.nz/cms/pdf-library/folder.2007-05-17.4547411750/MaritimeNZCommunication-to-NZOil-Gas-Sector-June2010.pdf>
- 53 <http://www.mfe.govt.nz/publications/rma/managing-environmental-effects-onshore-petroleum-development-activities/managing-environmental-effects-of-onshore-petroleum-development-activities-pdf.pdf>
- 54 http://www.edsconference.com/content/docs/2014_presentations/de%20Wit%20Barton%20NZ%20EEZ%20Regltry%20EDS%202014.pdf
- 55 <http://www.maritimenz.govt.nz/About-us/Fees-levies/OPL/>
- 56 <http://www.maritimenz.govt.nz/About-us/Fees-levies/OPL/>
- 57 Ministry for the Environment and Maritime New Zealand, 2006
- 58 <http://www.world-petroleum.org/index.php?/Environment/exploration-a-production-in-the-marine-environment.html>
- 59 <http://www.ogp.org.uk/global-insight/well-safety/>
- 60 <http://www.ogp.org.uk/global-insight/well-safety/>



14

Major marine developments



Contents

301	Introduction
302	Components of major marine developments
302	Large fixed structures
302	Dredging and disposal
303	Reclamation and declamation
304	Potential environmental effects
305	Landscapes, natural character and amenity
305	Marine ecosystems
306	Dredging and marine disposal
308	Coastal processes
308	Water quality
309	Relationship of tangata whenua with the marine environment
310	Public access
311	Reverse sensitivity
311	Marine heritage
312	Decision-making framework
312	Resource Management Act
313	New Zealand Coastal Policy Statement
313	Port development
314	Reclamation
315	Protecting indigenous vegetation and fauna
315	Protection outstanding natural character, features and landscapes
315	Protecting water quality
316	Coastal hazards
316	Managing public access
317	Cumulative effects
317	Elements of best practice
320	Case studies
320	Westhaven Marina development
322	Mangere sewage ponds restoration
323	Further reading and references
323	Endnotes

Introduction

New Zealand's geographic position means that the country must strongly rely on an effective port system to enable goods to be both brought into the country and exported. Close to 99 per cent of New Zealand's imports and exports are transported by sea, with over 31 million tonnes of exports and some 19 million tonnes of imports passing through the country's ports¹ Consequently, the infrastructure supporting the movement of goods is of great importance to New Zealand.

Coastal shipping services, operated by both local and international shipping companies, provide intra and interisland links and play a key role in the distribution of bulk cargos such as petroleum products and cement. There are around 30 global and regional shipping lines calling at New Zealand ports.² Six ports handle over 90 per cent of container volumes – Auckland, Tauranga, Napier, Wellington, Lyttelton and Otago.³ Large and essential marine infrastructure at these ports, such as wharves, jetties and terminals, supports commercial shipping.

Marinas, moorings, floating pontoons and boat ramps are necessary infrastructure to support recreational boating activity. New Zealand has one of the highest boat ownership rates in the world, and boats need to be stored, both on and off the water. Pressure for marinas has been greatest on the east coast of the northern part of the North Island where the larger proportion of New Zealanders live and holiday, and where the weather and coastline is particularly favourable for boating. In areas such as the Coromandel Peninsula, there have been high levels of conflict over marina proposals.

Canal developments create new coastal edges for waterfront houses and associated boat moorings. This kind of development is relatively new in New Zealand, with the first canal development being built at Pauanui on the Coromandel Peninsula during the early 1990s and others subsequently being developed at Whitianga, Gulf Harbour and Marsden Cove.

Wastewater treatment plants are large infrastructure developments which are generally located by the coast. This is because the wastewater system relies primarily on gravity to move sewage from urban areas to the treatment plant and then to release the treated water into a waterway. With these developments, there is a need for a range of supporting infrastructure, including pipes, outfalls and treatment ponds.

Other infrastructure which is commonly placed in the coastal area includes bridges, causeways, pipelines, airport runways, groynes and seawalls, all of which support human use of these environments. Despite the disparate uses for which these developments are intended, such projects often involve similar types of work. These include the placement and ongoing occupation and use of structures within the marine area, the dredging of the seabed with associated dredge material disposal, and the reclamation and declamation of land.

In the future, other types of marine infrastructure may start appearing in New Zealand's waters, including offshore wind farms and wind, wave and tidal energy devices.

These types of developments generally involve placing large structures in the marine environment. Another potential future use of the marine environment is for carbon sequestration. This involves CO₂ being separated from flue gases, and then being transported by pipes or vessels to a suitable site where the gas is stored in seabed rock.⁴ Depleted offshore oil and gas fields could possibly be used for this purpose.

Major marine developments associated with some specific activities are discussed in Chapter 10: Aquaculture and Chapter 13: Oil and gas exploration and mining.



Marsden Cove marina and canal development, Whangarei (Craig Potton)

Components of major marine developments

Large fixed structures

A wide variety of structures have been placed in the coastal marine area. These include relatively small objects such as moorings and navigational aids, as well as larger fixed structures that are rigidly attached to the seabed, such as wharves, breakwaters, seawalls, causeways and groynes. Some developments, such as ports and marinas, require the construction of a large number of structures within a confined location of the coastal marine area together with extensive development above mean high water springs.

A marina development is likely to include numerous piles and floating pontoons to create marina berths and fuelling facilities, breakwaters to protect the boats from wave damage, slipways, haul-out facilities and washdown areas for boat maintenance, boat ramps, and onshore facilities such as hard stand areas, administration and toilet blocks, shops, cafes and car parks.

Canal developments typically include seawalls around the edges of the canals, piles and floating pontoons to provide berths adjacent to the canal-side homes and may include bridges or causeways to provide pedestrian and/or vehicle access across the canals. Onshore structures typically include houses, apartments and commercial developments. A marina may be included as an integral part of a canal development.

Tidal energy can be exploited by building semi-permeable barrages across estuaries or by harnessing offshore tidal streams.⁵ In 2011, a 10 year consent was granted to construct a marine tidal turbine power station in the mouth of the Kaipara Harbour. The consent provided for 200 completely submerged marine tidal turbines with a maximum generating capacity of around 200MW to be located underwater, but it has yet to be constructed.

Dredging and disposal

Dredging of the seabed is commonly undertaken to increase, maintain or restore navigable water depths in the coastal marine area, particularly in association with ports, marinas and canal developments. There are two kinds of dredging: capital dredging which aims to physically change the depth of the seabed permanently; and maintenance dredging which aims to maintain a depth of the seabed. Figure 14.1 shows the different activities for which dredging consents have been granted in the Auckland region between 1994 and 2012.

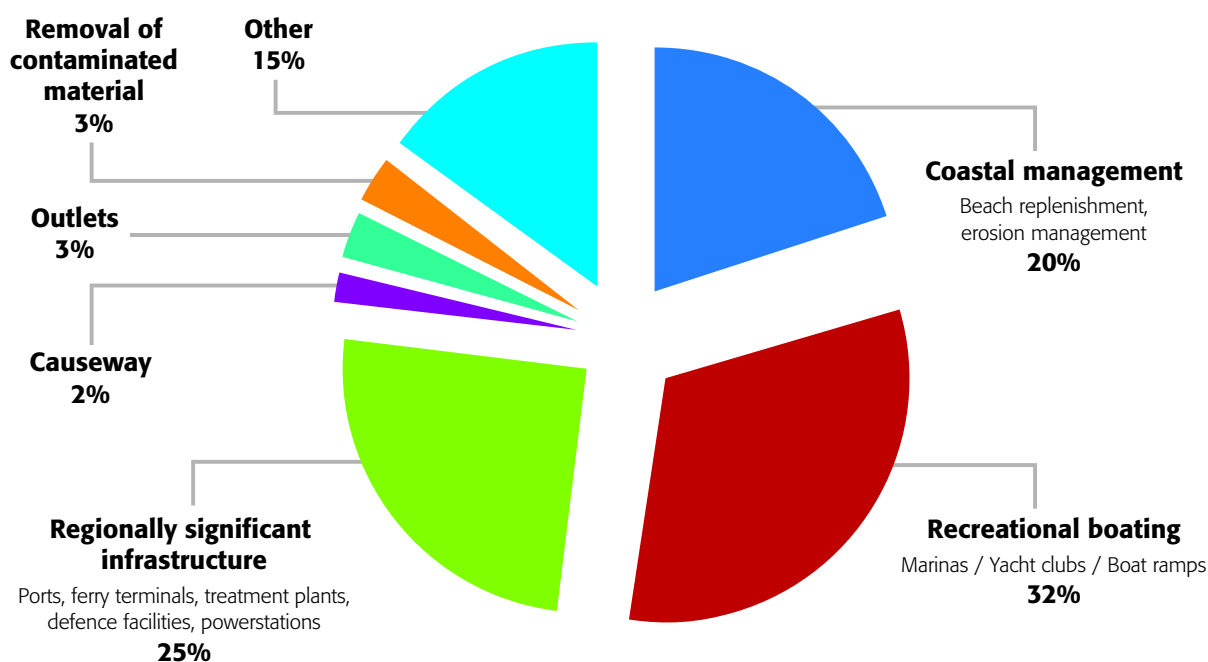


Figure 14.1 Categories of dredging in the Auckland region 1994 to 2012 (Kelly S and S Faire, 2012)

Marinas are often built in shallow coastal waters adjacent to land. Initial dredging is usually required to create a basin deep enough for boats to moor and navigate within. Such dredging is also commonly required to create a navigable channel to connect the marina basin with open water. In general, marinas and ports should not be located in places that trap sediment, because this creates a need for regular maintenance dredging.

Canal developments in New Zealand have, to date, been constructed on low-lying land adjacent to estuaries, inlets and harbours. Some areas, historically, were coastal wetlands. The canals are dug out of land above mean high water springs, lined with seawalls, and then flooded when the area is opened up to the sea. This approach has avoided the need for canals to be dug within the existing marine environment. Seabed dredging is usually required, however, to create a navigation channel between the canal development and the open sea.

To accommodate the growing size of international ships, capital dredging has also been undertaken at commercial ports to increase the navigable depths within major shipping channels and near wharves. This mirrors an international trend in deepwater port developments. A number of the larger ports are in the process of major development to support the next generation of container ships, including the Port of Tauranga and Port Otago. The Port of Tauranga, the country's largest port by volume exported, is currently dredging to deepen the shipping channel from 12.9 metres to 16.0 metres depth at low water. This is to provide sufficient draught for larger ships which carry many more containers.⁶

Maintenance dredging is often required to maintain access to ports, wharves, marinas and canal developments. This is particularly the case in enclosed harbours and sheltered bays, where there is a large quantity of sediment running off the land into waterways, and little energy from waves or currents to disperse the sediment before it settles onto the seabed.

Reclamation and declamation

Reclamation and declamation of coastal land alters the shape of the coastline to provide for development by adding or removing material from the coastal marine area. Reclamation has been undertaken extensively in many parts of the country to provide for extensions of the urban environment. Indeed, in cities such as Auckland, Wellington and Dunedin reclaimed land forms a significant part of the central business district. Approximately 10 per cent of the area of Auckland's Waitematā Harbour has been reclaimed since 1840. Today, there is a greater understanding of the negative environment impacts of reclamation, and little is undertaken for urban expansion. Reclamation is still undertaken, however, for port and marina developments. Some of these reclamations have included dredged material, such as at the Fergusson Container Terminal.⁷

Declamation has been an integral component of canal developments in New Zealand, where land is removed, allowing the sea to expand into new areas. Such projects tend to occur in estuaries, wetlands and other parts of the coastal marine area where the water is shallow and the new structures will be protected from the worst of the coastal weather.



Works under-way at Auckland's Viaduct Basin (Raewyn Peart)

Potential environmental effects

Major marine infrastructure can create positive economic and social benefits, through enabling transport and trade, and facilitating public access to, and enjoyment of, the marine environment. At the same time, because of the large scale of many of these projects, there can be significant adverse environmental effects on the marine environment which need to be addressed. These are summarised in Figure 14.2 and discussed in further detail below. Best practice ways of reducing these effects are discussed later in the chapter.

Area	Potential negative impact
Degradation of natural coastal landscapes	• Significant visual intrusion • Replacement of natural shapes and forms with hard lines and edges • Disruption of natural coastal patterns and processes • Domestication of the natural coastal landscape
Damage to marine ecosystems	• Direct loss of coastal habitat • Disruption of sensitive and/or ecologically productive ecosystems and transition zones • Damage to habitats of birds and intertidal fish and animals • Reduction in natural flushing, potentially leading to eutrophication • Shading areas of water resulting in the longer-term loss of marine plants • Removal of all marine life from parts of the seabed • Permanent loss of slow-growing, sensitive species which are unable to recover • Release of sediment into seawater which reduces light penetration, disrupts juvenile species and filter feeders, and smothers benthic communities over a wide area • Release of organic-rich sediments into seawater which can exacerbate algal blooms • Release of metals, chemicals, dioxins, organochlorines and polychlorinated biphenyls into the marine environment which can impact on the health and viability of marine species (such as where dredging of contaminated sediments is undertaken)
Changes to coastal processes	• Disruption of wave energy, currents and tidal flows • Alteration of sediment transport regimes causing coastal erosion or sediment accumulation • Alteration of seabed bathymetry • Alteration of particle sizes entering the marine environment, making an area more muddy than previously, and therefore changing the habitat
Degradation of water and sediment quality	• Discharge of sediment or pollutant-laden water which can cause sedimentation and contamination of the wider marine environment • Increased levels of rubbish, oil and antifoul paints entering the water through increased use of an area
Disrupting the relationship of tangata whenua with the marine environment	• Contamination of water and sediment which can render kaimoana unsafe to eat and reduce the size of harvestable populations • Loss of intertidal areas, which reduces the areas where kaimoana can be harvested • Loss of access to, or destruction of, culturally important sites
Degradation of marine heritage	• Damage to the integrity of archaeological sites, wāhi tapu and other sites of historical significance
Loss of public access and amenity	• Restriction or exclusion of public access to parts of the coastal environment • Loss of amenity through increased noise, light and traffic

Figure 14.2 Potential adverse environmental effects of major developments

Landscapes, natural character and amenity

The placement of large structures into the marine environment and changes to the natural form of the coastal edge through the construction of reclamations and seawalls can have a major impact on natural coastal landscapes. Such man-made elements can create a significant visual impact (depending upon perception), replace natural rounded shapes and forms with straight lines and hard edges, and disrupt natural coastal patterns and processes. Many of the impacts are irreversible in practical terms.

Large single structures, such as wharves and boat ramps, may be intrusive when seen from both the land and from the sea. They are also likely to bring with them increased activity in the form of cars, boats and people. But it is the cumulative effect of many structures along the coastline that can have the most detrimental impacts on natural coastal landscapes.

Ports and marinas, in particular, often require reclamations and numerous structures to be placed within the coastal marine area. Some are occupied by a high density of boats and supported by a wide array of facilities. Most ports have historically been developed in city centres or near major land-based infrastructure that supports the effective distribution of imports and exports. City ports are often dominated by a combination of container cranes, stacked containers and vessels. They are commonly accompanied by commercial or residential development. Natural landscapes in such locations are often already highly modified and urbanised, and this can reduce the landscape impacts of further expansion.

Canal developments also have a major impact on natural coastal landscapes, through the creation of canals, high-density housing, numerous boats and floating pontoons and commercial facilities. The marine area and coastal edges within these developments are often entirely artificially created.

Because of the major impact that these kinds of developments can have on natural coastal landscapes, and the irreversibility of such impacts, it is important that the developments which do occur are carefully located, and that where possible they are confined to areas where natural coastal landscapes have already been compromised, such as disused port areas.



Canal developments, such as the one shown here at Whitianga, include artificially created coastal edges and significant infrastructure including houses, gangways and floating pontoons (Raewyn Peart)

Marine ecosystems

Reclamation causes the direct loss of coastal habitat in the marine area that is reclaimed. It can also disrupt the sensitive ecosystems of transitional zones between the land and the sea which, in their undisturbed state, contain a diverse range of organisms and are very ecologically productive. These areas can provide important nursery areas for marine life and habitat for seabirds. Natural systems in these transitional zones can also be important for trapping sediment runoff and controlling erosion. For these reasons, reclamations and major structures should be located away from ecologically important intertidal and marine habitats such as salt marshes, shell banks, seagrasses, shellfish beds, mangroves and major fish habitats. They should also be located away from bird roosting, feeding and nesting sites.

Causeways can result in the direct loss or relocation of marine habitat through burial. They can also reduce wave action and the amount of flushing on the inland side of the structure, resulting in the build-up of sediment, and sometimes eutrophication. This can create new coastal wetland habitat, but if the trapped sediments are contaminated by land-based discharges, the resulting habitat can be of poor quality. The amount of flushing will be affected by the size of the culverts built under the causeway.⁹ Bridges allow much freer flushing to occur and therefore should be used instead of culverts where possible.

Construction of offshore wind farms would disturb the seabed in the immediate vicinity of the structures. However, the structures can also provide new habitat for marine life by creating shelter and surfaces on which vegetation and benthic communities can establish themselves. Large fixed structures such as wharves and breakwaters may shade areas of water. The impacts of

such shading will depend on the habitat affected, but most plants such as mangroves, seagrasses and seaweeds will deteriorate or die if shaded for significant periods.

One of the main concerns associated with carbon sequestration is the potential for leakage of CO₂ both during transportation and after storage. These leakages could be either abrupt or gradual. Their environmental impacts are not well understood, but is believed to be dependent upon the scale of the sequestration, and the location, geographical extent and prevailing conditions within the local marine environment.⁹



The causeway across the Moanaanuanu Estuary at Whangamatā has reduced the speed of water flowing out of the harbour, resulting in increased levels of sediment being deposited within the harbour, which in turn have supported the expansion of mangroves (Raewyn Peart)

Dredging and marine disposal

The main environmental impacts from dredging and marine disposal are related to the:¹⁰

- Direct disturbance and physical changes to the seabed
- Suspension of sediment and associated changes in water quality
- Release and remobilisation of contaminants on or in the seabed
- Changes in local hydrodynamics and settlement patterns of sediment

Capital dredging necessarily removes flora and fauna from the seabed and increases its depth. With the exception of deep-burrowing animals, the activity will completely remove all life from the seafloor for a period of time. The recovery rate of the benthic communities will depend on the particular nature of the ecosystem: some ecosystems dominated by opportunistic species may recover in only a few months, whereas those populated by slow-growing, sensitive species may take years, or may never recover.

Maintenance dredging usually involves frequent removal of material and the amount removed differs considerably depending on the site and type of development. For example, Pine Harbour maintains the channel by dredging 3,000 cubic metres per year, Westpark Marina dredges up to 10,000 cubic metres per year and the Lyttelton Port maintenance dredging has varied between 290,000 and 800,000 cubic metres per year over the past 10 years.

Dredging and disposal can also discharge sediment into the water column:

- As the seabed is disturbed by the dredge
- When sediment overflows from hoppers and barges carrying the dredged material
- As the sediment is dumped at a marine disposal site
- By resuspension of material at the disposal site

Some dredged material can be re-used, such as for beach renourishment or to fill a new reclamation. For example, Ports of Auckland disposes of dredgings into the Fergusson reclamation where the dredged material is mixed with cement to make mudcrete.¹¹ Where such options are not available, land disposal is typically costly and logistically difficult. For this reason, marine disposal is often the option of choice.

One of the significant risks of marine disposal is that the dredgings will be resuspended due to wave action and will be redistributed over a wide area. Where sediment is released into the water column, it can affect the clarity and turbidity of the water, potentially reducing light penetration. The suspended material may eventually settle on areas around the dredged site, blanketing the benthic animals and plants that live there. The effects of this will vary, depending on the resilience of the communities and the conditions that they are used to, but it may cause stress, reduced rates of growth or fatalities. Suspended sediment from maintenance dredging is likely to have less impact than that from capital dredging, due to the already disturbed nature of the site.¹²

Some animals are more sensitive to siltation than others. Shellfish, which have delicate feeding and breathing apparatus, may be affected by increases in sedimentation. Similarly, sediment can become trapped in the gills of young fish causing increased fatalities, and smothering of spawning or nursery areas for fish can result in the death of eggs and larvae. This could potentially reduce the level of recruitment for harvestable fisheries such as snapper. Lastly, smothering of intertidal areas may result in decreased availability of food for the birds and fish that feed there.

There have been concerns over the resuspension of sediment and impact on local ecosystems from disposal sites in the inner harbour areas. Historically, dredged material from the Pine Harbour Marina in the Waitematā Harbour has been disposed of locally. In 2009 the marina sought a resource consent to dispose of 3,000 cubic metres annually off Motukaraka Island near to the marina. Local concern about the impact of the disposed sediment on cockle, pipi and seagrass beds played a part in the consent being declined. The marina now has permission from Maritime New Zealand to dispose of the sediment off Great Barrier Island in deep water.¹²

In some areas where dredging is commonly undertaken, such as at the entrances to ports and marinas, the sediment on the seabed can often have an elevated level of contaminants. Dredging can remobilise contaminants accumulated in these areas and potentially redisperse them further away. The level of risk from this is based on the:¹⁴

- Degree of remobilisation
- Patterns of dispersal
- Sediment size and quality
- Bioavailability and toxicity of the contaminants
- Sensitivity of species in the downstream area

The potential for dredged sediment to cause toxic effects was assessed in 2012.¹⁵ This research reviewed monitoring and assessment data on sediment concentrations of five heavy metals (arsenic, chromium, copper, lead and zinc) in seven representative sites around the Auckland region (including two marinas, a port approach and a port area). The work highlighted that dredging material in this region ranged from low contamination to extremely contaminated. It concluded that the extent of contaminant dispersal is expected to be similar to sediment dispersal, being localised, but cautioned that care was needed with extremely contaminated sediments adjacent to sensitive areas.¹⁶



Maintenance dredging at the Westpark Marina removes up to 10,000 cubic metres of sediment a year (Raewyn Peart)

Coastal processes

Structures such as jetties, breakwaters, seawalls, reclamations and groynes may alter wave energy, currents and tidal flows. This, in turn, has the potential to change the transport of sediments along the coast, resulting in coastal erosion in some areas and higher levels of sedimentation and accretion in others. Reclamation inside harbours and inlets also reduces water space, and consequently the volume of water that moves in and out of the area on each tide, thereby affecting tidal flows and flushing. This is sometimes referred to as the tidal compartment.

The size, shape and orientation of structures relative to the shoreline and natural water flows will determine their hydrological and sedimentation effects.



Concerns have been raised that dredging the access channel to the Whangamatā Marina has changed the coastal dynamics of the area and degraded the Whangamatā surf break (Raewyn Peart)

Impact of Timaru breakwater

A notable example of how a structure can change sedimentation flows is the breakwater that was constructed as part of the Timaru Port in the late 1870s. This resulted in major erosion at Washdyke but also in the creation of a new beach at Caroline Bay. Before works to construct Timaru Port began, Caroline Bay was moderately deep, with narrow mixed sand. Now approximately 30,000 cubic metres of sand has accumulated annually in the shelter of the North Mole, and the shoreline has advanced seaward by more than 650 metres.¹⁷

Dredging can also alter the processes that govern the movement of sediments in the coastal marine area. Dredging for capital works has the potential to cause a significant change to the equilibrium of the system, depending on the circumstances. In some situations this can alter the bathymetry of the seabed, impact on the flow of seawater and have an effect on wave patterns, with consequential effects on areas such as surf breaks (this is an issue that was raised in connection with the marina at Whangamatā).

There are concerns that the construction and operation of marine tidal turbines could impact on local tides. For example, it was estimated that in the Bay of Fundy, tidal power plants could decrease local tides by 15 centimetres. However, the environmental impact assessments that have been completed indicate that impacts are site-specific and depend greatly upon local geography.¹⁸

Water quality

During the construction phase of a marine development, developers are generally required, as part of their resource consent, to use best practice preventative measures including silt curtains, settlement ponds, temporary retaining walls, temporary dry docking and new technologies. These measures are intended to reduce the risk of sediment or any potential contaminants being introduced into the wider marine environment. Compliance with these consent conditions is managed by unitary and regional councils.

After construction has been completed, the potential for degradation of water quality surrounding the structure will depend on what it is used for. Wharves, jetties and marinas typically attract large numbers of people and boats that, in turn, have the potential to create discharges into the marine area – including fuel spillage, sewage from boat toilets and rubbish. To help address this, Westhaven Marina has a ‘no discharge’ policy within the marina, which is discussed further in the first case study at the end of this chapter.

The parking and boat washing areas found at marinas may also be a source of contaminated runoff into the sea if they are not appropriately managed. In 2012 a project was undertaken by NIWA for the EPA which looked at the levels of copper leaching from antifoul paints. This research indicated that in many of New Zealand’s marinas copper concentrations could be above guidelines for the protection of marine life.¹⁹ Since then, the EPA has carried out a reassessment of antifouling paints²⁰ and the Ministry for Primary Industries has prepared draft antifouling and in-water cleaning guidelines.²¹

Clean marinas programme

The Clean Marinas programme is an industry-led programme that has been developed by the New Zealand Marina Operators Association. Its aim is to support marina operators, boatyards, contractors and recreational boaters to engage in environmentally sound practices. These include regular boat engine inspection and maintenance, proper waste disposal and reduction of discharges.²² There are a number of steps involved before a facility can gain 'Clean Marina' status, which include signing the Clean Marina Pledge, and having an assessment undertaken by an independent environmental expert. Clean Marina status has been awarded to the Marlborough Sounds Marinas, Opuā Marina, Westhaven Marina and Ashby's Boatyard. A number of other marinas have signed the Clean Marina Pledge.²³

When a marina or canal development site is located in a poorly flushing area, or the design of canals and structures prevent adequate flushing, the area can become a trap for sediment and pollutants, causing the water and sediment quality of the area to be affected.²⁴ An international report on canal estates highlighted that the majority of environmental problems arise from canal systems that are not designed to allow a certain amount of flushing, ultimately resulting in:²⁵

- Additional nutrient loading in the water
- An effect on biological diversity

The problem can be minimised through the use of wide canals which have good linkages, to allow a circular flow of water throughout the entire development. This helps to ensure that the development is well flushed by tidal action and that any contaminated runoff is collected and treated. Ongoing vigilant management is also required to ensure that spillages into the marine area are minimised through the adoption of good practice.



Westhaven Marina has been awarded Clean Marina status, which requires signing the Clean Marina Pledge and undergoing an independent environmental assessment (Raewyn Peart)

Relationship of tangata whenua with the marine environment

The impacts of development on the marine environment is a critical issue for coastal Māori. Gathering of kaimoana, both for sustenance and to provide for visitors, is an integral component of life. When those resources are compromised, such as when structures are placed over shellfish beds or in a place that obstructs access to them, the Māori way of life is also compromised. Activities which contaminate water and sediment can result in kaimoana being unsafe to eat and in a reduction in the abundance and diversity of marine life.

The impacts of dredging and reclamation are identified by the Te Awanui Tauranga Harbour Iwi Management Plan 2008 as significant cultural issues, in particular the impacts on traditional relationships in the area, disturbance to kaimoana and potential re-establishment of kaimoana.²⁶

Mahaanui Iwi Management Plan

The Mahaanui Iwi Management Plan 2013 identifies that 'dredging, reclamation, sedimentation and structures in the Whakaraupō/Lyttelton Harbour are affecting mahinga kai, including the potential effects of breakwaters on the ability of tidal flows to flush the harbour of sediment, and the resultant accumulation of sediment on kaimoana beds at Rāpaki'.²⁷ The iwi is concerned that dredging in Whakaraupō and the deposition of silt is increasing pressure on kaimoana resources of Te Ara Whānui o Makawhiua.

Coastal works such as reclamations and the construction of seawalls can result in the loss of intertidal areas, which reduces the areas where shellfish can be harvested. The Kāi Tahu ki Otago Natural Resource Management Plan 2005 identifies that such activities and structures restrict physical access to the foreshore and seabed for the collection of kaimoana.²⁸ Construction activities can also result in the destruction of culturally important sites such as wāhi tapu and traditional waka linkages and mooring places.

The location of wastewater treatment plants in the coastal environment has caused much concern among coastal Māori. A Cultural Impact Assessment of the Matata Wastewater Scheme prepared in 2014 identified significant concerns for Tangihia Hapū. This included the disposal of partially treated wastewater in a wāhi tapu area where the bones of tupuna were buried, and the location of the treatment plant in proximity of the Rangitihi Marae, both of which were considered to be culturally unacceptable.²⁹



The construction of seawalls, such as the one shown here at Waihi, can result in the loss of intertidal areas and places where kaimoana can be harvested (Raewyn Peart)

Public access

Structures built along and near the coastal edge can enhance public access (through boat ramps and jetties, as well as walkways around buildings) and also obstruct it (through restrictions to access both to the marine area and to areas within it). Port developments typically exclude public access to the coastal edge for safety and security reasons. However, some ports provide organised educational tours for school groups and interest groups. Marina and canal developments vary in the extent to which they provide for public access.

Auckland waterfront promenade

The waterfront walkway and cycleway development in Auckland will create 6 kilometres of continuous walkway and cycleway across the waterfront to encourage walking and cycling, increase access to the coastline, and provide sightlines to the water (e.g. Daldy Linear Park) while protecting, where appropriate, the commercial use of the water's edge.³⁰



Waterfront Auckland is currently developing a water's edge promenade and cycleway around Westhaven Marina (Waterfront Auckland)

Public access to marina berths is often excluded, particularly at night, for security reasons. The marina berths themselves are occupied by licencees, and are not freely available to other boat users, although some may be available to visiting boats for short-term periods. Some marinas incorporate public walkways, boat ramps, parking, picnic areas and cafes as part of the complex, and this enables members of the public to access and enjoy the coastal environment.

Public access is typically not available around the coastal edge of canal developments as privately-owned residential properties extend right down to the coastal edge. The waterways themselves are usually in public ownership and open to all users. Some canal developments also provide small public parks alongside sections of the canals.³¹

There has been considerable controversy over the extent to which members of the public should have access to private wharves and jetties. The general legal principle is that unless a coastal permit expressly limits public access, and such a limitation is necessary to give effect to the permit, the permit holder cannot exclude the public from the coastal marine area.³² The situation

in any specific case will be affected by the ownership of the seabed on which the structure is located and the conditions of the coastal permit authorising the use and occupation of the structure. Under the Marine and Coastal Area Act (discussed in Chapter 2: Marine management), no person owns or is capable of owning the common marine and coastal area,³³ but this area expressly excludes freehold titles currently held over the seabed.³⁴

Reverse sensitivity

The very nature of major marine developments, particularly ports and marinas, means that there are operational activities which result in issues beyond the immediate boundaries – such as noise and light from port operations. Adequate lighting is a necessary part of providing for a safe working port. In addition, many major marine developments require land-based transport links that can impact on the surrounding community. Such issues are often referred to as ‘reverse sensitivity’, which has been defined by the Environment Court³⁵ as ‘the legal vulnerability of an established activity to complaint from a new land use. It arises when an established use is causing adverse environmental impact to nearby land, and a new, benign activity is proposed for the land. The ‘sensitivity’ is this: if the new use is permitted, the established use may be required to restrict its operations or mitigate its effects so as not to adversely affect the new activity’.

Application of reverse sensitivity to expansion of Port Otago

During the council hearing on Port Otago’s Project Next Generation resource application in 2011 to extend the wharf (which included dredging), a number of residents from the Careys Bay and Deborah Bay communities submitted in opposition, based on the potential visual effects of the proposal. Their particular concerns related to the berthing of larger vessels, stacking of containers, and the use of container cranes. Some submitters expressed ‘genuine and deeply felt concern for the way both the Multi-purpose Wharf and the fishing jetty would intrude into Careys Bay, which some likened to a Cornish fishing village’. Evidence by Port Otago stated that it was essential that operations were integrated over the whole port site, without restrictions on operations in certain or specific areas, and that any restrictions could ‘undermine the purpose’ of the extension. The decision of the Commissioners appointed by the Otago Regional Council and Ministry of Conservation stated that ‘in deciding to grant the consents sought we are also mindful that the port, along with channel dredging, has been in existence for close to 150 years and no serious evidence of adverse effects occurring in that time, as a result of port operations, were brought to our attention’.

Marine heritage

Marine developments can negatively impact on marine heritage if reclamations or structures are built in the vicinity of, or over, archaeological sites, wāhi tapu and other sites of historical significance. As the location of many marine heritage sites is poorly recorded, research will often need to be undertaken in respect of individual marine development proposals to determine whether a historical site may be impacted upon. Sensitively designed marine developments can help to preserve and profile marine heritage.

Jellicoe historical precinct, Auckland waterfront

The 3.7 hectare Jellicoe Precinct at the Auckland waterfront profiles the historical and gritty experiences of a working waterfront through the adaptive reuse of structures such as warehouses, slipways, silos and shipping containers; the retention of artefacts like rail tracks, bollards and salvaged seawall stone; and the use of structures, public art, planting, play equipment and events that depict the history, culture and nature of the coastal environment.



Jellicoe Precinct, Auckland waterfront (*Waterfront Auckland*)

Decision-making framework

Resource Management Act

Under section 12 of the RMA, a range of activities in the coastal marine area are restricted. No person may undertake the following activities unless they are allowed by a national environmental standard, a regional coastal plan as well as a proposed regional coastal plan (if there is one), or a resource consent:³⁶

- Reclamation or drainage of the foreshore or seabed
- Construction, alteration or demolition of a structure in the coastal marine area
- Disturbance of the foreshore or seabed (e.g. by excavation or drilling) in a manner that is likely to have an adverse effect (except for lawfully harvesting a plant or animal)
- Deposition of substances that are likely to have an adverse effect on the foreshore or seabed
- Introduction of any exotic or introduced plant in, on, or under the foreshore or seabed
- Generally destroying, disturbing or damaging any part of the foreshore or seabed which is likely to have an adverse effect on historic heritage
- The occupation and use of space in the common marine and coastal area
- Carrying out activities within the coastal marine area

The regional coastal plan, required to be prepared by regional councils and approved by the Minister of Conservation, determines whether a major marine development in the coastal marine area will require consent, and what kind of consent will be required. In practice, all major developments in the marine area are likely to require resource consent, because of the potential significance of their effects. When considering a resource consent application, the consent authority must have regard to the actual and potential effects on the environment of allowing the activity (section 104) including the provisions of the NZCPS and applicable regional policy statement and regional coastal plan.



All major marine developments, including the construction of reclamations, canals, breakwaters and jetties as shown here in the Whangarei Harbour, will likely require a resource consent under the RMA (*Raewyn Peart*)

New Zealand Coastal Policy Statement

The NZCPS contains a number of policies relevant to major marine developments. These policies attempt to strike an effective balance between providing for activities in the coastal environment and ensuring that important natural resources and community and cultural values are protected in doing so. Policy 6 directs councils to recognise the benefits of the use and development of the coastal marine area, whilst at the same time discouraging activities that do not have a functional need to locate there.

NZCPS, Policy 6: Activities in the coastal environment

1. In relation to the coastal environment:

- a. recognise that the provision of infrastructure, the supply and transport of energy including the generation and transmission of electricity, and the extraction of minerals are activities important to the social, economic and cultural well-being of people and communities;
- b. consider the rate at which built development and the associated public infrastructure should be enabled to provide for the reasonably foreseeable needs of population growth without compromising the other values of the coastal environment;
- e. consider where and how built development on land should be controlled so that it does not compromise activities of national or regional importance that have a functional need to locate and operate in the coastal marine area;

2. Additionally, in relation to the coastal marine area:

- a. recognise potential contributions to the social, economic and cultural wellbeing of people and communities from use and development of the coastal marine area, including the potential for renewable marine energy to contribute to meeting the energy needs of future generations;
- d. recognise that activities that do not have a functional need for location in the coastal marine area generally should not be located there

The Department of Conservation has prepared a guidance note which provides detailed assistance to decision-makers on implementing Policy 6, which is specifically focused on coastal activities and addressing their impacts. Through this policy, the Department encourages local authorities to identify what activities have a functional need to be in the coastal marine area, and to provide for those activities in appropriate places by employing an efficient use of space.³⁷ There is a direction in Policy 6 for decision-makers to consider 'reasonably foreseeable needs' when determining the rate at which built development and associated public infrastructure should be provided for.

Port development

Policy 9 outlines the need to ensure development in the coastal environment does not adversely affect the efficient and safe operation of ports. It also requires councils to consider where, how and when to provide for the efficient and safe operation of ports in policy statements and plans.³⁸ While focused on maintaining the natural state of the environment, the NZCPS seeks to ensure that urban waterfront development does not adversely affect the ability of the country's national and international ports to operate and connect efficiently and safely with other ports and land-based transport modes, as well as to expand in a strategic manner.

NZCPS, Policy 9: Ports

Recognise that a sustainable national transport system requires an efficient national network of safe ports, servicing national and international shipping, with efficient connections with other transport modes, including by:

- a. ensuring that development in the coastal environment does not adversely affect the efficient and safe operation of these ports, or their connections with other transport modes; and
- b. considering where, how and when to provide in regional policy statements and in plans for the efficient and safe operation of these ports, the development of their capacity for shipping, and their connections with other transport modes.

The Department of Conservation's 'NZCPS 2010 Guidance Note Policy 9: Ports' is a valuable resource for any local authority incorporating provisions for port facilities into its planning documents and for submitters on such provisions. The guidance note recognises that a proactive and strategic approach to the management of activities and resources in and around ports is required in order to provide for their ongoing efficient operation and development.³⁹ In addition, identifying and understanding the reverse sensitivity issues, and controlling the establishment of activities in the vicinity of ports, are necessary to minimise the pressure on local authorities to limit port activities. Identifying a buffer zone around ports, and then including policies and rules in that zone that provide for the operation of a port, is one way to manage reverse sensitivity issues.

Port zones in district plans and regional coastal plans are one mechanism that can be used to address the needs of a port (and where, how and when to provide for the port) as part of an integrated and strategic approach to the management of the coastal environment.⁴⁰ The Port Zone provided in the Bay of Plenty Proposed Regional Coastal Plan 2014 provides for the current operational needs of the Port of Tauranga as a matter of priority while 'limiting the effects of those activities on cultural values and the environment'. It also provides for the 'future expansion and operational needs of the Port of Tauranga and its shipping channels' in 'appropriate locations, having regard to the potential adverse effects on the environment'.⁴¹

Reclamation

Reclamation is generally discouraged by Policy 10 of the NZCPS unless it results in regional benefits. The NZCPS directs that reclamation be avoided in the coastal marine area unless land is not available, the activity can only occur in the coastal marine area, there are no practicable alternative methods to provide for the activity, and the reclamation will provide significant regional or national benefit.⁴² The Bay of Plenty Proposed Regional Coastal Plan recognises that reclamation, which is identified in the 'Outline Development Plan Port of Tauranga', is appropriate in terms of specific Port Zone policies and Policy 10 of the NZCPS, 'provided that any adverse effects are appropriately managed, including by use of off-site mitigation'.⁴³

Particular care will be needed to determine which parts of a marine development need to be located at the land–water interface, and what the safety and efficiency effects would be of alternative locations.⁴⁴ Where a reclamation is considered suitable, the design and form of the structure should have regard to the effects of climate change and sea-level rise, and avoid the use of contaminated materials. De-reclamation of redundant land is encouraged only where it would help restore the natural character and provide for more public space.

NZCPS, Policy 10: Reclamation and de-reclamation

1. Avoid reclamation of land in the coastal marine area, unless:
 - a. land outside the coastal marine area is not available for the proposed activity;
 - b. the activity which requires reclamation can only occur in or adjacent to the coastal marine area;
 - c. there are no practicable alternative methods of providing the activity; and
 - d. the reclamation will provide significant regional or national benefit.
3. In considering proposed reclamations, have particular regard to the extent to which the reclamation and intended purpose would provide for the efficient operation of infrastructure, including ports, airports, coastal roads, pipelines, electricity transmission, railways and ferry terminals, and of marinas and electricity generation.
4. De-reclamation of redundant reclaimed land is encouraged where it would:
 - a. restore the natural character and resources of the coastal marine area; and
 - b. provide for more public open space.



The NZCPS recognises the importance of ports, such as Port Chalmers shown here, and the need to provide for their safe and efficient operation as well as future development (Raewyn Peart)

Protecting indigenous vegetation and fauna

An applicant for a major development will need to show that there will be no adverse effects on indigenous vegetation and fauna which are threatened, at risk or naturally rare. In relation to other indigenous species and ecosystems, an applicant must show that significant adverse effects have been avoided and other adverse effects have been avoided, remedied or mitigated. These matters are discussed further in Chapter 2: Marine management.

Marine biosecurity is an important issue for ports and marinas, because of the type and number of vessels arriving from overseas and from other New Zealand ports, which can be a significant source of non-indigenous marine organisms. This is discussed further in Chapter 5: Marine biosecurity and Chapter 15: Vessels. The Department of Conservation's policy guidance recommends that in policies and plans, providing for ports will have to be integrated with managing the risk of any release and spread of harmful aquatic organisms.⁴⁵

Protecting outstanding natural character, features and landscapes

The NZCPS requires adverse effects on outstanding natural features and landscapes and on areas of the coastal marine area with outstanding natural character to be avoided. This effectively means that major marine developments will not be able to proceed in locations identified as areas of outstanding natural features and landscapes in the coastal environment. They should instead be directed to localities where the natural character and landscape values of the area have already been compromised. This is supported by the recent Supreme Court decision that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes.⁴⁶

Protecting water quality

Policy 23, in addition to addressing the general management of discharges, specifically addresses what is required for managing discharges from ports and other marine facilities. A consent authority will consider whether the structure (both its construction and operation of activities on it) will have adverse effects on water quality, and where relevant, whether tidal flushing rates are adequate to ensure that pre-existing water quality is maintained or enhanced in the vicinity of the development.

The NZCPS directs decision-makers to have particular regard to avoiding significant adverse effects on ecosystems and habitats and minimising adverse effects on the life-supporting capacity of water as a result of discharges of contaminants.⁴⁷ Research in the Auckland region on the release of contaminants during dredging indicates that including policies that 'require adverse effects to be minimised within a specified mixing zone, and avoided beyond the mixing zone, would provide more guidance to applicants and decision makers'.⁴⁸



Councils need to consider whether discharges into the marine environment, such as the outfall from the Bell Island Sewerage Treatment Plant in Nelson shown here, will have significant adverse effects on ecosystems and habitats after reasonable mixing (Raewyn Peart)

NZCPS, Policy 23: Discharge of contaminants

1. In managing discharges to water in the coastal environment, have particular regard to:
 - a. the sensitivity of the receiving environment;
 - b. the nature of the contaminants to be discharged, the particular concentration of contaminants needed to achieve the required water quality in the receiving environment, and the risks if that concentration of contaminants is exceeded; and
 - c. the capacity of the receiving environment to assimilate the contaminants; and
 - d. avoid significant adverse effects on ecosystems and habitats after reasonable mixing;
 - e. use the smallest mixing zone necessary to achieve the required water quality in the receiving environment; and
 - f. minimise adverse effects on the life-supporting capacity of water within a mixing zone.
4. In managing discharges of stormwater take steps to avoid adverse effects of stormwater discharge to water in the coastal environment, on a catchment by catchment basis, by:
 - a. avoiding where practicable and otherwise remedying cross contamination of sewage and stormwater systems;
 - b. reducing contaminant and sediment loadings in stormwater at source, through contaminant treatment and by controls on land use activities;
 - c. promoting integrated management of catchments and stormwater networks; and
 - d. promoting design options that reduce flows to stormwater reticulation systems at source.
5. In managing discharges from ports and other marine facilities:
 - a. require operators of ports and other marine facilities to take all practicable steps to avoid contamination of coastal waters, substrate, ecosystems and habitats that is more than minor;
 - b. require that the disturbance or relocation of contaminated seabed material, other than by the movement of vessels, and the dumping or storage of dredged material does not result in significant adverse effects on water quality or the seabed, substrate, ecosystems or habitats;
 - c. require operators of ports, marinas and other relevant marine facilities to provide for the collection of sewage and waste from vessels, and for residues from vessel maintenance to be safely contained and disposed of; and
 - d. consider the need for facilities for the collection of sewage and other wastes for recreational and commercial boating.

Coastal hazards

The extent to which a major marine development will affect and be affected by coastal processes, including natural hazards, erosion and inundation, will also be considered by a consent authority. The NZCPS requires that applications avoid land use changes or development which would increase the risk of adverse effects from coastal hazards.

Managing public access

The NZCPS directs decision-makers to maintain and enhance public walking access to, along and adjacent to the coastal marine area, including by avoiding, remedying or mitigating any loss of public walking access resulting from subdivision, use or development.⁴⁹ As a result, applicants will need to carefully consider how public access to the coastal marine area can be provided for despite the development. There are specified circumstances when imposing restrictions is appropriate. Within some parts of a port, restricting public access along the waterfront may be necessary to



The impact of several smaller structures in the coastal marine area taken together can have the most effect on natural character values of the area. This is shown here at Mangonui where a coastal road and seawall, reclamation, wharf and several buildings overhanging the water collectively significantly reduce the natural character of the coastal fringe (Craig Potton)

protect public health and safety, or to ensure a level of security consistent with port activities. In such cases, applicants may need to identify opportunities to enhance or restore public access in other areas of the coastal marine area, or provide connecting access around the development, if such access cannot be provided in the development itself.⁵⁰

Cumulative effects

When considering applications to undertake development of structures in the coastal marine area, the consent authority will need to consider the cumulative effects of allowing the development, in recognition of the fact that it may well be the impact of several small structures in the coastal marine area taken together that has the most effect on the natural character or landscape values of the area.

Elements of best practice

Those managing the development of the coastal marine area must walk a fine line between promoting and facilitating the close bond that New Zealanders have with the sea and coastal activities and preventing overdevelopment that ruins the values for which it is enjoyed. The coastal environment is a finite and fragile resource, so the development of structures in it should be carefully planned and limited to those which are strictly necessary.

A first step in achieving best practice coastal management for large marine developments is to carefully plan for their location in a strategic manner. This can include delineating port management areas and marina areas in regional coastal plans, with provisions clearly indicating that developments proposed outside the delineated areas will not obtain consent. The recent Supreme Court decision that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes sets a clear precedent in support of this.⁵¹

There are a number of best practice elements that can be implemented to manage the adverse impacts of major marine developments and these are outlined in Figure 14.3.



Marinas and canal developments, such as the one shown here at Pauanui, need to be carefully designed to ensure there is a free flow of tidal water and good flushing (Raewyn Peart)

Potential adverse environmental effects	Best practice response
Degradation of natural coastal landscapes	• Locate reclamations and structures within the coastal marine area only if they are absolutely necessary and there is no practical alternative • Locate reclamations and structures where natural character and landscape values have already been compromised • Seek opportunities for restoring or enhancing areas that have been compromised • Design reclamations and structures to visually 'fit in' with the natural coastal environment – by minimising their size and where possible adopting natural shapes
Damage to marine ecosystems	• Locate marine developments away from ecologically important marine habitats such as salt marshes, shell banks, seagrasses, shellfish beds, mangroves, fish spawning areas and bird roosting, feeding and nesting sites • Locate marine developments away from any threatened, at-risk or rare species or ecosystems • Design marine developments to minimise impacts on natural coastal processes including sediment transport, tidal flows and local currents • Minimise shading of marine habitat in areas where mangroves, seagrasses or seaweeds are present • Locate and design marine developments to minimise the amount of capital and maintenance dredging required • Time dredging operations to minimise disturbance to sensitive marine animals • Follow the principles of disposal in the London Convention 1972 Annex III⁵² – including considering carefully the characteristics of the material being disposed, the disposal site and method (such as rate of disposal, dilution and dispersal characteristics, other materials already dumped in the area and water characteristics) and the possible effects of the disposal (on marine life, other uses of the sea and options for land-based disposal) • Dispose of dredgings on land as a first preference • Use clean sand dredgings to replenish local beaches • If there is no alternative to disposing of dredgings at sea, ensure that the disposal site is deep enough to avoid any resuspension of sediment in the coastal environment, preferably over the edge of the continental shelf • Minimise the number of dumping sites, by identifying a suitable site which can be used for dumping from a range of dredging activities within a region • Undertake frequent monitoring of the site to ensure that sediment is not dispersing outside the site boundaries
Changes to coastal processes	• Avoid reclamations unless there is no practicable alternative and the reclamation will provide significant regional or national benefit • Where reclamations occur, minimise their size and consider the shape of the reclamation and materials used to ensure they are compatible with the coastal environment • Carefully design reclamations and coastal structures in terms of size, shape and orientation to minimise impacts on natural seawater flows • Locate and design marine developments to minimise the amount of capital and maintenance dredging required • Where possible, accommodate dredged material within the reclamation, to avoid the need to dispose at sea

Potential adverse environmental effects	Best practice response
Degradation of water quality	- Carefully design marinas and canal developments to allow the free flow of tidal water and only site them in well-flushing areas - Where possible, avoid dredging in areas where there are contaminants concentrated in seabed sediment - Use the best available technology to reduce the amount of sediment released into the water column during construction - Adopt environmental management programmes to avoid discharges from marinas into the marine area from ongoing operations - Adopt voluntary certification programmes, such as the international Blue Flag accreditation⁵³
Disrupting the relationship of tangata whenua with the marine area	- Work closely with Māori to ensure that developments do not impact on wāhi tapu sites or other sites of significance to Māori - Ensure that the development does not obstruct access or adversely impact on kaimoana areas of importance to Māori - Ensure that development does not degrade water and sediment quality
Degradation of historic heritage	- Undertake sufficient research to identify any marine heritage sites in the vicinity of the proposed development area - Locate marine developments, where possible, away from any sites of heritage significance - Incorporate, where possible, heritage items into waterfront redevelopments
Public access and amenity impacts	- Make provision for public access around the coastal edge and to the marine area wherever possible - Provide public access to wharves and jetties unless exclusion can be justified for public health or safety reasons - Where public access needs to be excluded, provide for enhanced access elsewhere along the coastal marine area, close to the development site or around the development site to link up with coastal accessways - Apply no-complaints covenants to restrict new activities from complaining about adverse impacts from existing activities. For example, the Auckland District Plan Central Area Section 14.6.6.1 provides that building for accommodation in the Britomart Precinct will be a permitted activity where the site is subject to a no-complaints covenant in favour of the Ports of Auckland.⁵⁴

Figure 14.3 Best practice considerations for major marine developments



Marine structures should be designed to 'fit in' with the natural coastal environment by adopting natural shapes, such as this groyne at Kohimaramara Beach, Auckland, which has been designed to look like a rocky reef (*Raewyn Peart*)

Westhaven Marina development

Westhaven Marina, located in the centre of Auckland by the Harbour Bridge, is the largest marina in the southern hemisphere. St Mary's Bay became a popular place for yachting during the late 1800s and shortly after the turn of the century the Ponsonby Cruising Club established premises on the shores of the bay. A solid breakwater was constructed along the western end of the bay during the 1930s to provide better shelter for the yachts moored there, which were being damaged in storms. Several yacht clubs established premises on what became known as the Western Reclamation. By the 1980s the northern side of the marina was full and building started on the southern side to cater for ever increasing demand. Today there are more than 2,000 boats moored at the marina with berths provided from 10 to 30 metres long. Four yacht clubs and 50 commercial charter operators are also based there.⁵⁵

Waterfront Auckland has recently developed a plan for how Westhaven Marina is to be improved for the future. This plan comprehensively covers the land- and water-based development in the marina that is planned to occur over the next 30 years.⁵⁶ There are two drivers for the plan, being a desire to improve use of the public edge and also to ensure the best use of water space.

The vision for Westhaven Marina is 'to be the most successful marina in the Southern Hemisphere and home to a globally renowned marine industry. An iconic Auckland destination that is vibrant, accessible and attracts people to the sea'. The Westhaven Plan provides for a range of multiple uses, including boat parking, public access and improved environment. It is based on a number of design principles, as shown in Figure 14.4, and a range of measures to manage environmental impacts (discussed below).

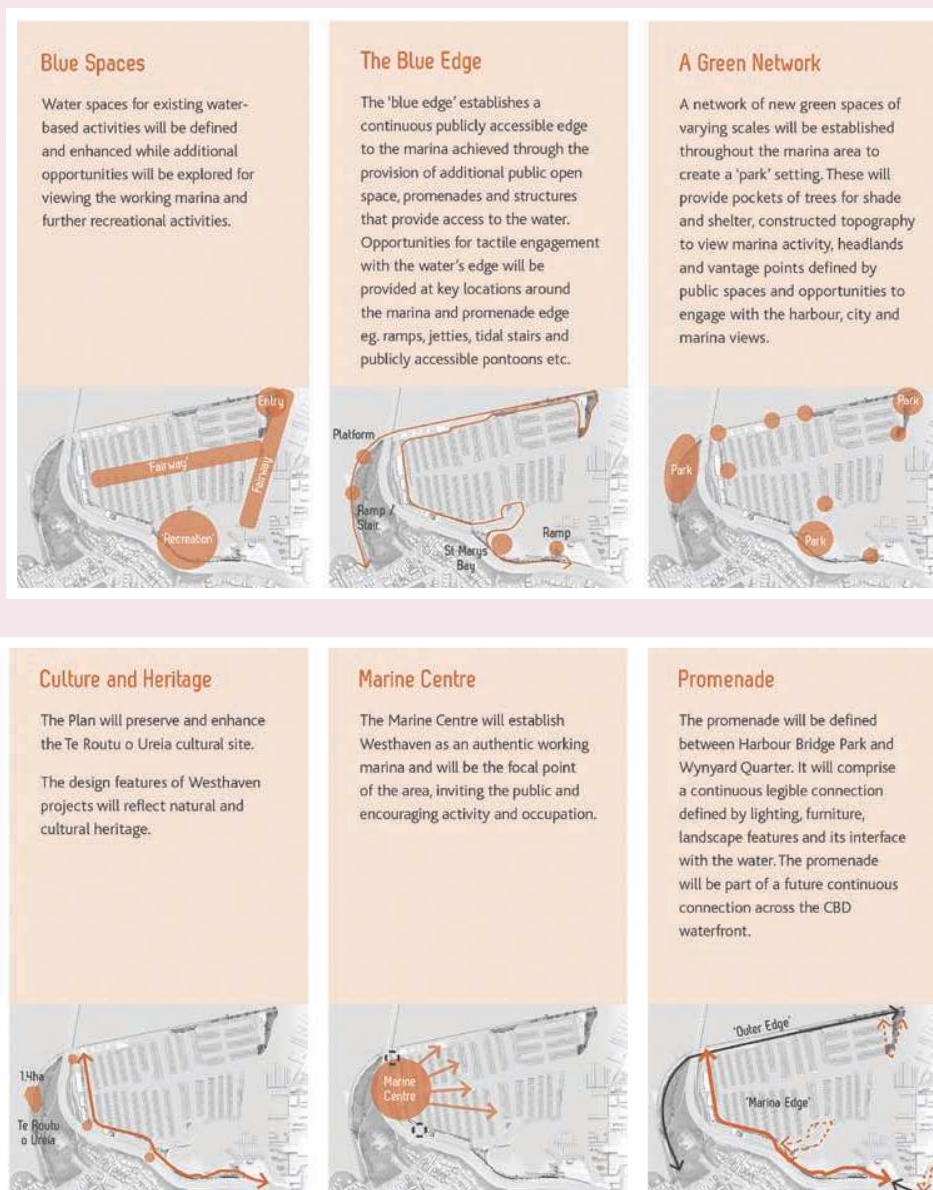


Figure 14.4 Westhaven Marina design principles (*Waterfront Auckland, 2013*)

Contaminated water has been entering the Westhaven Marina basin via a number of sources, including through the outfalls to the harbour and an outfall in St Mary's Bay. Waterfront Auckland is reducing and minimising any contamination of the water through marina-based activities, including:

- Mandatory sewerage removal system for any live-a-board vessel within Westhaven Marina so that vessels can remove black and grey water rather than discharge into the harbour
- Provision of sewerage reticulation in the recently completed Y Pier development for live-a-board vessels
- Stormwater filters to control the runoff from the motorways and car parks, removing hydrocarbons, sediments and gross pollutants
- Moving away from using tanalised piles towards using hardwood and then steel, which will be encased in sleeves
- Replacing all whaler boards to reduce the discharge of heavy metals on the water
- Replacing all galvanised through-bolts, which hold the berths and also rust, with pultron fibreglass rods and nylon nuts, resulting in no discharge of contaminants
- Discouraging boat owners from using coprous oxide antifouls and encouraging the use of Teflon as antifouls on appendages such as propellers
- In addition, the Government has banned tri-butyl tin antifouls which were commonplace until 1994 and the use of zinc as cathodic protection on vessels is reducing

Westhaven Marina and Z Pier are Blue Flag-certified, which is in part due to a number environmental solutions, including:

- Providing and promoting recycling facilities close to every pier, which include oil and oil filters
- Collecting and disposing of rubbish from the sea wall that drifts in from the harbour
- Implementing a 'no discharge' zone where sewage or pollutants are not allowed to enter the water
- Not allowing hull maintenance, in particular scrubbing or scraping, to occur on the marina
- Providing a berth specifically for the Watercare Harbour Clean-Up Trust



Westhaven Marina currently houses more than 2,000 boats, four yacht clubs and 50 commercial charter operators (Raewyn Peart)

Mangere sewage ponds restoration

The Mangere Wastewater Treatment Plant on Auckland's Manukau Harbour once supported the largest oxidation ponds in the world, covering more than 500 hectares of intertidal foreshore, and processing all domestic and industrial waste produced within a 40 kilometre radius.⁵⁷ With increasing volumes of waste from the fast-growing Auckland urban area, stress was placed on the plant, and this resulted in the need for a substantial upgrade.

With growing public awareness, and a need to future-proof the facility against the region's inevitable growth, Watercare Services Limited embarked upon a period of extensive public consultation. This resulted in Project Manukau, New Zealand's largest marine restoration effort.⁵⁸ The project, supported by the public, ran from 1998 till 2005 when the foreshore was re-opened to the public as a coastal walkway.

The primary objective of the project was the removal of the oxidation ponds and restoration of 13 kilometres of coastline. Oruarangi Creek and Mangere Lagoon, which had been closed off when the old sewage site was built, were reopened allowing them to flow into the sea for the first time in over 40 years.⁵⁹ The length of foreshore was reconnected to the sea, restoring it back to being part of the intertidal zone. Extensive areas of white shell beaches were constructed, extending along more than 13 kilometres of coastline, along with more than 270,000 native shrubs and trees which were planted with support from local community groups.

Water quality in the area has improved considerably, with a notable decline in bacteria levels, to the point where it is now deemed to be of swimming quality. Odours and midges (which had plagued the area whilst the ponds were in operation) disappeared.⁶⁰ Pipi beds re-established themselves as the estuarine ecosystems were restored. Previously destroyed fishing grounds have gradually replenished as aquatic life once again becomes established in the area. The native bush has provided habitat and food to an increasing number of wildlife and birds, particularly waders, which have been drawn back to the area by the expanse of beaches.

To further support the returning birdlife, purpose-built roosting sites were placed in the harbour. These have attracted native and migrating birds, turning the area into an internationally-recognised roosting and feeding area. The area is now inhabited by 80 bird species, including a breeding flock of 12 New Zealand dotterels of which there are only 2,000 left in the country. In peak season, the number of birds swell to an estimated 20,000 individuals, with half the world's wrybill population using Mangere Bridge to roost.⁶¹ An extensive network of paths has been created for the public to enjoy the natural surroundings and rich diversity of life that has been restored.



The restoration of the Mangere sewage ponds included the construction of extensive areas of shell beaches (Raewyn Peart)

Further reading and references

- Auckland Regional Council, 2004, *Auckland regional coastal plan*, Auckland Regional Council, Auckland, available at http://www.arc.govt.nz/plans/regional-policy-and-plans/operative-auckland-regional-plan-coastal/operative-auckland-regional-plan-coastal_home.cfm – especially chapters on structures (Chapter 12), reclamation and drainage (Chapter 13), and dredging (Chapter 15)
- Bay of Plenty Regional Council, 2014, *Proposed regional coastal environment plan*, Bay of Plenty Regional Council, Whakatane
- Beca Infrastructure Limited, 2012, *Auckland recreational boating study*, Prepared for Auckland Council, Auckland
- Brake L and R Peart, 2013, *Caring for our coast: An EDS guide to managing coastal development*, Environmental Defence Society, Auckland
- Department of Conservation, 2013a, *NZCPS 2010 Guidance note Policy 3: Precautionary approach*, available at <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/coastal-management/guidance/policy-3.pdf>
- Department of Conservation, 2013b, *NZCPS 2010 Guidance note Policy 6: Activities in the coastal environment*, available at <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/coastal-management/guidance/policy-6.pdf>
- Department of Conservation, 2013c, *NZCPS 2010 Guidance note Policy 9: Ports*, available at <http://www.doc.govt.nz/Documents/conservation/marine-and-coastal/coastal-management/guidance/policy-9.pdf>
- Environment Waikato, 2005, *Waikato regional coastal plan*, Environment Waikato, Hamilton, available at <http://www.ew.govt.nz/policy-and-plans/Regional-Coastal-Plan> – especially Appendix II: Decision-making criteria and considerations
- Kelly S and S Faire, 2012, *Coastal dredging in the Auckland region: Technical and policy review*, Prepared for Auckland Council, Auckland
- Nolan D (ed), 2012, *Environmental and resource management law*, LexisNexis, New Zealand
- NZCPS 2010 Implementation Steering Group, 2011, *New Zealand Coastal Policy Statement 2010 national implementation plan*, Department of Conservation, Wellington
- Port of Tauranga Limited, 2013, *Annual report 2013*, Port of Tauranga, Tauranga, available at <http://www.port-tauranga.co.nz/images.php?oid=4919>
- Otago Regional Council, 2011, *Decision of commissioners appointed by Otago Regional Council and Minister of Conservation in the matter of an application by Port Otago Limited for resource consents for Project Next Generation*, Otago Regional Council, Otago
- Tangihia Consultants and Associates Limited, 2014, *A cultural impact assessment of the Matata wastewater scheme*, Prepared for the Tangihia Hapu, Tangihia Consultants and Associates Limited, Whakatane
- Van Roon M and S Knight, 2004, *Ecological context of development: New Zealand perspectives*, Oxford University Press, Victoria – especially Chapter 9: Accommodating physical processes on exposed sandy coasts and harbour edges
- Waterfront Auckland, 2013, *Westhaven plan 2013*, Waterfront Auckland, Auckland

Endnotes

- 1 <http://www.transport.govt.nz/ourwork/sea/containerproductivitynzports/>
- 2 <http://www.treasury.govt.nz/economy/overview/2012/18.htm>
- 3 <http://www.transport.govt.nz/ourwork/sea/containerproductivitynzports/>
- 4 http://www.imo.org/blast/mainframe.asp?topic_id=1548
- 5 <http://www.oceanenergycouncil.com/ocean-energy/tidal-energy/>
- 6 <http://www.stuff.co.nz/waikato-times/business/8383777/Port-of-Tauranga-given-green-light-for-bigger-ships>
- 7 http://www.poal.co.nz/community_environment/dredging.htm
- 8 Van Roon M and S Knight, 2004, 112
- 9 http://www.imo.org/blast/mainframe.asp?topic_id=1548
- 10 Kelly S and S Faire, 2012
- 11 http://www.poal.co.nz/community_environment/dredging.htm

- 12 Kelly S and S Faire, 2012
- 13 <http://www.stuff.co.nz/auckland/local-news/eastern-courier/9119579/Dumping-moves-offshore>
- 14 Kelly S and S Faire, 2012
- 15 Kelly S and S Faire, 2012
- 16 Kelly S and S Faire, 2012
- 17 <http://www.geog.canterbury.ac.nz/research/staff/Hart%20et%20al%20Nat%20Histy%20chap20%20Coastal%20Systems.pdf>
- 18 <http://www.oceanenergycouncil.com/ocean-energy/tidal-energy/>
- 19 <http://www.aucklandcouncil.govt.nz/EN/planspoliciesprojects/reports/technicalpublications/Documents/monitoringresearchquarterly20121212.pdf>
- 20 http://www.epa.govt.nz/search-databases/HSNO%20Application%20Register%20Documents/APP201051_APP201051_Evaluation_and_Review_Final.pdf
- 21 <http://www.mpi.govt.nz/news-resources/consultations/draft-antifouling-and-in-water-cleaning-guidelines>
- 22 <http://www.cleanmarinas.org.nz/>
- 23 <http://www.cleanmarinas.org.nz/>
- 24 <http://www.goldcoast.qld.gov.au/documents/bf/coomera-river-catchment-study-guide.pdf>
- 25 http://www.ramsar.org/cda/en/ramsar-news-archives-2002-ramsar-topics-report-on/main/ramsar/1-26-45-87%5E21227_4000_0_
- 26 <http://www.boprc.govt.nz/media/30857/iwiplan-100208-teawanui.pdf>
- 27 <http://mkt.co.nz/mahaanui-iwi-management-plan/Mahaanui-IMP-web.pdf>
- 28 <http://ecan.govt.nz/publications/Plans/kai-tahu-ki-otago-natural-resource-mgmt-plan-05.pdf>
- 29 Tangihia Consultants and Associates Limited, 2014
- 30 Brake L and R Peart, 2013
- 31 Brake L and R Peart, 2013
- 32 *Hume v Auckland Regional Council* [2002] NZRMA 49
- 33 Marine and Coastal Area Act 2011, Section 11(2)
- 34 Marine and Coastal Area Act 2011, Section 9
- 35 *Hudson Bay Holdings Ltd v Auckland Council Decision* [2011] NZEnvC 14113
- 36 <http://www.legislation.govt.nz/act/public/1991/0069/latest/whole.html#DLM231949>
- 37 Department of Conservation, 2013b
- 38 NZCPS, Policy 9
- 39 Department of Conservation, 2013c
- 40 Department of Conservation, 2013c
- 41 Bay of Plenty Regional Council, 2014
- 42 NZCPS, Policy 10(1)
- 43 Bay of Plenty Regional Council, 2014
- 44 Department of Conservation, 2013c
- 45 Department of Conservation 2013c
- 46 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited and Ors*, SC 82/2013 [2014] NZSC 38
- 47 NZCPS, Policy 23
- 48 Kelly S and S Faire, 2012
- 49 NZCPS, Policy 19
- 50 Department of Conservation, 2013c
- 51 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited and Ors*, SC 82/2013 [2014] NZSC 38
- 52 <http://www.eoearth.org/view/article/151448/>
- 53 The Blue Flag certification is a voluntary eco-label awarded to more than 4,000 beaches and marinas in 48 countries across Europe, South Africa, Morocco, Tunisia, Brazil, Canada and the Caribbean. The certification sets 24 strict criteria dealing with water quality, environmental education and information and environmental management, among others. The certification is awarded seasonally and can be withdrawn if the marina fails to meet the criteria. The Landing and Westhaven Marinas in Auckland both currently hold a Blue Flag certification.
- 54 http://www.bellgully.co.nz/resources/pdfs/reverse_sensitivity.pdf
- 55 <http://www.westhaven.co.nz/westhaven/about/>
- 56 Waterfront Auckland, 2013
- 57 http://www.watercare.co.nz/SiteCollectionDocuments/AllPDFs/PDFs%20v2%20111010/Wastewater_history.pdf
- 58 http://www.watercare.co.nz/SiteCollectionDocuments/AllPDFs/Watercare_Coastal_Walkway_100214.pdf
- 59 http://www.aucklandcity.govt.nz/council/documents/technicalpublications/TR2009_112%20-%20Environmental%20Condition%20and%20Values%20of%20Manukau%20Harbour.pdf
- 60 https://www.ipenz.org.nz/heritage/conference/papers/Fitzmaurice_J.pdf
- 61 <http://www.forestandbird.org.nz/what-we-do/branches/central-auckland/bird-watching>

15

Vessels





Contents

327	Introduction
327	Potential environmental effects
328	Pollution
328	Oil and chemical spills
329	Sewage discharge
329	Antifouling
330	Dumping rubbish
330	Harm to marine animals and birds
331	Shoreline erosion
331	Greenhouse gas emissions
332	Decision-making process
332	International conventions
332	Resource management Act
333	Resource Management (Marine Pollution) Regulations
333	Local Government Act
333	The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act
333	Maritime Transport Act
335	Biosecurity Act
336	Elements of best practice
336	Case studies
336	Bryde's whale voluntary protocol for shipping in the Hauraki Gulf
337	<i>Rena</i>
339	Further reading and references
340	Endnotes

Introduction

Every day New Zealand's marine environment is being utilised by a wide variety of vessels. These include fishing boats, foreign cargo vessels, oil tankers, cruise ships and pleasure craft. Vessels transport the majority of New Zealand's import and export goods. They underpin the commercial fishing industry. They also enable extensive recreational activity along our coasts.

Each year there is a vast array of vessels visiting New Zealand waters, including:¹

- 1,300 fishing boats operating in coastal waters
- 770 foreign cargo vessels
- 90 oil tankers transporting over 10 million tonnes of oil
- 500,000 pleasure craft
- One million passengers crossing the Cook Strait
- 97 cruise liners with almost 140,000 passengers which make nearly 500 visits to New Zealand ports²

Potential environmental effects

The high level of vessel activity in New Zealand can negatively impact the marine environment. One of the key risks is biosecurity, where invasive species are introduced to the country on hulls of vessels or through the discharge of ballast water. This issue is discussed in Chapter 5: Marine biosecurity. The three other main factors considered in the following section are:

- Pollution
- Harm to marine animals and birds
- Shoreline erosion



Vessels, such as this car ferry operating in the Hauraki Gulf, can impact on the marine environment in a number of ways
(Raewyn Peart)

Pollution

Oil and chemical spills

Every year there are spills of diesel, petrol, oil and other toxic chemicals into the sea. For example, in 2006, over 80 spills were reported. Major oil spills can occur when vessels collide or run aground and hole their tanks, although the risk is reduced if vessels have double hulls. Spillages can also occur when oil cargoes are transferred. Recreational vessels can also release oil into the marine environment when oily water is pumped from bilges and when fuel tanks overflow during refuelling.

There have been several significant spills in New Zealand waters over the last two decades:³

- 1998: *Don Wong 529* in Stewart Island (400 tonnes of automotive gas oil)
- 1999: *Rotoma* in the Poor Knights Islands (~7 tonnes of oily bilge discharge)
- 2000: *Sea Fresh* in the Chatham Islands (60 tonnes of diesel)
- 2002: *Jody F Millennium* in Gisborne (25 tonnes of fuel oil)
- 2011: *Rena* in Tauranga (~350 tonnes of heavy fuel oil)

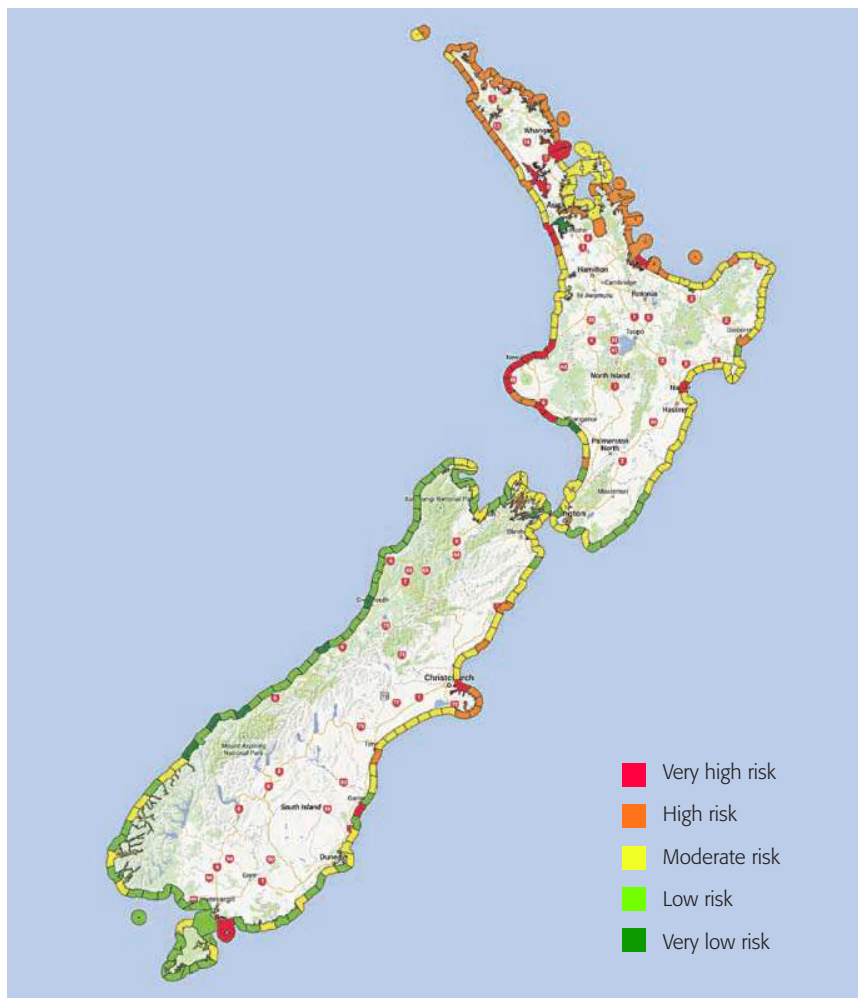


Figure 15.1 Map showing 20km coastal 'cells' that form the basis for an assessment of the environmental and social sensitivity to oil spills and the calculated risk associated with current vessel activity and associated movement of oil (*Maritime New Zealand*⁷)

Rena oil spill

The last big oil spill in New Zealand occurred in 2011, when the *M.V. Rena* struck Astrolabe Reef 12 nautical miles off Tauranga, and grounded. The vessel was carrying 1,368 shipping containers and 1,733 tonnes of heavy fuel oil at the time of the grounding. Around 350 tonnes of oil and 500 containers entered the marine environment. As a result of the disaster, a total of 2,410 dead birds were collected, of which 1,448 were oiled.⁴

Oil discharged into the marine area can have serious impacts on wildlife and habitats, depending on where the oil ends up. For example, oil spills in the open sea may cause little ecological damage as the oil is broken down naturally by micro-organisms (such as bacteria and fungi) into non-hazardous substances. Sunlight, wind and wave action also helps to break oil down. Conversely, major ecological impacts can occur if an oil spill moves onto the shoreline. For example, an oil spill in a sheltered environment such as an estuary means the oil can become trapped and seriously impact mangroves, worms, molluscs, juvenile fish and crustaceans, as well as contaminating shellfish beds.

Birds are particularly susceptible as oil damages the waterproofing on their feathers and, as a result, they can get cold and are prevented from swimming, floating or flying. The toxic components of oil can also cause serious or fatal illnesses and can contaminate the habitat and feeding grounds of coastal seabirds.

A regional risk profile has been prepared by Maritime New Zealand which highlights the levels of risk associated with oil spills from three different sizes of vessels. The 2010 risk profile is shown in Figure 15.1. The risk profile includes an assessment of the likelihood of a spill happening and the effects of the spill. Mapping of environmental sensitivity is combined with the risk assessment to calculate the probability of the potential impact of oil on the coastline.⁵

Sewage discharge

The Resource Management (Marine Pollution) Regulations require that untreated sewage not be discharged within 200 metres of a marine reserve, within 500 metres of land, a marine farm or a mātaihai reserve, or in water less than 5 metres deep. Outside of these limits, some vessels (such as fishing vessels) can discharge raw sewage into the sea from onboard toilets. This is only likely to constitute an issue in sheltered areas close to the shore where there is poor flushing, where there are recreational users of the marine area, or within the vicinity of marine farms or marine protected areas. It is, however, of significant concern to tangata whenua for spiritual and cultural reasons. While the environmental impact from the discharge of sewage from one boat is likely to be minor, the cumulative effects of the whole maritime sector can be significant.⁶

Antifouling

Antifouling coatings are commonly used to protect submerged surfaces and prevent or slow down the growth of fouling organisms. Historically, tributyltin was the main component of antifouling paints but the use of this was finally banned in 2008 due to the severe impact on marine organisms (such as causing sterility in some gastropods).

Today, antifouling coatings can be grouped into two main categories based on whether they rely on the release of toxic (biocidal) compounds to prevent biofouling. Toxic coatings release chemicals such as copper, iron and zinc that aim to prevent settlement or survival of marine organisms. Toxic-free coatings instead rely on their physical nature, such as having smooth surfaces, so organisms cannot attach. All toxic coatings must be registered and permitted for use as an 'antifoulant' by the EPA before they can be applied.

The chemicals in antifouling paints may contain harmful substances that pose a contamination risk if leached into the surrounding seawater. In fact, recent modelling by NIWA suggests that in many New Zealand marinas, copper concentrations are likely above the guidelines for protection of marine aquatic life.⁸



Exclusion zone for boating around *Rēna* wreck, Tauranga (Daniel Sharp)



Boat being antifouled (Daniel Sharp)

Dumping rubbish

Concerns regarding the impact of rubbish dumped in the marine environment have been increasing. A major threat to marine wildlife is plastic debris, mainly due to this being eaten or causing entanglement. It has been estimated that over a million birds and 100,000 marine animals, including mammals and turtles, die each year from plastic debris worldwide. A vast majority of plastic waste in the New Zealand marine environment comes from land; however rubbish discharged from vessels also has an impact. This issue is discussed further in Chapter 16: Marine tourism and recreation.

Harm to marine animals and birds

Ships moving through the water can hit marine mammals and fish and injure or kill them. Ship strike happens because dolphins and especially whales are moving much slower than vessels and spend considerable time on the surface, which can put them directly in the path of boats. The greatest number of marine mammal deaths from ship strike is caused by large ships travelling at more than 12 knots.

Bryde's whale ship strike

Ship strike is a particular problem in the Hauraki Gulf because it houses one of the busiest ports in the country and it is also important habitat for dolphins and the nationally critically threatened Bryde's whale. Between 1997 and 2007, 25 Bryde's whale deaths were recorded in the Gulf, of which 14 were found to have died from injuries consistent with collision with a large vessel.⁹ On average about two Bryde's whales a year are found dead from ship strike. The Hauraki Gulf Transit Protocol for Commercial Shipping is a voluntary measure agreed between the Ports of Auckland and the shipping industry. Vessels are asked to slow down to 10 knots while transiting the Gulf. If widely adopted, this should reduce the number of whale deaths caused by vessels. This is discussed further in the first case study below.

Vessels, including container ships, generate underwater noise, which can impact on the hearing system of marine mammals, both in enclosed waters and the open ocean. Human-produced noise has been shown to displace killer whales.¹⁰ Research has shown that the way marine mammals behave in response to noise varies considerably between the different species, as well as between individuals, sex and age classes. The context of the noise is also important.¹¹



Vessels can cause harm to marine mammals, such as the orca shown here in the Bay of Islands, through hull and propeller strike and the generation of underwater noise (Raewyn Peart)

Shoreline erosion

Vessels moving at fast speeds through coastal passages can create a large wake, which in turn can impact on the marine environment. This issue of shoreline erosion is particularly associated with fast ferries which can travel up to 40 knots, much faster than conventional vessels. This faster speed produces a longer-period wake that disturbs the seabed at greater depths than conventional shipping. This can result in changes to the marine habitat and the composition of the communities that live there. It can also alter the shape of the shoreline and result in accelerated coastal erosion. In confined areas such as the Marlborough Sounds, vessel wake has been associated with a reduction in marine species in the intertidal and shallow subtidal areas. Tangata whenua are concerned that the ferries could impact on kaimoana resources.¹²

Greenhouse gas emissions

Direct contributions to climate change through shipping activities and through the production of vessels and machinery is an increasing concern. The global shipping industry is believed to be responsible for around 1 billion tonnes of greenhouse gas emissions per year, which equates to approximately three per cent of the world's total emissions.¹³

There is now a major global effort being placed on reducing the impact of shipping on greenhouse gas levels. This will ultimately have an impact on New Zealand's shipping industry. For example, the European Commission is proposing that from 2018, large ships using European Union ports will need to report their verified emissions. Some of the major shipping lines entering New Zealand waters are investigating alternative power sources for their vessels. For example, using liquefied natural gas can potentially reduce CO₂ emissions by 20 per cent¹⁴ or powering container ships by kites can reduce fuel consumption by as much as 20 per cent.¹⁵



Powering container ships by kites can help reduce CO₂ emissions (SkySails¹⁶)

Decision-making processes

Management of marine pollution in New Zealand is undertaken under two international conventions and several Acts and Regulations.

International conventions

The International Convention for the Prevention of Pollution from Ships (MARPOL) is the main international convention covering prevention of pollution in the marine environment by ships from operational or accidental causes. The Convention was adopted on 2 November 1973 and has been updated by amendments through the years. It includes regulations aimed at preventing and minimising pollution from ships – both accidental and that from routine operations. As well as addressing such matters as accidental oil spills from tankers, the Convention controls pollution caused by the discharge of chemicals, harmful substances, sewage and garbage. In addition, the Convention on the Prevention of Marine Pollution by Dumping of Wastes or Other Matter prohibits the dumping of waste at sea except for a restricted group of materials. Even then, dumping is only permitted if there are no practical opportunities for re-use or recycling, and the effects of dumping on the marine environment will be minor. It is important that New Zealand is synchronised with the international maritime regulations, including MARPOL.¹⁷

Resource Management Act

The RMA addresses the discharge of pollutants into the marine area within the territorial sea under section 15. It includes provisions which:

- Prohibit the dumping of radioactive matter from any ship or offshore installation, and the storage of any radioactive waste in the sea¹⁸
- Require a resource consent to authorise the dumping of any waste from a ship or offshore installation or the incineration of any waste within the coastal marine area¹⁹
- Require regulations, a rule in a plan or a resource consent to authorise the discharge of a harmful substance or a contaminant from a ship or offshore installation into water, land or air²⁰

Rena consent application

In May 2014, the *Rena*'s owner lodged resource consent applications under the RMA to leave sections of the wreck and associated debris in place on Astrolabe Reef, and to authorise any future discharges of contaminants that may arise as a result. The Bay of Plenty Regional Council publicly notified these applications and 153 submissions were received. A Council hearing of the application has been set down for June 2015.²¹

Rules in regional coastal plans can control shipping within the coastal marine area. This includes matters such as vessel speed, discharges, dumping waste and noise. For example, the Regional Plan: Coast for Otago recognises that a major source of noise in the coastal marine area is from the operation of ships, both commercial and recreational, and provides a policy to manage the effects from this noise.



The RMA controls the discharge of pollutants into the territorial sea, including from ships such as the Interislander ferry shown here (Raewyn Peart)

Resource Management (Marine Pollution) Regulations

These regulations address pollution from vessels and offshore installations within the territorial sea. They are designed to implement the provisions of the international conventions. The provisions include the following:

- The incineration of any waste in a marine incineration facility is a prohibited activity
- The discharge of plastics, dunnage, lining and packaging materials from a ship is prohibited
- The discharge of garbage from any offshore installation is prohibited
- The discharge of other ship garbage is permitted so long as the particle size is 25 millimetres or less and the discharge occurs at least 3 nautical miles from low water mark and more than 0.27 nautical miles from an offshore installation

A rule may only be included in a regional coastal plan for discharges included under the regulations if it increases the distances seaward or increases the depth specified for any area, or increases the distances from a marine farm, marine reserve or mātaihai reserve. A rule in a regional coastal plan cannot relax the requirements in the regulation.

Monitoring and enforcement of discharges from ships within the territorial sea (12 nautical miles) is undertaken by regional councils. Councils also consider resource consents for the dumping of waste at sea within this zone. Such discharges and dumping within the EEZ are controlled by the EPA. This is described further in Chapter 3: EEZ legislation and the RMA is discussed further in Chapter 2: Marine management.

Local Government Act

Regional councils also control the movement of vessels under the Local Government Act. The legislation empowers the council to appoint a harbourmaster who generally controls vessels within the harbour limits.²² The harbourmaster or a council enforcement officer can give directions, for the purpose of ensuring navigation safety, that regulate the entry, departure, navigation and mooring of a vessel.²³ In addition, the council can make bylaws to regulate shipping, including controlling speed and navigational routes.²⁴ These bylaws are, however, constrained to those aimed at achieving navigation safety rather than environmental outcomes directly.

The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

The EEZ Act regulates the discharges of harmful substances from offshore structures and production facilities on board mining ships and the dumping of waste within the EEZ and continental shelf. This is discussed further in Chapter 3: EEZ legislation.

Maritime Transport Act

Marine protection rules made under this Act provide for both marine safety and pollution prevention within the coastal marine area and EEZ. They deal with a wide range of matters pertaining to vessels including the design, construction and operation of vessels, discharges of oil and waste, collision prevention and carriage of dangerous goods.

The Act provides for the protection of the marine environment from harmful substances.²⁵ This includes identifying harmful substances that are not to be discharged into the sea or on the seabed of the EEZ or continental shelf, and requiring notification of the arrival of a ship carrying oil or a noxious liquid substance or the transfer of such substances. New Zealand operates a voluntary code for commercial vessels carrying oil or other harmful liquid substances in bulk. The code, implemented in 1993, makes recommendations to ship owners



Regional councils control the movement of vessels within harbour limits, such as the Police vessel shown here on the Waitematā Harbour (Daniel Sharp)

and masters on how to navigate inshore waters in an effort to reduce the likelihood of an incident and subsequent pollution. Recommendations include keeping at least 5 nautical miles from the land, charted danger, or any outlying islands until reaching the position where alteration is required to make port or depart the coast. Specific routing instructions are available for approaches to 14 of the main commercial ports throughout the country. The relatively low levels of shipping traffic around New Zealand make the implementation of a formal mandatory routing system unnecessary.²⁶

In sensitive marine areas, the voluntary code can be superseded by a mandatory exclusion which provides the marine environment with the best protection from potential pollution or damage posed by shipping operations.²⁷ 'Areas to be avoided' are implemented under the UNCLOS and are recognised under international law. Vessels meeting certain criteria are prohibited from entering the waters within these areas. Within New Zealand, there are two such designated areas: the Three Kings Islands (a wildlife sanctuary off the northern tip of the North Island) and area off the north-east coast including the Poor Knights Islands.²⁸

Areas to be avoided by shipping

The Three Kings Islands were declared as an area to be avoided in 1995, and all vessels exceeding 500 gross tonnes are prohibited. The area stretching between Bream Head and Cape Brett and extending out to the Poor Knights Islands was identified as an area to be avoided in 2003. After two incidents when oily bilge water was released into the sea between the Poor Knights and the Tutukākā coast, threatening the area's world-renowned wildlife and marine ecology, the Poor Knights Islands ship restriction was set in place. A review by the Maritime Safety Association highlighted that the area has a high probability of an oil spill²⁹ due to vessels transporting oil to and from the Marsden Point Oil Refinery, and sailing to and from the Port of Auckland.³⁰ Within this area, any vessel greater than 45 metres in length is now prohibited. The only commercial ships exempt from this rule are fishing vessels that are engaged in fishing operations and barges under tow, provided the cargo is not oil or other harmful substances.

Maritime New Zealand has responsibility for marine oil spill response planning. The Marine Oil Spill Response Strategy 2006 (which is currently being updated) sets out the overarching framework for how Maritime New Zealand and its partners will respond to a marine oil spill. This is funded by an Oil Pollution Levy payable by those sectors whose activities raise the risk of an oil spill. Maritime New Zealand maintains a response capability to counter an oil spill of 3,500 tonnes. Where a larger incident occurs arrangements are in place to secure overseas assistance.³¹

Oil spill preparedness is funded by an industry levy, the Oil Pollution Levy, which is required under the Maritime Transport Act. The Levy is collected to run New Zealand's maritime oil pollution preparedness and response system. The Oil Pollution Levy applies to all commercial vessels over 100 gross tonnes and 24 metres or more in length (except those operating in freshwater), offshore oil installations, exploration wells and oil pipelines.³² The intention behind Maritime New Zealand's oil pollution preparedness and response is that the polluter pays. In circumstances where the polluter is not identified, the Levy is used to pay expenses from reserves set aside for that purpose.³³

The following process is followed to develop a new marine protection rule, or to amend an existing rule:

- Maritime New Zealand prepares a draft rule and undertakes informal consultation
- An invitation to provide written comment on the rule is advertised widely to the maritime community and general public
- Maritime New Zealand considers the written comments and prepares a revised rule which may be circulated to key stakeholders for comment before finalisation
- The Minister of Transport approves the rule and it is notified in the *Gazette* (section 448)

Figure 15.2 Process to develop a marine protection rule

Elements of best practice

Figure 15.3 provides a summary of the best practice responses for managing the impacts of vessels on the marine environment.

Potential environmental effects	Best practice response
Biosecurity risks	• Exchange ballast water in mid-ocean or operate a type-approved ballast water management system • Install an antifouling system that is designed for the ship’s operating profile to maintain the ship in a clean state throughout its docking cycle • Regularly inspect and maintain the antifouling system for niche areas (such as sea chests, bow thrusters, propeller shafts and intake gratings) of a ship
Pollution	• The design and construction of ships and operational procedures should aim to prevent oil and other hazardous substances being discharged into the marine environment • The routes for vessels transporting oil and other hazardous substances should avoid marine areas with high ecological values including marine reserves • The discharge of oily waste into the marine area in sensitive environments including estuaries, harbours and sheltered coastal areas should be avoided • Up-to-date oil spill contingency plans should be in place • The discharge of untreated sewage into harbours and near marine farms and marine protected areas should be avoided • Recreational vessels with seawater flushing toilets should have holding tanks installed • Avoiding washing or scrubbing boats on the foreshore or intertidal area • Provide sealed boat haul-out areas where discharges can be contained • Use more environmentally-friendly compounds in antifouling paint • Use land-based disposal of onboard generated waste where possible • Avoid the disposal of contaminated waste into the marine environment
Harm to marine animals and birds	• Where possible, redirect shipping routes away from marine mammal habitat • Reduce ships speed in areas where marine mammals are likely to be present to less than 10 knots • Reduce the speed of vessels in areas susceptible to ecological impacts
Shoreline erosion	• Reduce the speed of vessels in areas susceptible to wake-induced coastal erosion
Greenhouse gases	• Investigate alternative options for powering of vessels, both while at sea and shoreside

Figure 15.3 Best practice considerations for vessels



The speed of ships passing through areas where Bryde’s whales are likely, such as the Hauraki Gulf shown here, should be kept to less than 10 knots *(Raewyn Peart)*

Bryde's whale voluntary protocol for shipping in the Hauraki Gulf

The Hauraki Gulf is home to a small population of Bryde's whales, estimated at less than 200 individuals, and is one of only a few places in the world that can boast of a semi-resident population of a critically endangered whale species. The shipping passage through the Gulf is one of the busiest in New Zealand, used by over 1,500 ships per year berthing at the Port of Auckland, the country's premier cruise destination and primary port. On average, two Bryde's whales are killed every year as a result of vessel strike within the area, threatening the populations' long term survival.³⁴

Scientific research undertaken by the University of Auckland found that Bryde's whales spend 90 per cent of their time within 12 metres of the surface, which is the very part of the water column which ships hulls penetrate, with most large ships drawing between 10 and 12 metres.



Bryde's whale in the Hauraki Gulf, where ship strike threatens the resident population (Raewyn Peart)

The problem of vessels hitting whales also occurs overseas. Vessel strike on the Northern Atlantic Right Whale off the eastern seaboard of the USA and Canada was successfully reduced through the establishment of traffic separation schemes, vessel avoidance areas and speed restrictions.³⁵ Scientists investigated the utility of applying these approaches in the Hauraki Gulf. Because Bryde's whales are distributed throughout the Gulf, with no apparent regular seasonal changes in distribution, a traffic separation scheme was found unlikely to work. The most effective means of reducing the risk to whales was identified as being reducing the speed of vessels transiting the Gulf to 10 knots or less.

A collaborative process involving the Ports of Auckland, the shipping industry, the Department of Conservation, the University of Auckland, the Environmental Defence Society, council representatives and others was commenced in March 2012 to find ways to address the issue. The process was successful, just 18 months later, in developing a voluntary protocol aimed at reducing the number and impact of collisions between Bryde's whales and large vessels in the Hauraki Gulf.

Pre-planning and speed reduction are the critical factors for preventing Bryde's whale mortalities. The protocol, which was announced on 20 September 2013, comprises three primary approaches to reducing ship strike:³⁶

- **Passage planning** - Vessels are encouraged to use recommended shipping routes to enter and depart the port. Narrowing the area of the Gulf transited by ships reduces the risk of encountering a whale. By travelling at a maximum speed of 10 knots whilst in the Gulf, risks associated with an accidental impact are significantly reduced. There is good scientific evidence to suggest that if a whale is hit when the vessel is travelling at 20 knots, the chance of survival is less than 10 per cent, whereas if the vessel slows down to 10 knots, the chances of a whale surviving an impact rises to 75 per cent.
- **Keep watch** - When entering or leaving the gulf, especially during daylight hours, vessels are asked to have a dedicated observer equipped with binoculars. If a whale is sighted, vessels are asked to alter course and keep a 1000m clearance.
- **Reporting** - In the event of a sighting, the vessels should call in the sighting to Ports of Auckland using a pre-prepared script. Harbour control will then broadcast the message, with co-ordinates, to all vessels in the area so that they may alter course accordingly. The sighting information is also used for further statistical analysis.

The voluntary protocol has been notified to all shipping companies, and pilots boarding incoming ships to Auckland ensure that crew are aware of the protocol. A baseline assessment of the speed of ships transiting the Gulf, based on automatic tracking information which is electronically sent by ships via satellite, was undertaken in early 2013. This indicated that the additional time taken to transit the Gulf at 10 knots or less would, for most vessels, be less than one hour. The assessment is being repeated a year later to determine if speeds are reducing as a result of the protocol.

Ongoing work to promote continued uptake of the new measures are continuing.³⁷ Notification of the voluntary protocol to the International Maritime Organization is being considered. Additional research is also underway to count and map whales and other marine mammals in the Hauraki Gulf to build up a better understanding of their distribution.

Rena

Early in the morning of Wednesday, 5 October 2011, the 236 metre container vessel *Rena* grounded itself on Astrolabe reef. The reef lies 12 nautical miles off the Bay of Plenty coastline, and New Zealand's fifth largest city, Tauranga. As the vessel struck the reef, it wedged its bow section deeply into the rock, leaving the stern precariously afloat and two cargo holds flooded. On board *Rena* at the time were 1,386 containers and 1,733 tonnes of heavy fuel oil.³⁸ Initially the oil leaking from the damaged vessel appeared to be minimal, but by daylight it was obvious the disaster was on a larger scale than first thought. This vessel grounding ultimately resulted in New Zealand's largest marine oil spill response.³⁹



Debris starts to emerge from the submerged section of *Rena* (Maritime New Zealand)

A national level (Tier 3) oil spill response was initiated by Maritime New Zealand. Such a response is in co-operation with the vessel owner, regional council and wider industry, as well as the local community, iwi and any other relevant agencies. According to Maritime New Zealand Tier 3 events usually call for substantial resources to be mobilised, nationally and internationally, and for these resources to be integrated into a well-organised and co-ordinated response.⁴⁰ When the *Rena* response was in full swing there were around 200 people working in the Incident Command Centre and many hundreds cleaning up the beaches and wildlife. In total, around 8,000 volunteers registered to assist the clean-up and contributed over 18,000 hours of work.⁴¹

To date, the salvage operation has recovered over three-quarters of the containers and much of the fuel and oils from the *Rena* wreck. It has also removed the accommodation block and ensured that the bow is below the water line. However, a debris field still remains around the wreckage area.⁴² The owners of the vessel have now sought resource consent to leave sections of the wreck and debris in place on Astrolabe Reef. A council hearing of the application is set down for June 2015.

Astrolabe Reef is a jewel in the Bay of Plenty's marine environment, identified as a site of outstanding natural character in the Bay of Plenty Regional Policy Statement. After the *Rena* disaster, the proposed regional policy statement 'Variation One Coastal Policy', released by the Bay of Plenty Regional Council in 2013, downgraded the classification of the reef from outstanding to high natural character.⁴³ This decision was appealed to Environment Court by residents of Motiti Island and was upheld by the Court so that the original outstanding classification was reinstated in 2014.

A \$2.4 million fund was set up by Government in 2011 to help achieve the goals of the '*Rena* Long-term Environmental Recovery Plan'. This plan describes the environmental issues associated with the disaster and outlines the way these will be addressed through six key projects, as shown in Figure 15.4.⁴⁴ For the latest updates on these projects go to <http://www.renarecovery.org.nz/rena-recovery-projects.aspx>.

In late 2013 an 'Independent Review of Maritime New Zealand's Response to the *MV Rena* Incident' was carried out by former Secretary of Foreign Affairs and Trade Simon Murdoch.

The review reported that Maritime New Zealand had underplanned and underestimated the scale of the disaster.⁴⁵ In response to the review, an additional \$2.05 million of government funding was provided to Maritime New Zealand for the coming three years to implement the recommendations. At the same time, the '*Rena* Environmental Recovery Programme 2011–2013' report was released. This report found that there were few long-lasting impacts on the marine habitats in the Bay of Plenty from the *Rena* disaster, except for heightened levels of contaminants contained to Astrolabe Reef and the north-east area of Motiti Island.⁴⁶

- Monitoring programme – the aim of this programme was to assess the long-term environmental effects of the *Rena* grounding and monitor recovery. A focus of the programme was to assess kaimoana species and their marine habitats. Results have shown that the oil and other contaminants released from the *Rena* have had few long-lasting effects on the Bay of Plenty's maritime habitats.
- Shoreline programme – this monitored recovery of beaches and rocky shorelines from oil and debris from the *Rena*. In conjunction with the monitoring programme, it assessed how quickly the environment recovered from the different clean-up techniques used. No significant amounts of oil have been reported since August 2013.
- Wildlife programme – this assessed the effect on wildlife, with a particular focus on sensitive ecosystems, where there were endangered species. This programme is ongoing and due for completion in June 2015.
- Biosecurity programme – this aimed to survey high-risk sites and habitats to assess if new unwanted aquatic organisms had entered the environment, particularly where barge and support vessels were working. The overall conclusion from the surveys was that no high-risk species were detected during the surveys.
- Cultural impacts programme – This work provided an opportunity for tangata whenua to assess how their cultural values had been affected. Five cultural impact assessment reports have been completed and recommendations made. All reports supported total removal of the *Rena* wreck from Astrolabe Reef as being the only way to restore the area's mauri (life force). Maketu (Te Arawa) is also assessing the *Rena* impacts using a mātauranga (traditional ecological knowledge) approach.
- Mātauranga programme – this is focused on researching the effects of the *Rena* grounding using a mātauranga Māori approach. The work includes an assessment of mauri and research which will be undertaken by Maketu (Te Arawa) over 18 months.

Figure 15.4 Summary of the '*Rena* Long-term Environmental Recovery Plan' projects⁴⁷



The crane barge *Smit Borneo* working near the *Rena* wreckage (LOC)

Further reading and references

- David P, 2011, 'Marine pollution', in D Nolan (ed), 2012, *Environmental and resource management law*, 4th edition, LexisNexis, Wellington
- Environmental Protection Agency, 2012, *Preliminary risk assessment: Antifouling paints reassessment*, Environmental Protection Agency, Wellington
- Gillespie A, 2002, 'Environmental threats to cetaceans and the limits of existing management structures', *New Zealand Journal of Environmental Law*, 6, 97–138
- Harbison G R, S P Volovik, 1994, 'The ctenophore, *Mnemiopsis leidyi*, in the Black Sea: A holoplanktonic organism transported in the ballast water of ships', in *Proceedings of the National Oceanic and Atmospheric Administration Conference and Workshop on Nonindigenous Estuarine and Marine Organisms*, United States Government Printing Office, Washington, DC, 25–36
- Hauraki Gulf Forum, 2008, *Tikapa Moana – Hauraki Gulf State of the Environment Report*, Hauraki Gulf Forum, Auckland
- Ingram K, 2007, 'Ship collisions strike down whales', *Professional Skipper*, September/October, 38
- Kirk B and M Single, 2000, 'Coastal effects of new forms of transport: The case of the inter-island fast ferries', in P A Memon and H Perkins (eds), *Environmental planning and management in New Zealand*, Dunmore Press, Palmerston North
- Maritime New Zealand, n.d., *Oil and its impact on the marine environment*, available at <http://www.maritimenz.govt.nz/Environmental/Oil-and-oily-waste/Oil-biological-impact.asp>
- Maritime New Zealand, 1999, *New Zealand guidelines for sea disposal of waste*, Maritime New Zealand, Wellington, available at <http://www.maritimenz.govt.nz/Publications-and-forms/Environmental-protection/Sea-disposal-of-waste.pdf>
- Maritime New Zealand, 2006, *New Zealand marine oil spill response strategy*, Maritime New Zealand, Wellington, available at <http://www.maritimenz.govt.nz/Publications-and-forms/Environmental-protection/Oil-spill-response-strategy.pdf>
- Maritime New Zealand, 2012, *Maritime New Zealand statement of intent: 2012–2015*, Maritime New Zealand, Wellington
- Maritime New Zealand, 2014, *New Zealand marine oil spill response strategy 2015–2019*, Maritime New Zealand, Wellington
- Ministry for the Environment and Maritime New Zealand, 2006, *Environmental best practice guidelines for the offshore petroleum industry*, Maritime New Zealand, Wellington, available at <http://www.maritimenz.govt.nz/Publications-and-forms/Environmental-protection/Environmental-best-practice-guidelines-offshore-petroleum.pdf>
- Mulchay K and R Peart, 2012, *Wonders of the Sea: The protection of New Zealand's marine mammals*, Environmental Defence Society, Auckland
- New Zealand Coastal Society, 2014, *Rena: Lessons learned*, New Zealand Coastal Society, Wellington
- Oceana, 2009, *Toxic legacy: Long-term effects of offshore oil on wildlife and public health*, Oceana, Washington, DC
- Van Roon M and S Knight, 2004, *Ecological context of development: New Zealand perspectives*, Oxford University Press, Victoria,

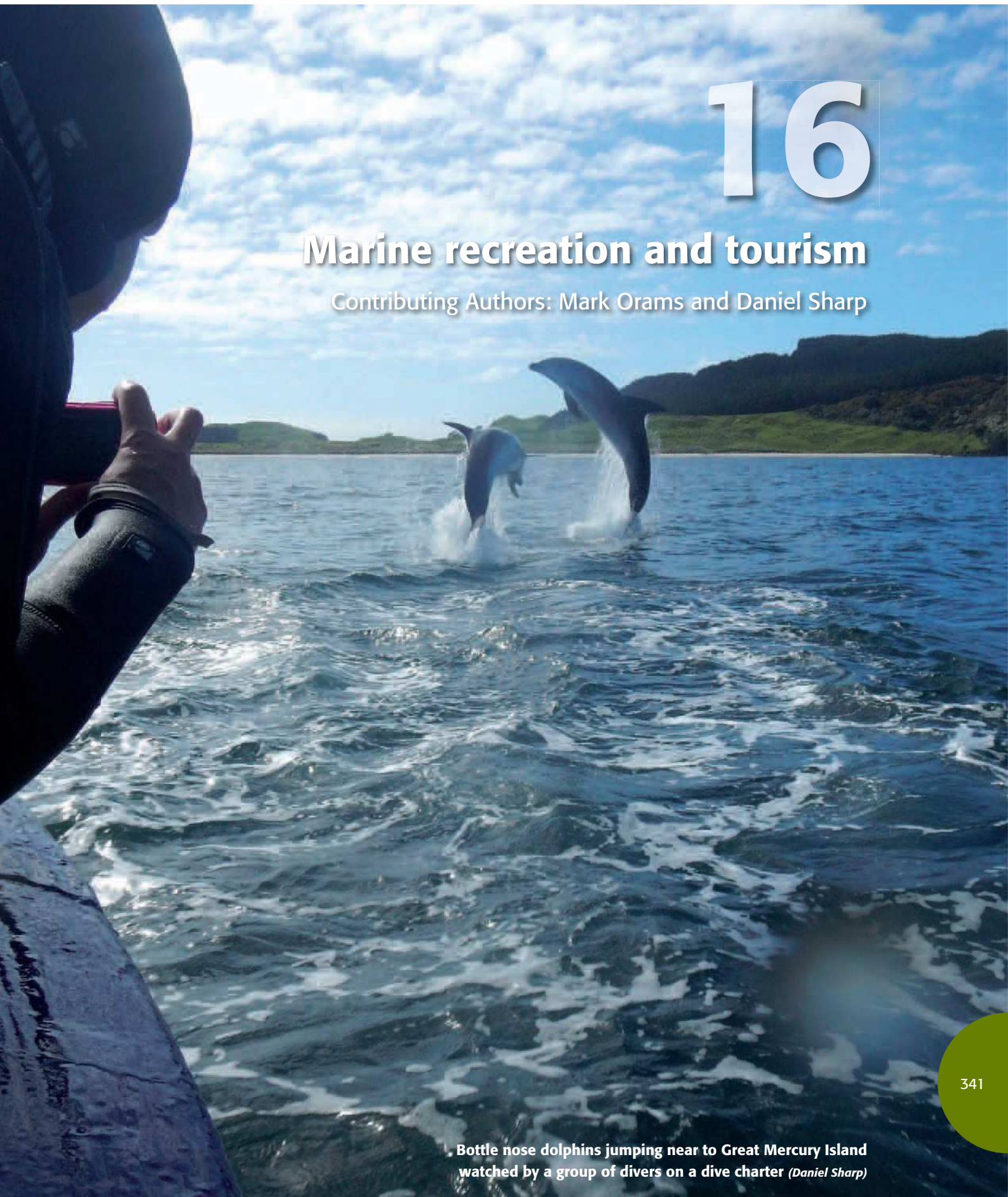
Endnotes

- 1 Maritime New Zealand, 2012
- 2 These reflect numbers between late 2010 and early 2011 during one cruise season.
- 3 <http://www.maritimenz.govt.nz/Environmental/Responding-to-spills-and-pollution/Past-spill-responses/Past-spill-responses.asp>
- 4 <http://www.maritimenz.govt.nz/Environmental/Responding-to-spills-and-pollution/Past-spill-responses/Rena-response.asp>
- 5 <http://www.maritimenz.govt.nz/Environmental/Marine-oil-spill-risk-assessment/>
- 6 <http://www.maritimenz.govt.nz/Environmental/Sewage-discharge/Sewage-discharge.asp>
- 7 <http://www.maritimenz.govt.nz/Environmental/Marine-oil-spill-risk-assessment/2010-results.asp>
- 8 <http://www.niwa.co.nz/news/are-antifouling-paints-harming-our-sealife>
- 9 Ingram K, 2007, 38; Haruaki Gulf Forum, 2008, 59
- 10 <http://www.orcaresearch.org/wp-content/uploads/2011/08/Visser-2007-Killer-whales-in-NZ-waters-SC-59-SM19.pdf>
- 11 <http://www.epa.govt.nz/Publications/Liz%20Slooten%20-%20noise%20conditions%20statement.pdf>
- 12 Van Roon M and S Knight, 2004, 114
- 13 http://ec.europa.eu/clima/policies/transport/shipping/index_en.htm
- 14 http://www.hamburgsud-line.com/hsdg/en/hsdg/qualityenvironment_1/shipofthefuture_1/engine_1/engine.html
- 15 <http://news.bbc.co.uk/2/hi/europe/7201887.stm>
- 16 http://www.skysails.info/fileadmin/user_upload/Power/Presse/EN_SkySails_Power_Wind_Power_Next_Level.pdf
- 17 [http://www.imo.org/About/Conventions/ListOfConventions/Pages/International-Convention-for-the-Prevention-of-Pollution-from-Ships-\(MARPOL\).aspx](http://www.imo.org/About/Conventions/ListOfConventions/Pages/International-Convention-for-the-Prevention-of-Pollution-from-Ships-(MARPOL).aspx)
- 18 RMA 1991, section 15(c)
- 19 RMA 1991, section 15(a)
- 20 RMA 1991, section 15(b)
- 21 <http://www.renaresourceconsent.org.nz/>
- 22 Maritime Transport Act 1994, section 33D
- 23 Maritime Transport Act 1994, section 33F
- 24 Maritime Transport Act 1994, section 33M
- 25 Maritime Transport Act 1994, Part 19
- 26 <http://www.maritimenz.govt.nz/Environmental/Environmental-requirements/Requirements-for-operators/Ship-routeing.asp>
- 27 <http://www.maritimenz.govt.nz/Rules/List-of-all-rules/Part190-marine-protection-rule.asp>
- 28 <http://www.maritimenz.govt.nz/Environmental/Environmental-requirements/Requirements-for-installations/Safety-zones.asp>
- 29 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=208356
- 30 http://www.marinenz.org.nz/documents/literature_review_on_the_poor_knights.pdf
- 31 <http://www.maritimenz.govt.nz/Environmental/Responding-to-spills-and-pollution/Responding-to-spills-and-pollution.asp>
- 32 <http://www.maritimenz.govt.nz/About-us/Fees-levies/OPL/>
- 33 <http://www.maritimenz.govt.nz/About-us/Fees-levies/OPL/>
- 34 Mulcahy K and R Peart, 2012
- 35 Mulcahy K and R Peart, 2012
- 36 http://www.poal.co.nz/news_media/2013_mediareleases/20130920_Whales.htm
- 37 <http://www.ifaw.org/australia/news/new-zealand%E2%80%99s-shipping-industry-steps-and-slows-down-whales>
- 38 New Zealand Coastal Society, 2014
- 39 Maritime New Zealand, 2014
- 40 Maritime New Zealand, 2014
- 41 New Zealand Coastal Society, 2014
- 42 New Zealand Coastal Society, 2014
- 43 http://www.nzherald.co.nz/bay-of-plenty-times/news/article.cfm?c_id=1503343&objectid=11286955
- 44 <http://mfe.govt.nz/publications/hazardous/rena-recovery/>
- 45 http://www.nzherald.co.nz/bay-of-plenty-times/news/article.cfm?c_id=1503343&objectid=11166706
- 46 http://sci.waikato.ac.nz/__data/assets/pdf_file/0008/180476/RENA-executive-summary.pdf
- 47 <http://www.renarecovery.org.nz/rena-recovery-projects.aspx>

16

Marine recreation and tourism

Contributing Authors: Mark Orams and Daniel Sharp



Bottle nose dolphins jumping near to Great Mercury Island watched by a group of divers on a dive charter (Daniel Sharp)



Contents

343	Introduction
344	Activities
345	Potential environmental effects
346	Swimming
346	Board sports
348	Diving and snorkelling
349	Boating
351	Recreational fishing
353	Interacting with marine mammals and seabirds
355	Special events and sporting contests
355	General coastal and beach activities
355	Holiday homes
356	Cruise ships
357	Decision-making framework
357	Resource Management Act
357	Resource Management (Marine Pollution) Regulations
358	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act
358	Marine Mammals Protection Act
359	Marine Mammals Protection Regulations
361	Conservation Act
362	Local and regional bylaws
362	Elements of best practice
366	Case studies
366	Dolphin Encounter Kaikōura
367	Dive! Tutukaka
368	Environment Southland's Deed of Agreement for cruise ships
369	Further reading and references
370	Endnotes

Introduction

In terms of its reputation and lifestyle, New Zealand is well known for its wide range of outdoor recreation activities (including tourism opportunities). These opportunities are supported by a relatively benign climate and a diverse range of outdoor settings which are in close proximity to each other. New Zealand also has a strong tradition of freedom of access to these settings and a culture which has valued outdoor recreation as an important part of the 'Kiwi way of life'. As an island nation, with a long and intricate coastline, the sea, beaches, estuaries, reefs and smaller islands are major recreation and tourism resources.

Tourism is one of the world's fastest-growing industries. In 2012, international tourism arrivals worldwide reached one billion, and are expected to further increase:¹

- Tourism is now one of New Zealand's largest export industries contributing nearly \$10 billion or 16 per cent of the nation's export earnings²
- International visitor arrivals to New Zealand doubled from 1993 to 2006, when there were 2.4 million arrivals, and reached 2.6 million in 2013³
- Domestic tourism (New Zealand residents travelling within New Zealand) is even more significant and combined with international tourism is responsible for \$24 billion of expenditure
- Tourism directly employs six per cent of the New Zealand workforce and indirectly employs a further 3.1 per cent
- In total, around one in 11 working New Zealanders is employed in the tourism industry⁴
- The growth of tourism is predicted to increase by four per cent every year for the next seven years⁵

Since the start of the 1970s, the use of marine ecosystems and adjacent coasts for recreation and tourism has grown enormously. There has been a continuous trend towards tourists seeking out more diverse and geographically remote coastal and marine tourism destinations.



Marine tourism has grown enormously over the past few decades, including that at Hot Water Beach, Coromandel Peninsula, shown here (Raewyn Peart)

Activities

There is a wide range of recreation and tourism activities that New Zealand's marine environment supports. Few locations now remain inaccessible due to the wide availability of new technologies.⁶ The 'Spectrum of Marine Recreation Opportunities',⁷ as shown in Figure 16.1, is a model that helps to categorise marine recreation and tourism experiences relative to their distance from shore and from human settlements. This model can be used to assist with understanding the range of experiences, and potential impacts that these activities are having or are likely to have on the environment.

Characteristics	Class 1: Easily Accessible	Class 2: Accessible	Class 3: Less Accessible	Class 4: Semi-remote	Class 5: Remote
Experiences	• Much social interaction • High level of services • Usually crowded • Noisy • Lots of activity	• Frequent contact with others • Some space/ times to escape from others	• Some contact with others • Locations/ times when no other people present • Quieter	• Infrequent contact with others • Peace and quiet • Close to nature	• Virtually no contact with others • Solitude • Tranquillity • Self-sufficiency
Environment	• Many human influences • Highly modified • Lower-quality natural environment	• Human structures and influences visible and close by • Environment quality variable	• Few human structures close by but some maybe visible • Higher environmental quality	• Limited evidence of human activity or structures • Very high-quality environment	• Isolated • Little to no evidence of human activity • Pristine environment
Locations	• Close to, or in, urban areas • Beaches, docks, piers, urban coastal parks and walkways/ cycleway • Close to roading, parking and mass transport options	• Intertidal zone and areas up to 100 metres offshore • Often requires walk of several hundred metres from car parking or mass transport	• 100 metres to 1 kilometre offshore • More than 20 kilometres from major urban area • More remote beaches, islands and coastal areas • Often requires boat for access or walk in >500 metres	• 1–50 kilometres offshore • Isolated coasts, islands or reefs difficult to access • Boats or overnight hiking needed to access	• >50 kilometres offshore • Coastal areas >100 kilometres from any significant human habitation
Examples of activities	• Sunbathing/ baking • People-watching • Swimming • Playing games • Eating • Social gatherings • Sightseeing • Special events (i.e. concerts)	• Swimming • Fishing • Boating • Surfing • Windsurfing/ kiteboarding	• Boating • Fishing • Snorkelling/ SCUBA • Nature-study • Surfing • Sailing • Sea-kayaking	• Coastal sailing • Remote coast hiking and camping • Live-aboard vessels (e.g. fishing, diving or surfing)	• Offshore blue water sailing • Live-aboard offshore vessels (e.g. for fishing or diving) • Remote coast/ reef sea-kayaking and surfing

Figure 16.1 Spectrum of marine recreation opportunities (Orams M B and M Lück, 2013)

In terms of patterns of use of marine areas for recreation and tourism, the model shows that both the intensity of use and the diversity of activities have an inversely proportional relationship with distance from both the shoreline (i.e. both intensity and diversity decrease further offshore) and places of human settlement. Thus urban beaches and near-shore locations are the most intensively used and offshore locations distant from cities and towns the least intensively used. This suggests that different management approaches will be required for the various types of marine environments as classified by proximity.

Research completed in 2012 showed that this pattern is generally true for New Zealand's marine environment with the notable exception of tourism 'hotspots', such as Milford Sound and the Poor Knights Islands, and locales that offer specific recreational experiences, such as the popular surfing breaks off the Raglan, Ahipara and South Taranaki coasts.⁸

Potential environmental effects

The growth and expanding influence of marine recreation and tourism is now well recognised in New Zealand. For example, NIWA noted in 2012 that ecotourism is a growing industry in the near-shore waters around New Zealand and is already starting to expand into the EEZ.⁹ Similarly, a study commissioned by the Department of Conservation also identified the increasing importance of the coast and near-shore coastal waters of New Zealand as resources for recreation and tourism.¹⁰

Historically, most of New Zealand's marine environment has been protected from intensive recreational and tourist use by its inaccessibility. This has changed over the past four decades and almost all parts of New Zealand's coastal environment are now accessible for recreation. While this has created opportunities for New Zealanders and visitors to the country, it has also resulted in increased environmental effects. While recreation and tourism may appear to be relatively benign and an inherently more sustainable use of natural resources than extractive activities such as fishing or deep-sea mining, it is irrefutable that coastal and marine recreation and tourism does have environmental impacts.¹¹ The continued growth in scale and distribution of this sector will, therefore, be accompanied by increasing environmental impacts.¹²

However, it is also important to recognise that there is huge potential for activities to be developed and managed in a way that have a positive impact on the very environment they rely upon.¹³ Marine and coastal recreation and tourism can also, potentially, foster a greater appreciation for the importance of marine environment and result in human actions that may benefit the ecosystem they depend on.

Figure 16.2 provides a summary of the main environmental impacts caused by different recreation and tourism activities in the marine environment. Each of these elements is discussed and explained in further detail below.



Marine tourism, such as that shown here in Milford Sound, Fiordland, can have a negative impact on the environment
(Raewyn Peart)

Potential impact	Swimming	Board sports	Diving and snorkelling	Boating	Fishing	Marine mammal and seabird interactions	Special events and sporting contests	General coastal and beach activities	Holiday homes	Cruise ships
Wildlife disturbance	■	■	■	■	■	■	■	■	■	■
Ocean noise				■	■	■	■			■
Ship strike				■	■	■	■			■
Trampling	■	■	■				■	■	■	
Over harvesting			■	■	■					
Marine pests			■	■	■		■			■
Anchor damage			■	■	■					
Rubbish	■	■	■	■	■	■	■	■	■	■
Wastewater/sewage				■	■	■			■	■
Climate change		■	■	■	■	■	■			■
Conflict between users	■	■	■	■	■	■	■	■	■	■
Impacts on near-shore environments	■	■	■	■	■	■	■	■	■	
Impacts on remote environments			■	■	■	■				■

Figure 16.2 Potential environmental impacts of recreation and tourism activities

Swimming

Swimming at the beach only became a popular recreational activity in New Zealand at the end of the 19th century. Prior to that, the marine area was perceived as dangerous; few Europeans could swim, and many people drowned. In the late 1880s medical authorities in New Zealand championed the health benefits of swimming. By the early 20th century, sea bathers were forming organisations to improve the image of swimming and provide amenities such as changing rooms at the beach. It was during this time that the increasing popularity of beach swimming saw a rise in surf lifesaving as both a voluntary service and competitive sport. As a consequence, beach swimming became safer and even more popular.¹⁴ Today, the beach lifestyle is an integral part of the national way of life and highly valued by New Zealanders and visitors, particularly during the summer months.

In terms of environmental impacts, swimming is generally a benign activity. However, there is an increasing demand for infrastructure such as shops and toilets to enable people to swim at the beaches. While people are swimming they can also disturb the local wildlife, trample on seabed habitats, leave rubbish behind and cause conflicts with other users such as fishers and surfers. Swimming provides many people with a great deal of pleasure and enjoyment and as such delivers some valuable positive physiological benefits.

Board sports

Surfing is believed to have originated from Hawai'i. The well-known Hawai'ian surfer Duke Kahanamoku toured New Zealand in 1915 giving surfing demonstrations in Auckland, Wellington and Christchurch.¹⁵ Locals were inspired and surfing became established in those areas. Surfing as a widespread recreational activity only became established in New Zealand in the 1960s. It was during that decade that the growth of the surfing and hippy culture really gained momentum, as it spread from California to Australia and then to New Zealand. Several New Zealand coastal towns become strongly associated with surfing, and the related

surfing and beach lifestyle, including Whangamatā, Mount Maunganui, Raglan, New Plymouth, Gisborne and Ahipara.

In 2013 the *New Zealand Herald* reported that more than 200,000 Kiwis and 30,000 tourists surf in New Zealand.¹⁶ It has become a way of life for many people living around the coast. As a consequence of the strong popularity of surfing, and threats to high-quality surf breaks from development, organisations such as the New Zealand Surf Break Protection Society have been advocating for legal protection of surf breaks. Some surf breaks now have statutory protection under the NZCPS. The idea of installing artificial surf reefs to promote and increase opportunities for surfing was popularised in New Zealand during the 1990s.



Board sports are very popular in New Zealand, including boogie boarding shown here (Daniel Sharp)

Windsurfing or boardsailing began in the early 1970s and, in a pattern typical of the invention of a new marine recreational activity, its popularity grew and the activity diversified. There is now general recreational windsurfing, wavesailing, slalom racing, speed sailing and Olympic competitions. Kitesurfing (also known as kiteboarding) has developed with similar speed since the early part of the 21st century and more recently stand up paddle boarding has grown rapidly. Competitive sporting events, recreational activities and commercial hire and tourism operations now exist for all of these marine recreational activities in New Zealand.

For many people it is hard to imagine that surfing might have negative impacts on the environment. However, there are a number of impacts, in particular accelerated erosion due to vehicle and foot traffic from surfers traversing over dunes and other sensitive coastal locations to access desired surf breaks. In addition, there are issues over increasing demand for infrastructure, disturbance of local wildlife, trampling on seafloor environments, leaving rubbish behind and competition with other existing recreational uses. Direct contributions to climate change through using vehicles and boats to access remote surf locations, and through the production of surfing equipment, are an increasing concern for people involved in this activity.

There are certainly psychological benefits from surfing with improvements to people's well-being, health and fitness and there are also proven economic gains from surfing. South Stradbroke Island in Australia is worth an estimated US\$20.7 million (NZ\$30 million) per year, via the 11,500 surfers who visit the island 64,000 times each year.¹⁷



Board sports, such as windsurfing in the Tauranga Harbour shown here, can impact on the marine environment (Lucy Brake)

The conflict between surfers and other users of the marine environment is an intense one that has played itself out in the Environment Court. One of the most well-known conflicts is the controversy over the Whangamatā Marina where local surfers were fighting to protect high-quality surf breaks from the damage that would be incurred by dredging to provide vessel access to a marina and safe harbour for recreational boaters and fishers.

There has also been much controversy over the artificial surf reefs which have been built along some parts of New Zealand's coast, for example:

- Mount Maunganui submerged reef, the first of its kind, is now set to be removed just a decade after it was constructed in the early 2000s. It was hoped that the reef would provide world-class surfing waves and attract new economic development into the area. But the reef has never functioned as it was intended and the Bay of Plenty Regional Council recently decided to remove the reef because it was altering coastal processes, changing currents and exacerbating risks to swimmers in the area¹⁸
- In 2006 construction began on an artificial reef 250 metres offshore from Ōpunake Beach in Taranaki. The work involved pumping sand into geotextile bags to create a wave-making reef. Work was never completed and in 2011 the South Taranaki District Council voted to restore the Ōpunake headland to its original state¹⁹

Diving and snorkelling

People have always had a keen interest in going beneath the sea. Ancient manuscripts contain depictions of early divers and centuries-old artefacts imply that people dove to gather materials for jewellery. In the 19th century people began to use diving bells, supplied with air from the surface, probably the first effective means of staying underwater for any length of time.²⁰ The invention of the SCUBA (self-contained underwater breathing apparatus) in 1943 changed the way we saw the underwater world, and increased the demand for ways that people could explore the marine environment. The Professional Association of Diving Instructors (PADI) is the world's largest recreational diver training organisation, with members teaching the vast majority of the world's recreational divers, issuing nearly one million certifications each year.²¹

New Zealand is a popular dive destination and is rated as having some of the top dive spots in the world. The protection offered by marine reserves has helped ensure that some key locations have remained attractive for diving. Jacques Cousteau, the famous marine explorer, rated the Poor Knights Islands (now a fully protected marine reserve) as one of the top 10 dive spots in the world. More recently, the marine reserve was named in CNN's top 50 dive sites, because it has some of the best subtropical underwater gardens on the planet. The *Rainbow Warrior*, a vessel deliberately sunk off the Northland coast to create an artificial reef and dive site, was named by the *Telegraph*, a major newspaper in the United Kingdom, as one of their top dive sites.²² The creation of artificial reefs, through the deliberate sinking of boats or other materials, is increasingly being used in New Zealand to create or enhance dive locations.

Snorkelling, where a mask and snorkel (usually also with fins for propulsion and safety) is used to view underwater, is also a very popular activity. It is an accessible and budget-friendly way for people to see and enjoy the marine environment and is commonly offered to a wide range of age groups at resorts and on charter boats throughout the world. Snorkelling in New Zealand is very popular, particularly over the summer months. With the country's long and varied coastline, there are many prime snorkelling spots offering people free and easy access. A number of marine reserves around the coast provide both diving and snorkelling opportunities. Examples include the Cape Rodney-Okakari Point (Goat Island) Marine Reserve, the Tāwharanui Marine Reserve and the Whanganui A Hei (Cathedral Cove) Marine Reserve.



Diving, such as that shown here near Leigh, can disturb marine life
(Raewyn Peart)

Most people who enjoy diving and snorkelling, have an inherent passion for the marine environment, and in general the sport of diving and snorkelling has minimal adverse impacts. These activities certainly have a number of positive benefits, including improved well-being and health, economic benefits for a local community through special dive sites and educational opportunities. New Zealand's marine reserves, such as the Poor Knights Islands, are a prime spot for divers and snorkellers to learn about the marine environment and how to protect it.

The easy access that divers and snorkellers have to the marine environment can provide positive benefits, including protection of resources, raising awareness and education. An example is the Project AWARE Foundation, established by PADI as a non-profit organisation dedicated to preserving the aquatic environment and improving environmental outcomes. The work of such organisations and groups is discussed further in the section on best practice below.

There are, however, a number of challenges facing the industry. Snorkelling and diving practices can disturb wildlife, and damage marine plants and animals, through trampling or crushing by a person's body, fins and equipment. Snorkellers, in particular, enjoy fish feeding which can change the natural behaviour and diet of fish species. This can be particularly problematic in fragile environments. The anchoring of dive boats can inflict potential damage on the seabed ecology. Areas of seagrasses and reefs are particularly vulnerable.

Rubbish can be left behind in marine environments by snorkellers and divers. Moving dive equipment or support vessels from one site to another can encourage the spread of pest plants and animals, thereby posing a biosecurity risk. Other negative effects include diver-induced behavioural changes (e.g. fish feeding), damage or destruction of subtidal benthic species (e.g. repeated dislodgement) and impacts on water quality through stirring up sediments and benthos. These activities can have an increasing impact on remote environments through improved access to offshore islands.

People will often snorkel or dive to gather shellfish and harvest fish and crayfish, and in some cases this can result in over-harvesting of resources, particularly those that are close to shore or vulnerable from gathering pressure. The recreational fishing rules put in place by the Ministry for Primary Industries,²³ such as gear restrictions, maximum catch limits and minimum size limits, are designed to help reduce the pressure on fish stocks from recreational harvesting. But illegal harvesting is made easier by the use of snorkelling and diving equipment, especially of shellfish like paua and kina. Direct contributions to climate change through using boats and production of dive equipment (because of the types of materials used in the production and the emissions from transportation) is an increasing concern for people involved in diving and snorkelling.

Boating

The development of plywood and marine glues after World War II encouraged a generation of New Zealanders to build their own boats and head out onto the water in launches, sailing dinghies and cruising yachts. From the 1960s the invention and use of a range of materials such as fibreglass for hulls, alloy for masts and dacron for sails transformed boating in New Zealand by making craft more seaworthy, reliable and affordable. During the past two decades, the wide availability of much improved navigational equipment such as the GPS chartplotter, has facilitated safe cruising further afield. Boats now provide the platform for a wide variety of marine recreational activities including fishing, diving, waterskiing/wakeboarding, sightseeing, racing and wildlife watching. Boats also provide a means of transport to previously inaccessible islands, reefs, harbours and beaches.²⁴



Dinghy racing, as shown here off Takapuna Beach, Auckland, serves to introduce youth to marine sports (Raewyn Peart)

Yachting and motorboating has continued to develop in New Zealand, both as a competitive sport and a recreational activity. There is a large cruising boating community throughout the country. A New Zealand Marine Industry Association assessment of boat ownership in 2010 estimated the number of boats in New Zealand to be above 470,000 (including 5,000 charter and tour boats).²⁵ In 2011 Maritime New Zealand commissioned market research on recreational boat ownership which indicated that 19 per cent of New Zealanders owned at least one boat at the time of the survey.²⁶

The relatively protected and safe harbours around the upper North Island (such as the Bay of Islands, Hauraki Gulf) and in the South Island (such as the Marlborough Sounds) are a boating paradise. Increasingly the offshore islands and remote cruising destinations are becoming more popular as technology allows people to travel further, more comfortably and safely in their boats. In addition, New Zealand (particularly the north-eastern part of the North Island) has become a favoured destination for blue-water yacht cruisers. These international tourism arrivals, while low in terms of volume, have had significant impacts on the coastal communities such as Ōpua, Russell, Whangarei and Auckland.

Whilst it has a long history, particularly in the Arctic and Northern Canada, where its use spans at least 5,000 years, sea kayaking has only recently become a widespread marine recreational activity. This growth mirrors that of boating more generally, as kayaks began to be constructed out of durable materials such as polyethylene plastic, which were mass-produced from moulds and therefore could be sold at affordable prices. Tourism New Zealand promotes kayaking as one of the best ways to explore New Zealand's coastline and inland waterways. The activity is popular with both private recreational kayakers and commercial tour operators who offer sea kayaking tours in many places such as the Nelson Lakes National Park, the Hauraki Gulf Marine Park, the Bay of Islands Marine Park, the Marlborough Sounds and the Coromandel Peninsula.

The use of personal water craft (jetskis) has also become a popular marine activity, with significant growth in both high-powered, manoeuvrable single-person craft and those that are suitable for multiple riders. They are used for activities as diverse as fishing, waterskiing, wakeboarding, surf riding and tow-in surfing (where surfers are towed in to surf large waves by jetskis). In addition, a wide diversity of watercraft are being designed and utilised in the sea including radio-controlled yachts and powerboats, waka-ama, dragonboats and other paddle-powered craft, submersibles, semi-submersibles, specialist wakeboarding boats, offshore powerboats (for racing), large expedition-designed and -equipped cruising boats, jetboats, and superyachts.

Boating has a number of positive impacts. Improving people's well-being and quality of life is most certainly one and another would be the economic value of the boating industry, especially for local communities in smaller regions. However, as recreational and tourism boating continues to increase in popularity, it is important that the impact this activity can have on the marine area is well understood.

Hull fouling is one of the greatest contributors to the domestic spread of invasive marine pests once they arrive in the country (discussed further in Chapter 5: Marine biosecurity).

Wharves, jetties and marinas typically attract large numbers of people and boats that, in turn, have the potential for discharges into the marine area, including fuel spillage, sewage from boat toilets, rubbish and leaching of antifouling paints which contain toxic chemicals. The parking and boat washing areas found at marinas may also be a source of contaminated runoff into the sea if they are not appropriately managed. Elevated heavy metal concentrations have been found in seabed sediments in the vicinity of marinas.²⁷

Increased noise, both from engines and propellers, and the boats themselves can have impacts on marine life, and of increasing concern is the potential impact from fish-finders and sonar depth sounders. The impact of reliable and relatively cheap marine navigation systems makes favoured habitats easily locatable and shared with others, so scallop beds, schooling fish, habitats suitable for crayfish and other good fishing spots are now found easily.

Waste and sewage management is a big concern for the boating community, particularly as access to remote and pristine locations is becoming much easier. The way that boat owners dispose of rubbish and fish waste can be a contentious issue. Regional councils have created rules to control the disposal of sewage from boats, because it can



The use of personal water craft, as shown here in Marlborough, has become a popular marine activity (Raewyn Peart)

significantly contribute to the contamination of coastal waterways, particularly in crowded anchorages. Larger boats tend to have onboard holding tanks which are discharged far out to sea. But the vast majority of the smaller boats have no such facility. So whilst there may be rules in place to prevent it happening, raw human sewage is often discharged directly into some of this country's most beautiful bays and harbours.

Other impacts from boating on the marine environment include wildlife disturbance, including the disturbance of flocking, resting or feeding birds, disruption of roost sites (e.g. shag roost sites in overhanging trees and on cliffs), disturbance of larger marine animals (e.g. marine mammals and sharks) and accidental deaths of seabirds (e.g. little blue penguins). Increasing ocean noise levels and ship strike as a result of recreational boating can impact on marine mammals (this is discussed further in Chapter 15: Vessels). Regulations prescribe good boat behaviour around marine mammals.

The impact of anchoring on seafloor communities is becoming an increasing concern. The Department of Conservation produced a report in 2006²⁸ which stated that one of the most direct adverse effects on seagrass beds is the damage caused by recreational boating activities (e.g. cutting by propellers, propeller wash, anchor and mooring damage and boat groundings) which may result in significant, localised impacts on the physical integrity of seagrasses. The potentially long-term negative impacts of recreational boating activities on seagrass habitats has been recognised internationally, and the cumulative impacts can lead to the large-scale loss of seagrass beds from heavily trafficked areas.²⁹

Other issues include the conflict of vessel traffic with other users. Swimmers in shallow areas are often concerned about their safety and it is for this reason that there are clear rules about the speed that boats can travel in certain areas.

The growth in recreational boating drives demand for marinas, boat ramps and other large marine infrastructure which can have very significant impacts on the marine environment. This is discussed in Chapter 14: Major marine developments, which includes a case study on Westhaven Marina in Auckland.



Boating, such as shown here in the Hauraki Gulf, can have a range of environmental impacts (Raewyn Peart)

Recreational fishing

Kaimoana, or food from the sea, has always been an important component of Māori culture. It is significant as a source of sustenance, as a good to be traded, and as part of the hospitality provided when hosting guests. Other New Zealanders have also used the sea as a source of food, and have fished to both supplement diets and for pleasure from the time that Europeans first arrived on these shores.

Recreational fishing remains extremely important and this activity takes many forms, including:

- Fishing from rocks or from human made structures
- Surfcasting from beaches
- Using spears to catch fish such as flounder in shallow tidal areas
- Using set nets
- Hook and line fishing
- Trolling using lures or live baits from boats
- Spea-fishing (either freediving or using diving gear)
- Rock lobster (crayfish) hunting
- Shellfish gathering (in intertidal areas, using dredges, via digging and via diving and snorkelling)
- Taking marine life from rocks, rock pools and the intertidal zone

Before the 1960s, most land-based fishing spots were reached using bicycles and walking. Four-wheel motorbikes, trail bikes, 4WD vehicles and helicopters have now allowed people to access difficult-to-reach and remote spots. Rapid developments in boating, diving, navigation and fishing technology have also allowed recreational fishers to access more remote fishing grounds, further from shore, and in deeper areas.

Big game fishing is now a major part of the recreational fishing and boating industry, with charter boats offering week-long trips to some of New Zealand's most remote fishing spots. New Zealand is well known around the world for its billfish (e.g. striped marlin), tuna fisheries, as well as the opportunities to fish for species such as snapper, kahawai, hāpuka, bluenose and trevalley.

Recreational fishing brings a number of positive impacts, including improving people's well-being and quality of life as well as the economic value of the recreational fishing industry. However, there are negative impacts from this activity. Recreational fishing allowances are set as part of the total allowable catch, which is a component of the QMS. The Ministry for Primary Industries sets and enforces recreational fishing rules for key species, such as snapper and trevalley, including bag and catch limits, potting and netting methods, types of equipment that can be used and the restricted or closed areas. These are all intended to constrain the catch within the QMS allowance and manage impacts of fishing on associated species.

Controlling and estimating the recreational fishing catch is challenging for a number of reasons, including the country's huge coastline; sparse population; large number of species targeted; large number of recreational fishing boats; the methods used; and the small research budgets available. Fishery compliance officers work around the coastline and are the point of contact for any phone calls to the Ministry about people taking too much or an undersized catch.

Surveys are periodically conducted to estimate the quantity and type of species caught. For some fish stocks, the recreational take represents a large proportion of the total catch,³⁰ such as with the blue cod fishery in the Marlborough Sounds and the snapper fishery in the Hauraki Gulf. There is a paucity of information in New Zealand about the impact of recreational fishing on the marine environment. According to researchers in 2010, data extrapolated from interviews with fishers suggests that the accidental catch of seabirds and threatened marine mammals in New Zealand's non-commercial fisheries is a 'problem that deserves more focused attention in the context of species conservation'. The Minister intends to focus on this issue when developing the next Inshore Finfish Annual Operational Plan.

In New Zealand's recreational fisheries, seabirds have been recorded being caught during hook and line fishing and in set nets.³¹ In July 2002, Southern Seabird Solutions was formed in an effort to develop and promote sustainable and responsible fishing practices in both the commercial and recreational sector. This work is included as a case study in Chapter 9: Management of fisheries.

Scientific research suggests that when recreational fishers target larger fish, they may be having a disproportionately negative effect on the fish population. This is due to the removal of the larger and older fish, which can produce a far greater number of eggs than smaller fish, and which have a higher reproductive success rate. This ultimately decreases the



Recreational fishing has always been an important part of the national culture, and many New Zealanders are introduced to the sport at a young age, as shown here (Daniel Sharp)



Recreational fishing, such as shown here at Bethells Beach, Auckland, can improve well-being and quality of life but also impacts on marine wildlife (Raewyn Peart)

ability of the species to reproduce, resulting in lower productivity over the longer term.³² Recreational fishing can disturb marine wildlife, increase ocean noise and result in ship strike on marine mammals. Many of the environmental impacts relating to boating discussed above are also relevant to the recreational fishing sector, including biosecurity, anchor damage, rubbish, sewage and greenhouse gas emissions. In addition, conflicts with other users of the marine environment are becoming increasingly contentious.

Interacting with marine mammals and seabirds

Because of its isolation and huge latitudinal variation (extending from the subtropics to the subantarctic) the extended islands of New Zealand have been important breeding locations for seabirds (New Zealand has been termed the 'sea-bird capital of the world') and for migratory marine mammals, such as the southern right whale. New Zealand has a high level of endemism in seabirds (e.g. the yellow-eyed penguin and Fiordland crested penguin), coastal birds (e.g. northern dotterel and Auckland Islands cormorant) and in marine mammals (e.g. the Hector's and Maui's dolphins and the New Zealand sea lion). Early human settlement of New Zealand made extensive use of these animals and significant industries were based on whales, fur seals and sea lions and a variety of seabirds (e.g. shearwaters or 'muttonbirds'). In many cases, these animals were hunted until their populations became mere remnants of their pre-human numbers.



Marine wildlife-based tourism, such as viewing orca shown here, is now the dominant form of tourism in the country (Mark Orams)

More recent protection of these creatures has created, in some cases, the opportunity to develop a new wildlife industry based on the idea of visiting and observing them as a tourist attraction.³³ Such activities have become extremely popular in New Zealand over the past three decades and marine wildlife-based tourism is now the dominant form of marine tourism in the country.

In New Zealand, this is a wide-ranging sector, which includes dolphin watching and swimming (snorkelling with dolphins), whale watching, remote underwater sea life watching, seabird viewing and, increasingly, shark diving encounters. There are a number of permits issued for companies to provide marine mammal and seabird interactions from land, boats and air-based platforms around the country. There are dolphin-based tours nationwide, and whale-based tours in the Hauraki Gulf (Bryde's whales) and Kaikōura (sperm whales). It is worthwhile noting that New Zealand no longer permits the holding of dolphins in captivity, however some pinnipeds (seals and sea lions) remain in zoos (such as Auckland Zoo).

The NIWA risk assessment of activities in the New Zealand EEZ prepared for the Ministry for the Environment in 2012 identified that tourists were seeking close surface and underwater encounters with mega-fauna or densely shoaling smaller species. The assessment also noted the trend for charter boats to carry camera systems or remotely operated vehicles capable of encountering and viewing spectacular underwater scenery in the deep ocean. These trends are likely to increase over the next 20–30 years.³⁴

Swimming with dolphins, shark dive encounters and similar activities all have the capacity to disturb surface and shallow subsurface species. The NIWA risk assessment noted that the area of habitat likely to be affected at any one time is very small and the risk to ecosystem functioning is probably negligible. However, the risks to individual protected species and key species may be higher and affected individuals or populations may take months to recover.³⁵

When the first marine mammal tourism permits were granted in the Bay of Islands, no one knew how many dolphins were actually present in the Bay, so it was not possible to make informed decisions about the level of pressure on them. The only information available was from sighting data provided by the dolphin-watching operators themselves.³⁶ More recently, research has provided clear evidence of a number of impacts from recreation and tourism on marine mammals, including:³⁷

- Disturbance to behaviour – there is a large amount of research demonstrating how tourist activity can disturb the behaviour of marine mammals. For example, the presence of tourist boats around dolphins in the Bay of Islands, the Hauraki Gulf, Mercury Bay, Bay of Plenty and Kaikōura, has been shown to reduce the time the dolphins spend foraging and resting. Instead, more time was spent milling and socialising. In addition, it took longer for the dolphins to return to a resting or foraging state, having been interrupted by tour boats, than if they were interrupted by other vessels.
- Negative impacts on successful rearing of young – there is some evidence to indicate that animals which are unable to forage and rest in the normal way may experience a reduced energy budget, which in turn makes it less likely that they will be able to breed successfully and rear healthy calves. Higher calf mortality can result. High calf mortality rates have been documented in bottlenose dolphins in the Bay of Islands and in Fiordland.³⁸
- Avoidance of habitat – the presence of tourist boats may cause some species to avoid the areas of their natural habitat where they are pursued. In general, dolphins will avoid boats either by diving vertically or by avoiding their path horizontally. A recent population study on bottlenose dolphins in the Bay of Islands has indicated that fewer dolphins are now present, possibly because they have begun to avoid the area which is frequented by large numbers of tour and recreational boats. In addition, fewer bottlenose dolphins are now present in Milford Sound during the busy summer tourism season.
- Varying degrees of impacts – some marine mammal species and populations are more susceptible to disruption by tourism than others. For example, the large pod sizes and night-time feeding behaviours of the dusky dolphin mean that the impacts of tourism are less than those on the smaller groups of bottlenose dolphins which forage during the day.

A lack of a robust scientific understanding about the size and range of marine mammal populations has contributed to the difficulty in managing the cumulative impacts of different tourism operations. For example, permits were granted for bottlenose dolphin interactions at multiple locations along the north-east coast of the North Island before it was fully understood that they were all targeting the same population of dolphins that ranged along the area. While there are similarities in the impact of marine mammal tourism across different species and locations, each scenario entails different challenges. In addition, in some locations there can be cumulative effects, when tourism impacts on marine mammals are combined with those created by other activities. For example, in Doubtful Sound, the decline in the bottlenose dolphin population is thought to be a result of a combination of factors including increasing boat tourism, historical commercial fishing activity and increased discharges of freshwater into the marine area from the Manapōuri Power Station outfall.³⁹

Ships moving through the water can hit marine mammals and fish and injure or kill them. Ship strike happens because dolphins and especially whales are moving much slower than vessels and spend considerable time on the surface, which can put them directly in the path of boats. The greatest number of whale deaths from ship strike is caused by large ships travelling at more than 10 knots. This is discussed further in Chapter 15: Vessels. The propellers of recreational boats can cause considerable damage when they strike dolphins, as evidenced by a number of animals which sport distinctive propeller scars on their fins and bodies.



Marine mammal tourism, as shown here in Akaroa Harbour, Banks Peninsula, can disturb the natural behaviours of the targeted animals (Raewyn Peart)

Special events and sporting contests

Many coastal locations are deliberately establishing special events to encourage visitation and the commensurate economic and social benefits to their venues. Seafood festivals, fishing contests, surfing competitions, sand-sculpting contests, beach volleyball championships, sailing regattas, multi-sport races, music concerts and wildlife festivals are examples of a multitude of deliberately staged events which encourage tourism from participators and spectators alike.⁴⁰

The use of the coast for special events has intensified in recent years. From the Auckland Anniversary Regatta to the seafood festival in Kaikōura and fishing events in Bay of Islands, the country's marine resources are being used as a basis for a wide range of events. Many of these are competitive sporting events, such as the State Ocean Swim Series, or for festivals and hospitably events, such as the Whitianga Scallop Festival which is one of New Zealand's most popular seafood events. Smaller communities often host local music festivals using the beach as a 'party' base. The Ōhope Beach concert at Mahy Reserve is just one example of where locals come together to enjoy local musicians and a family picnic at the beach.

One particular growth area is the spread of marine adventure-type challenges that are extreme in nature. Rowing or paddling across significant bodies of water (such as Cook Strait and Foveaux Strait), traversing remote coastal areas on foot, surfing off-shore ocean reef waves, solo boating voyages and deep diving (including breath-hold diving) are examples of the growing range of extreme activities which attempt to provide adventure, challenge and deliberately place participants at a higher perceived risk of injury or death. Such activities have been growing in popularity and profile and this trend is likely to continue.⁴¹

Major events and contents held on the coast play an important economic and social role in the community. The negative impacts that come with them, such as rubbish, trampling and disturbance of wildlife, can be well controlled by the managers of the event.

General coastal and beach activities

There are many other activities that do not fall into the above categories. These include the increasing popularity of walking along the coast which has seen major infrastructure investment such as the development of coastal walkways (e.g. paved pathways, boardwalks and trails) in places such as Napier, Ōrewa, Ōmaha, and Auckland's Tāmaki Drive.

Sunbathing and relaxing at the beach is hugely popular and there are many destinations around the country, both busy and peaceful, where people head to enjoy this activity. Sandy beaches close to areas of population are the most intensively used coastal locations. A good example is the beach at Mount Maunganui, which was voted as New Zealand's 2014 top beach by Tripadvisor users, and which is a very popular place for sunbathing, people-watching, playing games and picnicking. Additional activities which take place on the intertidal zone include walking dogs, land yachting, beach volleyball, beach cricket, touch and other ball sports, horse riding, motorbike and offroad vehicle use, exploring rock pools, jogging/running, cycling and other forms of exercise, kite flying, sand castle and sand sculpture creation, photography, painting, tai-chi, yoga and many others.

Many of the impacts of more general coastal and beach activities are fairly benign on an individual basis, such as trampling and rubbish generation. But when these are looked at from a cumulative perspective they can have a serious impact on the coastal and marine environment. Trampling of dune vegetation can contribute to coastal erosion through the destruction of the native sand-binding plants. Cars and motorbikes being driven on beaches and sand dunes can damage important native vegetation and fauna (such as spiders, shorebirds and skinks).

Holiday homes

New Zealand has a long tradition of coastal holiday dwellings. Traditionally these were rudimentary structures with little infrastructure and few services (e.g. no electricity, telephone, sewage and waste water services, or 'town' water supply). The stereo-typical Kiwi bach (or crib) was a small basic structure built close to a beach, with an outside 'long-drop' toilet and tank water collected from the roof. Because New Zealand's summer coincides with the public holiday period of Christmas and New Year, and the long school holiday break, lengthy periods of living at the bach (often shared by extended family groups and friends) over this holiday period became synonymous with the Kiwi summer lifestyle.

Over the past three decades, the growing population, urbanisation and improved transportation (both roads and vehicles) has seen coastal properties within two to three hours' drive of a city become highly sought after. This demand has led to an escalation of coastal property prices, a commensurate investment in coastal dwellings, structures and infrastructure, and a major transformation of the New Zealand coast. Traditional baches within two hours' drive of a city are now rare. Larger, more luxurious holiday

homes are now the norm and a number of coastal communities have become upmarket coastal resort-type developments, some complete with canal and marina front properties. Examples include Pauanui, Matarangi, Whangamatā, Ōmaha, Marsden Cove and Tutukākā.

What is clearly evident from this transition, and the financial inflation that has resulted, is how highly valued the coastal lifestyle is for New Zealanders. In a clear and almost uniform pattern, the monetary value of property increases the closer it is to the coast and waterfront, and properties with sea views attract premium prices. Proximity to the sea and all the opportunities it offers, both from a passive perspective (seeing, smelling and hearing the sea) and an active point of view (such as walking, swimming, fishing and boating), are hugely attractive to New Zealanders and highly valued by them.

The increase in the number of holiday homes around New Zealand's coast has resulted in a loss of natural character of the coastal environment.

This is particularly the case where infrastructure associated with holiday homes intrudes onto the coastal edge through such structures as seawalls, jetties and ramps (the impacts of such structures are discussed in Chapter 14: Major marine developments).



Holiday homes, such as those shown here at Tutukākā, have transformed the New Zealand coast (Raewyn Peart)



Holiday homes, such as these shown at Tongaporutu, can bring with them infrastructure that impacts on the natural character of the coastal edge (Raewyn Peart)

Cruise ships

New Zealand has seen a phenomenal growth in the cruise ship industry of over 250 per cent during the last five years, making cruise arrivals the third largest inbound market for New Zealand tourism.⁴² The New Zealand cruise season runs from September through to March. During the 2012–2013 cruise season, New Zealand received over 210,000 cruise passenger arrivals and more than 80,000 crew.⁴³ The cruise ship industry is an economic contributor to a number of New Zealand ports, particularly Auckland, Tauranga, Wellington, Akaroa, and Port Chalmers (Dunedin). Cruise ships also visit a number of remote New Zealand coastal locations such as White Island, Fiordland, Stewart Island (Rakiura) and New Zealand's subantarctic Islands.

Cruise passenger behaviour has been steadily shifting from organised tours to free and independent travellers. This change of behaviour means a shift in the demand for activities from these cruise passengers. Whilst most of these tend to be shore-based, increasingly there is demand for marine-based activities at stopover locations.

The NIWA risk assessment noted the increasing access of tourist ships into the subantarctic islands and the Ross Sea as remote cruising destinations. All vessels travelling to New Zealand (including pleasure craft) need to meet a number of requirements before and on arrival to ensure New Zealand's environment, economy and people are protected from imported pests and diseases.⁴⁴ These issues are discussed further in Chapter 15: Vessels.

Food scraps are either kept in bins on board cruise ships for discharge as quarantine waste ashore, under Ministry for Primary Industries supervision, or are burnt in incinerators whilst out at sea. All food-contaminated waste or plastics are discharged under Ministry supervision as quarantine waste for treatment. Where possible, cardboard, glass, waste cooking oil and scrap metal are sent

ashore under Ministry supervision for recycling. However not all New Zealand ports can handle waste recycling, in which case waste is either stored on ships until they arrive in Auckland, or discharged at Australian ports.

Waste fuel oil and sludge from cruise ships is retained on board in ships' tanks and only discharged at approved New Zealand ports through approved waste oil contractors. The contractors receive the waste oil from the ship, separate the oil and water at their plant, and reuse the oil for local industry consumption. Most cruise ships have sophisticated grey and black water treatment plants, or holding tanks with filters, which remove all or nearly all of the toxic waste and sewage. The treated liquid is pumped out at sea more than 12 miles off land. The residue is compacted and either incinerated at sea or disposed of ashore.

Ships accidentally striking marine mammals can be a concern for the cruise ship industry. This is discussed above in the section on marine mammal interactions.

Decision-making framework

Resource Management Act

The control of recreation and tourism activities in the coastal marine area falls to regional councils under the RMA.⁴⁵ The relevant sections are discussed in Chapter 2: Marine management.

The NZCPS includes objectives and policies which regional coastal plans must 'give effect to'. Of particular relevance to recreation is Objective 4 which seeks to maintain and enhance the public open space qualities and recreation opportunities of the coastal environment. Many of the policies in the NZCPS relate to the provision of open and free access to the coast (particularly Policies 18 and 19). Specific recognition is given to surf breaks in Policy 16 which requires the protection of listed surf breaks of national significance by (a) ensuring that activities in the coastal environment do not adversely affect the surf breaks and (b) avoiding adverse effects of other activities on access to, and use and enjoyment of, the surf breaks.

NZCPS, Policy 16: Surf breaks of national significance

Protect the surf breaks of national significance for surfing listed in Schedule 1, by:

- a. ensuring that activities in the coastal environment do not adversely affect the surf breaks; and
- b. avoiding adverse effects of other activities on access to, and use and enjoyment of the surf breaks.

Regional coastal plans generally include few controls on recreation and tourism activities in the coastal marine area where they do not involve occupation of space or non-temporary disturbance.

The Resource Management (Marine Pollution) Regulations

These regulations set out rules relating to discharges from ships (including any boat or craft used in navigation with or without any means of propulsion) in the coastal marine area. Treated sewage can be discharged into the coastal marine area provided it is the prescribed distance from a marine farm or mātaihai reserve. Untreated sewage may only be discharged from a ship if it is more than 500 metres from mean high water springs and in water depths of more than 5 metres, as well as the prescribed distance from a marine farm, marine reserve or mātaihai reserve.⁴⁶



Regulations apply to discharges from recreational vessels, such as those shown here at Motuarohia Island, Bay of Islands, including the discharge of sewage, plastic, oily water, food waste and other garbage (Raewyn Peart)

Discharge of plastics is prohibited, but other ship garbage (including food waste, paper, rags, glass, metal, bottles, and crockery) can be discharged if ground to a particle size of 25 millimetres or less and if the discharge occurs at least 3 nautical miles offshore. Discharge of oil or oily water is permitted from a ship only if the oil content does not exceed 15 parts per million.

Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

In general there are only a few recreational or tourism activities that occur in the EEZ because of its distance from the shoreline. Those that do take place there include cruise ships and offshore recreational boating. The NIWA risk assessment helped inform the development of the EEZ Act. This assessment concluded that there were no activities associated with offshore recreation and tourism that were considered to represent an extreme environmental risk and, as a consequence, should be prohibited.⁴⁷ However, the assessment did make a number of recommendations as to the status of certain tourism activities in the EEZ marine environment.

The following activities were considered to be of high environmental risk and were recommended to be categorised as discretionary:

- Tourist vessel activity impacting marine mammals (regulated under the Maritime Transport Act and the Marine Mammals Protection Act)
- The use of surface floodlights at night which may cause bird strikes (regulated under the Maritime Transport Act and the Wildlife Act)

In addition, surface activity and noise and subsurface activity were considered to be of moderate environmental risk and were recommended to be categorised as discretionary. The following activities were considered to be of low environmental risk and were recommended to be categorised as permitted:

- Operations near the seafloor likely to result in collisions with benthic fauna
- Subsurface lights and noise generated by camera systems, remotely operated vehicles and submersibles

The EEZ Act prohibits certain activities in the EEZ including structures, submarine pipelines, submarine cables, removal of non-living natural material, disturbance of the seabed or subsoil, and deposition of anything on the seabed, unless the activity is permitted by regulations or authorised by a specific marine consent.⁴⁸ The EEZ Act does not contain any explicitly permitted recreation/tourism activities and, as a consequence, all activities are considered discretionary and therefore require a specific marine consent.

Marine Mammals Protection Act

The Marine Mammals Protection Act protects all species of marine mammals within New Zealand waters. It is an offence to 'take' marine mammals, whether alive or dead, from their natural habitat or 'any other place' without a permit from the Department of Conservation. The term 'take' is interpreted to include any actions likely to harass, harm, injure, attract or to disturb a marine mammal.



Regulations under the Marine Mammals Protection Act require tourism operators, such as Dolphin Explorer shown here on the Hauraki Gulf, to obtain a permit from the Director-General of Conservation before offering marine mammal tours (Raewyn Peart)

Marine Mammals Protection Regulations

As a consequence of the growing interest in interacting with marine mammals in the wild in New Zealand, the Marine Mammals Protection Regulations 1992 were developed to regulate human contact with marine mammals by commercial operators, members of the public and others. Commercial marine mammal operations are prohibited unless a permit is issued by the Director-General of Conservation. Applications for permits are publicly notified and members of the public are able to make submissions.

The Regulations state that the Director-General of Conservation may not grant a permit unless satisfied there is substantial compliance with a number of criteria which include:⁴⁹

- The commercial operation should not have any significant adverse effect on the behavioural patterns of the marine mammals to which the application refers, having regard to, among other things, the number and effect of existing commercial operations
- That it should be in the interests of the conservation, management, or protection of the marine mammals that a permit be issued
- That the proposed operator, and such of the operator's staff who may come into contact with marine mammals, should have sufficient experience with marine mammals
- That the proposed operator, and such of the operator's staff who may come into contact with marine mammals, should have sufficient knowledge of the local area and of sea and weather conditions
- That the proposed operator, and such of the operator's staff who may come into contact with marine mammals, should not have convictions for offences involving the mistreatment of animals
- That the commercial operation should have sufficient educational value to participants or to the public

The Director-General has the final word over revoking or suspending any permits. In addition, the Director-General may place a moratorium on the issue of new permits when he or she has reasonable grounds to believe that it is necessary for the protection, conservation or management of any marine mammals.



Both recreational and commercial vessels coming into contact with marine mammals, such as the one shown here in the Bay of Islands, must comply with regulations including not causing the separation of an individual dolphin from its group (Raewyn Peart)

Moratoria on the issue of marine mammal permits

Moratoria have been placed on marine mammal tourism permits for sperm whale, dusky dolphin and New Zealand fur seal tourism at Kaikōura, New Zealand fur seal tourism in Abel Tasman, southern right whale tourism in the Auckland Islands and bottlenose dolphin tourism in the Bay of Islands. However, the Department of Conservation only acts to implement moratoria when there is hard evidence of a problem, something which has been difficult for scientists to provide.⁵⁰ As a result, it can take years for a moratorium to be put in place after potential problems are identified. For example, a moratorium on new permits was first proposed in the Bay of Islands in 1997, but was not implemented until 12 years later in 2009.⁵¹

There are also a number of regulations that commercial operators and any people coming into contact with a marine mammal are required to comply with. These include abandoning the interaction if the marine mammal is showing signs of becoming disturbed or alarmed and not causing the separation of individuals of marine mammals from their group. It is also illegal to feed marine mammals.

The Regulations also place restrictions on the speed that vessels may go when in the vicinity of any marine mammal. Extra requirements apply to whales, including that no person in the water shall be less than 100 metres from a whale and no vessel shall approach within 50 metres of a whale, unless authorised by the Director-General. There are also additional requirements that apply to dolphins or seals, including that no vessel shall proceed through a pod of dolphins and there should be a limit of the number of vessels able to make contact with a group of dolphins or seals.

Marine Mammals Protection Regulations 1992, Part 3 – Behaviour around marine mammals

18 Conditions governing commercial operations and behaviour of all persons around any marine mammal

Every commercial operation, and every person coming into contact with any class of marine mammal, shall comply with the following conditions:

- (a) persons shall use their best endeavours to operate vessels, vehicles, and aircraft so as not to disrupt the normal movement or behaviour of any marine mammal:
- (b) contact with any marine mammal shall be abandoned at any stage if it becomes or shows signs of becoming disturbed or alarmed:
- (c) no person shall cause any marine mammal to be separated from a group of marine mammals or cause any members of such a group to be scattered:
- (d) no rubbish or food shall be thrown near or around any marine mammal:
- (e) no sudden or repeated change in the speed or direction of any vessel or aircraft shall be made except in the case of an emergency:
- (f) where a vessel stops to enable the passengers to watch any marine mammal, the engines shall be either placed in neutral or be switched off within a minute of the vessel stopping:
- (g) no aircraft engaged in a commercial aircraft operation shall be flown below 150 metres (500 feet) above sea level, unless taking off or landing:
- (h) when operating at an altitude of less than 600 metres (2,000 feet) above sea level, no aircraft shall be closer than 150 metres (500 feet) horizontally from a point directly above any marine mammal or such lesser or greater distance as may be approved by the Director-General, by notice in the *Gazette*, from time to time based on the best available scientific evidence:
- (i) no person shall disturb or harass any marine mammal:
- (j) vehicles must remain above the mean high water spring tide mark and shall not approach within 50 metres of a marine mammal unless in an official carpark or on a public or private slipway or on a public road:
- (k) no person, vehicle, or vessel shall cut off the path of a marine mammal or prevent a marine mammal from leaving the vicinity of any person, vehicle, or vessel:
- (l) subject to paragraph (m), the master of any vessel less than 300 metres from any marine mammal shall use his or her best endeavours to move the vessel at a constant slow speed no faster than the slowest marine mammal in the vicinity, or at idle or 'no wake' speed:
- (m) vessels departing from the vicinity of any marine mammal shall proceed slowly at idle or 'no wake' speed until the vessel is at least 300 metres from the nearest marine mammal, except that, in the case of dolphins, vessels may exceed idle or 'no wake' speed in order to outdistance the dolphins but must increase speed gradually, and shall not exceed 10 knots within 300 metres of any dolphin:
- (n) pilots of aircraft engaged in a commercial aircraft operation shall use their best endeavours to operate the aircraft in such a manner that, without compromising safety, the aircraft's shadow is not imposed directly on any marine mammal.

19 Special conditions applying to whales

In addition to complying with the provisions set out in regulation 18, every commercial operation and every person coming into contact with whales shall also comply with the following conditions:

- (a) no person in the water shall be less than 100 metres from a whale, unless authorised by the Director-General:
- (b) no vessel shall approach within 50 metres of a whale, unless authorised by the Director-General:
- (c) if a whale approaches a vessel, the master of the vessel shall, wherever practicable,—
 - (i) manoeuvre the vessel so as to keep out of the path of the whale; and
 - (ii) maintain a minimum distance of 50 metres from the whale:
- (d) no vessel or aircraft shall approach within 300 metres (1,000 feet) of any whale for the purpose of enabling passengers to watch the whale, if the number of vessels or aircraft, or both, already positioned to enable passengers to watch that whale is 3 or more:
- (e) where 2 or more vessels or aircraft approach an unaccompanied whale, the masters concerned shall co-ordinate their approach and manoeuvres, and the pilots concerned shall co-ordinate their approach and manoeuvres:
- (f) no person or vessel shall approach within 200 metres of any female baleen or sperm whale that is accompanied by a calf or calves:
- (g) a vessel shall approach a whale from a direction that is parallel to the whale and slightly to the rear of the whale:
- (h) no person shall make any loud or disturbing noise near whales:
- (i) where a sperm whale abruptly changes its orientation or starts to make short dives of between 1 and 5 minutes duration without showing its tail flukes, all persons, vessels, and aircraft shall forthwith abandon contact with the whale.

20 Special conditions applying to dolphins or seals

In addition to complying with the conditions set out in regulation 18, any commercial operation and any person coming into contact with dolphins or seals shall also comply with the following conditions:

- (a) no vessel shall proceed through a pod of dolphins:
- (b) persons may swim with dolphins and seals but not with juvenile dolphins or a pod of dolphins that includes juvenile dolphins:
- (c) commercial operators may use an airhorn to call swimmers back to the boat or to the shore:
- (d) except as provided in paragraph (c), no person shall make any loud or disturbing noise near dolphins or seals:
- (e) no vessel or aircraft shall approach within 300 metres (1,000 feet) of any pod of dolphins or herd of seals for the purpose of enabling passengers to watch the dolphins or seals, if the number of vessels or aircraft, or both, already positioned to enable passengers to watch that pod or herd is 3 or more:
- (f) where 2 or more vessels or aircraft approach an unaccompanied dolphin or seal, the masters concerned shall co-ordinate their approach and manoeuvres, and the pilots concerned shall co-ordinate their approach and manoeuvres:
- (g) a vessel shall approach a dolphin from a direction that is parallel to the dolphin and slightly to the rear of the dolphin.

Conservation Act

Any marine recreation or tourism business or organisation wanting to use public conservation land or marine reserves for their activities needs to get permission from the Department of Conservation in the form of a concession. This is a legal requirement under Part 3B of the Conservation Act. Some of the relevant activities that require a concession include guided activities (such as snorkelling, kayaking and canoeing), water transport services, commercial education and instruction and running a sports event.



Marine recreation and tourism businesses operating within public conservation land and marine reserves, such as kayaking in the Ōkārito Lagoon shown here, require a concession from the Director-General of Conservation (Mark Orams)

Local and regional bylaws

Local and regional councils may enact bylaws to manage behaviour and activities in the public interest. Regional councils have the jurisdiction for managing the coastal marine area within their regional boundaries and may designate specific zones or areas, restrict certain activities or require specific permits to conduct activities. Regional councils thus have the ability to manage and control a range of marine recreation and tourism activities. They do this via mechanisms such as:

- Designating ski lanes
- Providing boat mooring areas
- Closing areas to boating
- Restricting boat speed
- Requiring safety equipment (such as lifejackets)
- Temporary closures of areas to allow for special events
- Restricting dogs and other animals in certain areas or at certain times
- Installing signs and navigation aids
- Installing and maintaining boat launching ramps, floating docks, wharves and other structures (such as swimming platforms)
- Replenishing sand on beaches and the construction of coastal pathways
- Controlling cars and motorbikes driving on sand dunes and beaches

Elements of best practice

The Department of Conservation has identified a number of recommended techniques and practices for managing visitor impacts on marine reserves including:⁵²

- Modifying visitor behaviour through education
- Regulating types of activities (such as commercial operations and boating activities)
- Encouraging minimal impact uses (such as boardwalks, interpretation trails and moorings to avoid anchoring)
- Reducing the use of specific sites
- Restricting access to sensitive sites
- Promoting compliance with the marine reserve regulations
- Identifying areas where damage can be repaired and undertaking restoration

More information can be found at the Department of Conservation's website.⁵³ Figure 16.3 provides some considerations for best practice to help promote positive environmental outcomes for each activity discussed in this chapter.

Activity	Best practice considerations
Swimming	• Identify areas for public swimming which are away from important habitats, especially shorebird feeding and nesting areas • Provide the minimal level of infrastructure (such as toilets, rubbish bins and car parking) to ensure impacts are reduced • Where access is over sensitive environments, such as dunes or wetlands, consider installing boardwalks or pathways to reduce impacts
Surfing, boardsailing, kiteboarding, stand up paddle boarding and surf skiing	• Provide the minimal level of land-based infrastructure (such as toilets, rubbish bins and car parking) to ensure impacts are reduced • Promote education as a way to modify user behaviour and attitudes towards the marine environment, such as that provided by local clubs and Surf Lifesaving New Zealand • Where access is over sensitive environments, such as dunes and wetlands, consider installing boardwalks or pathways to reduce impacts • Require certification for tourist operators to show environmental aspects have been addressed, for example Qualmark's Enviro-Gold is its top level in the Green rating for performers in Responsible Tourism⁵⁴
Diving and snorkelling	• Reduce the impacts on vulnerable species by rotating visits around a number of sites • Restrict the number of visitors to sensitive environments and habitats, for example at Golden Bay's Waikoropupu (Pupu Springs) diving is not permitted before midday and only four divers at a time are allowed to enter the water • Promote education as a way to modify diver and snorkelling behaviour, such as that provided by local clubs and diving organisations • Where access is over sensitive environments, such as dunes and wetlands, consider installing boardwalks or pathways to reduce impacts • Require certification for tourist operators to show environmental aspects have been given due consideration, for example Qualmark's Enviro-Gold is its top level in the Green rating for performers in Responsible Tourism⁵⁵ • In sensitive areas, control landing permits and only allow registered dive charter operators to land • Support diver awareness programmes, such as Project AWARE which is a non-profit organisation working in over 180 countries engaging the diving community to undertake voluntary work to care for the marine environment
Boating	• Install holding tanks in recreational vessels with seawater flushing toilets • Avoiding washing or scrubbing antifouling painted hulls on the foreshore or intertidal area • Provide sealed boat haul-out areas where discharges can be contained • Use more environmentally friendly compounds in antifouling paint • Minimise the production of onboard waste and use land-based disposal • Limit unnecessary use of any floodlights at night and ensure that those that are required are directed approximately vertically onto work surfaces • Reduce boat speed in areas where marine mammals are likely to be present and keep well clear of the animals • Reduce the speed of vessels in areas susceptible to ecological impacts from wake • In remote sensitive areas, control landing permits or enforce no public landing, such as at te Hauturu-o-Toi/Little Barrier Island in the Hauraki Gulf⁵⁶ • Identify public anchoring areas and install public mooring buoys in environments sensitive to anchor damage • Adopt the Clean Boating Programme recommendations for individual boaties, including guidance about simple household alternatives to harmful products to use while boating and ways to reduce the transfer of marine pests⁵⁷

Activity	Best practice considerations
Fishing	• Follow the Southern Seabird Solutions recommendations for mitigating impacts of recreational fishing on seabirds (discussed below) • Haul nets in as quickly as possible • Limit unnecessary use of any floodlights at night and ensure that those that are required are directed approximately vertically onto work surfaces • Reduce boat speed in areas where marine mammals are likely to be present and keep well clear of the animals • Reduce the speed of boats in areas susceptible to ecological impacts from wake • Identify public anchoring areas and install public mooring buoys in environments sensitive to anchor damage • Adopt the Clean Boating Programme recommendations while fishing, including the ways to reduce the spread of marine pests
Marine mammal and seabird interactions	• Ensure compliance with regulations governing interactions with marine mammals, so that the animals are not harassed • Educate the public about good behaviour around marine mammals • Restrict the number of marine mammal tourism trips to ensure that no significant adverse impacts on the populations occur • Provide rest times and exclusion areas for marine mammals to enable them to have a break and refuge from human interaction • Reduce boat speed in areas where marine mammals are likely to be present, keep a good look-out, and keep propellers well away from the animals • Exclude vehicles and members of the public from areas of the beach used by nesting seabirds
Cruise ships	• Exchange ballast water in mid-ocean or operate a type-approved ballast water management system • Reduce ships speed in areas where whales are likely to be present to less than 10 knots • Reduce the speed of vessels in areas susceptible to ecological impacts from wake • Reduce the impacts on vulnerable species by rotating any visiting cruise ships around a number of sites • Restrict the number of visitors able to access sensitive environments and habitats • Minimise the use of night illumination to reduce impacts on seabirds including limiting unnecessary use of any floodlights at night and ensuring that those that are required are directed approximately vertically onto work surfaces
Special events and sporting contests	• Identify areas for special events and sporting contests that are away from important habitats, especially shorebird feeding and nesting areas • Provide the minimal level of infrastructure (such as toilets and rubbish bins) appropriate for the level of activity to ensure the impacts from the event are reduced • Provide educational resources on registration that support good environmental outcomes
General coastal and beach activities	• Identify areas for public swimming which are away from important habitats, especially shorebird feeding and nesting areas • Provide the minimal level of infrastructure (such as toilets and rubbish bins) appropriate for the level of activities • Provide public education around caring for the environment
Holiday homes	• Concentrate development in areas which have already been impacted and avoid development in places with high natural character or landscape values • Where coastal areas are to be developed, design the development in a way that maintains natural character and landscape values, such as through setting buildings back from the coastal edge and away from sensitive habitats such as dunes and wetlands • Incorporate restoration of the coastal environment into the development

Figure 16.3 Best practice considerations for marine recreation and tourism

Reducing recreational catch of seabirds

In association with a national recreational fishing expert and TV programme, Southern Seabird Solutions has developed a DVD identifying tips that recreational fishers can use to reduce the risk of catching seabirds and how to release them safely if they are caught, including:⁵⁸

- Making a seabird release kit and keeping it on board
- Keeping bait and offal in covered containers on board
- Sinking bait beneath the surface as fast as possible
- Covering baited hooks when not in use
- Weighing down a burly bag with sinkers
- Using soft baits as seabirds tend to spit them out without injury

Clean Boating Programme

The Clean Boating Programme has developed a series of recommendations for environmentally-sound recreational boating practices in New Zealand. These include:

- Whenever possible, refuelling at an approved refuelling station, where spill-kits are available, and not transferring fuel to a boat in containers
- Fitting a holding tank for onboard toilets
- Having a waste container on the boat and carrying out what is carried in
- Using shoreside facilities whenever possible when cleaning boats
- Ensuring the engine is kept clean and well-maintained so that it does not leak oil or fuel
- Disposing of unused bait well offshore



The Clean Boating Programme has developed a range of recommendations which apply to recreational boating such as shown here in Tauranga Harbour (Daniel Sharp)

Dolphin Encounter Kaikōura

In the 1980s, Kaikōura was an economically depressed town with high unemployment levels. At this time a whale watch business called Nature Watch Charters was set up which was soon followed by a second enterprise known as Kaikōura Tours (subsequently renamed Whale Watch Kaikōura).⁵⁹ The interest in whale watching at Kaikōura rapidly increased. At the same time, Dolphin Mary Charters (subsequently named Dolphin Encounter) persuaded the Department of Conservation that viewing and swimming with dusky dolphins was a viable business, and it issued them a permit. This enabled many people to interact with dolphins for the first time. Previously, dolphins had only been widely accessible in captivity. With increasing tourist interest in the wild dolphins, the Department placed a moratorium on the issue of further permits until research into the impacts of tourism on the dolphins could be undertaken.

Research undertaken during the late 1990s found that the dolphins were accompanied by boats for much of the day (over 70 per cent of daylight hours), and that the boats changed dolphin behaviour. However, it was not possible to determine whether the changes in behaviour observed had a significant impact on the welfare of the dolphin population overall. The researchers suggested that a cautionary approach should be applied to any increase in dolphin tourism activity.⁶⁰

In 1999, as result of the research, the Department put in place a moratorium on the issue of any more dolphin swimming permits for a 10 year period. In addition, Dolphin Encounter and the other permit holders agreed to give the dolphins a two-hour 'time off' period during the middle of the day over the summer months. In anticipation of the moratorium expiring in 2009, the Department commissioned a three-year investigation into the effects of tourism activity on the dolphins.

Following this further research, the moratorium on the issue of additional permits was extended for another five years. The two-hour rest period for the dolphins during the summer months was made mandatory. A limit was placed on the number of swims which could be attempted on each trip and a prohibition was placed on dropping swimmers in front of approaching dolphin pods.⁶¹

The dolphin and whale encounters have transformed Kaikōura into a vibrant community bristling with new tourist accommodation, restaurants and souvenir shops. Marine mammals now underpin the town's economy.



Dolphin Encounter, operating in Kaikōura, initiated dolphin tourism in New Zealand (Raewyn Peart)

Dive! Tutukaka

Many international and domestic divers and tourists are keen to visit the Poor Knights Islands. These 11 million year old islands, lying 22 kilometres offshore, provide visitors with some breathtaking scenery both above and below the water, and the opportunity to see a huge variety of marine life. A marine reserve was established around the Poor Knight Islands in 1998 and the area is currently being assessed as a World Heritage site (part of the Whakarua Moutere Islands application extending along the north-eastern coastline of the North Island which is registered with UNESCO by the Department of Conservation).⁶²

Dive! Tutukaka, founded in 1999, operates out of the Tutukākā Marina in the Bay of Islands and focuses its visitor experience around the Poor Knights Islands. It is New Zealand's largest dive charter company, taking over 12,000 people to the Poor Knights Islands every year on five boats.⁶³ The enterprise has won a number of national and international awards for environmental and business practices, including New Zealand Tourism's Supreme Tourism Award in 2006, Rankers Travel awards and PADI Green Star awards.⁶⁴

The voluntary routing of ships around the Poor Knights Islands has been an important part of protecting this very special marine environment. After two incidents when oily bilge water was released into the sea between the Poor Knights and the Tutukākā coast, threatening the area's world-renowned wildlife and marine ecology, the voluntary routing was set in place. Poor Knights Islands is a mandatory area to be avoided by all vessels greater than 45 metres' length overall, with only a few exemptions to this rule.⁶⁵ This is discussed further in Chapter 15: Vessels.

The evolution of the dive company has gone hand in hand with advocating for marine protected areas. This company was a key player in advocating for the exclusion of recreational fishing from within the reserve. This was highly controversial, and it was the value for tourism that was advocated by Dive! Tutukaka that supported the case for the exclusion. Due to the company's frequent and numerous trips to the area, it is actively involved in ensuring that the Poor Knights are respected by others, and also reports violations to the shipping ban restrictions surrounding the island group.



Dive! Tutukaka is New Zealand's largest dive charter company and it takes over 12,000 people to the Poor Knights Islands every year (Daniel Sharp)

Environment Southland's Deed of Agreement for cruise ships

In 1997 a study into the environmental impacts of tourism on the inland waters of the Southland region identified cruise ships as presenting the greatest threat to the area. A key area of concern was the pollution resulting from the discharge of wastes into both water and air.⁶⁶ Environment Southland (Southland Regional Council) responded by introducing the Deed of Agreement which is a contract between the council and individual cruise ships. The Deed of Agreement for cruise ships wishing to enter the fiords was introduced through the Southland Regional Coastal Plan in 2001. Section 13 of the Plan deals with cruise ships operating in the internal waters of the Southland region, and cruise ships either require a resource consent or are required to sign the Deed of Agreement with Environment Southland.⁶⁷

The agreement places environmental obligations on all vessels visiting the fiords, including requirements for waste discharge.⁶⁸ Whilst within the Southland coastal management area, a 'zero discharge' policy applies to discharges to water covering both solid (food waste and packaging) and liquid wastes (sewage, grey water, bilge water and hazardous waste). Air discharges must be minimised, with a goal to reduce visible smoke emissions in the fiords. A suggested way of achieving this is to substitute marine gas oil for heavy fuel oils,⁶⁹ but ultimately it is up to the vessels to determine how they will meet the requirement. The taking on or exchanging of ballast water whilst in internal waters is strictly prohibited.⁷⁰

The Environment Southland Marine Fee was set up as part of the Agreement, where each ship entering the fiords is charged based on the vessel's gross tonnage. The fee enables the council to offset any costs arising from risks to the environment posed by the industry and to alleviate financial pressure on the council's coastal expenditure. Introduced in time for the 2001/2002 cruising season, the fee generated \$260,000 from 23 visits and the 2012/2013 season was estimated to generate \$1.9 million. By the end of the 2011/2012 season it was estimated that the cumulative total of fees taken was \$10 million.⁷¹ Part of the Marine Fee helps pay for a wide range of the council's coastal-based functions, including its marine biosecurity programme, supporting shoreline clean-up and coastal water quality science reporting and monitoring, all of which are impacted by the vessels entering Doubtful Sound.



Environment Southland has introduced a Deed of Agreement for cruise ships wishing to enter the fiords, including Doubtful Sound shown here (*Raewyn Peart*)

Further reading and references

- Beca Infrastructure Limited, 2012, *Auckland recreational boating study*, Prepared for Auckland Council, Auckland
- Birkeland C and P K Dayton, 2005, 'Uncovering secrets of our seamounts', *Trends in Ecology and Evolution*, 20(7), 356–358
- Booth K and M Mackay, 2007, *Tourism and recreation in New Zealand's natural environment: A bibliography and research synthesis*, Report prepared by Kay Booth and Associates for the Ministry of Tourism, Wellington, available at <http://www.srknowledge.org.nz/wp-content/uploads/2012/07/Booth-and-Mackay-2007.pdf>
- Cessford G, C Jones, M Orams and T Elson, 2012, *Developing a geodatabase for mapping recreational use and value indicator data of New Zealand's coastal/marine environment*, Report for the Department of Conservation, Wellington
- Constantine R, 1999, *Effects of tourism on marine mammals in New Zealand*, Science for Conservation: 106, Department of Conservation, Wellington
- McCrone A, 2001, *Visitor impacts on marine protected areas in New Zealand*, Science for Conservation 173, Department of Conservation, Wellington
- Ministry for Primary Industries, 2011, *The status of New Zealand's fisheries 2011*, Ministry for Primary Industries, Wellington
- Ministry for Primary Industries, 2013, *National plan of action – 2013 to reduce the incidental catch of seabirds in New Zealand fisheries*, Ministry for Primary Industries, Wellington
- Mulcahy K and R Peart, 2012, *Wonders of the sea: The protection of New Zealand's marine mammals*, Environmental Defence Society, Auckland
- NIWA, 2012, *Expert risk assessment of activities in the New Zealand Exclusive Economic Zone and extended continental shelf*, Prepared for the Ministry for the Environment, NIWA, Wellington
- Orams M B, 1999, *Marine tourism: Development, impacts and management*, Routledge, London
- Orams M B and M Lück, 2013, 'Marine systems and tourism', in A Holden and D Fennell (eds), *A handbook of tourism and the natural environment*, Routledge, London, 170–182
- Orams M B and M Lück, 2014, 'Coastal and marine tourism: Emerging issues, future trends and research priorities', in A L Lew, C M Hall and A M Williams (eds), *The Wiley-Blackwell Companion to Tourism*, Wiley-Blackwell, Chichester, 479–489
- Peart R, 2013, *Dolphins of Aotearoa: Living with New Zealand dolphins*, Craig Potton Publishing, Nelson
- Project AWARE Foundation, 2009, *AWARE – our world, our water*, Project AWARE Foundation, California
- Smith R A, 2007, 'Anchoring tourism to the coast: Innovative spatial and community strategies', in M Lück (ed), *Nautical tourism: Concepts and issues*, Cognizant Communication, New York, 25–36
- Statistics New Zealand, 2013, *Tourism Satellite Account: 2013*, Statistics New Zealand, Wellington
- Turner S and A Schwarz, 2006, *Management and conservation of seagrass in New Zealand: An introduction*, Science for Conservation 264, Department of Conservation, Wellington
- 2014, *Tourism highlights 2014 edition*, UNWTO, available at <http://mkt.unwto.org/publication/unwto-tourism-highlights-2014-edition>

Endnotes

- 1 United Nations World Tourism Organization, 2014
- 2 Statistics New Zealand, 2013
- 3 Statistics New Zealand, 2013
- 4 <http://www.tourismnewzealand.com/about-the-tourism-industry/>
- 5 http://www.tianz.org.nz/content/library/FINAL_NZTS2015_HR.pdf
- 6 Orams M B and M Lück, 2014
- 7 Originally developed by Orams in 1996 and further refined by Orams and Lück in 2013
- 8 Cessford G et al., 2012
- 9 NIWA, 2012

- 10 Cessford G et al., 2012
- 11 Orams M B and M Lück, 2014
- 12 Smith R A, 2007
- 13 Orams M B and M Lück, 2014
- 14 <http://www.teara.govt.nz/en/lifesaving-and-surfing>
- 15 <http://www.teara.govt.nz/en/lifesaving-and-surfing/page-4>
- 16 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=10912395
- 17 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=10912395
- 18 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=11239213
- 19 <http://www.stuff.co.nz/taranaki-daily-news/news/5456452/Money-for-surf-reef-a-wipeout>
- 20 <http://marinebio.org/oceans/scuba/>
- 21 Project AWARE Foundation, 2009
- 22 <http://www.telegraph.co.uk/travel/picturegalleries/7300359/The-worlds-top-ten-dive-sites.html?image=9>
- 23 <http://www.fish.govt.nz/en-nz/MFish+Apps/default.htm>
- 24 Orams M B, 1999, 17
- 25 Beca Infrastructure Limited, 2012
- 26 Beca Infrastructure Limited, 2012
- 27 <http://www.aucklandcouncil.govt.nz/SiteCollectionDocuments/aboutcouncil/planspoliciespublications/technicalpublications/TR2012033Antifoulingbiocidesinmarinas.pdf>
- 28 Turner S and A Schwarz, 2006
- 29 Turner S and A Schwarz, 2006
- 30 <https://www.mfe.govt.nz/publications/ser/enz07-dec07/html/chapter11-oceans/page3.html>
- 31 Ministry for Primary Industries, 2013
- 32 Birkeland C and P K Dayton, 2005
- 33 Orams M B, 1999, 25
- 34 NIWA, 2012
- 35 NIWA, 2012
- 36 Mulcahy K and R Peart, 2012
- 37 Mulcahy K and R Peart, 2012
- 38 Peart R, 2013
- 39 Mulcahy K and R Peart, 2012
- 40 Orams M B and M Lück, 2014
- 41 Orams M B and M Lück, 2014
- 42 <http://www.cruisewhitehaven.org.nz/about/>
- 43 <http://www.cruisewhitehaven.org.nz/about/>
- 44 <http://www.biosecurity.govt.nz/enter/ships>
- 45 RMA, section 30(1)(d)(vii)
- 46 Resource Management (Marine Pollution) Regulations 1998, section 11(2)
- 47 NIWA, 2012
- 48 EEZ Act, section 20
- 49 Marine Mammals Protection Regulations 1992, Regulation 6
- 50 Mulcahy K and R Peart, 2012
- 51 Mulcahy K and R Peart, 2012
- 52 McCrone A, 2001
- 53 <http://www.doc.govt.nz/Documents/science-and-technical/SFC173.pdf>
- 54 <http://www.tourismnewzealand.com/developing-your-tourism-business/qualmark/enviro-awards/>
- 55 <http://www.tourismnewzealand.com/developing-your-tourism-business/qualmark/enviro-awards/>
- 56 <http://www.doc.govt.nz/parks-and-recreation/places-to-visit/auckland/hauraki-gulf-islands/little-barrier-island-nature-reserve-hauturu-o-toi/plan-and-prepare/>
- 57 <http://www.biosecurity.govt.nz/files/biosec/camp-events/campaigns/clean-boating-guide.pdf>
- 58 <http://southernseabirds.org/resources/mitigation-tools-practices/recreational-anglers/>
- 59 Peart R, 2013
- 60 Peart R, 2013
- 61 Peart R, 2013
- 62 <http://www.worldheritagesite.org/alltentative.html>
- 63 <http://diving.co.nz/about>
- 64 <http://diving.co.nz/about/company/achievements/>
- 65 <http://www.lin.govt.nz/sites/default/files/docs/hydro/ntm/summary/annual/nz10.pdf>
- 66 <http://www.pce.parliament.nz/publications/all-publications/just-cruising-environmental-effects-of-cruise-ships-5>
- 67 <http://www.es.govt.nz/living-in-southland/along-the-coast/maritime/cruise-ships/>
- 68 <http://www.es.govt.nz/download.aspx?f=/media/21912/cruise-ships-factsheet.pdf>
- 69 <http://www.fmg.org.nz/sites/default/files/FMG%20March%202012%20newsletter.pdf>
- 70 <http://www.es.govt.nz/download.aspx?f=/media/7980/cruise-ship-deed-of-agreement-updated-august2012.pdf>
- 71 <http://www.es.govt.nz/living-in-southland/along-the-coast/maritime/cruise-ships/>

17

Catchment-based activities





Contents

373	Introduction
375	Non-point source pollution
375	Agriculture
375	Sedimentation
376	Nitrogen-nitrate and phosphate pollution
376	Forestry
376	Urban
377	Point source pollution
377	Agricultural and industrial enterprises
378	Mining
378	Wastewater treatment plants
379	Septic tanks
379	Stormwater and sewerage
380	Potential environmental effects
381	Cumulative impacts
381	Sedimentation
382	Heavy metals and other dissolved pollutants
384	Excess nutrients
384	Bacteria and viruses
385	Organic material
386	Decision-making process
386	National Policy Statement for Freshwater Management
388	New Zealand Coastal Policy Statement
391	Regional council planning
392	Integrated catchment management plans
394	Territorial authorities
395	Elements of best practice
397	Case studies
397	Auckland's Sustainable Catchments Programme
398	Tairua Harbour and Catchment Management Plan
400	Ōkahu Catchment Ecological Restoration Plan
401	Further reading and references
402	Endnotes

Introduction

Water flows in a constant cycle from the ocean to the sky and back again, often via a land mass, where it picks up other substances along the way back to the sea (as shown in Figure 17.1). Various substances, from both natural and human-influenced processes, are discharged from the land and transferred to the coastal marine environment as part of the natural water cycle.

The cycle can be thought of as starting when the sun's heat evaporates water from the oceans. The water vapour is carried along in air currents, which are forced up into cooler altitudes when they pass over a land mass. As the air cools, the water droplets condense and become visible as clouds. As the droplets become heavier, they eventually fall to the ground as rain (or snow).

Some rainwater is absorbed by plants and animals and is re-evaporated back into the atmosphere. Other rainwater flows back to the sea via streams, rivers, wetlands and estuaries (known generally as surface waters) taking with it dissolved material including nutrients and sometimes sediment. Yet other rainwater seeps down into underground rock and caves (known as groundwaters), where it may either flow underground to the sea, flow back up to join rivers or streams, or lie dormant underground. In addition, some rainwater falls onto urban areas where there are paved and other hard surfaces, and instead of percolating through the soil it is collected in stormwater drains and carried into the sea.

Human activity has greatly affected the amounts and types of substances that are transferred by water from the land into the sea, with corresponding effects on the health of marine ecosystems. Human-derived substances, and excess amounts of natural materials, impact on the marine environment by upsetting natural balances and introducing foreign or toxic elements into the ecosystem. Deforestation and intensification of land use has been a significant contributor to this problem, as has the destruction of wetlands, because vegetation fulfils an important role of filtering contaminants from water. Wetlands act as enormous sponges which absorb excess water during times of heavy rainfall, allow sediment to settle out, process and trap nutrients, and then slowly discharge cleaner water into rivers and estuaries.

In order of decreasing impact, land sources of increased sediment and contaminants in New Zealand's marine environment are generated mainly from agriculture, urban sewage, urban stormwater, industrial waste, agricultural waste, mining and forestry.¹



Water flows in a constant cycle from the ocean to the sky and back again, with much rainwater eventually ending up in the marine environment, such as at Doubtful Sound, Fiordland, shown here (Raewyn Peart)

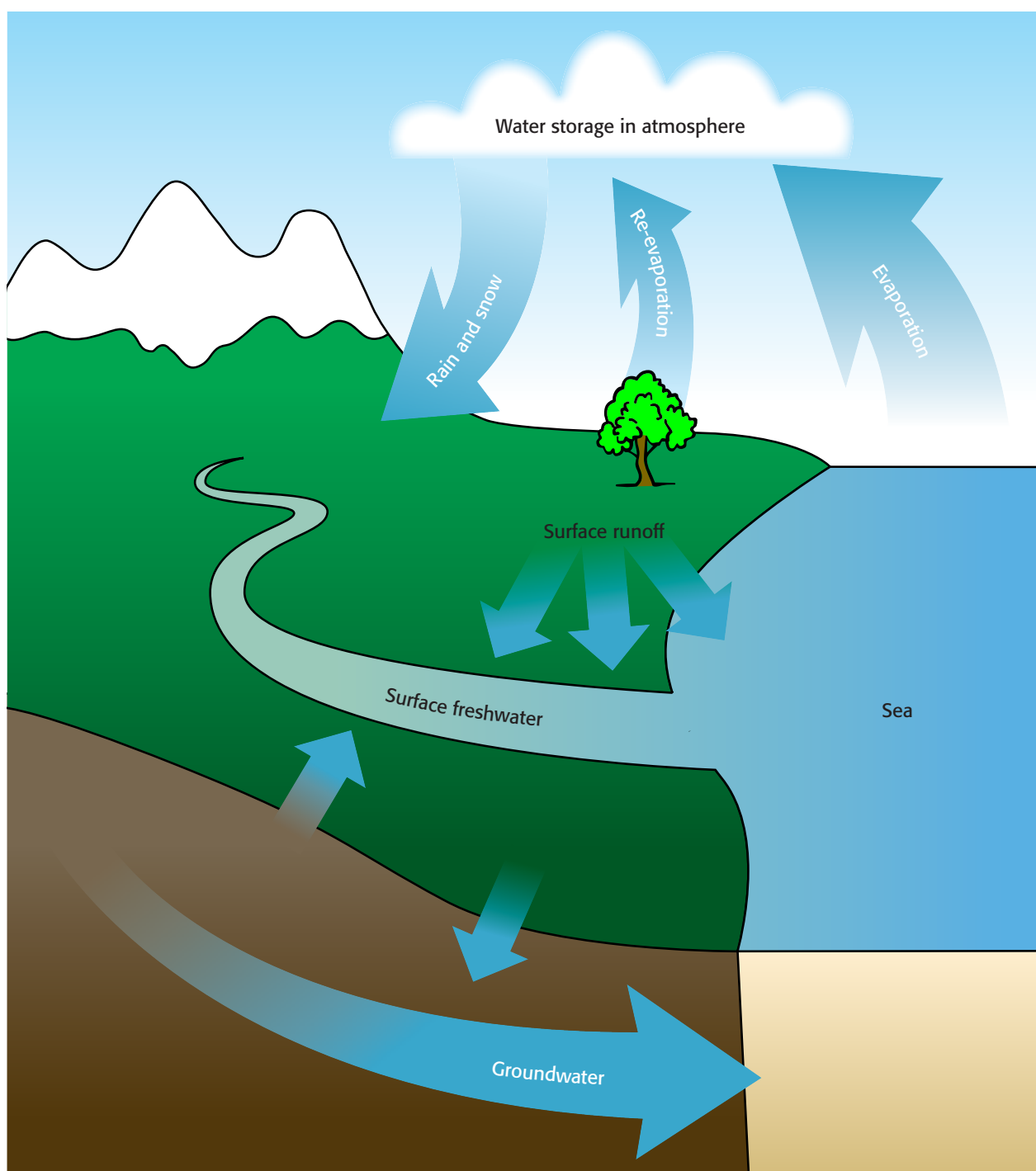


Figure 17.1 The water cycle

Catchment-sourced discharges can be divided into two types: point source and non-point source. Point source discharges refer to contaminants that enter the catchment from a single defined source such as a sewage treatment plant or a milking shed. Non-point source pollution refers to contaminants that enter the catchment from multiple diffuse sources, such as contaminant in runoff from pasture or urban areas. Point source and non-point source pollution are dealt with separately below.

Discharges into the coastal environment are also covered in depth in *'Caring for Our Coast: An EDS Guide to Managing Coastal Development'*, including an assessment of implementing the national policy on this issue and examples of good practice.

Non-point source pollution

Although pollution from point sources has declined in the last 30 years, contaminants from non-point sources continue to be a major problem for the health of New Zealand's marine environment. Agricultural activities, forestry and urban earthworks can generate considerable quantities of sediment, and lead to an increase in the nutrient and microbial contaminants within sediment. Runoff from urban areas is another significant issue, including vehicle-derived zinc, copper and lead washed off roads and zinc from unpainted iron roofs.

Agriculture

Agricultural pasture occupies more than a third of the land in New Zealand, and agricultural activity affects nearly all catchments.² Since European settlement, agriculture has changed the landscape of New Zealand dramatically, through the removal of indigenous forest and wetlands, and replacement with exotic grasses. Due to the highly erodible nature of many New Zealand soils, this has resulted in significantly increased quantities of sediment entering the marine environment. In addition, agricultural chemicals such as DDT, and nutrients from fertiliser and stock urine and faeces, can make their way into the sea.



Urine from cows, such as those shown here on the Hauraki Plains, can result in elevated nitrogen levels entering the marine environment (Raewyn Peart)

Sedimentation

Forests intercept rainwater which eventually evaporates back into the atmosphere. This is through leaves and branches trapping the water before it reaches the ground, and tree roots sucking rainwater out of the soil, which then evaporates from foliage through transpiration. When forest is replaced by short-rooted pasture grass, the volume of water running off the land increases, and it travels at a faster rate. This enables it to scour out the soil and transport greater volumes of sediment.

In the absence of tree roots to hold the soil together, greater amounts of sediment made up of inorganic matter (rock and soil) and organic matter (mostly animal faeces) can be washed from the land into water bodies and make their way to the sea. When paddocks extend right up to the edges of rivers and streams, with no riparian vegetative barrier, livestock can damage the soil which leads to a mobilisation of fine sediment and ultimately sedimentation in the channel.³

The increased quantities of water runoff from pastoral land travelling at faster speeds results in higher peak flows. This can cause stream bank erosion which generates additional sediment, and often widens channels and raises the level of channel beds.

Pastoral catchments have a sediment load that can be two to five times greater than that of a forested catchment.⁴ Although most deforestation in New Zealand took place more than 80 years ago, the effects of these changes on the landscape continue to be felt, as the hills and riparian areas continue to be used for livestock production. In recent years, some councils have provided incentives for farmers to replant riparian areas.⁵ In the past, there have been assistance programmes for soil stabilisation, but these are now less common.

Nitrogen-nitrate and phosphate pollution

Nitrogen can be a significant pollutant in both freshwater and marine environments. Phosphorus is naturally more common in marine environments so systems there tend to be nitrogen limited. Phosphorous is often more of a concern in freshwater, but the addition of both substances to the marine environment can cause problems.

Nitrogen, and to a slightly lesser extent, phosphorus, are some of the most significant by-products of agricultural activity. Nitrogen and phosphorus are essential parts of amino acids and nucleic acids, and consequently they are critical building blocks for all life on earth. Unfortunately when they appear in the marine environment in excess, they can be very harmful because they accelerate natural processes of growth and decay.

Along with sediments and microbes, nitrogen-nitrates (nitrogen dissolved in water) and phosphorus-phosphates (phosphorus dissolved in water) produced by agricultural activity find their way into the marine environment. These nutrients can come from a number of sources, the principal of which is animal waste. Nitrogen from livestock manure contributes around five times the amount of nitrogen to the land as nitrogenous fertilisers.⁶ The breakdown of organic nitrate and phosphate in faeces, and the oxidation of ammonia in urine, release large amounts of nutrients which frequently enter water bodies.

Nitrogen and phosphorus pollution is also derived from fertilisers, which are applied to agricultural pasture in order to promote grass growth. Total fertiliser use significantly increased in New Zealand between 1985 and 2004. The amount of nitrogen fertiliser used in New Zealand has increased about ten-fold since 1985 and has doubled since the mid-1990s.⁷ Furthermore, soil in agricultural pasture often has high natural levels of nitrogen due to the presence of clover. This is sown by farmers as a method of fixing nitrogen into the soil, and superphosphate is often applied to promote clover growth. As farming becomes more intensive, more nitrogen fertiliser is often used to increase yields, more is excreted in dung and urine, and consequently more nitrogen finds its way into the catchment.

Forestry

Although the effects of forestry on a catchment are generally short-term and localised, they can be significant. Tree-felling and track construction can increase the amount of sediment and nutrients (in particular nitrogen, together with potassium, phosphorus, magnesium and others) that are washed into the water because without the trees and roots to provide a barrier there is little to stop rainwater washing the soil into the waterways. The extent of this depends on the particular area, the type of soil and the amount of rain normally experienced. The worst effects are generally experienced for the first three to five years after felling. The impacts increase if large areas of forest are clear-felled at one time or in succession.

The negative effects of forestry can be mitigated by selective felling, clear-felling of only small areas within a catchment at a time and the retention of riparian planting, which protects the waterways from the majority of the runoff. In addition, careful track construction with cut-off drains and the like can help reduce runoff. Land in pasture produces greater levels of sedimentation and nutrient runoff than mature forest, so afforestation of an area previously used as pasture can return stream flows and sediment loads to normal levels within six to eight years of planting, and reduce runoff by 30 to 50 per cent within five to ten years.⁸

While water quality may benefit from forestation, there is a potential downside to reduced sediment loads in rivers, which can result in increased potential for coastal erosion because of less sediment reaching the coast.

Urban stormwater

Approximately 86 per cent of New Zealand's population lives in urban centres of more than 1,000 people.⁹ As a result of land clearance and urban development, most of these urban areas are covered in impermeable surfaces. For example, in the Auckland region only 19 per cent of the 10,000 kilometre-long stream network passes through native forest.¹⁰ In these paved and developed areas, rainwater cannot be absorbed by vegetation and soil, as it would be in rural areas. Therefore, to reduce the risk of flooding, towns and cities have stormwater systems that channel rainwater into gutters and pipes and eventually out into streams, lakes and coastal waters. Because a large number of New Zealand's urban centres are sited on estuaries and harbours, the stormwater many towns produce discharges directly to the coast.

Stormwater systems are often a significant source of pollutants into the catchment and coastal areas because the water that enters the pipe network picks up and carries with it a wide range of contaminants from the streets, industrial and construction sites, roofs and other surfaces. These substances include heavy metals (especially zinc, copper and lead – lead contamination has been falling

since lead was removed from petrol but zinc contamination continues to rise),¹¹ hydrocarbons from vehicles (these come from the road surface itself, exhaust emissions, brake linings and tyre wear), sediment and contaminated dust.¹²

Roof runoff is a major source of zinc contamination in catchments which have been urbanised for a long time, because of the large number of galvanised iron roofs in poor or unpainted condition. Zinc oxide used in tyres is another. In addition, urban earthworks have the potential to release large amounts of sediment into the catchment.¹³



Hard channelling of streams in urban areas, as shown here in Dunedin, can carry contaminated stormwater directly into the marine area (Raewyn Peart)

Point source pollution

Pollution of this type has generally declined over the last 20 to 30 years as treatment system technology has been upgraded and alternative methods of disposal have been developed. However, at specific sites in New Zealand it can still be a problem.

Agricultural and industrial enterprises

Meatworks, dairy factories, tanneries, fertiliser plants, stockyards, mills and many other types of agricultural and industrial enterprise all risk pollutants being discharged into the catchment. In the early years of European settlement, there were many such enterprises scattered across the country, all producing industrial or agricultural waste on a small scale. Some of these were connected to municipal treatment plants while others managed their own waste. In some cases, toxic waste was discharged directly into the marine environment.

Over the last 30 years, agricultural and industrial activity has changed, so that there are fewer, larger operations. For example, whereas in 1970 there were more than 200 dairy factories, there are now only about 30.¹⁴ These bigger businesses tend to deal with waste more effectively because of their relative financial strength and access to technology. Furthermore, the last 30 years have seen the implementation of stricter legal requirements and a shift in attitudes towards greater recognition of the importance of avoiding pollution, together with a greater interest in finding better technology to achieve this.

Agricultural enterprises generally now use two-pond treatment systems, often combined with land irrigation systems, to dispose of their effluent from dairy sheds and the like. Such systems pass the effluent through two separate ponds, each containing different naturally occurring types of bacteria, before discharging it into the catchment. This system removes 95 per cent of the organic matter from the effluent but may not greatly affect dissolved nutrients levels.

Land irrigation disposes of wastewater and nutrients to paddocks or forest rather than into the water. This is effective at protecting the catchment from some pollution, so long as the land is not overloaded. But if the capacity of the land to absorb the waste is exceeded, the nutrients will wash off into streams or leach into the groundwater, posing the same risk to marine ecosystems as they would have if they had been discharged directly into the catchment.¹⁵ Only some nitrogen pollution will be removed by plants, and only when they are growing. The remainder will pass into groundwater and eventually emerge in streams.

Mining

In the past, mining had a significant impact on water quality. Mines can discharge huge amounts of silt and sediment into the catchment, as well as large amounts of processing chemicals such as cyanide and heavy metals from the rock. Nowadays, mines have relatively effective treatment systems, which minimise the amount of silt and sediment being transferred into the catchment (e.g. by holding wastewater in pools to allow sediment to settle before releasing it into the river).

Gold mining gives rise to particular risks because when the mineral ore is exposed to air and water the metals can react chemically, dissolving and entering groundwater and streams. This can cause the catchment to become more acidic. The industry has methods to minimise the risk of this by transporting the rock to a treatment plant in a slurry, separating off the water, and then treating it to remove the heavy metals before discharge.

Mining often changes groundwater flows, and water flowing through exposed ore remnants or mine waste can acquire high levels of heavy metals, with such contaminated flows often continuing long after the mining has ended.

Wastewater treatment plants

Approximately 1.5 billion litres of domestic wastewater is discharged into the environment in New Zealand every day.¹⁶ Human effluent contains a number of substances that can be extremely damaging to the marine environment if released into the coastal waters untreated. The most significant pollutants are rotting organic matter, disease-causing micro-organisms, excess nutrients, ammonia and suspended solids.

Most of the household sewage in New Zealand is piped through treatment plants to remove pollutants before discharge: most towns of 5,000 or more residents have a reticulated sewage system which pumps everybody's wastewater into a common treatment plant and discharge point. There are more than 250 treatment plants in New Zealand, and although 24 of them discharge waste onto land, the rest discharge either into the ocean, rivers, streams or drains.

The plants vary in age and technological capability. Treatment systems use a number of stages in the treatment process, depending on how advanced they are. 'Primary' treatment means that suspended solids are filtered out of the effluent before it is discharged into the catchment, and 'secondary' treatment refers to the use of micro-organisms to remove organic matter and some nutrients, which are then decomposed in process facilities or oxidation ponds. Tertiary treatment further reduces nutrients and organic matter using oxidation ponds and/or various types of organic or chemical treatment. Ultra-violet treatment, and other technologies, are used by the most advanced plants in New Zealand to kill the pathogens that remain in the water.

Treatment plants are particularly designed for human wastes and effectively reduce organic pollution from this source. They are not generally designed for a broad range of industrial pollutants and may be quite ineffective in removing these. Industrial effluent is much more difficult to manage in treatment plants because it can be highly variable in loading and volume.

The solid material from treatment plants (biosolids) needs careful management. It may be suitable for use as a soil additive (but this may add to pollution risks) or it may need to be disposed of in a landfill site.



The Mangere Wastewater Treatment Plant, shown here, is one of the most advanced plants in the country, undertaking tertiary and ultra-violet treatment along with using other technologies (Raewyn Peart)

All treatment plants in New Zealand practise at least primary treatment, and a large number also use secondary and tertiary treatment. Only a small number use ultra-violet or equivalent systems for the reduction of pathogens in the final effluent. The treatment level is largely related to the sensitivity of the receiving waters. The effectiveness of wastewater treatment systems consequently varies. Some are ineffective because they are too small to cope with the volume of waste they receive. For example, some are put under pressure by new urban developments, by the increased amounts of effluent generated by an influx of visitors during holiday season, or by stormwater entering the systems and overloading them.

Septic tanks

Smaller settlements in New Zealand often do not have community wastewater treatment plants but rely on individual household septic tanks. Septic tanks are generally effective but can fail where the soil is unsuitable, or where the surrounding land area is not big enough for the soakage field. This is a particular problem in modern coastal developments, where large homes (rather than traditional baches) with modern conveniences requiring a large amount of water are built on small sections very close together. Overloading of the soakage fields, with resulting overland flow, is the most common problem. Septic tanks can also fail when they are inadequately maintained. Effluent that seeps untreated from septic tanks may get into streams or groundwater, and from there into the marine environment. Even with properly-operating systems, some nitrogen transfer to groundwater is inevitable.

Stormwater and sewage

Previous sections discussed the treatment of wastewater. This section considers environmental issues arising from the conveyance of wastewater to treatment plants.

Wastewater systems typically use gravity to move sewage discharged from houses and other premises along pipes towards the treatment works. When the sewage pipes reach the lowest point on land, gravity will no longer move the material, and pumping stations are required to pump the sewage up to the treatment plant. Many pumping stations are located near the coast, as this is often the lowest point to which sewage can be moved by gravity.

Pumping stations will usually have some holding capacity for sewage in the event that the pump fails, such as when there is an interruption in the electricity supply. When the pump stops for any reason, the sewage will keep flowing as a result of gravity and it will build up in the holding tanks at the station. When the holding capacity of the station is exceeded, raw sewage typically flows into the sea or a nearby river.



Wetlands and indigenous plantings assist with stormwater filtration at Wynyard Quarter, Auckland (Raewyn Peart)

Some regions use combined sewer and stormwater drains – this is common in old networks which have not yet been replaced with more modern systems that minimise the risk of pollution. In these areas pumping stations can overflow when there is heavy rain that quickly reduces their holding capacity. In other areas with separated sewers, stormwater infiltration through damaged pipes and poorly arranged or illegal connections can also result in inflows in excess of capacity. When this happens, raw sewage can be discharged into the marine area, although it will be diluted to some extent by the large amount of stormwater.

Stormwater infiltration can also cause major problems in sewage treatment plants by increasing the volume of liquid that they have to deal with, reducing the effectiveness of treatment, and increasing the risk that treatment systems will flood and allow effluent to escape. This is actually quite common as a result of faulty pipes, illegal cross-connections (where private properties illegally attach sewage pipes to the public stormwater drain), and old connections that have not been cut off.

Another way in which untreated urban sewage can enter the catchment is where sewer systems overflow into the stormwater network (exfiltration) and the natural environment. This generally happens as a result of failures in the pipes such as cracking followed by tree root invasion. It can also occur as a result of heavy rainfall. Pumped sewerage systems are becoming more common, and are resistant to infiltration, but failures can result in spills.

Auckland Central Interceptor Project

The Auckland Isthmus has had ongoing issues with major wastewater overflows into the marine environment resulting from a wastewater network in need of upgrading. Water quality at a number of Auckland beaches exceeds national guidelines for the concentration of enterococci bacteria. These bacteria indicate the presence of faecal material and disease-causing micro-organisms in coastal waters, high levels of which pose a significant risk to public health. In November 2011, more than a third of the region's most popular beaches failed water quality tests, and signs recommending 'no swimming' were erected for at least 10 beaches.¹⁷ There are now permanent health warnings in place at Cox's Bay, Meola Reef, Weymouth Beach, Little Oneroa Lagoon and Wairau Outlet.¹⁸

The growth and development predicted for the Auckland region, and the ageing infrastructure, means there is much concern about the capacity of the network in the future. Watercare Services is now focusing on delivering an \$800 million Central Interceptor Project which will see a large wastewater pipeline constructed to improve the wastewater network service for western and central Auckland. The interceptor, which could be up to 24 kilometres long, will reduce wastewater overflows, provide for population growth and mitigate the risk of pipe failures. The interceptor will be constructed in three phases between 2016 and 2030 to transport wastewater from central and west Auckland to the Mangere Wastewater Treatment Plant.¹⁹

Potential environmental effects

The amount of pollutants that are transferred from land into the oceans, and the effect that they will have on coastal marine ecosystems, varies depending not only on the amount of waste actually released and the type of waste, but also on the characteristics of the marine environment affected. As one would expect, transfer rates are much worse at times of more extreme wet weather, when more water is moving from the land to the sea.

Whether or not land-sourced pollutants remain in the coastal marine area depends on a number of factors, in particular the geographical makeup of the area and the winds and currents experienced by the locality. In harbours and estuaries, the amount of mixing that occurs and the time the inflowing water is retained in the area are important. Shallow inlets are more prone to retain sediment and pollutants than deeper ones, and estuaries and narrow inlets are more prone to retaining material than wide exposed beaches, which are subject to stronger currents that can carry inflowing substances out to sea. Many river estuaries have natural sea barriers broken only in high flows. Sediment and pollution retention behind these barriers is high.

New Zealand's particular geographic make-up means that there are large stretches of coastline that do tend to trap sediment and pollutants. As the Ministry for the Environment's 1997 State of the Environment report notes: 'Much of the coastline is made up of river-fed estuaries whose wide, shallow waters are permanently protected from ocean waves by sand or shingle bars or offshore islands'.²⁰ Where there are high river flows, the freshwater may pass along an open coast, floating over seawater for some distance before it is mixed with the seawater.

Cumulative impacts

Cumulative effects are referred to in the definition of 'effect' in section 3 of the RMA as effects that can build up over time or occur in combination with other effects. The term refers to the impact of a number of contaminants or pollutants on a particular ecological system, which in some cases may be greater than the sum of their individual impact. It also incorporates the concept of flow-on impacts. Pollutants are dangerous due to their cumulative effect, as chemicals tend to have poor solubility, their effects are long term and they can travel long distances regionally.²¹ Toxic chemicals can enter marine food chains, and can accumulate to high levels in top predators. Changes in the marine environment from land-based contaminants can be the result of a single large event, or many small ongoing events. These are often unnoticed and occur over a long time.



The effects of multiple activities within a catchment can cumulative to cause significant change in the marine environment, as has been the case with the Mahurangi Harbour in Auckland (*Raewyn Peart*)

Sedimentation

An increase in the rate that sediment is transferred into the receiving environment can bring about profound changes in the marine ecosystem. An excess of sediment can reduce water depths, water coverage and change the type of sediment found on the seafloor (such as the size of the particles). In addition, when excess sediment enters the receiving environment it can become suspended in the water, reducing light levels and the ability of marine creatures to feed, and affecting the clarity of the water and its turbidity (cloudiness of a fluid generally invisible to the naked eye). Research being undertaken by NIWA suggests that severe impacts can result from fine sediment being suspended in the water column, because of the way the particles interfere with light penetration of the water (or light attenuation).²² This can lead to impacts such as the inability of visual predators to see prey.



Elevated levels of sediment, as shown here in the Wairoa River feeding into the Kaipara Harbour, can bring about profound changes to the marine environment (*Raewyn Peart*)

Sediment can bring about changes in vegetation, be detrimental to fish habitats, degrade spawning areas and deplete invertebrate populations. These impacts can flow through the food web, particularly if a particular life stage of a species is affected. As most sediment is deposited in intertidal zones, it is the species that live in these areas that are often most affected by excess sedimentation.²³ This is particularly significant, because healthy intertidal zones support a particularly high diversity of species in comparison with other parts of the coastal area. For example, the Kaipara Harbour is the biggest estuary in the Southern Hemisphere, and is the place where most of the snapper on the west coast of the North Island originate. Sediment from the catchment threatens to overwhelm horse mussel beds and seagrass meadows, which are nursery grounds for snapper.²⁴

Some of the main environmental impacts from sediment suspended in the water or settling on the seafloor are identified in Figure 17.2 below.

Type	Impact
Direct	- Decreases oxygen concentrations and changes sediment properties such as grain size, chlorophyll and organic matter content²⁵ - Clogs the gills of fish, making it harder for them to breathe - Interferes with food intake and potentially affects growth and condition of fish and shellfish - Reduces visibility, making it harder for some fish to find food - Negatively affects microphytes or benthic primary producers, an important food source for many macrofaunal species - Smothers shellfish beds and cause cockles, pipi, mussels and scallops to die - Causes seafloor plants to die off - Reduces the hard surface area for kelp spores or baby paua to attach to
Indirect	- Reduces the food that is available for rocky reef ecosystems and larger animals - Reduces the growth and extent of kelp forests - Damages the entire productivity of the rocky reef ecosystem - Damages sensitive habitats that are important for part of a species life stage, such as nursery or juvenile areas - Reduces the abundance of many biologically important fish species - Reduces the stocks of recreationally and commercially important fisheries species (such as paua and crayfish)

Figure 17.2 Environmental impacts resulting from increased sediment in the marine environment²⁶

Sediment impacts in Tasman Bay

Tasman Bay in the South Island regularly suffers from murky seabed conditions which are believed to have come from sediment carried into the marine area by floods of the Motueka River, which then gets resuspended in the water by tidal currents in the bay. In addition, ongoing trawling and dredging resuspends the sediment. Research has found that scallops on the seafloor temporarily stop feeding when in highly turbid water, whereas those in clearer water in the same vicinity continue to feed. In addition, the growth of scallops was shown to be adversely affected by high levels of suspended sediments over the long term.²⁷

There are now concerns that coral-like bryozoan colonies further north of the Motueka River, which are important snapper and tarakihi nursery habitats, are under increasing stress from the sedimentation. With the seabed in the area becoming largely covered in sediment, there are few hard surfaces for the bryozoans to settle on. Ultimately this is reducing the overall productivity of the tarakihi and snapper fisheries as well as decreasing the area's biodiversity.²⁸

Heavy metals and other dissolved pollutants

Many heavy metals and other pollutants find their way into the marine area from roads, from industrial and agricultural activity, and from domestic households. Such toxic substances have also been used directly in the marine environment. In the 1950s, 'booster' agents such as DDT (dichloro-diphenyl-trichloroethane), organomercurial, organolead and arsenical compounds were used on underwater surfaces to discourage antifouling organisms.

In particular, heavy metals, chemicals, hormones, dioxins, organochlorines, PCBs (polychlorinated biphenyls) and DDT cause problems. The problem may not just be mortality of marine organisms, but disruption of growth or reproduction, which may occur

at much lower than lethal concentrations. Only a small proportion of heavy metal and pollutant contaminants are dissolved in the water column; the majority remain attached to particles of sediment, which settle in the inner parts of harbours and estuaries. In these areas, where wave and current movements are likely to be too weak to move the polluted sediments around, they can build up to toxic levels.²⁹

Scientific studies since the 1950s have shown that a high body burden of organochlorine pesticides (such as DDT) in animals can cause ecological damage, including abnormalities in the reproduction and development of fish-eating birds. Concerns have also been voiced about the impact that such chemicals are having on some marine mammals, including that they may be damaging their health, affecting their fertility, reproduction, immune system and growth.³⁰



Very high DDT levels have been detected in Hector's dolphins such as these shown here in Akaroa, Banks Peninsula (Raewyn Peart)

DDT and Hector's dolphins

DDT was used in the 1950s and 1960s to control agricultural pests. During this period, it was applied over wide areas of land, often being mixed into fertilisers and top-dressed onto pastures. The use of DDT in New Zealand was restricted in the 1970s and banned in 1989. Today, the residual DDT concentrations in the country's pasture soils are relatively low.³¹ However, there is increasing evidence that DDT from the soils is still washing into the marine environment and having a negative impact on the marine ecosystem. The near-shore habitat of New Zealand's threatened Hector's dolphins has exposed this species to a variety of pollutants and contaminants such as organochlorines and heavy metals. Hector's dolphin tissue has been found to contain high levels of organochlorines such as DDT, PCBs and dioxins.³² The DDT levels detected in Hector's dolphins are very high on a global scale and second only to the highly contaminated Canadian beluga whales.³³

Organochlorines are known to have detrimental effects on the reproductive success of some marine mammal species, in some cases resulting in sterility, and they may suppress the immune system in some species.³⁴ Persistent organic pollutants such as DDT, PCBs and dioxins are known to be particularly soluble in fatty tissues, and because the body is unable to excrete these substances, they accumulate over each animal's lifetime. In mammals, they are also passed onto the young in their mother's milk. When tissues of stranded and bycaught common dolphins in New Zealand have been analysed, traces of PCBs and a range of pesticides including DDT have been found.

Once these substances enter the food chain, they travel up the different trophic levels. The pollutants can have effects higher up the food chain as shellfish and other organisms are eaten by fish, birds and marine mammals. Marine organisms can be affected by the toxicity when they ingest the sediment or take up toxins that have dissolved into the water column. The effects that such toxins have on marine life varies widely but can be very serious. For example, even small amounts of lead in the water are very toxic to marine life.

Auckland sediment contaminant monitoring

The Auckland Council has been running the Regional Sediment Contaminant Monitoring Programme since 1998. The information this programme collects provides important data for Auckland's state of the environment reporting, management of stormwater quality, resource consenting, policy development and public education. Recent analysis of this data has shown that spatial patterns remain similar to those previously reported. The highest concentrations of contaminants were found in muddy upper estuarine areas where runoff from extensively urbanised and industrialised catchments is entering the coastal marine area, particularly in the Tāmaki Estuary and central Waitematā Harbour. Lowest contaminant concentrations are found in estuaries with rural or forested catchments and on open coastal beaches.³⁵

Excess nutrients

Nutrients, in particular nitrogen and phosphorus, occur naturally in aquatic ecosystems and are essential elements for growth. This is why they are used in large quantities in fertiliser to promote grass growth on agricultural pasture. However, where excess amounts of nutrients reach the catchment they can upset the balance of the ecosystem and cause serious problems.

When there are excess nutrients in the water, they serve to fertilise microscopic algae, leading to abnormal increases in their growth and subsequent decay. Their decay stimulates aerobic (oxygen-consuming) bacteria, which in turn causes severe reductions in water clarity and oxygen levels, potentially leading to the deaths of fish and other marine life. When the plants decay, the nutrients are released back into the water, stimulating algal growth and allowing the cycle to start again. This phenomenon is known as eutrophication. The consequent reduction in oxygen levels can result in 'dead zones', such as occur at Chesapeake Bay in the USA during mid-summer periods.³⁶

The Firth of Thames' nitrogen overload

In 2012, the Waikato Regional Council published a report into the water quality of 17 rivers in the Hauraki region using data gathered between 2000 and 2009. This showed that the Piako, Waitoa and Waihou Rivers were in poor to intermediate condition, being oxygen-depleted and murky. The report showed these rivers had concentrations of nitrogen and phosphorus five to seven times above guidelines.³⁷ As a consequence, the Firth of Thames is considered to be probably more productive today than it was in previous times, but water quality (in terms of nutrient levels, colour and clarity) is likely to have diminished.³⁸ More recently long-term monitoring has shown evidence of oxygen depletion as well as nutrient runoff-driven acidification of the seawater in the Firth of Thames.³⁹

The 'State of the Hauraki Gulf 2014' report confirmed that riverine inputs of nitrogen to the Firth of Thames now exceed loads from oceanic sources, and that the nitrogen loads entering the Firth from Hauraki Plains rivers are significant. The report noted that this is consistent with the Plains having one of the highest stocking rates for cows in the country. Long-term studies indicate that nitrogen inputs cause 'seasonal elevation in carbon dioxide, corresponding sags in oxygen and increasing acidification'.⁴⁰

Bacteria and viruses

Animal and human faecal matter contains large numbers of bacteria, which are produced by the gut. Although many are harmless, some of these bacteria can cause serious health problems in humans if they reach water used for recreation and shellfish gathering. Shellfish contaminated with sewage may contain pathogens such as norovirus, hepatitis A, shigella, vibrio and salmonella. These can cause dysentery, gastro-enteritis or other illnesses within a few hours or days.⁴¹

Coliform bacteria are common in human and animal faecal waste – but can come from other sources. The amount of coliforms per hundred millilitres of water is used to indicate the likely presence of faecal matter in water. A more precise test for animal and human waste is that for a particular coliform, *Escherichia coli*, which comes from warm-blooded animals alone. Viruses can also be found in human and animal faecal matter and the main class is the enteric viruses, which cause illnesses such as polio, meningitis and hepatitis.

Where levels of viruses or bacteria in coastal waters are relatively high, they can pose a risk to swimmers and other contact water users. They can also concentrate in shellfish, which eat by filtering food through the water, and can therefore be dangerous to humans who consume the shellfish.



Cox's Bay in Auckland is permanently closed to swimming and shellfish collecting due to a high level contamination from sewage (Raewyn Peart)

Organic material

Agricultural runoff, urban wastewater and industrial waste transfer organic material to the catchment. This can include faeces from humans and animals; proteins, vegetables and sugars from food preparation; and cleaning soaps. When effluent enters a healthy aquatic environment where there is plenty of oxygen dissolved in the water, aerobic (oxygen-using) bacteria, which occur naturally in the water, will eat the organic material, using up oxygen in the water to do so. Consequently their numbers grow beyond what would normally be expected in a healthy aquatic ecosystem.

This use of oxygen can have very serious effects on the health of the ecosystem, as fish and plants need oxygen levels to be maintained in order to survive. Even an occasional reduction of the oxygen level can have the effect of removing the marine species that rely on a continuous supply at a high level. If all the oxygen is used up, anaerobic bacteria (non-oxygen-using bacteria) will take over, decomposing the waste material and producing gases such as methane hydrogen sulphide and carbon dioxide, making the water toxic.⁴²

Thus, the potential harm caused by organic waste can be measured by its potential to remove oxygen from the water. There is a standard measure known as the 'biochemical oxygen demand' which is the amount of dissolved oxygen needed by aerobic bacteria to consume the waste (over a 5 day period where the water temperature is 20 degrees centigrade, measured in terms of the number of grams of organic material per cubic metre). Thus, if waste deposited in the catchment has a biochemical oxygen demand loading that is too high, too much oxygen will be removed from the water by the aerobic bacteria, jeopardising plant and animal life.

The surface sediment of the seabed is an important ecological zone and the home of some specialised forms of life. It is normally oxygenated. Excess sediments and low oxygen in the water above can cause it to become anoxic and unable to support its normal inhabitants.



Urban wastewater pipes, such as that shown here at Mangawhai, Northland, can transport a range of contaminants into the marine environment (Raewyn Peart)

Decision-making process

The RMA provides the framework for the control of activities that result in land-sourced pollutants entering the marine area. The RMA adopts a catchment-based approach to environmental management, with the boundaries of regional councils largely coinciding with the boundaries of water catchments and including the coastal marine area. This is based on the understanding that many different activities within a catchment will cumulatively impact on the quality of lakes and rivers and ultimately on what is discharged into the sea. Effectively managing these impacts requires an integrated and catchment-focused approach.

Under section 15 of the RMA, the discharge of contaminants into water (or onto land where there is a possibility that the contaminants will run off into water) is prohibited unless the activity is expressly allowed by a national environmental standard, a rule in a regional plan as well as a rule in any proposed regional plan, a resource consent or regulations.

National Policy Statement for Freshwater Management

The National Policy Statement for Freshwater Management (Freshwater NPS) sets out national policy for freshwater management within New Zealand under the RMA.

Freshwater NPS water quality objectives

- *Objective A1: To safeguard: a) the life-supporting capacity, ecosystem processes and indigenous species including their associated ecosystems of fresh water; and b) the health of people and communities, at least as affected by secondary contact with freshwater, in sustainably managing the use and development of land, and of discharges of contaminants.*
- *Objective A2: The overall quality of fresh water within a region is maintained or improved while: a) protecting the significant values of outstanding freshwater bodies b) protecting the significant values of wetlands and c) improving the quality of freshwater in water bodies that have been degraded by human activities to the point of being over-allocated.*

The Freshwater NPS provides a framework for achieving its objectives. Regional councils are required to develop freshwater objectives (the intended environmental outcome), to set limits (the maximum amount of resource use which will allow a freshwater objective to be met), and then to establish methods to avoid overallocation (resource allocation beyond a limit or resource use where a freshwater objective is not being met) for all freshwater management units (a water body, multiple water bodies or a part of a water body determined by the regional council as the appropriate scale for freshwater management).

National freshwater objectives set out the intended environmental outcome for a freshwater management unit. Part CA – National Objectives Framework sets out the approach for establishing freshwater objectives. Regional councils must:

- Identify the values for each freshwater management unit, including two compulsory values ‘ecosystem health’ and ‘human health for recreation’ as well as any other values
- Identify the attributes applicable to each value, including the attributes listed in Appendix 2 as well as other attributes
- For those attributes listed in Appendix 2 (currently phytoplankton, total nitrogen, total phosphorus, periphyton, nitrate toxicity, ammonia toxicity, dissolved oxygen, *E. coli* and cyanobacteria), assign an attribute state at or above the minimum acceptable state
- Formulate freshwater objectives, by reference to the relevant numeric attribute state for those attributes listed in Appendix 2 or, for other attributes, in numeric terms where practicable or narrative terms

Appendix 1 describes national values and uses for freshwater, including the two compulsory values ‘ecosystem health’ and ‘human health for recreation’. Appendix 2 identifies some ‘attributes’ relevant to the two compulsory values. It contains attribute tables which set out for each attribute the unit in which it is to be measured, four attribute states, numeric and narrative descriptions of those states and the national bottom line. Appendix 2 is currently incomplete. For example, there are a number of attributes relevant to ecosystem health that are not yet specified, including dissolved inorganic nitrogen, dissolved reactive phosphorus, dissolved oxygen, temperature, macro-invertebrate community index, and water clarity. Regional councils will need to continue to set objectives for these attributes for each freshwater management unit.

Freshwater limits specify the maximum amount of resource use available which will allow the objectives to be met. For example, nitrogen-leaching limits for agricultural land use may be included to achieve a dissolved inorganic nitrogen objective.

Regional policy statements, regional plans and district plans are required to 'give effect to' the Freshwater NPS. Regional councils must implement it by 31 December 2025, or by 31 December 2030 if meeting the earlier date would be impracticable or would result in lower-quality planning.

There are a wide range of methods which may be used to ensure freshwater objectives and limits are achieved. This will require rules controlling land use changes, such as urban development and agricultural intensification. In areas where objectives and limits are not being achieved, this will require management of existing land use activities. For example, the preparation and implementation of farm environmental management plans and nutrient budgets can enable a range of environmental impacts from a farming operation to be identified and managed. Regional plans generally require stock to be excluded from freshwater bodies and the coastal marine area. Another mechanism which can be used to control non-point source discharges at a catchment level is a 'cap and trade' scheme which allocates tradable pollution allowances to individual land users.

Forestry generally causes a large amount of sediment runoff during the harvesting cycle which may have a significant impact on freshwater bodies and the coastal environment. Regional plans should include controls that ensure forestry operations are managed to minimize sediment runoff. This should include requiring riparian areas to be retained and restricting the proportion of a catchment being harvested at any one time. In highly erodable areas, clear-felling forestry may be inappropriate.

Estuaries, which lie at the interface between the freshwater and coastal environment, are not freshwater bodies under the definition in the Freshwater NPS. However, the policy statement directs regional councils to 'improve the integrated management of freshwater ... including the interactions between freshwater ... and the coastal environment' and make or change regional policy statements 'to provide for the integrated management of the effect of the use and development of ... land and fresh water on coastal water'. Therefore an important consideration when setting freshwater objectives and limits is the characteristics of the receiving coastal environment. In particular limits which protect the values of freshwater may not protect the values of estuaries, for example nitrogen limits. Wetlands are freshwater bodies and the Freshwater NPS specifically provides for those.⁴³



The Freshwater NPS requires councils to set limits for freshwater quality, and these will impact on the state of freshwater entering the marine environment, such as at Port Jackson, Coromandel Peninsula, shown here (Raewyn Peart)

New Zealand Coastal Policy Statement

The NZCPS applies within the coastal environment – this includes the coastal marine area and all land with a coastal influence (generally defined by the first ridgeline but can include coastal systems such as wetlands and dunes extending further inland). Regional policy statements, regional plans and district plans must ‘give effect’ to the objectives and policies contained within the NZCPS.

A number of policies relate to catchment-based activities that may impact on the marine area, in particular Policy 21 (set out in Chapter 2: Marine management), 22 and 23.



Estuaries, such as the one shown here at Ngunguru, Northland, are not included in the Freshwater NPS, but regional councils are required to provide for the integrated management of the use of land and freshwater on coastal waters (*Raewyn Peart*)

NZCPS, Policies 21, 22 and 23: Water quality

Policy 21: Enhancement of water quality

Where the quality of water in the coastal environment has deteriorated so that it is having a significant adverse effect on ecosystems, natural habitats, or water-based recreational activities, or is restricting existing uses, such as aquaculture, shellfish gathering, and cultural activities, give priority to improving that quality by:

- a. identifying such areas of coastal water and water bodies and including them in plans;
- b. including provisions in plans to address improving water quality in the areas identified above;
- c. where practicable, restoring water quality to at least a state that can support such activities and ecosystems and natural habitats;
- d. requiring that stock are excluded from the coastal marine area, adjoining intertidal areas and other water bodies and riparian margins in the coastal environment, within a prescribed time frame; and
- e. engaging with tangata whenua to identify areas of coastal waters where they have particular interest, for example in cultural sites, wāhi tapu, other taonga, and values such as mauri, and remedying, or, where remediation is not practicable, mitigating adverse effects on these areas and values.

Policy 22: Sedimentation

1. Assess and monitor sedimentation levels and impacts on the coastal environment.
2. Require that subdivision, use, or development will not result in a significant increase in sedimentation in the coastal marine area, or other coastal water.
3. Control the impacts of vegetation removal on sedimentation including the impacts of harvesting plantation forestry.
4. Reduce sediment loadings in runoff and in stormwater systems through controls on land use activities.

Policy 23: Discharge of contaminants

1. In managing discharges to water in the coastal environment, have particular regard to:
 - a. the sensitivity of the receiving environment;
 - b. the nature of the contaminants to be discharged, the particular concentration of contaminants needed to achieve the required water quality in the receiving environment, and the risks if that concentration of contaminants is exceeded; and
 - c. the capacity of the receiving environment to assimilate the contaminants; and
 - d. avoid significant adverse effects on ecosystems and habitats after reasonable mixing;
 - e. use the smallest mixing zone necessary to achieve the required water quality in the receiving environment; and
 - f. minimise adverse effects on the life-supporting capacity of water within a mixing zone.
2. In managing discharge of human sewage, do not allow:
 - a. discharge of human sewage directly to water in the coastal environment without treatment; and
 - b. the discharge of treated human sewage to water in the coastal environment, unless:
 - i. there has been adequate consideration of alternative methods, sites and routes for undertaking the discharge; and
 - ii. informed by an understanding of tangata whenua values and the effects on them.
3. Objectives, policies and rules in plans which provide for the discharge of treated human sewage into waters of the coastal environment must have been subject to early and meaningful consultation with tangata whenua.
4. In managing discharges of stormwater take steps to avoid adverse effects of stormwater discharge to water in the coastal environment, on a catchment by catchment basis, by:
 - a. avoiding where practicable and otherwise remedying cross contamination of sewage and stormwater systems;
 - b. reducing contaminant and sediment loadings in stormwater at source, through contaminant treatment and by controls on land use activities;
 - c. promoting integrated management of catchments and stormwater networks; and
 - d. promoting design options that reduce flows to stormwater reticulation systems at source.
5. In managing discharges from ports and other marine facilities:
 - a. require operators of ports and other marine facilities to take all practicable steps to avoid contamination of coastal waters, substrate, ecosystems and habitats that is more than minor;
 - b. require that the disturbance or relocation of contaminated seabed material, other than by the movement of vessels, and the dumping or storage of dredged material does not result in significant adverse effects on water quality or the seabed, substrate, ecosystems or habitats;
 - c. require operators of ports, marinas and other relevant marine facilities to provide for the collection of sewage and waste from vessels, and for residues from vessel maintenance to be safely contained and disposed of; and
 - d. consider the need for facilities for the collection of sewage and other wastes for recreational and commercial boating.

Policy 21 requires 'priority' to be given to improving water quality in areas of the coastal environment where water quality has deteriorated so that it is having a significant adverse effect on ecosystems, natural habitats, or water-based recreational activities, or is restricting existing uses such as shellfish gathering. This policy also directs regional councils to identify such areas of coastal water and water bodies and include them in their plans. For example, the proposed Auckland Unitary Plan includes the spatial identification of degraded water areas. These were drawn from a report on marine degraded areas informed by benthic health, sediment chemistry and water quality monitoring programmes (see Figure 17.3).⁴⁴



The NZCPS sets out what local authorities need to do to effectively manage stormwater discharges, such as the one shown here at North Shore, Auckland, including adopting an integrated catchment management approach (Raewyn Peart)

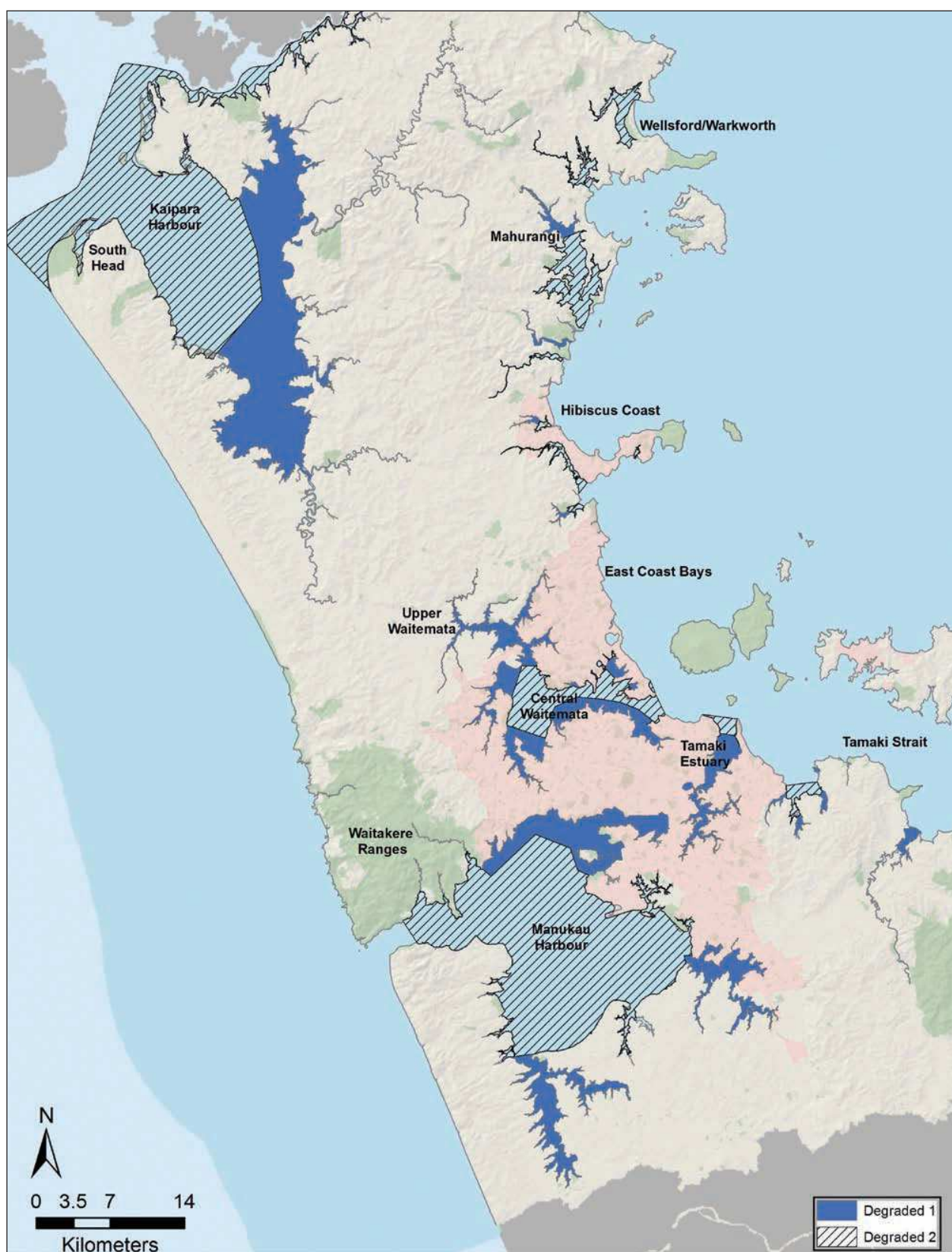


Figure 17.3 Map showing the marine degraded areas in Auckland (*Auckland Council*)

Through Policy 21, regional councils must require that stock be excluded from the coastal marine area, adjoining intertidal areas and other water bodies and riparian margins in the coastal environment, within a prescribed time frame. The 'Australian and New Zealand Guidelines for Fresh and Marine Water Quality' promulgate a water quality management framework that could be useful for the purpose of implementing Policy 21 of the NZCPS. These water quality guidelines contain 'trigger values'. If concentrations of contaminants are below these values there is unlikely to be any concern. If concentrations are above, further investigation is required to develop site-specific criteria and if concentrations are above the site-specific criteria, action is required. These guidelines provide an important reference for water quality management in New Zealand, particularly in regards to toxic contaminants, and they offer methods for setting limits on pollutant concentrations in coastal and marine environments.⁴⁵

Policy 22 requires that subdivision, use or development will not result in a significant increase in sedimentation in the coastal marine area, or other coastal water. This policy requires councils to collect baseline information with which to assess the current levels of sedimentation in coastal waterways and the marine area. They are also required to identify the likely environmental impacts of current sedimentation levels, monitor changes in sedimentation levels and impacts, and ensure that there is no significant increase in sedimentation in coastal waterways and the marine area. Councils will need to deal with sediment generation on a catchment basis to address all activities that contribute to sediment loadings in runoff and stormwater systems.

Policy 23 requires particular regard to be given to avoiding significant adverse effects on ecosystems and habitats after reasonable mixing when managing discharges to water in the coastal environment. This particular policy arose from the recommendation of the Board of Inquiry that water quality policies in the former NZCPS be strengthened. This was to ensure that decision-makers who are considering discharges to water in the coastal environment have proper regard to how to prevent or minimise adverse effects, particularly on ecosystems, habitats and the life-supporting capacity of water.⁴⁶

Policy 23 provides clear direction that discharges of untreated human sewage into coastal water should not be occurring, including those from wastewater system overflows. In addition, the discharge of treated sewage into the coastal environment should only be permitted where there has been adequate consideration of alternatives and the effects on tangata whenua values has been understood and considered. There must also be early and meaningful consultation with tangata whenua when developing plan provisions which provide for such activity.

Policy 23(4) sets out what local authorities need to be doing to effectively manage stormwater discharges. This includes adopting an integrated catchment management approach which links the management of stormwater networks with that of other activities within the catchment, reducing contaminant and sediment loadings at their source through such actions as placing controls on land use activities, adopting good design, and requiring stormwater treatment. In addition this subpolicy requires that cross-contamination between stormwater and sewage systems is addressed. The Department of Conservation has prepared some brief guidance on implementing Policy 23. This makes it clear that when considering the discharge of treated sewerage into the coastal environment, councils will need to consider the location, frequency, duration, volume, level of treatment and the extent to which the discharge is the best practicable option (within a relevant time frame).⁴⁷

Regional council planning

Regional councils have responsibility for controlling the use of land for the purpose of maintaining coastal water quality and coastal marine ecosystems⁴⁸ and controlling discharges into water bodies.⁴⁹ This includes managing the negative effects of sedimentation and pollutants which are derived from land-based sources and which end up being discharged into the marine area. Regional councils prepare regional policy statements and regional plans to address the issue. Some regional councils prepare separate plans which specifically address issues such as sedimentation and discharges from dairy farms to help them effect their obligations under the RMA. In practice it has proved easier to deal with point source pollution under the RMA than non-point source pollution.

There are two main approaches that regional plans take to avoiding or minimising impacts of land use on the marine area:

- One method employed by a number of regional councils is to classify coastal receiving waters and to apply appropriate water quality or sediment quality standards to those waters. This is a more 'effects-based' approach, in that any consequent land use activity could be expected to at least consider the effects of the activity on the relevant receiving water quality or sediment quality standard. This allows innovation on behalf of the consent holder as to what site management practices or controls could be implemented. Water quality standards which are linked to rules will have a more direct effect on outcomes than standards linked to policies.

- Another method is to focus more on the land use activity itself and apply methods to control or regulate these types of activities. For example, if a rural catchment is identified as having a negative impact on the ecological health of the marine area because of the high levels of sedimentation, and the main causes are identified as forestry and farming activity, the regional plan might include rules to mitigate these effects, such as requiring stock to be kept out of waterways and preventing harvesting in riparian areas.

These methods can be combined – such as those relating to land uses being linked to water quality standards.

Non-point source pollution poses more of a challenge for regulators because it is not caused by isolated and identifiable acts which are generally more easily controlled by the consent process. Regional councils are responsible for identifying the environmental effects of both urban and agricultural runoff, and implementing regional solutions and management. To discharge this duty, councils may produce technical guidelines for the management of stormwater systems and sediment control programmes. They may also set up incentive programmes and form community-based groups to encourage farmers, forestry companies and other users to alter the way they use their land to minimise the effects of runoff. There are opportunities to undertake stronger actions on these issues, including through new regional rules prohibiting intensification in some areas, requiring farm environmental management plans, and requiring planting/harvesting plans for forestry.



Regional councils are responsible for identifying the effects of runoff from forestry, farming and urban activities, such as those shown here at Kawakawa Bay, Auckland, and implementing solutions (Raewyn Peart)

Integrated catchment management plans

Catchment management plans are a tool to integrate the management of natural resources and the protection of community values. Such a management approach is an established concept used around the world. However, in the past they have tended to focus on the planning and management of a catchment. In recent years the focus has widened to include the coastal-receiving environment for that catchment.⁵⁰

Harbour and catchment management planning focuses on managing land and water together to achieve long-term outcomes which are beneficial to the wider environment. Such a tool integrates multiple stakeholder concerns, issues and objectives into one overarching plan, as highlighted by Figure 17.4. Such types of planning documents are non-statutory but have the capacity to inform and support statutory documents such as district and regional plans.



Figure 17.4 The wide range of issues addressed by the Whangamatā Harbour and Catchment Management Plans (Waikato Regional Council)⁵¹

Māori have long been advocates for planning, particularly in regards to coastal issues.⁵² There are a number of tools, including *taiapūre* and *mātaītai*, which are increasingly being used as part of managing a wider catchment. These tools are discussed in Chapter 4: Kaitiakitanga. More recently *iwi* have initiated catchment planning (see the third case study).

The success of catchment management planning lies in the engagement of the community and key agencies to support and implement the actions needed.⁵³ This is discussed further in the case study on Tairua Harbour and Catchment Management Plan below.

There are a number of best practice elements for catchment management planning that have been identified by research, including:^{54,55}

- Political leadership and multi-agency co-operation
- Cross-sector collaboration
- Inclusion of *iwi* and community aspirations and drivers
- Robust science
- Improved capacity building and action on the ground
- Provision of adequate resourcing for both planning and implementation phases
- Good governance and clear institutional frameworks
- Monitoring and evaluation of outcomes that leads to adaptive management
- The presence of a strong catchment manager or champion

A summary of the outcomes from the 10 year Integrated Catchment Management for the Motueka River programme identified a number of issues that were addressed during the programme⁵⁶ as shown in Figure 17.5.

Topic	Issues addressed
Allocation of scarce water resources among competing land and instream uses	• How do local land uses affect the availability of ground and surface water over the entire catchment? • What is the most defensible way to plan for the allocation of water resources among competing alternative uses? • How much water is required to maintain instream values? • How do economic returns affect landowners' and recreational uses of water?
Managing land uses in harmony with freshwater resources	• How is sediment mobilised to reach rivers, and what impacts does it have? • What information is needed to best manage river gravel allocation? • Why has the Motueka catchment trout population declined then partially recovered? • How can water quality be maintained or improved with intensifying land use? • Are there some fundamental solutions to water quality contamination? • Is riparian revegetation the 'silver bullet' for improving water quality?
Managing land and freshwater resources to protect and manage marine resources	• What are the risks to marine farming from activities on land? • What are the factors that increase or decrease the production and values of fish and shellfish? • Where are the faecal bacteria affecting marine farms coming from? • What are the relative effects of river flows (especially floods) on scallop and mussel production compared to activities like dredging and marine farming?
Integrative tools and processes for managing cumulative effects	• What are preferred development pathways to ensure continued sustainable management at catchment scale? • What are the opportunities for using modelling to predict cumulative effects of land use scenarios? • How can models help decision-makers balance environmental impacts alongside social, economic and cultural benefits when planning for further development?
Building human capital and facilitating community action	• What methods would best motivate environmental stewardship by catchment and community groups? • What methods can we use to promote effective interaction between scientists, resource managers, and the community? • How can iwi build their capability in the resource management process? • What are some 'off the wall' examples of innovative and enduring community engagement that we could use to facilitate community action?

Figure 17.5 Research issues addressed by the Integrated Catchment Management for the Motueka River
(Fenemor A, 2013)

Territorial authorities

Under section 31 of the RMA, territorial authorities are responsible for controlling the effects of land use outside the coastal marine area, and this includes downstream effects on the marine environment. Territorial authorities can prepare district plans to address these issues. Whereas regional councils tend to focus on catchment-wide issues, territorial authorities focus more on site development issues such as earthworks and the construction of impervious areas.

District plans can indicate the amount of impervious area (building or paving) that a property can have, and property owners who wish to exceed this limit must apply to the territorial authority for consent to do so. Territorial authorities and their local network operators also have responsibility for stormwater infrastructure in their area, such as maintenance and replacement of stormwater pipes, maintaining stormwater pumping systems and outfalls, construction of silt traps and detention ponds.



Constructed wetlands, as shown here in the Whisper Cove development at Snells Beach, Auckland, can help reduce the contaminant load in stormwater before it reaches the marine environment (Raewyn Peart)

Territorial authorities undertake monitoring and mapping to identify any issues associated with localised flooding and overland flow, as well as any pollution-related issues. The Local Government Act requires them to undertake assessment of the water and sanitary services in their district, including wastewater and stormwater, to assess amongst other things whether they pose a danger to public health.

Elements of best practice

Potential environmental effects	Elements of best practice
Agriculture	
Runoff from pasture causes increased sedimentation of the marine environment	• Exclude stock from waterways, wetlands and the coastal marine area • Create riparian buffer zones using native flora along waterways to protect the catchment from sediment, organic material and nutrients washed from pasture by the rain • Construct runoff management systems such as wetlands or pools along subsurface farm drains, where sediment and contaminants can settle • Restore old wetlands to create natural obstacles to sediment runoff • Construct farm tracks on sites where gradients are low • Reduce stocking densities or retire land on steep or highly erodible slopes • Develop and implement environmental farm plans • Provide financial incentives to encourage farmers to replant riparian zones with indigenous vegetation, and to retire land from productive use in particularly erodible areas
Nitrogen-nitrate and phosphate pollution derived from urine and fertilisers used to promote pasture growth encourage growth of algae and other marine vegetation, leading to eutrophication, oxygen depletion and ocean acidification	• Avoid using more fertiliser than can be taken up by the crop or pasture – soil nitrogen testing can indicate exactly how much nitrogen fertiliser a field requires, and therefore help avoid overfertilising • Ensure that catchments are not farmed overintensively, to avoid generating more pollution than the catchment can cope with • Use stand-off pads and barns for stock during wet seasons
Organic matter (animal faeces) enters the waterways, stimulating the activity of aerobic bacteria which deplete oxygen levels in the water	• Prevent stock from accessing waterways and the coastal edge • Have adequate farm treatment systems to cope with the volume and strength of effluent produced • Ensure land irrigation is not overused to avoid effluent sprayed onto fields running off
Forestry	
Felling of trees and construction of tracks results in increased runoff into the catchment	• Undertake tree planting, felling and track construction in plantation forests so that a riparian buffer is retained to protect the catchment from sedimentation • Avoid clear-felling on highly erodible soils • Avoid clear-felling large areas of land at one time, or in succession, in the same catchment • Install cut-off drains at regular intervals on tracks so that water runoff can be managed and directed away from waterways • Include in regional plans rules restricting the planting and harvesting of plantation trees from within a specified distance from rivers, streams and the coastal margin, and if necessary seasonally to avoid high rainfall times • Include in the regional plan rules controlling the harvesting of trees on steep or erodible slopes
Burning to clear land for forestry or after harvesting, and mechanical clearing of the plantation, cause intensive soil disturbance leading to increased runoff into the catchment	• Maintain riparian buffer zones along waterways to prevent large amounts of sediment from entering the catchment • Avoid highly invasive mechanical and burning techniques to clear land with a high erosion potential • Carefully manage the disposal of slash

Potential environmental effects	Elements of best practice
Urban stormwater	
In periods of heavy rain stormwater increases the volume of sewage so that wastewater treatment plants cannot treat it all	• Replace combined sewage/stormwater systems with separate networks • Monitor and maintain sewage pipes to prevent leakage
Earthworks generate increased sediment	• Prepare catchment management plans to protect natural channels, restore wetlands and provide stormwater retention ponds • Require an erosion and sediment control plan as part of building consent applications and include strict conditions in resource consents to require adequate sediment detention methods to be used. These may include chemical sediment control (flocculation), silt fences, decanting earth bunds or sediment retention ponds. • Undertake major building works at times of the year when rainfall is unlikely to be heavy • Retain existing vegetation wherever possible and effect revegetation as soon as possible • Stabilise disturbed areas and minimise the size of the area disturbed as much as possible • When considering zoning for plans, pay particular attention to the potential for the cumulative effects of earthworks in an area to cause significant sedimentation of the receiving marine environment
Urban sewage	
Wastewater treatment works fail to remove organic matter, nutrients and micro-organisms from the effluent before it is discharged into the catchment	• Use the best available technology (tertiary with ultra-violet pathogen treatment or similar) in wastewater treatment plants to ensure that the water discharged into the coastal marine area contains minimal amounts of organic matter, micro-organisms and nutrients • Plan the construction of wastewater treatment works so that they have the capacity to deal effectively with all the waste generated by the community, and with future or seasonal growth in the area. • Ensure sufficient capacity to avoid overflows of raw sewerage into the marine environment
Septic tanks	• Construct septic tanks only in areas where the soil is suitable • Construct septic tanks only where there is an adequate amount of land surrounding the property to cope with the amount of effluent generated • Adequately maintain septic tanks and soakage fields to ensure that they do not fail • Use the latest technology in designing effective septic tank systems
Sewage enters the stormwater system and from there enters the catchment untreated	• Prevent illegal connections • Include stormwater control measures such as permeable paving in new developments and modifications to existing urban areas • Adequately maintain piping • Use swales and planted areas as buffers to limit and filter stormwater runoff • Restore old wetlands to filter and limit the flow of stormwater to the sea
Industrial effluent	
Industrial processes result in the discharge of effluent into the catchment	• Use the best available technology to minimise pollution, and instigate programmes for the reduction of waste • Use constructed or restored wetlands as an alternative treatment and disposal system; they can be effective at removing nutrients as well as organic material
Heavy metal contamination in stormwater	• Phase in a prohibition on the use of unpainted zinc roofs • Reduce vehicle use through good urban design and provision of public transport, biking trails and pedestrian friendly routes • Require copper-free brake linings for vehicles and more chemical-friendly tyres
Figure 17.6 Best practice considerations for catchment activities	

Auckland's Sustainable Catchments Programme

The purpose of the Auckland Council's Sustainable Catchments Programme is to provide an integrated catchment planning and implementation framework to protect the long-term health of receiving environments through iwi and community interventions. With a byline of 'Weaving Science and Community into Action' the programme uses a range of tools to foster land use practice change including riparian restoration, education, creative engagement, farm plans, incentives, research and walkways.

The programme currently works in seven identified priority catchments across Auckland. These are: Mahurangi, Whangateau, Kaipara (Hoteo), Greater Tamaki, Henderson – Huruheru (Project Twin Streams), South Waitemata (Meola, Oakley and Motions) and Manukau (Papakura/Pahurehure). Two of these, the Mahurangi Action Plan and Project Twin Streams, are 10 years old this year.

Figure 17.7 provides some highlights of the programme.



Project Twin Streams, Auckland (Raewyn Peart)

Key area	Highlight
Regenerating blue-green areas	- More than 780,000 plants have been planted by Project Twin Streams since 2004. Over half a million of these have been planted by around 78,000 volunteers - Co-hosting planting days with Mahurangi College students and Whangateau Harbour Care to demonstrate riparian restoration best practice at sites in Whangateau and Mahurangi
Enhanced fisheries	- Annual shellfish monitoring workshops and field work in Whangateau with Mahurangi College students and Whangateau Harbour Care - Planting of riparian species providing stream shade and habitat for aquatic species along the Henderson Creek, Opanuku, Oratia, Swanson, Paremuka and Waikumete Streams, at La Rosa Reserve, downstream of Miranda Reserve and on the Meola and Oakley Creeks - Hoteo catchment work specifically targets areas which drain into high-value seagrass beds in the Kaipara Harbour located at the mouth of the Hoteo River
Manawhenua integration	- Trial site of the Ngāti Whatua holistic approach and management on the Anamata Stream - Mahurangi Catchment Strategic Plan written with and endorsed by Ngāti Whatua o Kaipara
Community ownership	- Community organisations manage work programme in Project Twin Streams catchments - Mahurangi community actively involved in planning implementation of the Mahurangi Action Plan strategy - Groups in Whangateau planning public land projects with council
Active land management	- Over 100 kilometres of stream and wetlands have been fenced and around 900 hectares of land have been protected since 2004 through grants across Mahurangi, Hoteo and Whangateau. The grants for fencing and riparian planting are provided to applicants in priority subcatchments to support land use management changes. - Ongoing active land management along the five Project Twin Streams catchments and in the South Waitemata catchment with multiple agencies and groups involved - Over 50 farm plans written to support rural land management
Knowledge generation	- Over 100 creative engagement art projects have been completed by the Project Twin Streams community - Erosion and Sediment Control Plans have been produced for the Hoteo, Mahurangi and Whangateau catchments to guide planning and implementation - Community open day to share science in the Hoteo catchment attended by over 100 people

Figure 17.7 Achievements of Auckland's Sustainable Catchments Programme

Tairua Harbour and Catchment Management Plan

Sedimentation of Tairua Harbour, on the east coast of the Coromandel Peninsula, has long been a concern for the local community and management agencies. A report by the Hauraki Catchment Board in 1977 stated that it seemed 'fairly certain that the tidal compartment is under attack' from the greatly increased sediment flow that had occurred over the previous 100 years. It continued, 'there is an urgent need for a comprehensive catchment scheme aimed ... at a plan of land use which will conserve native forest, and will create an appropriate compromise between 'production' and 'protection' for the purpose of controlling the entry of sediment into the river system'.⁵⁷

The Waikato Regional Council worked closely with the local community and stakeholders during the late 2000s to identify issues related to the health and well-being of the Tairua catchment and harbour and the preservation of environmental and community values. It was agreed that an integrated Tairua Harbour and Catchment Management Plan was needed to provide an assessment of current pressures and issues in the Tairua Harbour and catchment and to provide a practical strategy to alleviate these.⁵⁸ In addition, relevant matters identified in the Coromandel Peninsula Blueprint (a regional strategy for managing growth and development) would be addressed within the Plan. More detailed site-specific plans are to be developed with landowners and land managers relating to specific works and issues on private properties.

During the drafting of the Plan, consultation was undertaken with a range of stakeholder groups, agencies and landowners in an effort to ensure the issues, outcomes and works areas focus on what is needed by the community. Key current and potential issues for the Tairua Harbour and catchment identified by the Plan are:⁵⁹

- Declining aquatic and terrestrial habitat
- Evidence of a general decline in water quality, particularly in peak summer period
- The perceived and actual impact of forestry activities
- Potential for pine to pasture land conversion
- Mangrove expansion impacting on recreational use and other significant habitats
- Salt water paspalum expansion and impact on habitat and hydrology
- Impact of plant and animal pests
- Sedimentation rates in the harbour and surrounding streams
- Decline in fish numbers and species diversity in the harbour

Figure 17.8 provides an example of the way that issues are identified, methods to respond to these and the expected outcomes.



The Waikato Regional Council is preparing a harbour and catchment plan for Tairua to address sedimentation and decline in habitats within the harbour (Raewyn Peart)

Issue 1: Coastal land and water management (including mangrove management)

Key concerns	Potential methods	Expected outcomes
• Sedimentation of harbour and decreased access even during high tide • Increased use of harbour and sometimes conflicting users • Access to harbour and coastal environment • Expansion of mangroves (last decade) into recreational area and other significant habitats such as seagrass • Rapid expansion of saltwater paspalum • Decline in coastal habitats • A reduction in number of fish caught in the harbour • Shellfish toxins leading to gathering and consumption restrictions	• Review of dredging consent and possible increased dredging • Land management practices implemented up stream to alleviate sediment input • Preservation of coastal habitats through restoration works including saltwater paspalum control, planting and retirement • Ongoing Beachcare works • Appropriate mangrove removal consents gained and implemented • Saltwater paspalum control consent sought and implemented • Enforcement of navigation safety bylaws • Re-establishment of harbour committee with multiple stakeholders	• Improved harbour access • Improved habitat • Historical recreation access point regained • Line held on mangrove spread • Intact dunes systems providing buffers for coastal erosion, landscape values and habitat • Better communication between agencies and harbour users

Figure 17.8 Example of an issue and associated key concerns, potential methods to manage and expected outcomes from the Tairua Harbour and Catchment Management Plan (Waikato Regional Council⁶⁰)

An essential part of the Plan is implementation and monitoring. A detailed implementation plan provides associated actions for achieving outcomes. Where needed, these should be able to feed into different annual and long-term planning processes in order to allocate the required funding. Monitoring is also important, and should identify changes over time and enable comparisons to be made.



Expected outcomes from the Tairua Harbour and Catchment Management Plan include improved harbour access and habitat (Raewyn Peart)

Ōkahu Catchment Ecological Restoration Plan

The Ōkahu Bay catchment lies within the Tamaki Estuary of the Auckland Harbour. It is a small embayment that is now home to a diverse group of communities and is an important tourism destination within Auckland City. Historically it was the tribal base or papakainga of Ngāti Whātua Ōrākei,⁶¹ and remains an important location for the hapū.

More recently, Ōkahu Bay has suffered from a number of environmental impacts, including:⁶²

- Severely degraded ecological values through human colonisation and past grazing practices
- Highly modified urban ecology following lengthy human occupation
- Contamination from being the primary receiving environment for stormwater runoff from the wider catchment
- Halting of the natural coastal processes interacting between land and sea from the construction of Tāmaki Drive, including frequent flooding episodes
- Impacts of tidal influences on groundwater and resultant impacts on groundwater levels and Ōkahu Bay urupā and burials

In an effort to manage some of these impacts, Ngāti Whātua Ōrākei prepared the 'Ōkahu Bay Ecological Restoration Plan', which in addition implements the 'Whenua Rangatira Reserves Management Plan' and the 'Ngāti Whātua Ōrākei Heritage and Resource Management Kaupapa, Strategy and Policy 2010–2011'.⁶³

Visionary Statement

'Waters fit to swim in at all times, with thriving marine eco-systems that provide sustainable kaimoana resources to a Ngāti Whātua Ōrākei community who have strong daily presence in and on the bay as users and kaitiaki.'

The Plan has been designed in a highly practical manner to promote, develop and enhance Ōkahu Bay as the public face or gateway to the whenua rangatira while respecting its existing cultural and spiritual value to the tangata whenua and enhancing its relationship with the Waitematā.⁶⁴ Priorities include brokering strategic relationships with agencies, expanding the work of Ōkahu Rākau Bush Care and extending Ko Te Pūkākī (ecological restoration project) throughout the catchment.

Both quantitative measures (testing the effect of stress associated with anthropogenic activities on ecology) and qualitative research (where the Ngāti Whātua Ōrākei and the Ōrākei Community define ecological health and social indicators) are being used to monitor the outcomes from the Plan.⁶⁵ Their work has included the first hapū-based mussel reef restoration programme in New Zealand.



Ōkahu Bay is a popular recreational destination for a wide range of activities (Raewyn Peart)

Further reading and references

- Auckland Council, 2012, *Marine sediment contaminants: Status and trends assessment 1998 to 2010*, Auckland Council Technical Report TR2012/041, Auckland Council, Auckland
- Auckland Regional Council, 2001, *Auckland regional plan: Sediment control*, Auckland Regional Council, Auckland
- Board of Inquiry, 2009, *Proposed New Zealand Coastal Policy Statement (2008) Board of Inquiry report and recommendations, Volume 1: Findings, recommendations and recommended NZCPS (2009)*, Wellington
- Brake L and R Peart, 2013, *Caring for our coast: An EDS guide to managing the coastal environment*, Environmental Defence Society, Auckland
- Brake L and R Peart, 2013, *Treasuring our biodiversity: An EDS guide to the protection of New Zealand's indigenous habitats and species*, Environmental Defence Society, Auckland
- Bryant S, B Handyside and M Dunphy, 2007, *Forestry operations in the Auckland region: A guideline for erosion and sediment control*, Auckland Regional Council Technical Publication 223, Auckland Regional Council, Auckland, available at <http://www.aucklandcouncil.govt.nz/EN/planspoliciesprojects/reports/technicalpublications/Pages/technicalpublications201-250.aspx>
- Carbines M, J Walker, M Cameron and M Vaughan, 2013, *Identification of degraded marine receiving environments*, Auckland Council working report WR2013/009, Auckland Council, Auckland
- Environment Canterbury, 2007, *Erosion and sediment control guideline: A better way of managing earthworks and the environment*, Environment Canterbury, Christchurch, available at <http://ecan.govt.nz/publications/General/FullErosionandSedimentControlGuideline.pdf>
- Feeney C, W Allen, A Lees and M Drury, 2010, *Integrated catchment management: A review of literature and practice*, Report prepared for the Ministry for the Environment, Environmental Communications Limited, Auckland
- Fenemor A, 2013, *A summary of outcomes and selected formal publications from the Integrated Catchment Management (ICM) research programme: 2000–2011*, Landcare Research, Auckland
- Hauraki Gulf Forum, 2004, *Hauraki Gulf state of the environment report 2004*, Hauraki Gulf Forum, Auckland, available at <http://www.arc.govt.nz/environment/coastal-and-marine/hauraki-gulf-forum/hauraki-gulf-state-of-the-environment-report.cfm>
- Hauraki Gulf Forum, 2008, *Hauraki Gulf state of the environment report 2008*, Hauraki Gulf Forum, Auckland, available at <http://www.arc.govt.nz/environment/coastal-and-marine/hauraki-gulf-forum/hauraki-gulf-state-of-the-environment-report.cfm>
- Hauraki Gulf Forum, 2011, *State of our Gulf – state of the environment report 2011*, Hauraki Gulf Forum, Auckland, available at www.aucklandcouncil.govt.nz
- Ministry for the Environment – <http://www.mfe.govt.nz/issues/water/water-quality.html>
- Ministry for the Environment – <http://www.mfe.govt.nz/issues/waste/>
- Ministry for the Environment, 1997, *State of the environment report 1997*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/ser/ser1997/index.html>
- Ministry for the Environment, 2003, *Sustainable water management: A handbook for smaller communities*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/publications/waste/wastewater-mgmt-jun03/html/part1-section2.html>
- Ministry for the Environment, 2007, *State of the environment report 2007*, Ministry for the Environment, Wellington, available at <http://www.mfe.govt.nz/environmental-reporting/soe-reports/>
- Ministry for the Environment, 2012, *Recreational water quality in New Zealand*, Ministry for the Environment, Wellington
- Morrison M A, M L Lowe, D M Parsons, N R Usmar and I M McLeod, 2009, 'A review of land-based effects on coastal fisheries and supporting biodiversity in New Zealand', *New Zealand Aquatic Environment and Biodiversity Report No. 37*, Ministry of Fisheries, Wellington, available at <http://www.fish.govt.nz/en-nz/Publications/NIWA+review+of+land-based+effects+on+coastal+fisheries+and+ecosystems.htm>
- Ngāti Whātua Ōrākei, 2012, *Ōkahu bay ecological restoration plan*, Compiled by Richelle Kahui-McConnell, Ngāti Whātua, Auckland
- Schwarz A, R Cole, R Budd, R Taylor, J Hewitt, L Hunt, J Shima, N Phillips, 2006, 'Impacts of terrestrial runoff on the biodiversity of rocky reefs', *New Zealand Aquatic Environment and Biodiversity Report No. 7*, Ministry of Fisheries, Wellington
- Waikato Regional Council, 2011, *Tairua Harbour and catchment management plan*, Waikato Regional Council Technical Report 2011/40, Waikato Regional Council, Hamilton
- Williamson B, G Lewis, G Mills and B Vant, 2003, 'Contaminants on the coast', in J R Goff, S L Nichol and H L Rouse (eds), *The New Zealand coast: Te tai o Aotearoa*, Dunmore Press, Palmerston North

Endnotes

- 1 Ministry for the Environment, 1997
- 2 Ministry for the Environment, 2007
- 3 <http://www.crc.govt.nz/publications/Consent%20Notifications/upper-waitaki-submitter-evidence-maf-technical-paper-review-riparian-buffer-zone-effectiveness.pdf>
- 4 Ministry for the Environment, 1997, 40
- 5 See for example <http://www.trc.govt.nz/publications/regional+plans/guidelines/farming/MARGINS.HTM>
- 6 Ministry for the Environment, 2007
- 7 Ministry for the Environment, 2007
- 8 Ministry for the Environment, 1997, 48
- 9 Ministry for the Environment, 2007, 266
- 10 <http://www.aucklandcouncil.govt.nz/EN/environmentwaste/naturalenvironment/Documents/streamsideplantingguide.pdf>
- 11 Hauraki Gulf Forum, 2004, Chapter 5
- 12 <http://www.mfe.govt.nz/environmental-reporting/freshwater/river/nutrients/sources.html>
- 13 Auckland Regional Council, 2004, quoted in Hauraki Gulf Forum, 2004, 74
- 14 www.biotechlearn.org.nz
- 15 Ministry for the Environment, 1997, 42
- 16 <http://www.mfe.govt.nz/issues/waste/wastewater/>
- 17 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=10779897
- 18 http://www.aucklandcouncil.govt.nz/EN/environmentwaste/researchmonitoring/beach_water_quality_safeswim/Pages/latest_monitoring.aspx
- 19 http://www.watercare.co.nz/SiteCollectionDocuments/AllPDFs/PDFs%20v2%20111010/Wastewater_history.pdf
- 20 Ministry for the Environment, 1997, 27
- 21 <http://www.austlii.edu.au/au/journals/AUIntLawJl/2003/5.pdf>
- 22 <http://www.niwa.co.nz/freshwater-and-estuaries/freshwater-and-estuaries-update/freshwater-update-55-october-2012/multiple-effects-of-fine-suspended>
- 23 Hauraki Gulf Forum, 2004, 67
- 24 Morrison M A et al, 2009
- 25 <http://www.aucklandcity.govt.nz/council/documents/technicalpublications/TP211%20Effects%20of%20suspended%20sediment%20concentrations%20on%20suspension%20and%20deposit%20feeding%20marine%20macrofauna.pdf>
- 26 <http://www.fish.govt.nz/en-nz/Environmental/Land-based+Effects+on+Fisheries+and+Ecosystems/Suspended+Sediment+and+Sedimentation.htm>
- 27 <http://www.fish.govt.nz/en-nz/Environmental/Land-based+Effects+on+Fisheries+and+Ecosystems/Suspended+Sediment+and+Sedimentation.htm>
- 28 <http://www.fish.govt.nz/en-nz/Environmental/Land-based+Effects+on+Fisheries+and+Ecosystems/Suspended+Sediment+and+Sedimentation.htm>
- 29 Hauraki Gulf Forum, 2004, 57
- 30 <https://www.mfe.govt.nz/issues/managing-environmental-risks/hazardous/about/organochlorines/what-are.html>
- 31 <https://www.mfe.govt.nz/laws/trade/china/documents/pops-in-pastoral-farming-and-horticulture-part1.pdf>
- 32 http://www.fish.govt.nz/NR/rdonlyres/2C2AB64F-ABA8-4F6B-954F-BFE76C102C38/0/hectors_discussion_document_april_07.pdf
- 33 http://www.fish.govt.nz/NR/rdonlyres/2C2AB64F-ABA8-4F6B-954F-BFE76C102C38/0/hectors_discussion_document_april_07.pdf
- 34 http://www.fish.govt.nz/NR/rdonlyres/2C2AB64F-ABA8-4F6B-954F-BFE76C102C38/0/hectors_discussion_document_april_07.pdf
- 35 <http://www.aucklandcouncil.govt.nz/EN/planspoliciesprojects/reports/technicalpublications/Documents/monitoringresearchquarterly201404.pdf>
- 36 http://resources.ccc.govt.nz/files/Water_2166_DissolvedInorganicNitrogenInEstuary-docs.pdf
- 37 http://www.nzherald.co.nz/nz/news/article.cfm?c_id=1&objectid=10801492
- 38 Hauraki Gulf Forum, 2011
- 39 <http://www.seachange.org.nz/PageFiles/144/John%20Zeldis%20-%20Sea%20Change%20presentation%20to%20SWG3.pdf>
- 40 <http://www.aucklandcouncil.govt.nz/EN/environmentwaste/naturalenvironment/Pages/stateoftheaurakigulf.aspx>
- 41 http://www.foodsmart.govt.nz/elibrary/collecting_shellfish_keeping.htm
- 42 Ministry for the Environment, 2003
- 43 National Policy Statement for Freshwater Management 2014, Objective A2(b)
- 44 Carbines M et al, 2013
- 45 <http://www.mfe.govt.nz/publications/water/anzecc-water-quality-guide-02/index.html>
- 46 Board of Inquiry, 2009, 9
- 47 <http://www.doc.govt.nz/conservation/marine-and-coastal/coastal-management/nz-coastal-policy-statement/policy-statement-and-guidance/sewage-discharges/>
- 48 RMA, section 30(1)(c)
- 49 RMA, section 30(1)(f)
- 50 <http://www.aucklandcity.govt.nz/council/documents/technicalpublications/TR2009-092.pdf>
- 51 <http://www.waikatoregion.govt.nz/Services/Regional-services/River-and-catchment-management/Looking-forward-to-a-healthier-harbour-in-Whangamata/>
- 52 Feeney C, W Allen, A Lees and M Drury, 2010
- 53 Waikato Regional Council, 2011
- 54 <http://www.aucklandcity.govt.nz/council/documents/technicalpublications/TR2009-092.pdf>
- 55 Feeney C et al., 2010
- 56 Fenemor A, 2013
- 57 <http://www.waikatoregion.govt.nz/PageFiles/18177/Tairua%20-%20Healthy%20catchment%20%20healthy%20harbour%20newsletter.pdf>
- 58 Waikato Regional Council, 2011
- 59 Waikato Regional Council, 2011
- 60 <http://www.waikatoregion.govt.nz/PageFiles/18177/Tairua%20-%20Healthy%20catchment%20%20healthy%20harbour%20newsletter.pdf>
- 61 <http://www.justice.govt.nz/tribunals/waitangi-tribunal/resources/teaching-aids/resource-kits/orakei>
- 62 Ngāti Whātua Ōrākei, 2012
- 63 Ngāti Whātua Ōrākei, 2012
- 64 Ngāti Whātua Ōrākei, 2012
- 65 <http://www.aucklandcouncil.govt.nz/EN/AboutCouncil/representativesbodies/haurakigulf/forum/Documents/hgf2014okahucatchmentrichellekahuimconnell.pdf>



18

Conclusion

Conclusion

New Zealand is blessed with a very large, varied and productive marine environment. With its estuaries, fiords, reefs, soft sediments, seamounts and hydrothermal vents, it is home to a huge variety of marine life and significant mineral and petrochemical resources. The marine area is of enormous value to New Zealanders in terms of economic, social and cultural values. It is of particular importance to Māori who have a long and particularly close historical association with the sea and harvesting its bounty. It is therefore critical that we manage it well.

Decisions affecting the country's marine environment are made within a complex and overlapping legislative framework which has developed over the last 50 years. The most recent addition is the EEZ Act, passed into law in 2012, which applies an environmental management framework to a range of deep-sea activities. This Guide has sought to demystify this framework by clearly describing how decisions are made and by whom. It has also identified how members of the public and other parties can become involved in the various decision-making processes.

Many industries are dependent on accessing the marine environment and its resources and these have been described in detail in Part 2 of the Guide. These activities can potentially cause adverse effects on the marine environment, but in many cases these can be addressed through the application of best practice approaches. The Guide identifies a number of ways in which this can be achieved.

With the implementation of best practice in respect of marine and catchment activities, supported by proactive marine spatial planning and a comprehensive network of marine protected areas, New Zealanders should be able to continue to enjoy their oceans for generations to come.



Blue maomao, Poor Knights Islands, Northland (Daniel Sharp)

Index



Great Mercury Island, Coromandel (Raewyn Peart)

Entries in **bold** refer to figures.

A

acidification 14, 384
acoustic alarms 191
acoustic monitoring, passive 265
adaptive management
 in EEZ Act 54–5
 in minerals mining 265, 269–70
 and precautionary approach 37
 and sand mining 242–3, 247
Admiralty Bay 211
adverse effects
 avoiding significant 242, 315, 391
 on existing interests 72
 requirement to avoid 30–1, 33–4, 36–7
agriculture, discharges from 375–7, 385, 391–2
Akaroa Marine Reserve 164
algae 11
 and excess nutrients 384
 non-indigenous 95
amenity values 34–5, 52, 212, 230, 239
anchoring, impact on seafloor 351
animal waste 375–6, 384–5
antifouling paints
 in aquaculture 216
 copper leaching from 308
 and minerals mining 265
 and sand mining 244
 on vessels 329
Antipathella fiordensis 11
ANZEEC (Australia and New Zealand Guidelines for Fresh and Marine Water Quality) 268
Aotearoa Fisheries Limited 81, 91n3, 196
aquaculture
 in areas of significant conservation value 126
 best practice for **220–2**
 in coastal plans 29
 and ecosystem health 140
 environmental impacts of 209–14
 Government adviser on 24
 guidance for monitoring 223
 land-based 203
 locations of **204**, 212
 Māori ownership of 26, 79, 81
 monitoring environmental change around 225
 as pest pathway 100, **210**
 planning process for 216–18
 production and revenue of **203**
 protection for existing rights holders 220
 species farmed 208–9
Aquaculture Ecological Guidance Package 209
aquaculture law reforms 82, 216–17
Aquaculture Management Areas 216, 228n49
Aquaculture New Zealand 173, 207
 codes of practice 222

Aquaculture Settlement Areas 82
Aquaculture Steering Group 80
Aquaculture Strategy and Five-year Action Plan to Support 203
aquarium trade 100
Aquatic Environment and Biodiversity Annual Review 185, 192
archaeological sites 26, 43, 304, 311
The Area 20
Areas of Ecological Importance 65, 289, **290**
areas of particular concern 146
ascidians *see* sea squirts
Asian clam 96, 98, 107
Asian paddle crab 96, 98
Astrolabe Reef 332, 337–8
Auckland
 marine degraded areas in **390**
 waterfront promenade 310
Auckland Central Interceptor Project 380
Auckland Council, and Hauraki Gulf Forum 38
Auckland Islands 13, 42, 182, 360
Auckland Regional Sediment Contaminant Monitoring Programme 383
Auckland Sustainable Catchment Programme **397**
Auckland Unitary Plan
 combining district and regional plans 166n1
 marine degraded areas in 389
 public process in **154**

B

bacteria
 environmental impact of 384
 and oxygen depletion 385
ballast water
 import standards for 105–6
 as pest pathway 96, 100
Banks Peninsula, marine mammal sanctuary 42
barnacles, and aquaculture 210
bathymetry 237, 255, 269, 308
Bay of Islands, marine mammal tourism in 354, 360
Bay of Plenty Regional Council 90, 111, 332, 337, 348
beaches
 general activities on 355
 replenishing 90, 231, 306, 362
 urban 345, 355
beach lifestyle 346–7, 356
beach seining 178
benthic habitats *see also* seabed
 and aquaculture 221
 in ecological backbone 142
 and fishing methods 175, 177–9, 189, 192–3
 and minerals mining 267, 269
 reducing impact on 192–3
 and sand mining 236, 243
benthic protection areas 124, 130, **193**, 267, 269
best available information 40, 52–3, 58, 61, 189
big game fishing 352
bilge water
 contaminated 210
 oily 334, 367
billfish 180, 352
biochemical oxygen demand 385
biodiversity
 and aquaculture 215
 area-based protection of 118–20, 128, 130
 and canal developments 309
 in EEZ Act 58
 in Fisheries Act 40
 highest levels of 10
 in marine consent process 70
 in NZCPS 30–1, 242, 287
 threats to 95
biodiversity offsetting 45–6, 244, 265
biofouling
 and aquaculture 210
 portable treatment facility for 112
 preventing 106
 and ultrasound 108
 use of term 99
biogenic habitat 62, 261, 263
biogenic reefs 8, 128, 236
biogeographic regions 128–31, **129–30**, 140, 155
biological diversity *see* biodiversity
bioregionalisation 142
biosecurity
 and aquaculture 210
 and marine infrastructure 315
 and minerals mining 259–60
 in NZCPS 32
 and oil and gas exploitation 286
 public involvement in 158
 and *Rena* disaster 338
 and vessels 327
Biosecurity Act 1993
 consultation in 158
 functions under 100
 jurisdiction of 41
 pathways in 99
 and regional councils 26
 threatened species in 71
Biosecurity Response Knowledge Base 108–9
biosolids 378
bird baffle 191
birds *see* seabirds
bird strike
 and marine recreation 358
 and minerals mining 265
 and sand mining 235
bivalve molluscs, beds of large **63**
black-backed gulls 42, 175
blowout preventer 280, 282, 295
blowouts 282

blue cod 352
 Blue Flag certification 324n53
 blue-green areas 397
 bluenose 177, 352
 blue-water yacht cruisers 350
 BMSY (below maximum sustainable yield) 181
 Board of Inquiry
 and cross-boundary activities 72
 and nationally significant proposals 166n3
 public involvement in 153
 and resource consents 159
 on water quality 391
 boardsailing see windsurfing
 boardwalks 355, 362
 boating
 in marine reserves 122
 recreational 349–51, 362, 365
 solo voyages 355
 boat owners
 and marine biosecurity 108
 numbers of 350
 bottlenose dolphins 13, 158, 354
 bottom-dwelling animals 8
 bottom trawling 175, 179–80, 267, 269
 brachiopod beds **64**
 brown algae 11
 Bryde's whale 13, 211, 265, 330, 336, 353
 bryozoans 11
 and biogenic reefs 8
 impact of fishing on 177, 179
 non-indigenous 95
 and seamounts 9
 bryozoan thickets **62**
 burial at sea 57
 bycatch
 incentives to minimise 182–3
 of non-target species 174
 reducing 190–2
 and trawling 179
 bycatch management measures 187

C

cable protection zones 127–8
 cables
 in EEZ Act 57, 67
 on the high seas 20
 calcareous gravel 8
 calcareous tube-worm thickets **62**
 canal developments 301–3, 305, 309–10, 319
 Canterbury Basin 277
 Cape Rodney-Okakari Point Marine Reserve 118, 122, 134, 348
 capital dredging 90, 165, 302–3, 306
 carbon sequestration 45, 282, 285, 301, 306
 Careys Bay 311
 Caroline Bay 308
 catchment management
 and aquaculture 142
 best practice for **395–6**
 integrated 316, 389, 391, 393, **394**, 397
 and natural character 33
 catchment management plans 392, **393**, 396
 Catlins Coast 8, 42, 158
Caulerpa taxifolia 96, 98, 107
 causeways 301–2, 305–6
 caution, favouring 52–4, 59, 247
 Cawthron Institute 205–7, 209, 225
 chaetopteridae worm fields **62**
 Chapman, Val 133
 Chatham Islands, reefs near 7–8
 Chatham Rise
 and carbon sequestration 45
 fish harvest from 171
 oceanic meeting point over 6
 phosphate mining on 253, 257, 267–70, **268**
 seamount habitats of 9
 chemical pollutants 215
 Chinese mitten crab 96, 98, 107
 Chinook salmon see king salmon
 Clean Boating Programme 365
 Clean Marinas programme 309
 Clifford and Cloudy Bay, marine mammal sanctuary 42
 climate change
 in marine consent process 71
 and marine recreation 347, 349
 potential impacts of 45
 and shipping activities 331
 closed systems, removing sand from 237, 240, 243
 coastal and marine classification system 128
 coastal cells **328**
 coastal environment
 natural character of 33–4
 preservation of 28–9
 coastal erosion
 and climate change 45
 and marine infrastructure 304, 308
 and sand mining 237–8
 and vessels 331
 coastal hazards 30, 45, 316
 coastal land
 management of 23, 26
 reclamation and declamation of 303, 314
 see also reclamation
 coastal marine area
 activities permitted in 29
 large fixed structures in 302, 305
 limits of 24
 shipping control within 332
 coastal permits 26
 coastal processes
 and artificial reefs 348
 impact of marine infrastructure on 304, 308, 316
 natural adjustment for 37

 and sand mining 235, 237, 244
 coastal shipping 301
 cobalt crust mining 255, 257–8, **263**, **266**
 cockles 12, 145, 177, 307
 Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations 65, 265, 287–8
 coliform bacteria 384
 commercial fishing
 and costs of research 184
 and ecosystem health 140
 in EEZ 130
 environmental impacts of **180**
 general spatial pattern of inshore trawling **172**
 legal definition of 181
 management framework for **183**
 Māori customary rights to 83
 Māori ownership of 79
 and Māori resource protections 87–9
 and marine reserves 24, 42, 122
 and minerals mining 261
 monitoring of 187
 and oil and gas exploitation 284
 overview of 171
 structure of 173
 sustainability measures for 182–3
 commercial shipping
 Oil Pollution Levy on 293
 voluntary code for 333–4, 336
 common dolphins 13, 179, 187, 191, 198, 383
 common marine and coastal area 44, 85–6
 Conservation Act 1987 23
 and Hauraki Gulf 39
 and kaitiakitanga 86
 and marine recreation 361
 strategies and plans under 155
 conservation management plans 152, 155
 Conservation Services Programme 187
 conservation values, areas of significant 120–1, 126–7
 contaminants
 areas trapping 309, 380
 discharge of 35, 219, 315–16, 321, 332, 386, 389
 remobilising in dredging 307, 315
 sources of increased 373
 continental shelf
 in EEZ Act 51–2
 impact of sand mining on 236
 jurisdictional powers over 19–20
 mineral rights on 40
 Convention on Biological Diversity 55, 117
 Convention on the Prevention of Marine Pollution by Dumping of Wastes or Other Matter 332
 Cook Strait 8, 327
 copper
 in antifouling agents 216, 308, 329
 in urban runoff 376

corals
 black 9, 11, 42
 cold-water 259
 soft 177, 179
 stony 42, **62**, 269
 use of term 11

core sampling 256

Coromandel Peninsula
 marina proposals on 301
 mussel farming on 203

Coromandel Peninsula Blueprint 398

crab, invasive species of 95–6

Craft Risk Management Standard for Biofouling on Vessels Arriving in New Zealand 106, 265

crayfish
 commercial fishing of 171
 at Goat Island 118, 133
 review of stocks 185

cross-boundary activities 72

Crown Minerals Act 1991
 classification of activities 264, 287
 and marine reserves 122
 public consultation under 158
 and sand mining 232, 241

cruise ships 90, 356–7, 368

cultural heritage, marine 43

cultural values, protection of 119

cumulative effects
 in EEZ Act 58
 of land-based pollutants 381
 management of 141
 in marine consent process 70–1
 of marine infrastructure 317
 in NZCPS policy 37

cumulative impacts
 of minerals mining 265, 268
 of sand mining 236, 238

customary fishing rights 83, 86, 89, 217, 246

customary marine title groups
 in EEZ Act 56, 61, 67, 81
 in marine consent process 69, 72
 and resource consents 86

customary marine title order 91n7

customary rights, protected 28, 44, 72, 86, 91n8

D

Danish seining 178

DDT 375, 382–3

dead zones 384

Decision-Making Committee *see also* Trans-Tasman Resources Ltd
 in EEZ Act 51–3
 and marine consents 160

deemed values 182–3

deep-sea mining **259**, **260**, 265

deep-sea vents 59, 262

deepwater fisheries 171, 179, 184

Deepwater Group Limited 173, 191–2

Deepwater Horizon disaster 282, 293

deepwater port developments 303

deforestation 373, 375

demersal fishes 284

Department of Conservation
 on biodiversity offsetting 45
 function of 23–4
 guidance on aquaculture planning 218
 guidance on marine infrastructure 313, 315
 and kaitiakitanga 86
 and marine mammals 42, 124
 and marine reserves 122
 and MPI observer programme 187
 policies and strategies 155
 on precautionary approach 37
 and seismic surveying 65

dewatering 266

diffusers 266–7

dioxins 304, 382–3

discharge management plan 68, **291**

discharges
 from minerals mining 264, 266
 from oil and gas exploitation 287–8
 from ports 315
 power to regulate 25, 57, 59
 within catchments 26, 32, 374

discretionary activities
 and marine consent process 67
 under EEZ Act 51, 60

district plans
 consents required under 29
 and Freshwater NPS 387
 and land-based pollutants 394
 and NZCPS 29
 port zones in 314
 public participation in 153
 and RMA 26, 38, 51
 zoning in 140

Dive! Tutukaka 367

diving
 deep 355
 and ecosystem health 140
 for fish 176–7, 349
 in marine reserves 122
 recreational 348–9, 367

dogs
 restricting 362
 walking 355

dolphin dissuasive devices 191

Dolphin Encounter Kaikōura 366

dolphins
 impact of fishing on 42, 175, 178–9
 impact of sand mining on 239
 reducing bycatch of 191
 regulations on contact with 361

species of 13

dolphin tourism 353–4, 366

Doubtful Sound, dolphins at 354

downwelling 14

dredging
 in Auckland region **302**
 in coastal marine area 302–3
 discharge of organic material from 104
 environmental impact of 175, 179, 237, 306–9
 public opposition to 165

drilling lubricant 285

drilling platforms 281, 285–6, 294

drilling rigs *see* oil rigs

drill pipes 280

drillships 280, 286

dumping
 from minerals mining 264
 from oil and gas exploitation 288
 power to regulate 57, 59
 from vessels 332

dune vegetation, trampling on 355

dusky dolphins 13, 211, 354, 360, 366

E

earthworks 29, 375, 377, 394, 396

East Cape 7–8, 64

East Coast Basin 277

Eastern Sea Farms Limited 206

ecological backbone 139, 142–3

economic wellbeing 52, 54

ecosystem health 140, 386

ecosystems
 rare and vulnerable 71
 resilience of 118

ecosystems approach 14, 141, 188–9

ecosystems-based management 139

ecosystem services 117–18, 139, 142

ecotourism 140, 345

eelgrass 11, 31

eels
 farming 209
 longfin 14
 stock assessment 184

EEZ (Exclusive Economic Zone)
 benthic protection areas in 124, 130
 Biosecurity Act in 100
 declaration of 171
 ecotourism in 345
 hydrocarbon deposits in 277
 jurisdictional powers over 19–20
 sand mining in 243
 size of 5
 trawling in 179, 193

EEZ Act
 activities restricted under **57**
 and climate change 45
 cross-boundary activities under 72

- differences with RMA 51
 - and discharges from vessels 333
 - and EPA 25
 - kaitiakitanga in 81
 - on marine consent requirements 70–1, 160
 - and marine recreation 358
 - minerals mining under 264
 - on oil and gas exploitation 287–8, 291
 - purpose and principles of 51–3, 55–6
 - regulations under 37, 58–60, 66–7
 - and sand mining 232, 242, 245, 247
 - status of activities under **60**
 - time periods for consent hearings 69
 - EEZ Amendment Act 57
 - EEZ Permitted Activities Regulations 61, 266
 - EEZ Regulations, preparation of 157
 - effluent 377–80, 385, 395–6
 - emergency response procedures 68, 266, 291–2
 - Emissions Trading Scheme 25
 - endemic species 9–10, 12, 194
 - endemism 64, 265, 294, 353
 - environment, EEZ Act definition of 52
 - Environment Act 1986 24
 - environmental assessment, initial 61, 66
 - Environmental Codes of Practice 209, 222
 - environmental compensation 71
 - Environmental Defence Society, and Akaroa Marine Reserve 164
 - environmental health, benchmarking of 118
 - Environmental Legal Assistance Fund 165
 - environmental management plans 266, 387, 392
 - environmental performance objectives 55, 247
 - environmental principles, in Fisheries Act 40
 - environmental protection, favouring 53–4, 59
 - Environmental Protection Authority Act 2011 25
 - Environmental Reporting Bill 25
 - environmental standards, national 156–7
 - Environment Court
 - on adaptive management 54
 - appeals to 126, 154–5, 165
 - on aquaculture 219
 - enforcement orders from 163
 - and private plan changes 162
 - and resource consents 159
 - on reverse sensitivity 311
 - on sand mining 235, 237, 240, 242–3
 - and surfing 348
 - Environment Southland 108, 368
 - EPA (Environmental Protection Authority) 25
 - and cross-boundary activities 72
 - decision-making criteria of **70**
 - and discharges from vessels 333
 - imposing consent conditions 71
 - information principles for 52–4
 - Māori Advisory Committee for 81
 - in marine consent process 51, 67–70, 72–3
 - and minerals mining 270
 - notification of 66–7
 - and oil and gas exploitation 288, 291
 - and resource consents 159
 - Escherichia coli* 384, 386
 - estuaries 6–7
 - in Freshwater NPS 387
 - land-sourced pollutants in 380
 - European shore crab 96, 98, 107
 - European Union, marine spatial planning in 140
 - eutrophication 14, 304–5, 384, 395
 - evidence, briefs of 70
 - exclusion areas
 - for cables 120
 - for mining 235, 259, 261, 269
 - Exclusive Economic Zone and Continental Shelf (Environmental Effects - Discharge and Dumping) Regulations 2014 288
 - Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 see EEZ Act
 - exfiltration 380
 - existing activities 59, 319
 - existing interests
 - Māori 54, 56, 81
 - in marine consent process 68–72
 - in regulation development 58
 - Exotic Pests and Diseases Hotline 107–8
 - exploration
 - in minerals mining **256**
 - in oil and gas exploitation 279–80
 - explosions 57
 - extended continental shelf, size of 5
- F**
- farms see agriculture; aquaculture
 - Fergusson reclamation 306
 - ferries, fast 331
 - ferromanganese crust mining see cobalt crust mining
 - fertilisers
 - contamination from 375–6
 - DDT in 383
 - phosphate in 254
 - Finfish Aquaculture Environmental Code of Practice 222
 - finfish farming 207–8, 210, 212, **215**, 216
 - finfish feed 207, 210, 215–16
 - Fiordland
 - rocky reefs near 7
 - spread of *Undaria* to 99
 - Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 42–3, 126
 - Fiordland marine area 43, 126
 - Fiordland marine biosecurity partnership 108
 - Fiordland Marine Guardians 42–3, 108, 126
 - fiords 9, 43, 368
 - Firth of Thames
 - aquaculture in 208, 214, 224
 - nitrogen overload in 384
 - fish
 - divers and swimmers feeding 349
 - endemic 12
 - gills of 260, 382
 - reducing bycatch of 192
 - Fisheries (Kaimoana Customary Fishing) Regulations 1998 83, 87
 - Fisheries (South Island Customary Fishing) Regulations 1999 83, 87
 - Fisheries 2030 strategy 184
 - Fisheries Act 1996
 - activities managed under 181
 - caution requirements in 53
 - consultation rights under 151
 - fisheries plans under 183–4
 - kaitiakitanga in 80
 - and Māori resource protection 87–8
 - Māori rights under 82–3
 - and marine mammal protection 42
 - and Ministry for Primary Industries 24
 - and regional councils 25
 - restricting areas under 124
 - sustainability measures under 39–40, 157
 - Fisheries Inshore New Zealand Limited 173
 - fisheries management, levels of complexity in **188**
 - fisheries planning structure **184**
 - fisheries plans 152, 183–4
 - fisheries quota, Māori control of 81
 - fisheries resources
 - impact of aquaculture on 212, 214
 - impact of oil and gas exploitation on 284
 - Māori customary protection of 120
 - fisheries spillover 118–19
 - fishery compliance officers 352
 - fish farms 24, 37, 207, 211, 215, 221–2; see also aquaculture
 - fishfinders 350
 - fishing activities
 - areas closed to **124**
 - best practice for 188, **189–90**
 - categories of 40
 - environmental impact of 174–5, 186–7
 - and marine mammals 42
 - and protected species 42
 - reporting illegal 163
 - fishing down 174
 - fishing industry see commercial fishing
 - fishing methods
 - effects of specific 176–9, **180**
 - restrictions on 88, 120, 127, 157, 182
 - sketches of **176**
 - fishing nets 174–5
 - fish spawning areas 243, 265, 294, 318
 - fish stocks
 - assessment of 184–5
 - identification of 181
 - management of 182

- mining impact on 260
- recreational impact on 352
- setting target sizes for 186
- FLA1 quota management area 181
- flatfish 171, 178, 181
- flat oyster (*Tiostrea chiensis*) 207
- floating pontoons 301–2, 305
- Florida Keys National Marine Sanctuary 140
- flushing 213, 305, 308–9, 329
- Food and Agriculture Organization International Guidelines 269
- food gathering, customary 83
- food pellets *see* finfish feed
- food waste 220, 288, 356–8, 368
- food webs 14
- foreshore and seabed, Māori rights in 84–5
- Foreshore and Seabed Act 2004 84
- Forest and Bird 56, 164, 242
- forestry
 - runoff from 375–6, 387
 - and Tairua Harbour 398
- freshwater limits 386
- Freshwater NPS (National Policy Statement for Freshwater Management) 29, 32, 386–8
- fur seals 13
 - and aquaculture 211
 - and fishing methods 175, 177, 179
 - and tourism 360
- fur seal sanctuaries 84

G

- garbage 288, 332–3, 357
- geological features, protection of 119
- geothermal energy 64
- gill netting 174, 176, 178, 180
- glass eels 209
- Global Industry Response Group 295, **296**
- Goat Island Marine Discovery Centre 120
- gold mining 378
- Great Barrier Reef Marine Park 119, 140
- Great South Basin 277
- greenhouse gas emissions 45, 277, 284, 331, 353
- Greenpeace of New Zealand Inc. v The EPA* [2013] NZHC 3842 68
- green seaweeds 7, 10–11
- Greenshell Mussel Industry Environmental Code of Practice 222
- grey-faced petrels 42
- groundwater 373, 377–9, 400
- Guardians of Fiordland's Fisheries and Marine Environment *see* Fiordland Marine Guardians
- Guardians of Paku Bay Association 165

H

- habitat definitions 61
- habitat-forming species *see* biogenic habitat
- hake 171, 179, 182, 268
- hāpuku

- farming 208
- longlines for 177
- hard substrate 62, 258–9, 263, 267, 269
 - artificial 265–6
- harmful aquatic organisms *see also* marine pests
 - and aquaculture 210, 219
 - controlling spread of 32, 104
 - in NZCPS 32
 - pathways for 99–101
- harmful substances, in Maritime Transport Act 41; *see also* contaminants
- Harvest Strategy Standard 2008 182, 186
- Hauraki Gulf
 - cable protection zones in **127**
 - and carbon sequestration 45
 - marine protected areas in 131
 - marine spatial plan for 139, 141, 144–5
 - see also* SeaChange
 - mussel beds of 224
 - protocol for shipping in 336
 - rocky reefs in 7
 - sand mining in 240
 - ship strike in 330
 - whale-based tourism in 353
- Hauraki Gulf Forum 125
 - membership of 38
 - state of the environment reports 145
- Hauraki Gulf Marine Park **38**, 39
- Hauraki Gulf Marine Park Act 2000 38–9, 125, 145
- Hauraki Gulf Transit Protocol for Commercial Shipping 330
- hazardous substances 25, 333, 335
- Health and Safety in Employment (Petroleum Exploration and Extraction) Regulations 2013 292
- Health and Safety in Employment Act 1992 292
- health warnings, permanent 380
- heavy metals
 - and dredging 307
 - environmental impact of 382–3
 - land-based sources of 376, 378
 - and marine infrastructure 321
 - and sand mining 245
- Hector's dolphin
 - and aquaculture 211
 - and DDT 383
 - impact of fishing on 178, 180, 189
 - reducing bycatch of 191
 - and sand mining 238
 - threatened status of 31
- helicopter overflight 239
- hepatitis 384
- Heritage New Zealand Pouhere Taonga 26
- Heritage New Zealand Pouhere Taonga Act 2014 43, 86
- heritage orders 43, 154
- High Court
 - appeals to 67, 72, 155, 159, 164–5

- and customary fishing rights 86
- judicial review by 124, 151, 153
- High Hazards Unit 292
- high seas 19–20, 257
- historic heritage, preservation of 26, 28
- hoki 171, 179, 268
- holiday homes 355–6, 364
- horse-mussel beds 8
- hull fouling 99, 108, 112, 210, 350
- human sewage 328, 384–5, 389, 391
- hydrothermal vents 9–10, **64**, 142, 254, 262
- hyperbenthos 267

I

- impact assessment 65, 68–9, **109**, 160
- impervious area 394
- import health standards 100, 105, 158, 210
- indigenous fauna 28, 30, 219, 315
- indigenous finfish, farming 208
- indigenous habitats 35
- indigenous vegetation 28, 30–1, 219, 315, 395
- individual transferable quotas 182
- industry, water pollution from 377, 385
- information, adequacy of 69–70, 246
- information principles 52–4, 56, 160
- inshore fisheries 171
- inshore sand mining 230–2, 234–5
- Intergovernmental Panel on Climate Change 45
- International Association of Oil and Gas Producers 295–6
- International Convention for the Control and Management of Ships' Ballast Water and Sediments 2004 106
- International Maritime Organization 25, 106, 336
- International Oil Pollution Prevention Certificate 291
- International Seabed Authority 20, 255, 257, 266
- International Union for Conservation of Nature and Natural Resources 30–1, 268
- intertidal areas, loss of 307, 309–10
- invertebrates, in marine habitats 7–8, 10–11
- iron sand 231–2, 234, 241, 245
- irrigation 377
- IUCN World Commission on Protected Areas 131
- iwi
 - and biosecurity 109
 - and Department of Conservation 86
 - legislative recognition of 81
 - marine farm space allocated to 82
- iwi and hapū
 - coast areas of special significance 88
 - consultation under RMA 56, 153–4
 - and establishment of taiāpure 162
 - harvesting seabirds 42
 - involvement in commercial sectors 79
 - marine management role of 26, 81, 86
 - and marine protected areas 119, 122
 - in marine spatial planning 143
 - notification under EEZ Act 61, 67, 69, 81

and oil and gas exploitation 287
and South Taranaki Bight sand mining 246
iwi for 187

J

jack-up rigs 280, 285
Japanese kelp (*Undaria pinnatifida*) 95, 97, 99, 210
Jellicoe historical precinct 311
jellyfish 11
jetskis 350
judicial review 68, 154, 158, 163–4

K

kahawai 12, 175, 178, 352
Kaikōura, whale-based tourism at 353, 360, 366
Kaikōura (Te Tai o Marokura) Marine Management Act 2014 84, 126, 131
Kaikōura Marine Area 84
Kaikōura Marine Strategy 131
kaimoana areas 142, 240, 286, 309, 338
Kaipara Harbour
oyster seed from 206
sand mining in 236, 238–40, 242
sedimentation in 382
tidal energy turbines in 45, 302
Kāi Tahu ki Otago Natural Resource Management Plan 309
kaitiakitanga
in coastal environment 26
and commercial ventures 79
description of 79, **80**
legislative provisions for 81–3, 86
Kapiti Coast, biogenic reefs off 8
Kapiti Marine Reserve 122
Kapuni field 281
kayaking 350, 361
kelp 7, 14
kelp forests 118, 133, 382
Keppel Islands 119
Kermadec Islands 10, 12, 24, 102–3, 130, 208
killer whale see orca
kina 11–12, 118, 177, 349
kina barrens 118, 133, 175
kingfish, yellowtail 177, 208
king salmon 203–4, 207
kitesurfing 347

L

landscapes, natural 36, 239, 246, 304–5, 318; see also outstanding natural landscapes
land-sourced pollutants 380, 386
land use
intensification of 373, 387, 392, 394
managing impact of 391–2
and water quality 32
leathery sea squirt (*Styela clava*) 97, 99, 104, 210
lichens 11

lifecycles 14
Limeburners Bay 231
ling 171, 177, 179, 268
London Convention 55, 318
London Protocol 291
Long Bay-Ōkura Marine Reserve 120
longlining 175–8, 180, 190, 192
Lyttelton Port 110, 301, 306, 309

M

Maari hydrocarbon field 281, 288
mackerel, jack 171, 175, 179, 191
macro-algae beds **64**
Mahaanui Iwi Management Plan 309
mahi tapu 80
Mahurangi Action Plan 397
maintenance dredging 302–3, 306, 318
Managing Our Oceans discussion document 74
manawhenua 120, 144, 397
manganese nodules mining 253, 255, 257–8
mitigation and management strategies for **266**
system for **262**
Mangawhai-Pākiriri sand mining 232, 235, 237, 240
Mangere sewage ponds restoration 322
Mangere Wastewater Treatment Plant 380
mangrove expansion 398
mangrove habitats 7–8
mangrove removal, unlawful 163
Māori Advisory Committee, in EEZ Act 56, 81
Māori Commercial Aquaculture Claims Settlement Act 2004 82
Māori Fisheries Act 1989 83
Māori Fisheries Act 2004 81
Māori Heritage Council 26
Māori language (te reo Māori) 69, 80
Māori values, and sand mining 240
Māori see also iwi and hapū; tangata whenua
marina developments 103, 165, 302–3
marinas
and biosecurity 32, 101, 107, 110–11
dredging for 303
environmental impact of 305, 309, 350
pressure for 301
and public access 310
and recreational boating 351
marine activities
classification of 51
competition between 139–40, 142
marine adventure challenges 355
Marine and Coastal Area (Takutai Moana) Act 2011 44, 85, 311
marine biodiversity offsetting 45
marine biosecurity
legislative framework for 100
national leadership for 95
Marine Biosecurity Porthole 96–7
marine consent
activities required for 57, 59
conditions on 71, 73
exercising 72
lapse or cancellation of 73
for sand mining 242
marine consent applications
administering 51
case law on completeness 68
hearings for 69–70, 72, 161
matters taken into account **70**
notification and consultation in 56, 69, 81
marine consent process **67**, 68–9
public involvement in **160–2**
marine disposal 292, 306
marine ecosystems
disruption of 175
impact of marine infrastructure on 305
vulnerable 62, 189, 193–4
marine energy projects 45
marine environment
impact of land management on 79
interconnection of 139–40
recreational users' awareness of 349
marine farms see aquaculture
Marine Fee 368
marine habitats
diversity of 6–7, 10
protection of 23
marine heritage 43, 304, 311, 319
Marine High Risk Site Surveillance programme 96–7, 111
marine infrastructure 301
best practice for 317, **318–19**
in EEZ Act 67
environmental impact of **304**, 305–6, 308–10
and holiday homes 356
and public access 310–11
and swimming 346
Marine Invasive Taxonomic Service 97, 107
marine investigations 107
marine mammal observers 65, 265
marine mammals 13
harm to 163
and heavy metals 383
human contact with 353–4, 358–60
impact of aquaculture on 211
impact of fishing on 175, 177–8, 182, 352–3
impact of oil and gas exploitation on 285, 288–9
and minerals mining 260, 265, 268
reducing bycatch of 191
and sand mining 238, **239**
and ship strike 330, 351
voluntary operational procedures for 191–2
marine mammal sanctuaries 124
establishment of 23, 42
and oil and gas exploitation 288–9
public consultation on 158

- Marine Mammals Impact Assessment 65
- Marine Mammals Protection Act 1978 42, 289
 - establishing reserves under 124
 - and Hauraki Gulf 39
 - Ministers responsible under 23
 - population management plans under 155
 - and tourism 358–9
- Marine Mammals Protection Regulations 1992 359–61
- marine management agencies **18, 22, 23**
- marine management areas 266
- marine management legislation **27–8**
- marine non-indigenous species 95, **96**
- Marine Oil Spill Response Strategy 334
- marine parks 120, 125, 144
- marine pests
 - and Biosecurity Act 41
 - default lead decision-maker for **102–4**
 - examples of 96–7
 - identifying and responding to incursions 107–8
 - impacts of **97–8**
 - management of **105**
 - and minerals mining 259
 - and oil and gas exploitation 286
 - pathways for 99–100, **210**
 - preventing arrival 105–6
 - public role in surveillance 108
 - and recreational boating 350
 - surveys for 101
 - use of term 95
- marine pollution *see also* land-sourced pollutants
 - management of 25, 332–3
 - from vessels 106
- marine protected areas 117
 - benefits of **117, 118–20**
 - guidelines for selecting **131**
 - management tools for 128
 - mechanisms to create **120–1**
 - network of 128, 131–2
 - policy on 128, 130
 - public involvement in 155
 - special legislation for 126
- Marine Protected Areas: Classification, Protection Standard and Implementation Guidelines 130
- Marine Protection Areas Policy Statement and Implementation Plan 155
- marine protection planning forums 155
- marine protection plans, development of 131
- Marine Protection Rules
 - and Maritime Transport Act 41, 158
 - and oil and gas exploitation 291
 - and pollution prevention 41
 - process of developing **334**
 - public participation in 158
- marine protection tools 128
- marine recreation
 - activities for **344, 345**
 - best practice for 362, **363–4**
 - environmental impact of 345, **346**
- marine reserves 122
 - benefits of 119–20
 - and biodiversity 118–19
 - and customary fisheries rights 86
 - establishment of 23, 41
 - extent of 140
 - judicial review of decisions on 163
 - process to establish **122–3**
 - protection levels of 128
 - public involvement in creation 155, 158
 - reporting offences within 163
 - visitor impact on 362
- Marine Reserves Act 1971 41
 - authorities under 23–4
 - creation of reserves under 122
 - review of 123
- Marine Reserves Regulations 1993 122
- marine space, tools for allocating **217**
- marine spatial planning 139
 - application of 140–1
 - development of 142
 - differences from land 140
 - ecosystem-based 139
 - and EEZ Act 58
 - elements of 142–3
 - potential benefits of 141
- marine structures
 - abandonment of 214
 - illegal 163
- marine title, customary 72, 86, 91n7
- maritime boundaries, treaties on 20
- maritime jurisdictions **19**
- Maritime New Zealand 25
 - and marine consent applications 69, 81
 - and oil spill response 292–3
 - and *Rena* disaster 337
- Maritime Transport Act 1994
 - discharges under 57, 291–2
 - on harmful substances 41
 - and Maritime New Zealand 25
 - and oil spill response 292–3
 - and pollution from vessels 333
- Marlborough Sounds
 - aquaculture in 203–4, 206, 211, 214, 216, 218–20
 - marine pests in 99, 111
 - vessel wake in 331
- MarLIN method 61
- MARPOL (International Convention for the Prevention of Pollution from Ships) 55, 261–3, 332
- mātaihai reserves
 - and catchment management 393
 - establishment of 83–4, 87
 - in Hauraki Gulf 125
 - and protected areas 120
 - public involvement in 158
- Matata Wastewater Scheme 310
- mātauranga Māori 143, 338
- Māui A hydrocarbon field 281
- Māui's dolphin 13
 - and fishing methods 178
 - listed as threatened 31
 - protection measures for **289**
 - reducing bycatch of 191
 - and sand mining 238–9, 242
- maximum sustainable yield 181
- May Fisheries Assessment Plenary Report 185
- McCallum Bros 237, 240
- mean high water springs 48n5
- methane hydrates 255, 258–9, 263
- Milford Sound 345, 354
- Mimiwhangata Marine Park 125
- mineral extraction, phases of **232**
- mineral mining 253, 256, 265
- mineral resources
 - allocation of 28, 241, 253, 264, 287
 - in RMA 28
 - sites of 253, **254**
 - types of 254, **255**
- minerals mining
 - best practice for **265**
 - components of 262
 - environmental impact of **259, 260, 261–3**
 - exploration tools for **256**
 - methods of 255–9
- Minerals Programme for Minerals 241, 264
- minerals programmes, public consultation on 158
- mining 262; *see also* minerals mining; sand mining
 - in marine reserves 122
 - pollution from 378
 - and RMA 28
- mining rights allocation
 - for minerals mining 264
 - for oil and gas exploitation 287
 - for sand mining 241
- Minister for Primary Industries 23
 - duties of 24
 - fisheries management advice for 185–7
 - and marine mammal protection 42
 - role in taiāpure and rāhui 88
 - and sustainability measures 39
- Minister for the Environment, and EPA 25
- Minister of Conservation
 - duties of 23–4
 - and Hauraki Gulf Forum 38
- Minister of Energy and Resources 22, 24, 264
- Minister of Transport 24–5, 41, 121, 334
- Ministry for Primary Industries
 - and aquaculture 205, 209
 - best practice standards for 182
 - biosecurity surveillance programme **107, 108**

establishment of 24
 and fisheries sustainability measures 157
 and marine biosecurity 95–6, 100
 and marine protected areas 128
 and marine reserves 23
 observer programme 187
 research commissioned by 184
 Science Working Groups 184
 surveillance programme 101
 Sustainable Farming Fund 207
 Ministry for the Environment 24–5
 Ministry of Business, Innovation and Employment 24, 225, 241, 277
 mixing zone 315–16, 389
 mobile offshore drilling units 280
 mooring areas 362
 moraines, glacial 146, **147**
 moratoria 216, 359–60, 366
 morse-code leatherjacket 12
 Motiti Island 337
 motor vehicles, driving on beaches 355
 Motueka River, catchment management for 393, **394**
 Mount Maunganui
 artificial reef at 348
 beach at 355
 Mount Maunganui Mātaihai Reserve 89
 mud, in oil drilling 280, 282
 Muriwhenua report 83
 mussel farming 205–6
 conversion of 219
 environmental impact of 210–14, **213**
 locations of 203
 off-shore 206
 and *Pyura doppelganger* 109
 structures for 214
 mussels
 giant vent 10
 green-lipped 88–9, 98, 203, 205, 214, 224
 mussel spat 109, 205

N

National Fisheries Plans 183–4
 national importance, matters of 28, 86, 159
 National Pest Management Plan of Action 100, 110
 National Plans of Action 186
 National Policy Statement for Freshwater Management 29, 153, 386
 National Policy Statement on Renewable Electricity Generation 45, 153
 national significance, matters of 25, 38–9
 native rock oyster 206
 natural character
 and aquaculture 212, 218, 220–1
 in NZCPS 33–5
 and oil and gas exploitation 287
 protection of 28, 44
 and sand mining 235, 240, 242, 244, 246
 natural currents, alteration by aquaculture 214
 natural features 33–4, 36, 125, 246; *see also* outstanding natural features
 natural gas 253, 277, 282, 331; *see also* oil and gas exploitation
 natural resources, in Conservation Act 23
 Nature Watch Charters 366
 Newbury principles 71
 New Zealand Biodiversity Strategy 2000 37, 117, 128
 New Zealand Conservation Authority 23
 New Zealand fairy tern 13, 31
 New Zealand King Salmon Company Limited 218, 225
 New Zealand Marina Operators Association 309
 New Zealand marine area
 accessibility for recreational use 345
 importance of 404
 international obligations in 55
 Māori ownership of resources in 80
 productivity of 14
 size of 5
 topography of **5**, 6
 New Zealand maritime jurisdictions 19–20, 21
 New Zealand Petroleum and Minerals 24
 and minerals mining 264
 and oil and gas exploitation 277, 287
 and sand mining 241
 New Zealand Surf Break Protection Society 347
 New Zealand Threat Classification System 30–1, 71
 Next-Generation Sequencing 225
 Ngāi Tahu *see also* Kāi Tahu
 and Akaroa Marine Reserve 164
 and Fiordland marine area 126
 fisheries claims by 83
 Ngāi Tahu Fisheries Settlement Limited 81
 Ngā Motu/Sugar Loaf Islands Marine Protected Area 119, 125–6
 Ngāti Whatua, and Auckland catchment management 397, 400
 Ngā Whenua Rāhui kawenata 44
 night setting 191
 Ninety Mile Beach, mussel spat from 205, 214
 nitrogen, overload of 384
 nitrogen-nitrate pollution 376–7, 395
 NIWA (National Institute of Water and Atmospheric Research)
 and marine biosecurity 96–7
 and sensitive environments 61
 non-commercial fishing *see also* recreational fishing
 Māori customary 83, 182
 in marine reserves 83, 122
 non-point source pollution 374–7, 391–2
 non-target species
 fishing impact on 174–5
 reducing impact on 196
 and specific fishing methods 177–8

Northern Pacific seastar 96, 107
 Northland, rocky reefs near 7
 Northland Plateau 9
 Northland Regional Council 104, 108–10, 112
 nutrient recycling 14, 118
 nutrients, excess 14, 378, 384
 NZCPS (New Zealand Coastal Policy Statement)
 and aquaculture 212, 218–20
 and climate change 45
 and coastal environment management 29–30
 development of 23
 and land-based pollution 388–9, 391
 and marine biosecurity 104
 and marine infrastructure 312–16
 and marine recreation 357
 objectives and policies 30–7
 and oil and gas exploitation 287
 public involvement in 153
 and sand mining 241, **242**

O

ocean currents **5**, 6, 9, 140, 260
 oceanic meeting point 6
 ocean ranching 227n20
 offal, batching 190–1
 offshore installations, pollution from 25, 288, 291–2, 333
 offshore islands, marine recreation on 349
 offshore sand mining 232, **234**, 243
 Ōhope Beach 355
 oil
 seabed extraction of 253, 277
 see also petroleum
 waste 357
 oil and gas exploitation
 amenity impacts of 286
 best practice for 293–6
 discharge and dumping activities **288**
 environmental impacts of 282–6, **283–4**
 health and safety management 292
 minimising environmental impacts **294**
 stages of **279**, 280–1
 Oil Pollution Levy 41, 293, 334
 oil rigs 280, **281**
 decommissioning of 281–2, 285
 towing 99, 286
 oil spill preparedness 41, 68, 293, **296**, 334–5
 oil spill response 292–3
 controls regulating 37, 334
 and *Rena* disaster 337–8
 oil spills
 environmental and social sensitivity to **328**
 environmental impacts of 282, **283**
 from minerals mining 259, 265
 from vessels 328–9
 Ōkahu Bay 400
 OMV New Zealand Limited 288

Operational Guidelines for New Zealand's Harvest Strategy Standard 186
 Ōpunake Beach 348
 orange roughly 12, 171, 174, 179
 orca 13, 245, 330, 353
 organic material, disposal of 32, 103–4, 385
 organochlorines 304, 382–3
 Oruarangi Creek and Mangere Lagoon 322
 Ostreid herpesvirus 96, 207
 outdoor recreation 343
 outstanding natural features, in NZCPS 28, 36, 218, 315
 outstanding natural landscapes
 and aquaculture 212, 218–19
 in NZCPS 36–7, 317
 and oil and gas exploitation 287
 and sand mining 242
 oxidation ponds 322, 378
 oxygen depletion 14, 213, 215, 384–5, 395
 oyster farming 206–7, 212, 214
 Oyster Industry Code of Practice 222
 oyster racks 213–14, 221

P

Pacific Community, Secretariat of 265
 Pacific Ocean 5–6, 255
 Pacific oysters 95, 203–4, 206, 214
 PADI (Professional Association of Diving Instructors) 348–9
 Pākinī Beach 232
 Paku Bay 165
 Parengarenga 232
 pāua
 diving for 177
 farming 203–4, 212–13
 Paua Industry Council Limited 173
 Pauanui 301, 356
 PCBs (polychlorinated biphenyls) 382–3
 peer reviews, independent 69
 penguins 13, 178
 permitted activities
 aquaculture excluded from 216
 under EEZ Act 51, 57, 59–61
 Permitted Activity Regulations 56
 pesticides 383
 pest management 100–1
 petroleum *see also* oil and gas exploitation
 from continental shelf 20
 under EEZ Act 37, 51, 66, 287–8
 petroleum basins 277, **278**
 phosphate mining 253, 256, 261
 phosphate pollution 376
 phosphorite nodules 254, 267
 phosphorus, excess 376, 384
Phyllochaetopterus socialis 62
 phytoplankton 10
 impact of aquaculture on 213–15
 and minerals mining 260
 productivity of 14
 Pilot Bay 90, 99, 111
 Pine Harbour, dredging at 306
 Pine Harbour Marina 307
 pinnipeds 13, 353
 pipelines
 in EEZ Act 57
 on the high seas 20
 pipi 12, 90, 307, 322, 382
 plankton 10
 plastics, disposal of 330, 333, 356, 358
 point source pollution 374, 377–8
 policy statements
 national 152–3
 regional *see* regional policy statements
 pollutants *see* contaminants
 pollution allowances, tradable 387
 pollution hotline 163
 pollution incidents
 preventing 25
 reporting 163
 Poor Knights Islands
 Rotoma spill near 328
 shipping to avoid 334, 367
 tourism at 345, 348–9, 367
 population management plans 42, 152, 155
 Port Companies Act 1988 90
 Port Gore 218, 220
 Port of Tauranga
 dredging at 78, 90, 303
 Port Zone for 314
 Port Otago 303, 311
 ports 301
 capital dredging at 303
 and coastal development 313
 discharges from 389
 environmental impact of 305
 port zones 314
 post-activity report 66–7
 potting 175–7, 180, 352
 Pouto 238
 precautionary approach
 and EEZ Act 53
 and minerals mining 242–3
 NZCPS on 30, 37
 and oil and gas exploitation 287
 predator nets 211, 216
 pre-hearing dispute resolution 69
 pre-operational baseline monitoring 55, 247
 private plan changes 127, 162
 prohibited activities 51, 60, 287–8, 333
 Project AWARE Foundation 349
 Project Manukau 322
 Project Twin Streams 397
 propeller wash 351

prospecting
 for minerals resources 255
 and sand mining 232
 under Crown Minerals Act 40
 under EEZ Act **66**
 protected customary rights groups
 in marine consent process 69, 72
 and resource consents 86
 under EEZ 56, 61, 66–7
 protected customary rights order 91n8
 protected species
 areas frequented by 265
 harm to 163
 impact of fishing on 175, 187
 legislation identifying 42
 working groups on 185
 pseudofaeces 213
 public access
 managing 316–17
 and marine infrastructure 304, 310
 as matter of national importance 28
 NZCPS on 37
 public involvement, opportunities for 151, **152–3**, **159**
 public liability insurance 71
 publicly-initiated processes 162
 pumping stations 379–80
 purse seining 42, 175–6, 178, 180, 192
Pyura doppelgangera 98, 108, **109**

Q

QMS (Quota Management System)
 and aquaculture 217–18
 establishment of 171
 and fish stocks 181
 impact on Māori 83
 and mussel spat 205
 and non-target species 174
 recreational fishing in 352
 qualitative environmental performance objectives 55, 247
 Queensland Marine Fish Habitats Offset Policy 46
 quota management areas 87, **181**, 182, 261, 284

R

radioactive matter 332
 rāhui 86–7, 120
Rainbow Warrior 348
 rangatiratanga, and kaitiakitanga 80
 reasonably foreseeable needs 313
 receiving waters 379, 391
 reclamation
 and areas of significant conservation value 126
 environmental impact of 305
 in NZCPS 30, 314
 at Tairua Marina 165
 under RMA 28

- recreation, human health for 386
 - recreational fishing 351–3
 - and Akaroa Marine Reserve 164
 - in Fiordland 42
 - in Hauraki Gulf 144–5
 - illegal 349
 - Māori customary 83
 - and Māori customary protections 84, 89
 - in marine parks 125
 - in mātaihai reserves 89
 - and sustainability measures 157
 - in taiāpure reserves 84
 - and total allowable catch 181–2
 - recreational fora 187
 - recreational significance, areas of 243, 294
 - recreational vessels 99, 328, 335, 363; *see also* boating, recreational
 - red seaweeds 11
 - reefs, artificial 347–8
 - regional coastal plans
 - and aquaculture 208, 216–17
 - areas of conservation value in 126–7
 - formation of 26
 - and marine heritage sites 43
 - and marine infrastructure 312, 314, 317
 - and marine pollution 332–3
 - and marine protected areas 128
 - and marine recreation 357
 - and NZCPS 29
 - and oil and gas exploitation 287
 - public involvement in 153
 - and sand mining 241
 - and spatial planning 141
 - regional councils
 - and coastal shipping 333
 - dealing with offences 163
 - and land-based pollution 387, 389, 391–2
 - and marine biosecurity 95, 100, 104, 110
 - and marine environment 25–6
 - and marine recreation 362
 - and oil spill response 292
 - and RMA 29
 - and waste from boating 350–1
 - regional parks 26
 - regional plans
 - breaches of rules 163
 - land-based pollution in 387, 391–2
 - regional policy statements
 - land-based pollution in 387, 391
 - and NZCPS 29, 32, 34–6
 - public involvement in 153
 - and RMA 25–6, 38, 72
 - regulations
 - criteria to take into account **58**
 - public participation in making **156**
 - regulatory rounds 185–6
 - Reinga-Northland Basin 277
 - re-injection 281
 - relocation of animals 266
 - remotely operated vehicles (ROVs) 279, 353, 358
 - Rena disaster 89, 293, 328, 332, 337–8
 - Rena Long-term Environmental Recovery Plan **338**
 - renewable energy 45, 140, 313
 - replacement consent 71
 - 'Requirements for Vessels Arriving in New Zealand' 101
 - Research and Science Information Standard 186
 - reserve land 44, 125
 - Reserves Act 1977 26, 44
 - resource consent
 - failure to notify 163
 - in Hauraki Gulf 38
 - joint application with marine consent 72
 - and marine biosecurity 104
 - public involvement in 159
 - renewing 220
 - for sand mining 241
 - Resource Management (Marine Pollution) Regulations 329, 333, 357
 - Resource Management Act 1991 (RMA)
 - and aquaculture 216
 - and climate change 45
 - and EEZ Act 51–2, 72–3
 - environmental compensation under 71
 - and Hauraki Gulf Act 38
 - Māori-related provisions in 29, 81–2
 - and marine biosecurity 104
 - and marine consent 72
 - and marine heritage sites 43
 - and marine infrastructure 312
 - and marine pollution 332, 386–7
 - and marine recreation 357–8
 - maritime limits of jurisdiction 37
 - national environment standards under 157
 - and oil and gas exploitation 287
 - policies and plans for 153–4
 - public participation under 151, 159
 - purpose and principles of 28
 - and sand mining 241
 - spatial planning in 141
 - and statutory agencies 23–6
 - Response Prioritisation Tool 108
 - reverse sensitivity 311
 - Revive our Gulf initiative 224
 - Rhode Island Ocean Special Area Management Plan 146
 - rhodolith beds **63**
 - Rio Declaration on Environment and Development 37, 53
 - riparian restoration 397
 - risk analysis, in import health standards 105
 - rock lobster *see* crayfish
 - Rock Lobster Industry Council 173
 - rock pool fish 12
 - rocky reefs 6–7, 12
 - roof runoff 377
 - roost sites 351
 - rubbish
 - and marine infrastructure 304, 308, 321
 - and marine recreation 346–7, 349
 - from vessels 330
 - runoff
 - and aquaculture 212
 - contaminants in 374–5, 383
 - into estuaries 6
 - managing 321, 376, 392
 - nutrient 267, 376
 - in NZCPS 388, 391
 - in regional plans 26
- ## S
- Sabella* *see* Mediterranean fan worm
 - salmon farms 207
 - conversion of mussel farms into 219
 - environmental impact of 211
 - freshwater 227n20
 - monitoring environment around 225
 - sand mining
 - allocation of rights 241
 - best practice for 243, **244**
 - case law on sustainability 237
 - cessation due to erosion 238
 - environmental impacts of **235**, 236–7
 - marine consents for 52
 - and natural character and amenity 239–40
 - prospecting and exploration **233**
 - sand plume *see* sediment plume
 - Sanford Limited 196, 205, 220
 - scallops
 - impact of dredging on 179
 - movement of 12
 - and sedimentation 382
 - science, and fishing industry 190
 - science working groups 184–5
 - scientific information, certainty of 242
 - scientific research
 - on the high seas 20
 - under EEZ Act **66**
 - SCUBA 348
 - seabed
 - customary rights in 84
 - disturbance of 57, 66
 - seabed dredging 303
 - seabed habitats
 - and anchoring 351
 - and aquaculture 212–13, 215
 - fishing impact on 175
 - impact of oil and gas exploitation 285
 - and sand mining 236
 - seabed mining
 - active 232, 234, 239

- environmental impacts of 235–6, 238–40, 259–60
 - in marine reserves 42
- seabirds 13
 - and aquaculture 211
 - bycatch of 175, 190–1, 352, 365
 - and fishing methods 177–8
 - interacting with 353
 - and minerals mining 259–60, 265, 268
 - and oil and gas exploitation 285
 - protection of 23, 42, 186–7
 - and recreational boating 351
 - reporting harm to 163
 - and sand mining 239
- SeaChange - Tai Timu Tai Pari 39, 131, 141, 144
- sea cucumbers 12, 171, 213
- seafloor cutting/fragmentation 256
- seafloor massive sulphides mining 256–7, 262
 - mitigation and management strategies for **266**
 - system for **261**
- Seafloor Mining Tool 257
- seafood
 - international demand for 203
 - precision harvesting of 196
- Seafood New Zealand 173
- seagrass beds
 - in ecological backbone 142
 - excluding trawling from 193
 - impact of dredging on 307
 - and recreational boating 351
 - and sedimentation 382
- seagrasses 11, 14
- sea horses 12
- sea kayaking 350
- Sea Lion Exclusion Device 191, **192**
- sea lions 13
 - early industries based on 353
 - impact of fishing on 175, 179, 182, 191
 - and oil and gas exploitation 284–5
 - and sand mining 243
- Sealord 81, 83, 196
- seamounts 6, 9
 - fishing-restricted areas over 124
 - invertebrates on 12
 - mining near 263
- sea pens 9, 12, **63**
- SeaSketch 145
- sea slugs 12
- sea squirts 10, 95, 99, 109
- sea urchins *see* kina
- seawalls, construction of 309–10
- seaweed beds 7
- seaweeds
 - endemic 10
 - impact of fishing on 179
- sediment
 - areas trapping 309, 380
 - resuspension of 266, 307, 318
 - sources of increased 373
- sedimentation
 - environmental impact of 381, **382**
 - in NZCPS 388, 391
 - and pasture land 375–6
 - regional council responsibilities for 391
 - in Tairua Harbour 398–9
- sediment cycle 237
- sediment plume
 - and minerals mining 260–2, 265, 267
 - and sand mining 235–6, 240, 245–6
- sediment runoff 305, 387, 395
- sediment–seawater interface 8
- seine netting 128, 175, 178, 180
- seismic streamers 285
- seismic surveying
 - and Areas of Ecological Importance **290**
 - and minerals mining 255
 - in oil and gas exploitation **279**, 280, 285, 288–9
 - restrictions on 42
 - in sand mining 235, 239
 - under EEZ Act 65–6
- semi-submersibles 280, 350
- sensitive environments
 - description of **62**, 63–4
 - identifying 61
- septic tanks 379, 396
- set net prohibitions 191
- sewage
 - contamination by 316, 384, 389
 - discharge 329, 350–1, 357, 391
 - treatment 266, 357, 374, 378–80
- sharks 171, 174, 178, 180, 196, 351
- shearwaters 42, 178, 353
- shellfish 11–12
 - impact of pollution on 212
 - Māori customary resource protection for 88
 - and sedimentation 307
 - sewage contaminating 384
- shellfish farming 212–15, 221, 223
- shellfish gathering
 - illegal 163
 - methods of 177
- Shell Todd Oil Services Limited 281, 286, 288
- shipping *see also* commercial shipping; vessels
 - areas to be avoided by 127, 334
 - control within coastal marine area 332
 - New Zealand's reliance on 301
- ship strike 330
 - and cruise ships 357
 - in Hauraki Gulf 336
 - and minerals mining 259–60, 266, 268
 - and recreational boating 351
 - and recreational fishing 353
 - and sand mining 235, 239
 - and tourism 354
- shipwrecks 43, 146, 235, 246
- siltation 202, 214, 307
- Sir Peter Blake Marine Education and Recreation Centre 120
- sliding baseline problem 143
- slime layers 99
- sludge 357
- slurry pipes 256
- SNA1 quota management area 181–2
- snapper
 - habitats over lifecycle 14
 - juvenile 192, 196, 236
 - longlining for 177
 - and marine farms 214
 - and marine reserves 119, 133–4
 - quota management areas for **181**
 - recreational fishing of 352
- snorkelling 348–9, 351, 353, 361, 363
- soft sediments 8
- Solwara 1 257
- sonar depth sounders 350
- South African brown mussel 286
- southern blue whiting 179, 185
- Southern New Zealand dotterel 31
- Southern Ocean 6
- southern right whale 13, 211, 265, 268
- southern royal albatross 13
- Southern Seabird Solutions 191, 195, 352, 365
- South Taranaki Bight
 - hydrocarbons in 288
 - sand mining in 232, 234, 245–7
- spat, transportation of 100, 205, 210
- species of concern 65–6
- sperm whales 9, 353, 360–1
- Spirits Bay 8, 10, 63
- sponge gardens **63**, 142, 193
- sponge hotspots **63**
- sponges
 - and biogenic reefs 8
 - endemic 10
 - impact of fishing on 177, 179
 - on seamounts 12
- sports events 355, 361, 364
- spudding 280
- squid 171, 175, 177, 179–80, 192
- squid jigging 176–7
- Stakeholder Consultation Process Standard 186
- stakeholder working groups 145–6
- stargazer 12
- State of the Hauraki Gulf reports 144, 384
- Stewart Island
 - aquaculture at 203, 225
 - biogenic reefs off 8
 - cruise ships to 356
 - oil spills at 328
- St Mary's Bay 320–1

- stormwater
 - expenditure on infrastructure 26, 394
 - management of 392
 - in NZCPS 316, 388–9, 391
 - overloading sewage system 379–80
 - urban 376–7
 - stormwater filters 321
 - Strategy for Managing the Environmental Effects of Fishing 186
 - structural complexity 194
 - structured learning by doing 37, 54
 - Subantarctic Islands 13, 24, 102–3, 130, 356
 - subdivision 26, 29, 33, 242, 316, 388, 391
 - Submarine Cables and Pipelines Protection Act 1996 121, 127
 - subsoil, disturbance of 57, 66
 - substrate type 128
 - suction dredging 234–5, 256
 - Sugar Loaf Islands Marine Protected Area Act 1991 126
 - superphosphate 267, 376
 - support vessels
 - for oil and gas exploitation 281, 285
 - for seabed mining 239
 - Supreme Court
 - on adaptive management 54–5
 - on adverse effects 37
 - on aquaculture 218
 - on coastal plan changes 315, 317
 - on RMA 52
 - surf breaks
 - and dredging 308
 - in NZCPS 30, 34, 357
 - and oil and gas exploration 294
 - protection of 347
 - and sand mining 235, 242–3
 - surfing 345–8, 350, 355, 357, 363
 - surveillance
 - for new marine pests 107
 - under Biosecurity Act 100–1
 - sustainability rounds 185–6
 - sustainable management
 - in EEZ Act 51
 - and kaitiakitanga 80
 - policy guidelines for 72
 - in RMA 28
 - swimming
 - recreational 346
 - and recreational boating 351
- T**
- Taharoa iron sand mining site 232, 235
 - taiāpure reserves 84, 87–8
 - Akaroa harbour as 164
 - and catchment management 393
 - in Hauraki Gulf 125
 - public involvement in 158, 162
 - tailings, deposition on seabed 236, 256, 260, 265
 - Tairua Harbour and Catchment Management Plan 393, 398, **399**
 - Tairua Marina 165
 - Tāmaki Drive 355, 400
 - Tāmaki Estuary 383, 400
 - tangata tiaki/kaitiaki 87
 - tangata whenua *see also* Māori
 - fisheries rights of 83
 - and fisheries sustainability measures 157
 - and kaitiakitanga 80
 - management practices of 87–8
 - in monitoring programmes 265
 - perspective on *Rena* disaster 89
 - spiritual relationship to natural resources 38, 125, 304, 309–10, 319, 329
 - and traditional reserves 125
 - and water quality 388–9, 391
 - taonga 28, 80–1, 86, 388
 - Taranaki Basin, hydrocarbon deposits in 277
 - target fish populations 174–5
 - Tasman Bay
 - calcareous gravels at 8
 - Perna perna* in 286
 - sedimentation in 382
 - Tauranga Harbour 89–90, 99, 111
 - Tauranga Moana groups 89
 - Tawharanui Marine Park 125, 155, 348
 - Te Awanui Tauranga Harbour Iwi Management Plan 309
 - technical reviews, independent 69
 - Te Korowai o Te Tai o Marokura 131
 - Te Ohu Kai Moana 81
 - territorial authorities
 - and land-based pollutants 394–5
 - and marine management 26
 - and national environmental standards 157
 - and RMA 29
 - territorial sea
 - extent of 6
 - jurisdictional powers over 19
 - Te Tapuwae O Rongokako Marine Reserve 119
 - Te Waewae Bay, marine mammal sanctuary 42
 - Te Whanganui A Hei (Cathedral Cove) Marine Reserve 119
 - therapeutants 215–16
 - threatened species 30, 58, 70–1, 86, 268
 - Three Kings Islands 10, 334
 - tidal compartment 308, 398
 - tidal energy 302, 308
 - tikanga Māori 69, 80, 91n8
 - Timaru breakwater 308
 - titanomagnetite iron sand 231–3, 241
 - toilets
 - on beaches 346
 - onboard 329, 365
 - tolerance, assessments of 61
 - Tom Bowling Bay 10
 - Tonga Island Marine Reserve 119
 - Tori lines 191
 - total allowable catch, setting 181
 - total allowable commercial catch 83, 156–7, 163, 182–3
 - tourism
 - best practice for **363–4**
 - environmental impact of 345, **346**
 - and interaction with wildlife 353–4
 - iwi involvement in 79
 - and marine conservation 119, 142
 - special events for 355
 - visitor arrivals to NZ 343
 - tourism opportunities 116, 119, 343
 - track construction 376, 395
 - transitional zones 305
 - Trans-Tasman Resources Ltd
 - mining consent application 253
 - prospecting permit 232–3
 - Trans-Tasman Resources Ltd Marine Consent Decision
 - and adaptive management 55
 - on consent conditions 71
 - and environmental impacts 235
 - existing interests in 246–7
 - and information 54, 56
 - on RMA 72
 - and sand mining 243
 - summary of **245–6**
 - and timeframes 68
 - trapping 176–7, 180
 - trawling
 - advanced methods of 196
 - and benthic habitats 192–3
 - bycatch of 174, 191
 - environmental impact of 175, 179
 - prohibition of 128
 - Treaty of Waitangi
 - and conservation 86
 - in EEZ Act 56, 72, 81
 - fisheries under 83, 162, 171
 - in RMA 29, 81
 - tree-felling 376
 - trevally 175, 352
 - tributyltin 329
 - Trident Systems 190
 - trigger indicators or values 55, 247, 391
 - trolling 176–7, 180, 351
 - tuatua 236
 - tube worms
 - and aquaculture 210
 - and biogenic reefs 8
 - and hydrothermal vents 10
 - Tui hydrocarbon field 281
 - turtles 42, 175, 178, 180, 330
 - Tutukākā Marina 367

two barrier rule 282

U

ultrasound, and biofouling 108

UNCLOS (United Nations Convention on the Law of the Sea) 19–20, 55, 334

underwater noise

and aquaculture 211

and marine mining 239, 260

and oil and gas exploitation 284–5

and recreational boating 350

from vessels 330

undue adverse effects test 216–18

United Kingdom, marine spatial planning in 141

United Nations Commission on the Limits of the Continental Shelf 20

United Nations General Assembly Resolution on Sustainable Fisheries 194

University of Auckland, and marine reserves 122, 133–4

upwelling 14, 254

uranium 232, 268

urban areas

runoff from 375–6

sewage from 301, 322, 373, 380, 396

urban waterfront development 313

urchin barrens *see* kina barrens

urupa sites 240, 286, 400

V

vessels *see also* cruise ships; shipping

best practice for managing impact **335**

environmental effects of 327–31

numbers visiting NZ 327

pollution caused by 328, 332–3

waste from 356–7

vibrations 57, 233, 259, 261–3

viruses 384

volcanoes, underwater 6, 258

W

Waaka, Lance 90

wading birds 7

wāhi tapu

legislative framework for 43

Māori relationship to 28, 86

and marine infrastructure 309–11

mining impact on 240

oil and gas exploitation impact on 286

register of 26

and RMA 81

Waikato North Head 234

Waikato Regional Council 110, 223, 384, 398

Waikawa Bay 111

Wairarapa coast, biogenic reefs off 8

Waitangi Tribunal

and fisheries resources 83

and foreshore and seabed policy 84

Waitematā Harbour 303, 383

waste recycling 357

wastewater

conveyance of 379

urban 385

wastewater treatment plants 26, 301, 310, 378–9, 396

water, oceanic bodies of 6

water column

and finfish farming 215

impact of dredging and disposal on 306

impact of minerals mining on 259, 262, 265

nutrient flow in 97

sediment suspended in 179, 381

and shellfish farming 212–13

toxins dissolved in 383

water cycle 373, **374**

water quality

in aquaculture 219

in Hauraki region 384

and marine infrastructure 308, 315

and marine recreation 349

and minerals mining 266, 268

in NZCPS 30, 32, 389, 391

regional council responsibility for 391–2

water treatment, on cruise ships 357

water treatment systems 377–8

water turbidity 239, 259, 306, 381

Wednesbury test 166n15

weighted lines 191

West Coast North Island Marine Mammal Sanctuary 42, 288, **289**

Westhaven (Te Tai Tapu) Marine Reserve 127

Westhaven Marina 308–10, 320–1

design principles of **320**

Westpark Marina 306–7

wetlands

access to recreational areas over 363

and agriculture 375

declamation in 303

fencing 397

in freshwater NPS 387

and water cycle 373

whakapapa 80

Whakarua Moutere Islands 367

whales 13

beaked 268

entanglement for 211

regulations on contact with 361

ship strike on 354

whale watching 353, 366

Whangamatā, Harbour and Catchment Management Plans 393

Whangamatā Marina 308, 348

Whangamatā sea squirt 99

Whanganui Inlet 127

Whenua Rangitira Reserves Management Plan 400

Wildlife Act 1952 42

and Minister of Conservation 23

reserves established under 127

Wildlife Act 1953 23

wildlife management reserves 103, 127

wild species

and aquaculture 214–15, 221

commercial fishing of 171

Wilson Bay 214

wind energy 140

wind farms, offshore 305

windsurfing 347, 363

WorkSafe New Zealand 292

World Trade Organization Sanitary and Phytosanitary Agreement 105

XYZ

xenophyophore beds **62**

yachting 320–1, 350

zero discharge policy 368

zinc

and antifouling coatings 329

and dredging 307

in finfish feed 216

in urban runoff 376–7

zoning 29, 140, 266

zooplankton 9–10

Zostera capricorni 11

About the authors...



Lucy Brake is an award-winning environmental writer with a post graduate qualification in environmental science and journalism. She has an extensive background in natural resource management both within government and private consultancy in New Zealand and overseas. Her beautiful children, Tyler and Zach, have a passion for this country's very special marine environment and she dedicates this book to them.



RAEWYN PEART is Policy Director for the Environmental Defence Society. Raewyn was for some years a practising resource management lawyer before moving into the environmental policy field. For the past 15 years Raewyn has been involved in investigating and writing about a range of environmental challenges facing New Zealand including landscape protection, coastal development and marine management. Raewyn's last book focused on the challenges of protecting marine mammals and is titled *Dolphins of Aotearoa: Living with New Zealand Dolphins*. Raewyn is also a keen photographer, sailor, snorkeler and snowboarder.



New Zealand has jurisdiction over a very large, varied and productive marine environment. It is of enormous importance to New Zealanders in terms of economic, social and cultural values. With increasing interest in developing marine resources it is critical that we manage this marine area well.

Sustainable Seas demystifies a complex and overlapping legislative framework by clearly describing how decisions affecting the marine environment are made and by whom. It identifies how users of the marine environment, members of the public and others can participate constructively and effectively in these decision-making processes. Many activities can potentially cause adverse effects on the marine environment, but these can often be addressed through the application of best practice approaches. By bringing together environmental and development interests in oceans management this publication identifies a number of ways in which this can be achieved.

Sustainable Seas identifies the potential benefits of marine spatial planning and marine protection and describes their application in practice. With the implementation of such approaches, coupled with best practice applied to marine and catchment activities, New Zealanders should be able to enjoy their oceans for generations to come.

Sustainable Seas is an invaluable resource for all those who have an interest in New Zealand's marine environment including members of the fishing, aquaculture, shipping and mining sectors, tourism operators, marine developers, recreational users, conservationists, catchment and marine managers, scientists and students.

